skin friction drag and pressure drag due to flow separation. So our result in this example clearly overestimates the skin friction drag coefficient for the airfoil. But wait a minute! In actuality, the boundary layer over a body always starts out as a *laminar* boundary for some distance from the leading edge, and then transists to a turbulent boundary layer at some point downstream of the leading edge. The skin-friction drag is therefore a combination of laminar skin friction over the forward part of the airfoil, and turbulent skin friction over the remaining part. Let us examine this situation.

4.12.3 Transition

In Section 4.12.1 we assumed that the flow over a flat plate was all laminar. Similarly, in Section 4.12.2 we assumed all turbulent flow. In reality, the flow *always* starts out from the leading edge as laminar. Then at some point downstream of the leading edge, the laminar boundary layer becomes unstable and small "bursts" of turbulence begin to grow in the flow. Finally, over a certain region called the *transition region,* the boundary layer becomes completely turbulent. For purposes of analysis, we usually draw the picture shown in Figure 4.43, where a laminar boundary layer starts out from the leading edge of the flat plate and grows parabolically downstream. Then at the *transition point,* it becomes a turbulent boundary layer growing at a faster rate, on the order of $x^{4/5}$ downstream. The value of x where transition is said to take place is the *critical* value x_{cr}. In turn, x_{cr} allows the definition of a *critical Reynolds number* for transition as

$$\mathrm{Re}_{x_{cr}} = \frac{\rho_\infty V_\infty x_{cr}}{\mu_\infty} \tag{4.89}$$

Transition is discussed in more detail in Section 15.2. Volumes of literature have been written on the phenomenon of transition from laminar to turbulent flow. Obviously, because τ_w is different for the two flows—as clearly illustrated by comparing the results of Examples 4.8 and 4.9—knowledge of where on the surface transition occurs is vital to an accurate prediction of skin friction drag. The location of the transition point (in reality, a finite region) depends on many quantities as discussed in Section 15.2. However, if the critical Reynolds number is given to you (usually from experiments for a given type of flow over a given

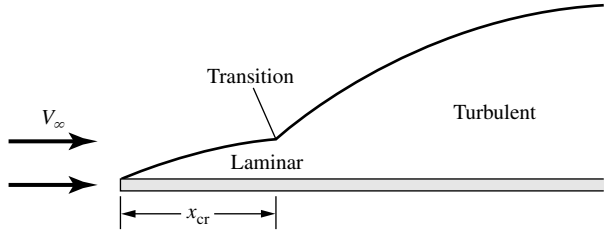


Figure 4.43 Transition from laminar to turbulent flow. The boundary layer thickness is exaggerated for clarity.

body), then the location of transition, x_{cr}, for that type of flow can be obtained directly from the definition, Equation (4.89).

For example, assume that you have an airfoil of given surface roughness (one of the factors that affect the location of transition) in a flow at a free-stream velocity of 50 m/s and you wish to predict how far from the leading edge the transition will take place. After searching through the literature for low-speed flows over such surfaces, you may find that the critical Reynolds number determined from experience is approximately $\text{Re}_{x_{cr}} = 5 \times 10^5$. Applying this "experience" to your problem, using Equation (4.89), and assuming the thermodynamic conditions correspond to standard sea level, where $\rho_\infty = 1.23\ \text{kg/m}^3$ and (from Section 1.11) $\mu_\infty = 1.789 \times 10^{-5}$ kg/(m)(s), you find

$$x_{cr} = \frac{\mu_\infty \text{Re}_{x_{cr}}}{\rho_\infty V_\infty} = \frac{(1.789 \times 10^{-5})(5 \times 10^5)}{(1.23)(50)} = 0.145 \text{ m}$$

Note that the region of laminar flow in this example extends from the leading edge to 14.5 cm downstream from the leading edge. If now you double the freestream velocity to 100 m/s, the transition point is still governed by the critical Reynolds number, $\text{Re}_{x_{cr}} = 5 \times 10^5$. Thus,

$$x_{cr} = \frac{(1.789 \times 10^{-5})(5 \times 10^5)}{(1.23)(100)} = 0.0727 \text{ m}$$

Hence, when the velocity is doubled, the transition point moves forward one-half the distance to the leading edge.

In summary, once you know the critical Reynolds number, you can find x_{cr} from Equation (4.89). However, an accurate value for $\text{Re}_{x_{cr}}$ applicable to your problem must come from somewhere—experiment, free flight, or some semi-empirical theory—and this may be difficult to obtain. This situation provides a little insight into why basic studies of transition and turbulence are needed to advance our understanding of such flows and to allow us to apply more valid reasoning to the prediction of transition in practical problems.

EXAMPLE 4.10

For the NACA 2412 airfoil and the conditions in Example 4.7, calculate the net skin friction drag coefficient assuming that the critical Reynolds number is 500,000.

■ Solution

Consider Figure 4.44, which shows a flat plate with a laminar boundary layer extending from the leading edge over the distance x_1 to the transition point (region 1), and a turbulent boundary extending over the distance x_2 from the transition point to the trailing edge (region 2). The critical Reynolds number is

$$\text{Re}_{x_{cr}} = \frac{\rho_\infty V_\infty x_{cr}}{\mu_\infty} = 5 \times 10^5$$

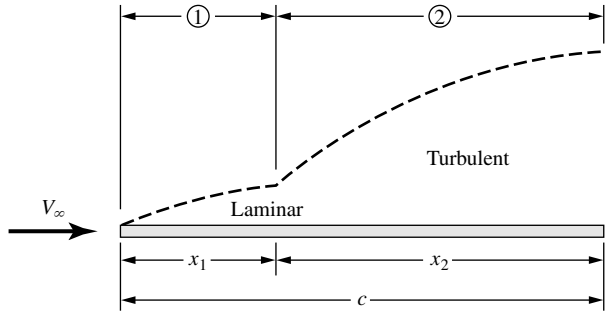


Figure 4.44 Laminar (region 1) and turbulent (region 2) flow over a flat plate.

where in Figure 4.44, $x_{cr} = x_1$. Hence

$$\text{Re}_{x_{cr}} = \frac{\rho_\infty V_\infty x_1}{\mu_\infty} = 5 \times 10^5$$

The Reynolds number based on chord length is given as

$$\text{Re}_c = \frac{\rho_\infty V_\infty c}{\mu_\infty} = 3.1 \times 10^6$$

Thus

$$\frac{\text{Re}_{x_{cr}}}{\text{Re}_c} = \frac{5 \times 10^5}{3.1 \times 10^6} = 0.1613 = \frac{(\rho_\infty V_\infty x_1/\mu_\infty)}{(\rho_\infty V_\infty c/\mu_\infty)} = \frac{x_1}{c}$$

This locates the transition point relative to the chord length, that is, in Figure 4.44, we have

$$\frac{x_1}{c} = 0.1613$$

Because the Reynolds number in the equations for skin friction drag coefficient is always based on length measured *from the leading edge,* we can not simply calculate the turbulent skin friction drag coefficient for region 2 by using Equation (4.88) with a Reynolds number based on x_2. Rather, we must carry out the following procedure.

Assuming all turbulent flow over the entire length of the plate, the drag (on one side of the plate) is $(D_{f,c})_{\text{turbulent}}$, where

$$(D_{f,c})_{\text{turbulent}} = q_\infty S (C_{f,c})_{\text{turbulent}}$$

As usual, we are dealing with the drag per unit span, hence $S = c(1)$.

$$(D_{f,c})_{\text{turbulent}} = q_\infty c (C_{f,c})_{\text{turbulent}}$$

The turbulent drag on just region 1 is $(D_{f,1})_{\text{turbulent}}$:

$$(D_{f,1})_{\text{turbulent}} = q_\infty S (C_{f,1})_{\text{turbulent}}$$

Here, $S = (x_1)(1)$:

$$(D_{f,1})_{\text{turbulent}} = q_\infty x_1 (C_{f,1})_{\text{turbulent}}$$

Thus, the turbulent drag just on region 2, $(D_{f,2})_{\text{turbulent}}$, is

$$(D_{f,2})_{\text{turbulent}} = (D_{f,c})_{\text{turbulent}} - (D_{f,1})_{\text{turbulent}}$$
$$(D_{f,2})_{\text{turbulent}} = q_\infty c(C_{f,c})_{\text{turbulent}} - q_\infty x_1(C_{f,1})_{\text{turbulent}}$$

The laminar drag on region 1 is $(D_{f,1})_{\text{laminar}}$

$$(D_{f,1})_{\text{laminar}} = q_\infty S(C_{f,1})_{\text{laminar}} = q_\infty x_1(C_{f,1})_{\text{laminar}}$$

The total skin-friction drag on the plate, D_f, is then

$$D_f = (D_{f,1})_{\text{laminar}} + (D_{f,2})_{\text{turbulent}}$$

or,
$$D_f = q_\infty x_1(C_{f,1})_{\text{laminar}} + q_\infty c(C_{f,c})_{\text{turbulent}} - q_\infty x_1(C_{f,1})_{\text{turbulent}} \tag{4.90}$$

The total skin-friction drag coefficient is

$$C_f = \frac{D_f}{q_\infty S} = \frac{D_f}{q_\infty c} \tag{4.91}$$

Combining Equations (4.90) and (4.91):

$$C_f = \frac{x_1}{c}(C_{f,1})_{\text{laminar}} + (C_{f,c})_{\text{turbulent}} - \frac{x_1}{c}(C_{f,1})_{\text{turbulent}} \tag{4.92}$$

Since $x_1/c = 0.1613$, Equation (4.92) becomes

$$C_f = 0.1613(C_{f,1})_{\text{laminar}} + (C_{f,c})_{\text{turbulent}} - 0.1613(C_{f,1})_{\text{turbulent}} \tag{4.93}$$

The various skin friction drag coefficients in Equation (4.93) are obtained as follows. The Reynolds number for region 1 is

$$\text{Re}_{x_1} = \frac{\rho_\infty V_\infty x_1}{\mu_\infty} = \frac{x_1}{c}\left(\frac{\rho_\infty V_\infty c}{\mu_\infty}\right) = \frac{x_1}{c}\text{Re}_c = 0.1613(3.1 \times 10^6) = 5 \times 10^5$$

(Of course, we could have written this down directly because $x = x_1$ is the transition point, *determined* from the critical Reynolds number that is given as 5×10^5.) From Equation (4.86) for laminar flow, with the Reynolds number based on x_1, we have

$$(C_{f,1})_{\text{laminar}} = \frac{1.328}{\sqrt{\text{Re}_{x_1}}} = \frac{1.328}{\sqrt{5 \times 10^5}} = 0.00188$$

The value of $(C_{f,c})_{\text{turbulent}}$ has already been calculated in Example 4.8, namely,

$$(C_{f,c})_{\text{turbulent}} = 0.00372 \quad \text{(for one side)}$$

From Equation (4.88) with the Reynolds number based on x_1,

$$(C_{f,1})_{\text{turbulent}} = \frac{0.074}{\text{Re}_{x_1}^{1/5}} = \frac{0.074}{(5 \times 10^5)^{0.2}} = 0.00536$$

Inserting these values into Equation (4.93), we have

$$C_f = 0.1613(0.00188) + 0.00372 - 0.1613(0.00536) = 0.003158$$

Taking into account both sides of the flat plate,

$$\text{Net } C_f = 2(0.003158) = \boxed{0.0063}$$

From the data in Figure 4.11, the measured airfoil drag coefficient is 0.0068, which includes *both* skin friction drag and pressure drag due to flow separation. The result from Example 4.10, therefore, is qualitatively reasonable, giving a skin friction drag coefficient slightly less than the measure total drag coefficient. However, our calculated result of $C_f = 0.0063$ is for a critical Reynolds number of 500,000 for transition from laminar to turbulent flow. We do not know what the critical Reynolds number is for the experiments on which the data in Figure 4.11 are based. In Example 4.10, the assumption of $\text{Re}_{x_{cr}} = 500{,}000$ is very conservative; more likely the actual value is closer to 1,000,000. If we assume this higher value of $\text{Re}_{x_{cr}}$, what does it do to the calculated result for C_f? Let us take a look.

EXAMPLE 4.11

Repeat Example 4.10, but assuming the critical Reynolds number is 1×10^6.

■ Solution

$$\frac{x_1}{c} = \frac{1 \times 10^6}{3.1 \times 10^6} = 0.3226$$

Which, as we could write down immediately, is twice the length from Example 4.10 because the critical Reynolds number is twice as large. Equation (4.93) becomes

$$C_f = 0.3226(C_{f,1})_{\text{laminar}} + (C_{f,c})_{\text{turbulent}} - 0.3226(C_{f,1})_{\text{turbulent}} \tag{4.94}$$

For region 1, we have

$$(C_{f,1})_{\text{laminar}} = \frac{1.328}{\sqrt{\text{Re}_{x_1}}} = \frac{1.328}{\sqrt{1 \times 10^6}} = 0.001328$$

The value of $(C_{f,c})_{\text{turbulent}}$ is the same as before:

$$(C_{f,c})_{\text{turbulent}} = 0.00372$$

Once again, for region 1 assuming turbulent flow, we have

$$(C_{f,1})_{\text{turbulent}} = \frac{0.074}{(\text{Re}_{x_1})^{1/5}} = \frac{0.074}{(1 \times 10^6)^{1/5}} = 0.004669$$

Substituting the above results in Equation (4.94), we have

$$C_f = 0.3226(0.001328) + 0.00372 - 0.3226(0.004669) = 0.002642$$

Since this result is for one side of the plate, the net skin friction drag coefficient is

$$\text{Net } C_f = 2(0.002642) = \boxed{0.00528}$$

Note: Comparing the results from Examples 4.10 and 4.11, we see that an increase in $\text{Re}_{x_{cr}}$ from 500,000 to 1,000,000 resulted in a skin friction drag coefficient that is 16 percent smaller. This difference underscores the importance of knowing where transition takes place on a surface for the calculation of skin friction drag.

Also, comparing the calculated results for skin friction drag coefficient with the measured total drag coefficient of 0.0068, from Example 4.10 the calculated $C_f = 0.0063$ would imply that the pressure drag due to flow separation is about 7.4 percent of the total drag. The result from Example 4.11 of $C_f = 0.00528$ would imply that the pressure drag due to flow separation is about 22 percent of the total drag.

Is this breakdown between skin friction and pressure drag quantitatively reasonable? An answer can be found in the recent results of Lombardi *et al.* given in Reference 92. Here, the authors calculated both the skin friction drag coefficient and the total drag coefficient for an NACA 0012 airfoil using an accurate computational fluid dynamic technique. More details of their calculations are given in Section 20.4. For a Reynolds number based on chord length of 3×10^6, and including a model for transition, they calculated a total drag coefficient of 0.00623 and a skin friction drag coefficient of 0.00534, indicating that the pressure drag due to flow separation is 15 percent of the total drag. For a streamlined body, this drag breakdown is reasonable; the drag on a streamlined two-dimensional shape is mostly skin friction drag, and by comparison the pressure drag is small. For example, it is reasonable to expect 80 percent of the drag to be skin friction drag and 20 percent to be pressure drag due to flow separation.

This is not to say that pressure drag due to flow separation is unimportant; quite the contrary, as the body becomes less streamlined (more like a blunt body), the pressure drag becomes the dominant factor. We need to take a closer look at this phenomenon.

4.12.4 Flow Separation

Pressure drag on an airfoil is caused by the flow separation. For a completely attached flow over an airfoil, the pressure acting on the rear surface gives rise to a force in the forward direction which completely counteracts the pressure acting on the front surface producing a force in the rearward direction, resulting in zero pressure drag. However, if the flow is partially separated over the rear surface, the pressure on the rear surface pushing forward will be smaller than the fully attached case, and the pressure acting on the front surface pushing backwards will not be fully counteracted, giving rise to a net pressure drag on the airfoil—the pressure drag due to flow separation.

What flow conditions are conducive to flow separation? To help answer this question, consider the flow over the NASA LS(1)-0417 airfoil at zero angle of attack, as shown in Figure 4.45. The streamlines move smoothly over the airfoil—there is no flow separation of any consequence. A computational fluid dynamic solution of the variation of pressure coefficient over the upper surface of the airfoil is shown at the bottom of Figure 4.45. Starting at the stagnation point at the leading edge, where for incompressible flow $C_p = 1.0$, the flow rapidly expands around the top surface. The pressure decreases dramatically, dipping to a minimum pressure at a location about 10 percent of the chord length downstream of the leading edge. Then as the flow moves farther downstream, the pressure

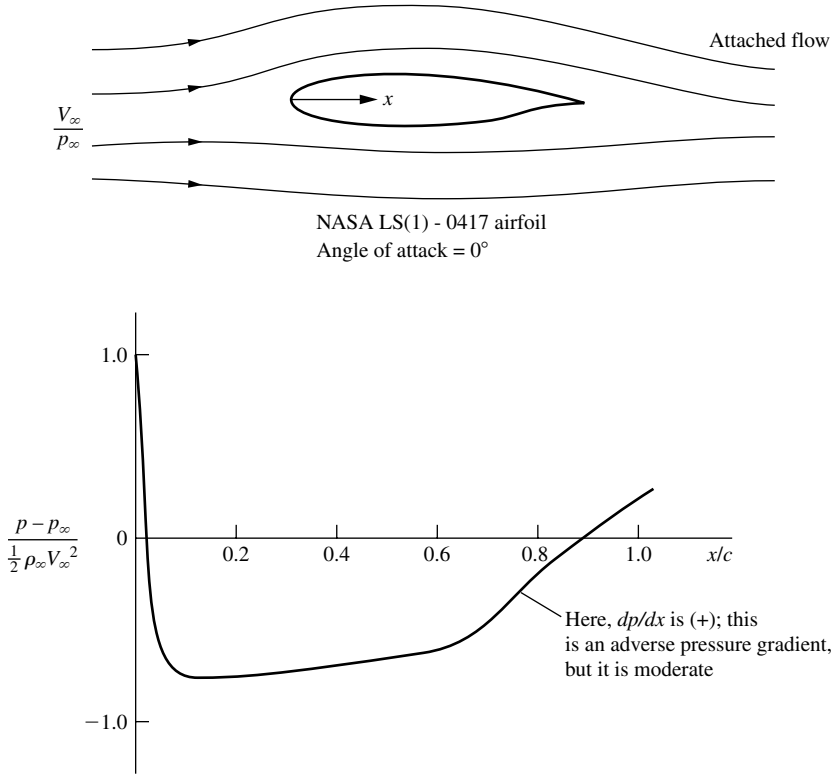
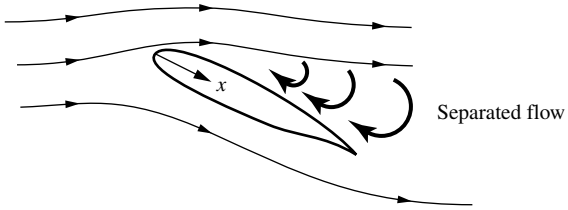


Figure 4.45 Pressure distribution over the top surface for attached flow over an airfoil. Theoretical data for a modern NASA low-speed airfoil, from NASA Conference Publication 2046, *Advanced Technology Airfoil Research,* vol. II, March 1978, p. 11 (*Source:* After McGhee, Beasley, and Whitcomb).

gradually increases, reaching a value slightly above free-stream pressure at the trailing edge. This region of increasing pressure is called a region of *adverse* pressure gradient. By definition, an adverse pressure gradient is a region where the pressure *increases in the flow direction,* that is, in Figure 4.45, the region where dp/dx is *positive*. For the conditions shown in Figure 4.45, the adverse pressure gradient is moderate; that is, dp/dx is small, and for all practical purposes the flow remains attached to the airfoil surface except for a small region near the trailing edge (not shown in Figure 4.45).

Now consider the same airfoil at the very high angle of attack of 18.4 degrees, as shown in Figure 4.46. First, assume we have a purely inviscid flow with no flow separation—a purely artificial situation. A numerical solution for the inviscid flow gives the results shown by the dashed curve in Figure 4.46. In this artificial situation, the pressure would drop precipitously downstream of the leading edge to a value of C_p almost -9, and then rapidly increase downstream, recovering



NASA LS(1) - 0417 airfoil
Angle of attack = 18.4°

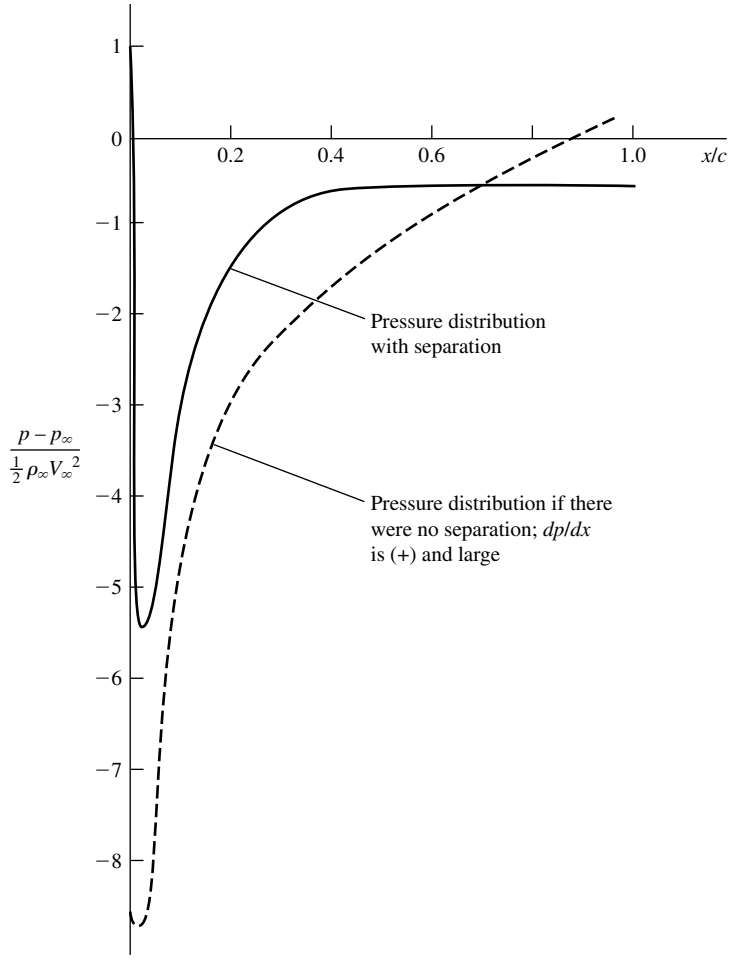


Figure 4.46 Pressure distribution over the top surface for separated flow over an airfoil. Theoretical data for a modern NASA low-speed airfoil, from NASA Conference Publication 2045, Part 1, *Advanced Technology Airfoil Research,* vol. 1, March 1978, p. 380. (*Source:* After Zumwalt and Nack).

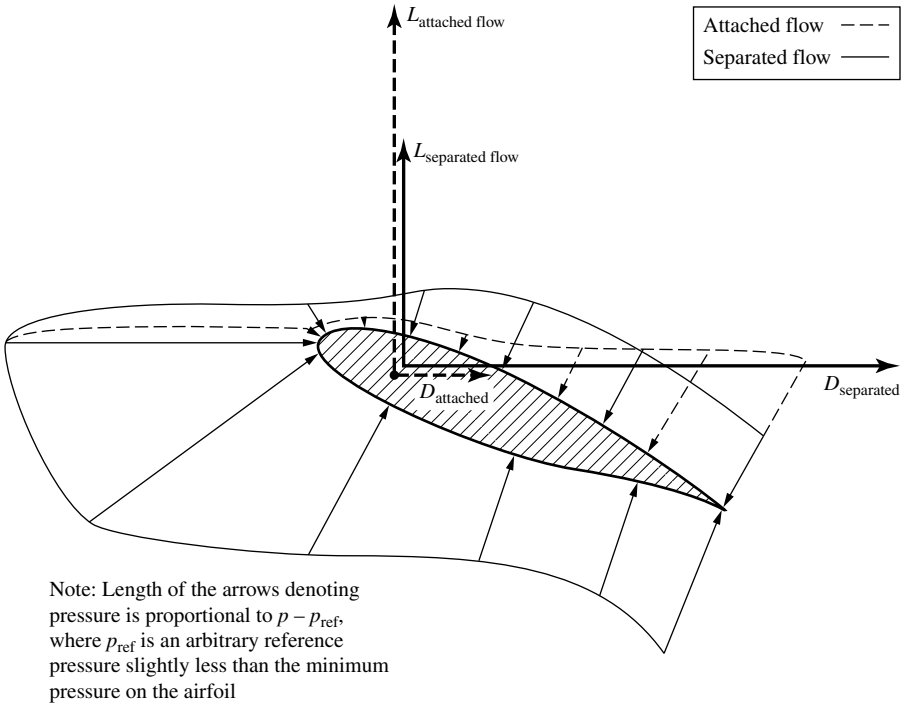


Figure 4.47 Qualitative comparison of pressure distribution, lift, and drag for attached and separated flows. Note that for separated flow, the lift decreases and the drag increases.

to a value slightly above p_∞ at the trailing edge. In this recovery, the pressure would increase rapidly, in contrast to the case shown in Figure 4.45. The adverse pressure gradient would be severe; that is, dp/dx would be large. In such a case, the *real* viscous flow tends to separate from the surface. In this real separated flow, the *actual* surface pressure distribution is given in Figure 4.46 by the *solid* curve, obtained from a computational fluid dynamic viscous flow calculation using the complete Navier-Stokes equations (see Chapter 15). In comparison to the dashed curve, the actual pressure distribution does not dip to as low a pressure minimum, and the pressure near the trailing edge does not recover to a value above p_∞.

It is important to visualize and compare the pressures acting on the surface of the airfoil for the case shown in Figure 4.46, this comparison is sketched in Figure 4.47. Here the airfoil at a large angle of attack (thus with flow separation) is shown with the real surface pressure distribution symbolized by the solid arrows. Pressure always acts normal to the surface. Hence the arrows are all locally perpendicular to the surface. The length of the arrows is representative of the magnitude of the pressure. A solid curve is drawn through the base of the arrows to form an "envelope" to make the pressure distribution easier to visualize. However, if the flow were *not* separated, that is, if the flow were attached, then the pressure distribution would be that shown by the dashed arrows (and the dashed envelope).

The solid and dashed arrows in Figure 4.47 qualitatively correspond to the solid and dashed pressure distribution curves, respectively, in Figure 4.46.

The solid and dashed arrows in Figure 4.47 should be compared carefully. They explain the two major consequences of separated flow over the airfoil. The first consequence is a loss of lift. The aerodynamic lift (the vertical force shown in Figure 4.47) is derived from the net component of the pressure distribution in the vertical direction in Figure 4.47 (assuming that the free stream relative wind is horizontal in this figure). High lift is obtained when the pressure on the bottom surface is large and the pressure on the top surface is small. Separation does not affect the bottom surface pressure distribution. However, comparing the solid and dashed arrows on the top surface *just downstream of the leading edge,* we find the solid arrows indicating a higher pressure when the flow is separated. This higher pressure is pushing down, hence reducing the lift. This reduction in lift is also compounded by the geometric effect that the position of the top surface of the airfoil near the leading edge is approximately horizontal in Figure 4.47. When the flow is separated, causing a higher pressure on this part of the airfoil surface, the direction in which the pressure is acting is closely aligned to the vertical, and hence, almost the full effect of the increased pressure is felt by the lift. The combined effect of the increased pressure on the top surface near the leading edge, and the fact that this portion of the surface is approximately horizontal, leads to the rather dramatic loss of lift when the flow separates. Note in Figure 4.47 that the lift for separated flow (the solid vertical vector) is smaller than the lift that would exist if the flow were attached (the dashed vertical vector).

Now let us concentrate on that portion of the top surface *near the trailing edge*. On this portion of the airfoil surface, the pressure for the separated flow is now *smaller* than the pressure that would exist if the flow were attached. Moreover, the top surface near the trailing edge geometrically is inclined more to the horizontal, and, in fact, somewhat faces in the horizontal direction. Recall that the drag is in the horizontal direction in Figure 4.47. Because of the inclination of the top surface near the trailing edge, the pressure exerted on this portion of the surface has a strong component in the horizontal direction. This component acts towards the left, tending to counter the horizontal component of force due to the high pressure acting on the nose of the airfoil pushing toward the right. The net pressure drag on the airfoil is the difference between the force exerted on the front pushing toward the right and the force exerted on the back pushing toward the left. When the flow is separated, the pressure on the back is lower than it would be if the flow were attached. Hence, for the separated flow, there is *less* force on the back pushing toward the left, and the *net* drag acting toward the right is therefore *increased*. Note in Figure 4.47 that the drag for separated flow (the solid horizontal vector) is larger than the drag that would exist if the flow were attached (the dashed horizontal vector).

Therefore, two major consequences of the flow separating over an airfoil are:

1. A drastic loss of lift (stalling).

2. A major increase in drag, caused by pressure drag due to flow separation.

Why does a flow separate from a surface? The answer is addressed in detail in Section 15.2. In brief, in a region of adverse pressure gradient the fluid elements moving along a streamline have to work their way "uphill" against an increasing pressure. Consequently, the fluid elements will slow down under the influence of an adverse pressure gradient. For the fluid elements moving outside the boundary layer, where the velocity (and hence kinetic energy) is high, this is not much of a problem. The fluid elements keep moving downstream. However, consider a fluid element deep inside the boundary layer. Its velocity is already small because it is retarded by friction forces. The fluid element still encounters the same adverse pressure gradient because the pressure is transmitted without change normal to the wall, but its velocity is too low to negotiate the increasing pressure. As a result, the element comes to a stop somewhere downstream and reverses its direction. Such reversed flow causes the flow field in general to separate from the surface, as shown at the top of Figure 4.46. This is physically how separated flow develops.

4.12.5 Comment

In this section we estimated skin friction drag on an airfoil by using the model of a flat plate at zero angle of attack, and calculating the skin friction drag for the airfoil using the formulas for the flat plate such as Equations (4.86) for laminar flow and Equation (4.88) for turbulent flow.[5] How reasonable is this? How close does the flat plate skin friction drag on a flat plate come to that on an airfoil? How close does the local shear stress distribution over the surface of the flat plate resemble that on the airfoil surface?

Some answers can be obtained by comparing the relatively exact computational fluid dynamic calculations of Lombardi et al. (Reference 92) for the viscous flow over an NACA 0012 airfoil at zero angle of attack with that for a flat plate. The variation of the local skin friction coefficient, defined as $c_f = \tau_w/q_\infty$, as a function of distance from the leading edge is given in Figure 4.48 for both the airfoil and the flat plate. They are remarkably close; clearly, for the purpose of the present section the modeling of the airfoil skin friction drag by use of flat plate results is reasonable.

With this, we end our discussion of airfoil drag at low speeds. Although the main thrust of this chapter is the low-speed inviscid potential flow over airfoils with the consequent prediction of lift, the present section provides some balance by exploring the effects of viscous flow on airfoil behavior, and the consequent production of drag. More aspects of the real flow over airfoils are given in Section 4.13.

[5] In 1921, Walter Diehl's article in NACA TR111, entitled "The Variation of Aerofoil Lift and Drag Coefficients with Changes in Size and Speed," suggested that the airfoil drag coefficient varies in the same manner as that for a flat plate at zero angle of attack. He did not say that the airfoil drag coefficient equals that for a flat plate, but rather has the same Reynolds number variation. Diehl's suggestion, however, appears to be the first effort to use flat plate data in some fashion to estimate airfoil drag.

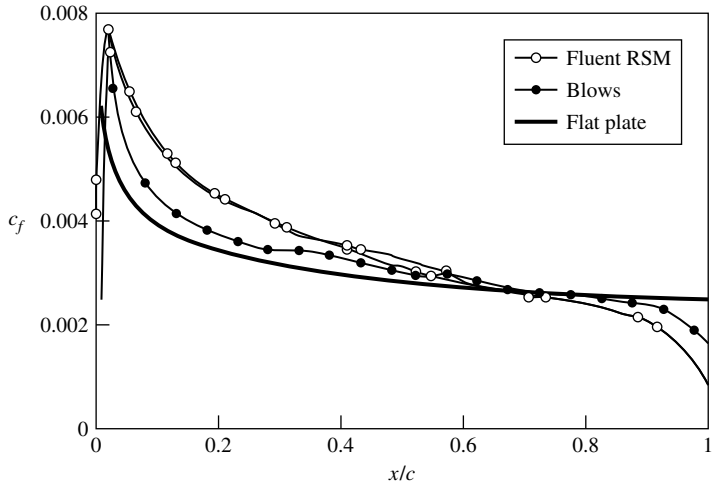


Figure 4.48 Local skin-friction coefficient distributions over an NACA 0012 airfoil, compared with that for a flat plate.

4.13 APPLIED AERODYNAMICS: THE FLOW OVER AN AIRFOIL—THE REAL CASE

In this chapter, we have studied the inviscid, incompressible flow over airfoils. When compared with actual experimental lift and moment data for airfoils in low-speed flows, we have seen that our theoretical results based on the assumption of inviscid flow are quite good—with one glaring exception. In the real case, flow separation occurs over the top surface of the airfoil when the angle of attack exceeds a certain value—the "stalling" angle of attack. As described in Sections 4.3 and 4.12, this is a viscous effect. As shown in Figure 4.9, the lift coefficient reaches a local maximum denoted by $c_{l,\max}$, and the angle of attack at which $c_{l,\max}$ is achieved is the stalling angle of attack. An increase in α beyond this value usually results in a (sometimes rather precipitous) drop in lift. At angles of attack well below the stalling angle, the experimental data clearly show a *linear* increase in c_l with increasing α—a result that is predicted by the theory presented in this chapter. Indeed, in this linear region, the inviscid flow theory is in excellent agreement with the experiment, as reflected in Figure 4.10 and as demonstrated by Example 4.6. However, the inviscid theory does not predict flow separation, and consequently the prediction of $c_{l,\max}$ and the stalling angle of attack must be treated in some fashion by viscous flow theory. Such viscous flow analyses are the purview of Part 4. On the other hand, the purpose of this section is to examine the *physical* features of the real flow over an airfoil, and flow separation is an inherent part of this real flow. Therefore, let us take a more detailed look at how the flow field over an airfoil changes as the angle of attack is increased, and how the lift coefficient is affected by such changes.

The flow fields over an NACA 4412 airfoil at different angles of attack are shown in Figure 4.49. Here, the streamlines are drawn to scale as obtained from

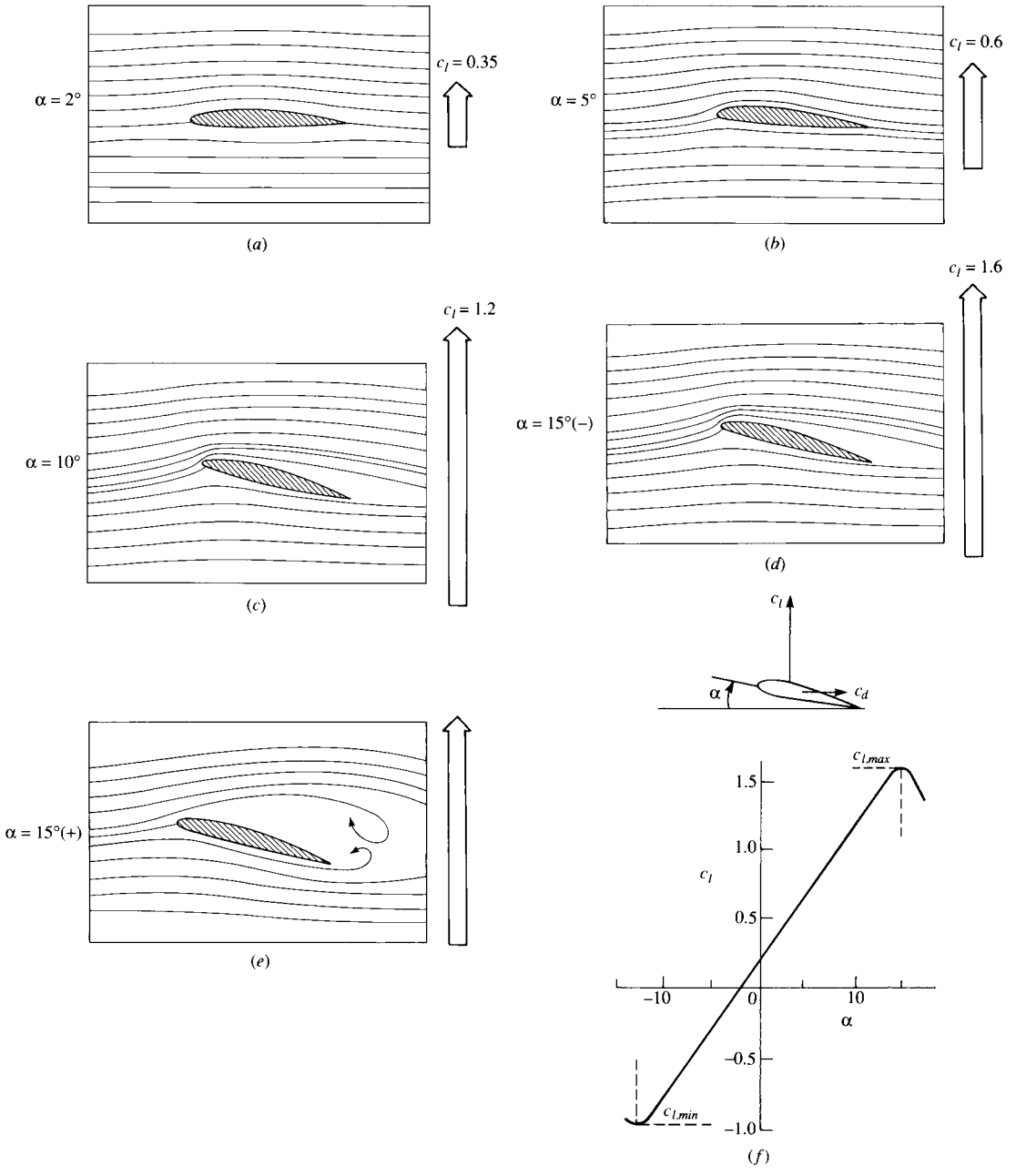


Figure 4.49 Example of leading-edge stall. Streamline patterns for an NACA 4412 airfoil at different angles of attack. *(The streamlines are drawn to scale from experimental data given by Hikaru Ito in Reference 50.)* $\mathrm{Re} = 2.1 \times 10^5$ and $V_\infty = 8$ m/s in air. The corresponding experimentally measured lift coefficients are indicated by arrows at the right of each streamline picture, where the length of each arrow indicates the relative magnitude of the lift. The lift coefficient is also shown in part (*f*).

the experimental results of Hikaru Ito given in Reference 50. The experimental streamline patterns were made visible by a smoke wire technique, wherein metallic wires spread with oil over their surfaces were heated by an electric pulse and the resulting white smoke creates visible streaklines in the flow field. In Figure 4.49, the angle of attack is progressively increased as we scan from Figure 4.49*a* to *e*; to the right of each streamline picture is an arrow, the length of which is proportional to the value of the lift coefficient at the given angle of attack. The actual experimentally measured lift curve for the airfoil is given in Figure 4.49*f*. Note that at low angle of attack, such as $\alpha = 2^\circ$ in Figure 4.49*a*, the streamlines are relatively undisturbed from their freestream shapes and c_l is small. As α is increased to 5°, as shown in Figure 4.49*b*, and then to 10°, as shown in Figure 4.49*c*, the streamlines exhibit a pronounced upward deflection in the region of the leading edge, and a subsequent downward deflection in the region of the trailing edge. Note that the stagnation point progressively moves downstream of the leading edge over the bottom surface of the airfoil as α is increased. Of course, c_l increases as α is increased, and, in this region, the increase is linear, as seen in Figure 4.49*f*. When α is increased to slightly less than 15°, as shown in Figure 4.49*d*, the curvature of the streamlines is particularly apparent. In Figure 4.49*d*, the flow field is still attached over the top surface of the airfoil. However, as α is further increased slightly above 15°, massive flow-field separation occurs over the top surface, as shown in Figure 4.49*e*. By slightly increasing α from that shown in Figure 4.49*d* to that in Figure 4.49*e*, the flow quite suddenly separates from the leading edge and the lift coefficient experiences a precipitous decrease, as seen in Figure 4.49*f*.

The type of stalling phenomenon shown in Figure 4.49 is called *leading-edge stall;* it is characteristic of relatively thin airfoils with thickness ratios between 10 and 16 percent of the chord length. As seen above, flow separation takes place rather suddenly and abruptly over the entire top surface of the airfoil, with the origin of this separation occurring at the leading edge. Note that the lift curve shown in Figure 4.49*f* is rather sharp-peaked in the vicinity of $c_{l,\max}$ with a rapid decrease in c_l above the stall.

A second category of stall is the *trailing-edge stall*. This behavior is characteristic of thicker airfoils such as the NACA 4421 shown in Figure 4.50. Here, we see a progressive and gradual movement of separation from the trailing edge toward the leading edge as α is increased. The lift curve for this case is shown in Figure 4.51. The solid curve in Figure 4.51 is a repeat of the results for the NACA 4412 airfoil shown earlier in Figure 4.49*f*—an airfoil with a leading-edge stall. The dot-dashed curve is the lift curve for the NACA 4421 airfoil—an airfoil with a trailing-edge stall. In comparing these two curves, note that:

1. The trailing-edge stall yields a gradual bending-over of the lift curve at maximum lift, in contrast to the sharp, precipitous drop in c_l for the leading-edge stall. The stall is "soft" for the trailing-edge stall.
2. The value of $c_{l,\max}$ is not so large for the trailing-edge stall.
3. For both the NACA 4412 and 4421 airfoils, the shape of the mean camber line is the same. From the thin airfoil theory discussed in this chapter, the

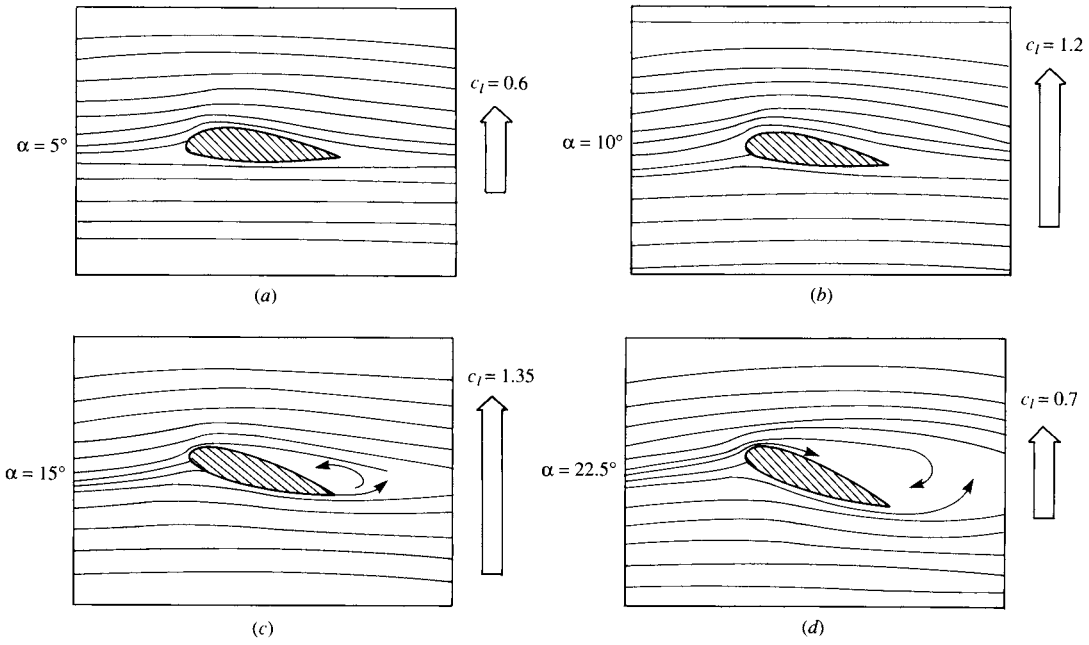


Figure 4.50 Example of trailing-edge stall. Streamline patterns for an NACA 4421 airfoil at different angles of attack. *(The streamlines are drawn to scale from the experimental results of Hikaro Ito in Reference 50.)* $\text{Re} = 2.1 \times 10^5$ and $V_\infty = 8$ m/s in air.

linear lift slope and the zero-lift angle of attack should be the same for both airfoils; this is confirmed by the experimental data in Figure 4.51. The only difference between the two airfoils is that one is thicker than the other. Hence, comparing results shown in Figures 4.49 to 4.51, we conclude that the major effect of thickness of the airfoil is its effect on the value of $c_{l,\max}$, and this effect is mirrored by the leading-edge stall behavior of the thinner airfoil versus the trailing-edge stall behavior of the thicker airfoil.

There is a third type of stall behavior, namely, behavior associated with the extreme thinness of an airfoil. This is sometimes labeled as "thin airfoil stall." An extreme example of a very thin airfoil is a flat plate; the lift curve for a flat plate is shown as the dashed curve in Figure 4.51 labeled "thin airfoil stall." The streamline patterns for the flow over a flat plate at various angles of attack are given in Figure 4.52. The thickness of the flat plate is 2 percent of the chord length. Inviscid, incompressible flow theory shows that the velocity becomes infinitely large at a sharp convex corner; the leading edge of a flat plate at an angle of attack is such a case. In the real flow over the plate as shown in Figure 4.52, nature addresses this singular behavior by having the flow separate at the leading edge, even for very low values of α. Examining Figure 4.52*a*, where $\alpha = 3°$, we observe a small region of separated flow at the leading edge. This separated

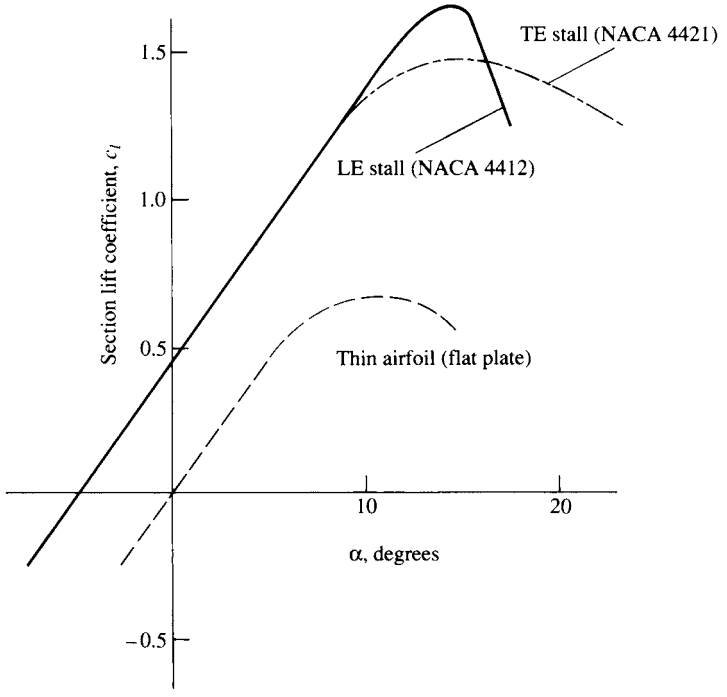


Figure 4.51 Lift-coefficient curves for three airfoils with different aerodynamic behavior: trailing-edge stall (NACA 4421 airfoil), leading-edge stall (NACA 4412 airfoil), thin airfoil stall (flat plate).

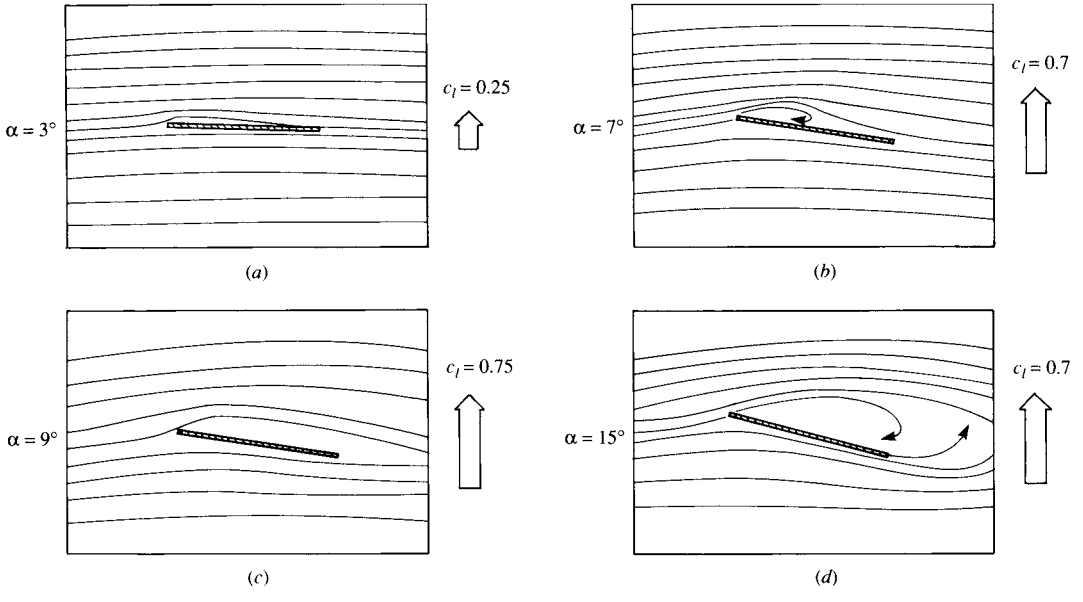


Figure 4.52 Example of thin airfoil stall. Streamline patterns for a flat plate at angle of attack. *(The streamlines are drawn to scale from the experimental data of Hikaru Ito in Reference 50.)*

flow reattaches to the surface further downstream, forming a *separation bubble* in the region near the leading edge. As α is increased, the reattachment point moves further downstream; that is, the separation bubble becomes larger. This is illustrated in Figure 4.52*b* where $\alpha = 7^\circ$. At $\alpha = 9^\circ$ (Figure 4.52*c*), the separation bubble extends over almost the complete flat plate. Referring back to Figure 4.51, we note that this angle of attack corresponds to $c_{l,\max}$ for the flat plate. When α is increased further, total flow separation is present, such as shown in Figure 4.52*d*. The lift curve for the flat plate in Figure 4.51 shows an early departure from its linear variation at about $\alpha = 3^\circ$; this corresponds to the formation of the leading-edge separation bubble. The lift curve gradually bends over as α is increased further and exhibits a very gradual and "soft" stall. This is a trend similar to the case of the trailing-edge stall, although the physical aspects of the flow are quite different between the two cases. Of particular importance is the fact that $c_{l,\max}$ for the flat plate is considerably smaller than that for the two NACA airfoils compared in Figure 4.51. Hence, we can conclude from Figure 4.51 that the value of $c_{l,\max}$ is critically dependent on airfoil thickness. In particular, by comparing the flat plate with the two NACA airfoils, we see that *some* thickness is vital to obtaining a high value of $c_{l,\max}$. However, beyond that, the amount of thickness will influence the type of stall (leading-edge versus trailing-edge), and airfoils that are very thick tend to exhibit reduced values of $c_{l,\max}$ as the thickness increases. Hence, if we plot $c_{l,\max}$ versus thickness ratio, we expect to see a local maximum. Such is indeed the case, as shown in Figure 4.53. Here, experimental data for $c_{l,\max}$ for the NACA 63-2XX series of airfoils is shown as a function of the thickness ratio. Note that as the thickness ratio increases from a small value, $c_{l,\max}$ first increases, reaches a maximum value at a thickness ratio of about 12 percent, and then decreases

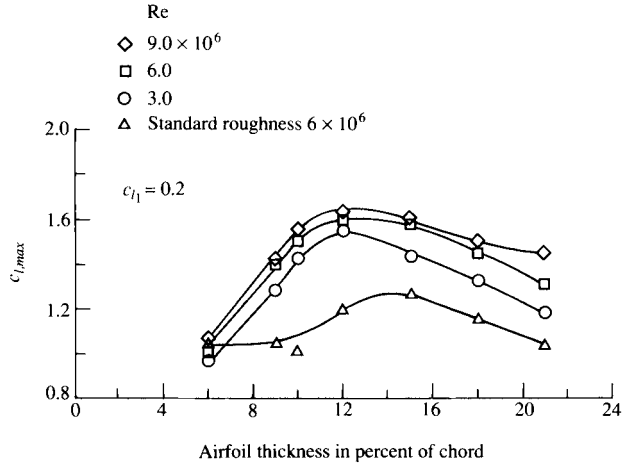


Figure 4.53 Effect of airfoil thickness on maximum lift coefficient for the NACA 63-2XX series of airfoils. *(Source: Abbott and von Doenhoff, Reference 11.)*

at larger thickness ratios. The experimental data in Figure 4.53 is plotted with the Reynolds number as a parameter. Note that $c_{l,\max}$ for a given airfoil is clearly a function of Re, with higher values of $c_{l,\max}$ corresponding to higher Reynolds numbers. Since flow separation is responsible for the lift coefficient exhibiting a local maximum, since flow separation is a viscous phenomenon, and since a viscous phenomenon is governed by a Reynolds number, it is no surprise that $c_{l,\max}$ exhibits some sensitivity to Re.

When was the significance of airfoil thickness first understood and appreciated? This question is addressed in the historical note in Section 4.14, where we will see that the aerodynamic properties of thick airfoils even transcended technology during World War I and impacted the politics of the armistice.

Let us examine some other aspects of airfoil aerodynamics—aspects that are not always appreciated in a first study of the subject. The simple generation of lift by an airfoil is not the prime consideration in its design—even a barn door at an angle of attack produces lift. Rather, there are two figures of merit that are primarily used to judge the quality of a given airfoil:

1. *The lift-to-drag ratio* L/D. An efficient airfoil produces lift with a minimum of drag; that is, the ratio of lift-to-drag is a measure of the aerodynamic efficiency of an airfoil. The standard airfoils discussed in this chapter have high L/D ratios—much higher than that of a barn door. The L/D ratio for a complete flight vehicle has an important impact on its flight performance; for example, the range of the vehicle is directly proportional to the L/D ratio. (See Reference 2 for an extensive discussion of the role of L/D on flight performance of an airplane.)
2. *The maximum lift coefficient* $c_{l,\max}$. An effective airfoil produces a high value of $c_{l,\max}$—much higher than that produced by a barn door.

The maximum lift coefficient is worth some additional discussion here. For a complete flight vehicle, the maximum lift coefficient $C_{L,\max}$ determines the stalling speed of the aircraft as discussed in the Design Box at the end of Section 1.8. From Equation (1.47), repeated below:

$$V_{\text{stall}} = \sqrt{\frac{2W}{\rho_\infty S C_{L,\max}}} \tag{1.47}$$

Therefore, a tremendous incentive exists to increase the maximum lift coefficient of an airfoil, in order to obtain either lower stalling speeds or higher payload weights at the same speed, as reflected in Equation (1.47). Moreover, the maneuverability of an airplane (i.e., the smallest possible turn radius and the fastest possible turn rate) depends on a large value of $C_{L,\max}$ (see Section 6.17 of Reference 2). On the other hand, for an airfoil at a given Reynolds number, the value of $c_{l,\max}$ is a function primarily of its shape. Once the shape is specified, the value of $c_{l,\max}$ is what nature dictates, as we have already seen. Therefore, to increase $c_{l,\max}$ beyond such a value, we must carry out some special measures. Such special measures include the use of flaps and/or leading-edge slats to increase

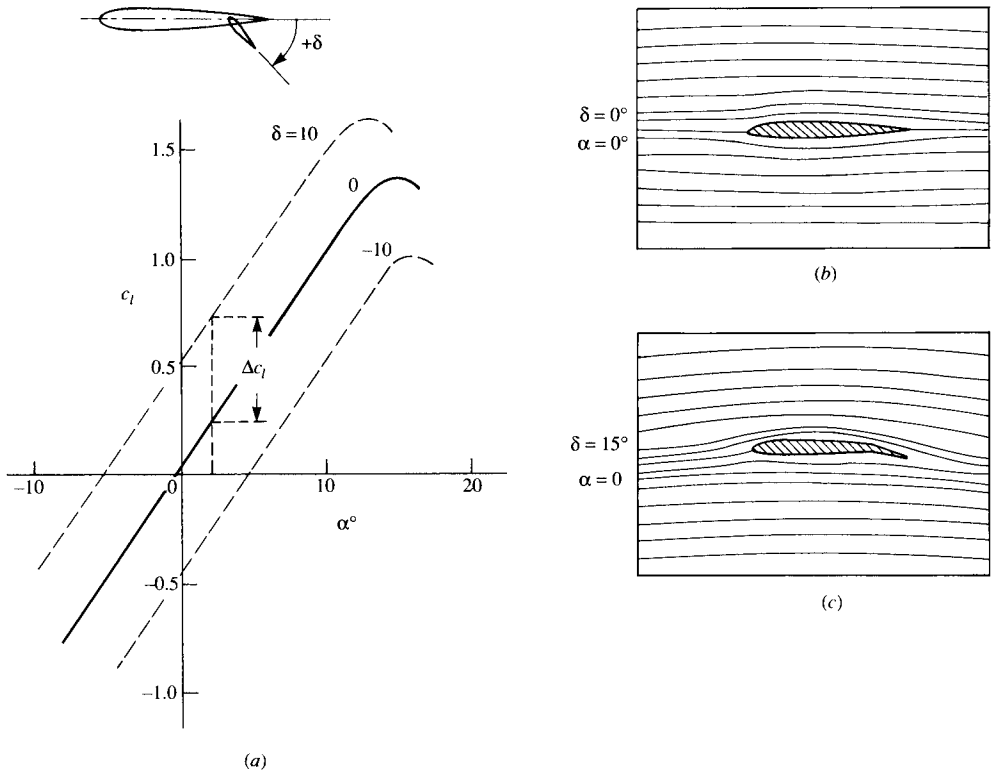


Figure 4.54 Effect of flap deflection on streamline shapes. *(The streamlines are drawn to scale from the experimental data of Hikaru Ito in Reference 50.)* (*a*) Effect of flap deflection on lift coefficient. (*b*) Streamline pattern with no flap deflection. (*c*) Streamline pattern with a 15° flap deflection.

$c_{l,\max}$ above that for the reference airfoil itself. These are called *high-lift devices,* and are discussed in more detail below.

A trailing-edge flap is simply a portion of the trailing-edge section of the airfoil that is hinged and which can be deflected upward or downward, as sketched in the insert in Figure 4.54*a*. When the flap is deflected downward (a positive angle δ in Figure 4.54*a*), the lift coefficient is increased, as shown in Figure 4.54*a*. This increase is due to an effective increase in the camber of the airfoil as the flap is deflected downward. The thin airfoil theory presented in this chapter clearly shows that the zero-lift angle of attack is a function of the amount of camber [see Equation (4.61)], with $\alpha_{L=0}$ becoming more negative as the camber is increased. In terms of the lift curve shown in Figure 4.54*a*, the original curve for no flap deflection goes through the origin because the airfoil is symmetric; however, as the flap is deflected downward, this lift curve simply translates to the left because $\alpha_{L=0}$ is becoming more negative. In Figure 4.54*a*, the results are given for flap deflections of $\pm 10°$. Comparing the case for $\delta = 10°$ with the no-deflection

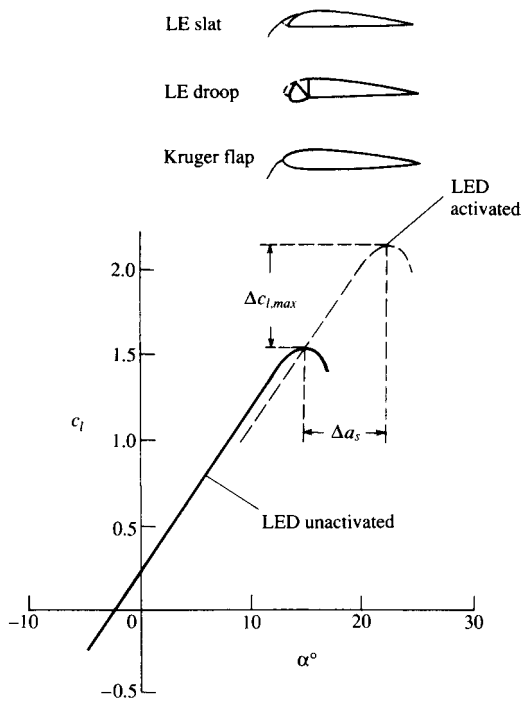


Figure 4.55 Effect of leading-edge flap on lift coefficient.

case, we see that, at a given angle of attack, the lift coefficient is increased by an amount Δc_l due to flap deflection. Moreover, the actual value of $c_{l,\max}$ is increased by flap deflection, although the angle of attack at which $c_{l,\max}$ occurs is slightly decreased. The change in the streamline pattern when the flap is deflected is shown in Figure 4.54*b* and *c*. Figure 4.54*b* is the case for $\alpha = 0$ and $\delta = 0$ (i.e., a symmetric flow). However, when α is held fixed at zero, but the flap is deflected by 15°, as shown in Figure 4.54*c*, the flow field becomes unsymmetrical and resembles the lifting flows shown (e.g., in Figure 4.49). That is, the streamlines in Figure 4.54*c* are deflected upward in the vicinity of the leading edge and downward near the trailing edge, and the stagnation point moves to the lower surface of the airfoil—just by deflecting the flap downward.

High-lift devices can also be applied to the leading edge of the airfoil, as shown in the insert in Figure 4.55. These can take the form of a leading-edge slat, leading-edge droop, or a leading-edge flap. Let us concentrate on the leading-edge slat, which is simply a thin, curved surface that is deployed in front of the leading edge. In addition to the primary airflow over the airfoil, there is now a secondary flow that takes place through the gap between the slat and the airfoil leading edge. This secondary flow from the bottom to the top surface modifies

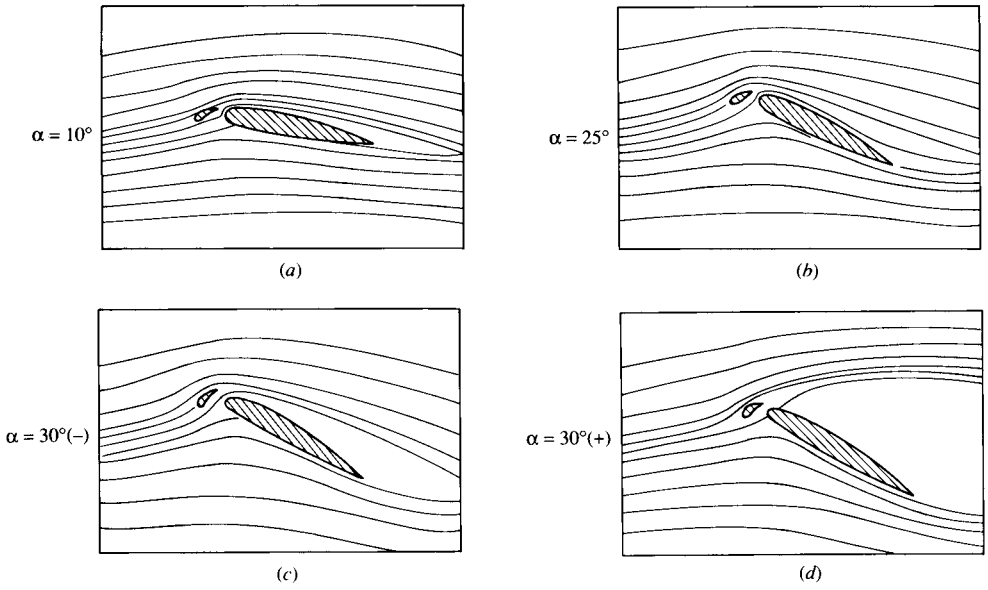


Figure 4.56 Effect of a leading-edge slat on the streamline pattern over an NACA 4412 airfoil. *(The streamlines are drawn to scale from the experimental data in Reference 50.)*

the pressure distribution over the top surface; the adverse pressure gradient which would normally exist over much of the top surface is mitigated somewhat by this secondary flow, hence delaying flow separation over the top surface. Thus, a leading-edge slat increases the stalling angle of attack, and hence yields a higher $c_{l,\max}$, as shown by the two lift curves in Figure 4.55, one for the case without a leading-edge device and the other for the slat deployed. Note that the function of a leading-edge slat is inherently different from that of a trailing-edge flap. There is no change in $\alpha_{L=0}$; rather, the lift curve is simply extended to a higher stalling angle of attack, with the attendant increase in $c_{l,\max}$. The streamlines of a flow field associated with an extended leading-edge slat are shown in Figure 4.56. The airfoil is in an NACA 4412 section. (*Note:* The flows shown in Figure 4.56 do not correspond exactly with the lift curves shown in Figure 4.55, although the general behavior is the same.) The stalling angle of attack for the NACA 4412 airfoil without slat extension is about 15°, but increases to about 30° when the slat is extended. In Figure 4.56*a*, the angle of attack is 10°. Note the flow through the gap between the slat and the leading edge. In Figure 4.56*b*, the angle of attack is 25° and the flow is still attached. This prevails to an angle of attack slightly less than 30°, as shown in Figure 4.56*c*. At slightly higher than 30° flow separation suddenly occurs and the airfoil stalls.

The high-lift devices used on modern, high-performance aircraft are usually a combination of leading-edge slats (or flaps) and multielement trailing-edge flaps. Typical airfoil configurations with these devices are sketched in Figure 4.57. Three

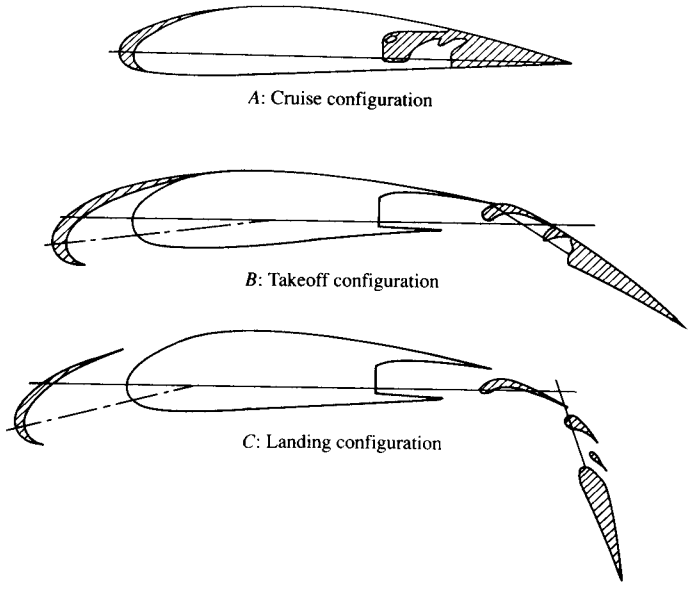


Figure 4.57 Airfoil with leading-edge and trailing-edge high-lift mechanisms. The trailing-edge device is a multielement flap.

configurations including the high-lift devices are shown: *A*—the cruise configuration, with no deployment of the high-lift devices; *B*—a typical configuration at takeoff, with both the leading- and trailing-edge devices partially deployed; and *C*—a typical configuration at landing, with all devices fully extended. Note that for configuration *C*, there is a gap between the slat and the leading edge and several gaps between the different elements of the multielement trailing-edge flap. The streamline pattern for the flow over such a configuration is shown in Figure 4.58. Here, the leading-edge slat and the multielement trailing-edge flap are fully extended. The angle of attack is 25°. Although the main flow over the top surface of the airfoil is essentially separated, the local flow through the gaps in the multielement flap is locally attached to the top surface of the flap; because of this locally attached flow, the lift coefficient is still quite high, on the order of 4.5.

With this, we end our discussion of the real flow over airfoils. In retrospect, we can say that the real flow at high angles of attack is dominated by flow separation—a phenomenon that is not properly modeled by the inviscid theories presented in this chapter. On the other hand, at lower angles of attack, such as those associated with the cruise conditions of an airplane, the inviscid theories presented here do an excellent job of predicting both lift and moments on an airfoil. Moreover, in this section, we have clearly seen the importance of airfoil *thickness* in determining the angle of attack at which flow separation will occur, and hence greatly affecting the maximum lift coefficient.

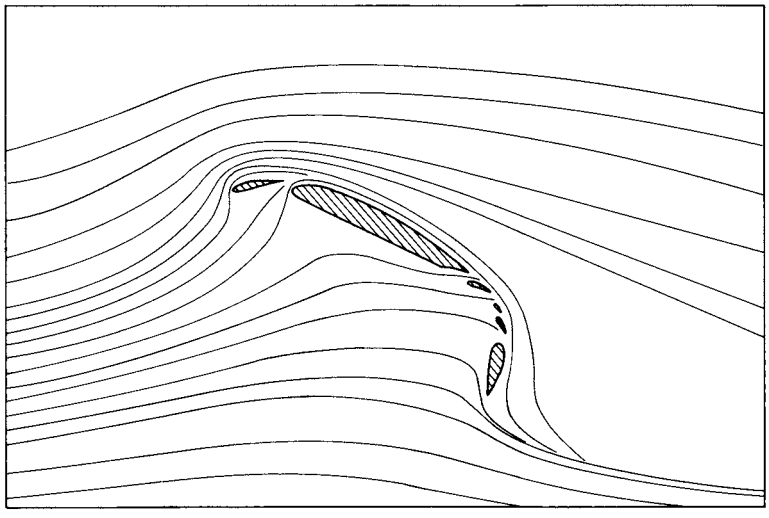


Figure 4.58 Effect of leading-edge and multielement flaps on the streamline pattern around an airfoil at angle of attack of 25°. *(The streamlines are drawn to scale from the experimental data of Reference 50.)*

4.14 HISTORICAL NOTE: EARLY AIRPLANE DESIGN AND THE ROLE OF AIRFOIL THICKNESS

In 1804, the first modern configuration aircraft was conceived and built by Sir George Cayley in England—it was an elementary hand-launched glider, about a meter in length, and with a kitelike shape for a wing as shown in Figure 4.59. (For the pivotal role played by George Cayley in the development of the airplane, see the extensive historical discussion in chapter 1 of Reference 2.) Note that right from the beginning of the modern configuration aircraft, the wing sections were very thin—whatever thickness was present, it was strictly for structural stiffness of the wing. Extremely thin airfoil sections were perpetuated by the work of Horatio Phillips in England. Phillips carried out the first serious wind-tunnel experiments in which the aerodynamic characteristics of a number of different airfoil shapes were measured. (See section 5.20 of Reference 2 for a presentation of the historical development of airfoils.) Some of Phillips airfoil sections are shown in Figure 4.60—note that they are the epitome of exceptionally thin airfoils. The early pioneers of aviation such as Otto Lilienthal in Germany and Samuel Pierpont Langley in America (see chapter 1 of Reference 2) continued this thin airfoil tradition. This was especially true of the Wright brothers, who in the period of 1901–1902 tested hundreds of different wing sections and planform shapes in their wind tunnel in Dayton, Ohio (recall our discussion in Section 1.1 and the models shown in Figure 1.7). A sketch of some of the Wrights' airfoil sections

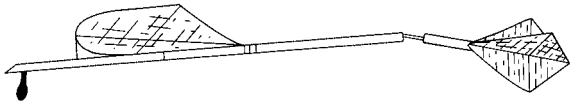


Figure 4.59 The first modern configuration airplane in history: George Cayley's model glider of 1804.

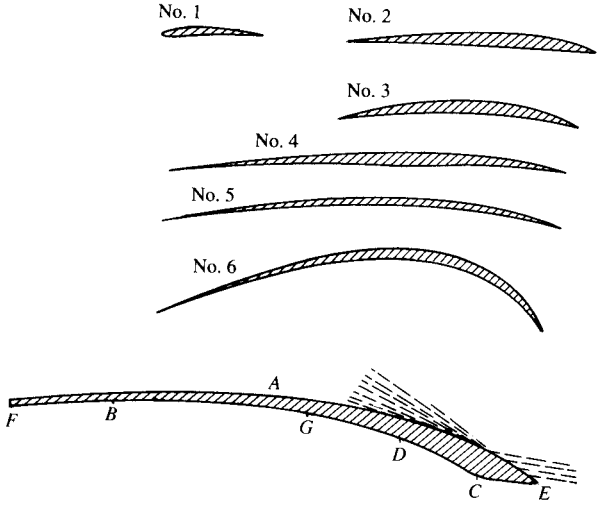


Figure 4.60 Double-surface airfoil sections by Horatio Phillips. The six upper shapes were patented by Phillips in 1884; the lower airfoil was patented in 1891. Note the thin profile shapes.

is given in Figure 4.61—for the most part, very thin sections. Indeed, such a thin airfoil section was used on the 1903 Wright Flyer, as can be readily seen in the side view of the Flyer shown in Figure 4.62. The important point here is that *all* of the early pioneering aircraft, and especially the Wright Flyer, incorporated very thin airfoil sections—airfoil sections that performed essentially like the flat plate results discussed in Section 4.13, and as shown in Figure 4.51 (the dashed curve) and by the streamline pictures in Figure 4.52. Conclusion: These early airfoil sections suffered flow-field separation at small angles of attack and, consequently, had low values of $c_{l,\max}$. By the standards we apply today, these were simply very poor airfoil sections for the production of high lift.

This situation carried into the early part of World War I. In Figure 4.63, we see four airfoil sections that were employed on World War I aircraft. The top three sections had thickness ratios of about 4 to 5 percent and are representative of the type of sections used on all aircraft until 1917. For example, the SPAD XIII (shown in Figure 3.50), the fastest of all World War I fighters, had a thin airfoil section like the Eiffel section shown in Figure 4.63. Why were such thin airfoil

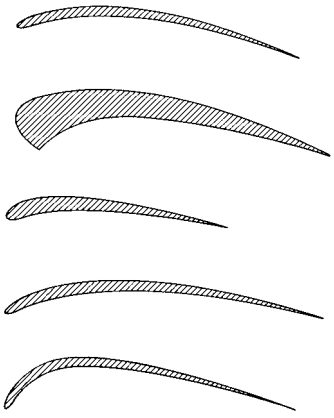


Figure 4.61 Some typical airfoil shapes tested by the Wright brothers in their wind tunnel during 1902–1903.

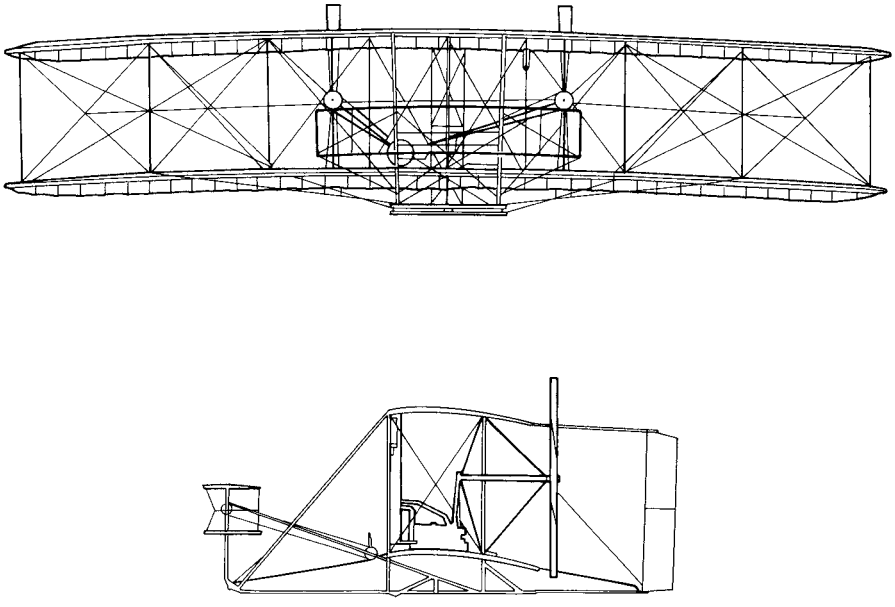


Figure 4.62 Front and side views of the 1903 Wright Flyer. Note the thin airfoil sections. *(Courtesy of the National Air and Space Museum.)*

sections considered to be the best by most designers of World War I aircraft? The historical tradition described above might be part of the answer—a tradition that started with Cayley. Also, there was quite clearly a mistaken notion at that time that *thick* airfoils would produce high drag. Of course, today we know the opposite

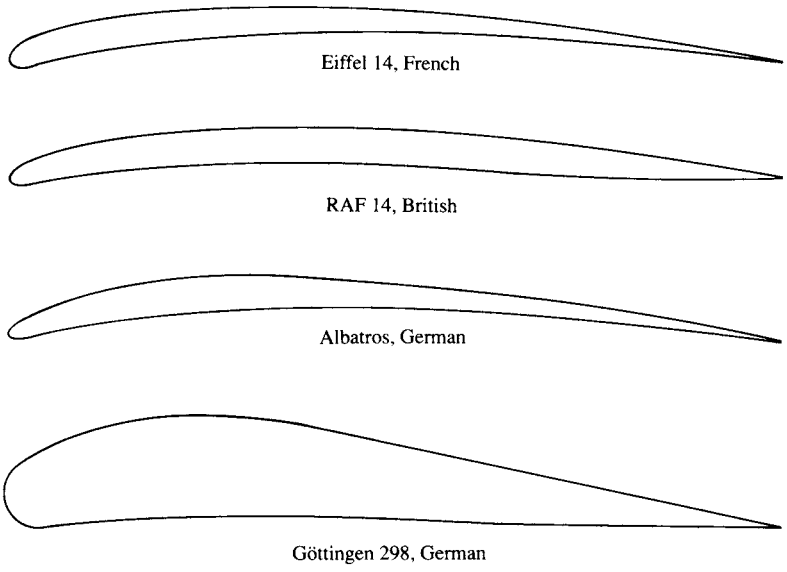


Figure 4.63 Some examples of different airfoil shapes used on World War I aircraft. *(Source: Loftin, Reference 48.)*

to be true; review our discussion of streamlined shapes in Section 1.12 for this fact. Laurence Loftin in Reference 48 surmises that the mistaken notion might have been fostered by early wind-tunnel tests. By the nature of the early wind tunnels in use—small sizes and very low speeds—the data were obtained at very low Reynolds numbers, less than 100,000 based on the airfoil-chord length. These Reynolds numbers are to be compared with typical values in the millions for actual airplane flight. Modern studies of low Reynolds number flows over conventional thick airfoils (e.g., see Reference 51) clearly show high-drag coefficients, in contrast to the lower values that occur for the high Reynolds number associated with the flight of full-scale aircraft. Also, the reason for the World War I airplane designer's preference for thin airfoils might be as simple as the tendency to follow the example of the wings of birds, which are quite thin. In any event, the design of all English, French, and American World War I aircraft incorporated thin airfoils and, consequently, suffered from poor high-lift performance. The fundamentals of airfoil aerodynamics as we know them today (and as being presented in this book) were simply not sufficiently understood by the designers during World War I. In turn, they never appreciated what they were losing.

This situation changed dramatically in 1917. Work carried out in Germany at the famous Göttingen aerodynamic laboratory of Ludwig Prandtl (see Section 5.8 for a biographical sketch of Prandtl) demonstrated the superiority of a thick airfoil section, such as the Göttingen 298 section shown at the bottom of Figure 4.63. This revolutionary development was immediately picked up by the famous designer Anthony Fokker, who incorporated the 13-percent-thick Göttingen 298 profile in his new Fokker Dr-1—the famous triplane flown by the "Red Baron," Rittmeister

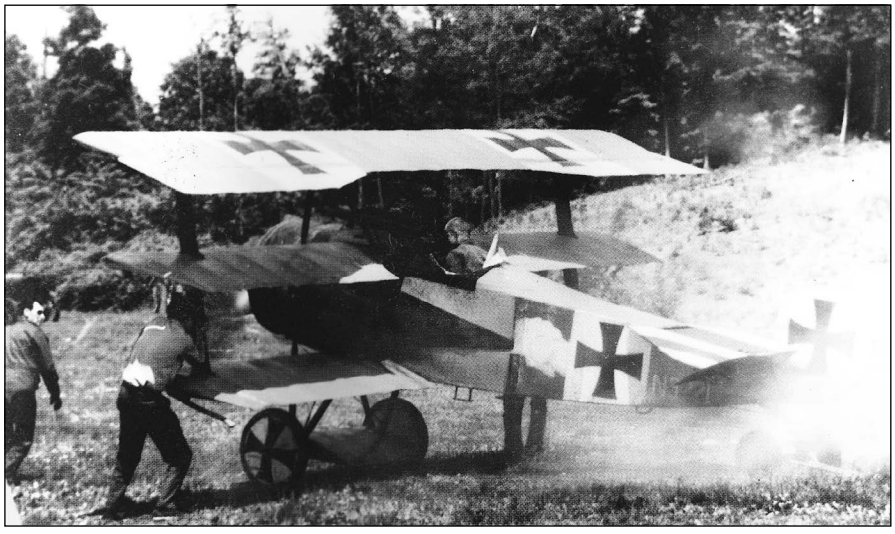


Figure 4.64 The World War I Fokker Dr-1 triplane, the first fighter aircraft to use a thick airfoil. *(Source: Loftin, Reference 48.)*

Manfred Freiher von Richthofen. A photograph of the Fokker Dr-1 is shown in Figure 4.64. The major benefits derived from Fokker's use of the thick airfoil were:

1. The wing structure could be completely internal; that is the wings of the Dr-1 were a cantilever design, which removed the need for the conventional wire bracing that was used in other aircraft. This, in turn, eliminated the high drag associated with these interwing wires, as discussed at the end of Section 1.11. For this reason, the Dr-1 had a zero-lift drag coefficient of 0.032, among the lowest of World War I airplanes. (By comparison the zero-lift drag coefficient of the French SPAD XIII was 0.037.)
2. The thick airfoil provided the Fokker Dr-1 with a high maximum lift coefficient. Its performance was analogous to the *upper* curves shown in Figure 4.51. This in turn, provided the Dr-1 with an exceptionally high rate-of-climb as well as enhanced maneuverability—characteristics that were dominant in dog-fighting combat.

Anthony Fokker continued the use of a thick airfoil in his design of the D-VII, as shown in Figure 4.65. This gave the D-VII a much greater rate-of-climb than its two principal opponents at the end of the war—the English Sopwith Camel and the French SPAD XIII, both of which still used very thin airfoil sections. This rate-of-climb performance, as well as its excellent handling characteristics, singled out the Fokker D-VII as the most effective of all German World War I fighters. The respect given by the Allies to this machine is no more clearly indicated than by a paragraph in article IV of the armistice agreement, which lists war material



Figure 4.65 The World War I Fokker D-VII, one of the most effective fighters of the war, due in part to its superior aerodynamic performance allowed by a thick airfoil section. (*Source:* Loftin, Reference 48.)

to be handed over to the Allies by Germany. In this article, the Fokker D-VII is *specifically* listed—the only airplane of any type to be explicitly mentioned in the armistice. To this author's knowledge, this is the one and only time where a breakthrough in airfoil technology is essentially reflected in any major political document, though somewhat implicitly.

4.15 HISTORICAL NOTE: KUTTA, JOUKOWSKI, AND THE CIRCULATION THEORY OF LIFT

Frederick W. Lanchester (1868–1946), an English engineer, automobile manufacturer, and self-styled aerodynamicist, was the first to connect the idea of circulation with lift. His thoughts were originally set forth in a presentation given before the Birmingham Natural History and Philosophical Society in 1894 and later contained in a paper submitted to the Physical Society, which turned it down. Finally, in 1907 and 1908, he published two books, entitled *Aerodynamics* and *Aerodonetics*, where his thoughts on circulation and lift were described in detail. His books were later translated into German in 1909 and French in 1914. Unfortunately, Lanchester's style of writing was difficult to read and understand; this is partly responsible for the general lack of interest shown by British scientists in

Lanchester's work. Consequently, little positive benefit was derived from Lanchester's writings. (See Section 5.7 for a more detailed portrait of Lanchester and his work.)

Quite independently, and with total lack of knowledge of Lanchester's thinking, M. Wilhelm Kutta (1867–1944) developed the idea that lift and circulation are related. Kutta was born in Pitschen, Germany, in 1867 and obtained a Ph.D. in mathematics from the University of Munich in 1902. After serving as professor of mathematics at several German technical schools and universities, he finally settled at the Technische Hochschule in Stuttgart in 1911 until his retirement in 1935. Kutta's interest in aerodynamics was initiated by the successful glider flights of Otto Lilienthal in Berlin during the period 1890–1896 (see chapter 1 of Reference 2). Kutta attempted theoretically to calculate the lift on the curved wing surfaces used by Lilienthal. In the process, he surmised from experimental data that the flow left the trailing edge of a sharp-edged body smoothly and that this condition fixed the circulation around the body (the Kutta condition, described in Section 4.5). At the same time, he was convinced that circulation and lift were connected. Kutta was reluctant to publish these ideas, but after the strong insistence of his teacher, S. Finsterwalder, he wrote a paper entitled "Auftriebskrafte in Stromenden Flussigkecten" (Lift in Flowing Fluids). This was actually a short note abstracted from his longer graduation paper in 1902, but it represents the first time in history where the concepts of the Kutta condition as well as the connection of circulation with lift were officially published. Finsterwalder clearly repeated the ideas of his student in a lecture given on September 6, 1909, in which he stated:

> On the upper surface the circulatory motion increases the translatory one, therefore there is high velocity and consequently low pressure, while on the lower surface the two movements are opposite, therefore there is low velocity with high pressure, with the result of a thrust upward.

However, in his 1902 note, Kutta did not give the precise quantitative relation between circulation and lift. This was left to Nikolai E. Joukowski (Zhukouski). Joukowski was born in Orekhovo in central Russia on January 5, 1847. The son of an engineer, he became an excellent student of mathematics and physics, graduating with a Ph.D. in applied mathematics from Moscow University in 1882. He subsequently held a joint appointment as a professor of mechanics at Moscow University and the Moscow Higher Technical School. It was at this latter institution that Joukowski built in 1902 the first wind tunnel in Russia. Joukowski was deeply interested in aeronautics, and he combined a rare gift for both experimental and theoretical work in the field. He expanded his wind tunnel into a major aerodynamics laboratory in Moscow. Indeed, during World War I, his laboratory was used as a school to train military pilots in the principles of aerodynamics and flight. When he died in 1921, Joukowski was by far the most noted aerodynamicist in Russia.

Much of Joukowski's fame was derived from a paper published in 1906, wherein he gives, for the first time in history, the relation $L' = \rho_\infty V_\infty \Gamma$—the

Kutta-Joukowski theorem. In Joukowski's own words:

> If an irrotational two-dimensional fluid current, having at infinity the velocity V_∞ surrounds any closed contour on which the circulation of velocity is Γ, the force of the aerodynamic pressure acts on this contour in a direction perpendicular to the velocity and has the value
>
> $$L' = \rho_\infty V_\infty \Gamma$$
>
> The direction of this force is found by causing to rotate through a right angle the vector V_∞ around its origin in an inverse direction to that of the circulation.

Joukowski was unaware of Kutta's 1902 note and developed his ideas on circulation and lift independently. However, in recognition of Kutta's contribution, the equation given above has propagated through the twentieth century as the "Kutta-Joukowski theorem."

Hence, by 1906—just 3 years after the first successful flight of the Wright brothers—the circulation theory of lift was in place, ready to aid aerodynamics in the design and understanding of lifting surfaces. In particular, this principle formed the cornerstone of the thin airfoil theory described in Sections 4.7 and 4.8. Thin airfoil theory was developed by Max Munk, a colleague of Prandtl in Germany, during the first few years after World War I. However, the very existence of thin airfoil theory, as well as its amazingly good results, rests upon the foundation laid by Lanchester, Kutta, and Joukowski a decade earlier.

4.16 SUMMARY

Return to the road map given in Figure 4.7. Make certain that you feel comfortable with the material represented by each box on the road map and that you understand the flow of ideas from one box to another. If you are uncertain about one or more aspects, review the pertinent sections before progressing further.

Some important results from this chapter are itemized below:

A vortex sheet can be used to synthesize the inviscid, incompressible flow over an airfoil. If the distance along the sheet is given by s and the strength of the sheet per unit length is $\gamma(s)$, then the velocity potential induced at point (x, y) by a vortex sheet that extends from point a to point b is

$$\phi(x, y) = -\frac{1}{2\pi} \int_a^b \theta \gamma(s)\, ds \tag{4.3}$$

The circulation associated with this vortex sheet is

$$\Gamma = \int_a^b \gamma(s)\, ds \tag{4.4}$$

Across the vortex sheet, there is a tangential velocity discontinuity, where

$$\gamma = u_1 - u_2 \tag{4.8}$$

The Kutta condition is an observation that for a lifting airfoil of given shape at a given angle of attack, nature adopts that particular value of circulation around the airfoil which results in the flow leaving smoothly at the trailing edge. If the trailing-edge angle is finite, then the trailing edge is a stagnation point. If the trailing edge is cusped, then the velocities leaving the top and bottom surfaces at the trailing edge are finite and equal in magnitude and direction. In either case,

$$\gamma(\text{TE}) = 0 \tag{4.10}$$

Thin airfoil theory is predicated on the replacement of the airfoil by the mean camber line. A vortex sheet is placed along the chord line, and its strength adjusted such that, in conjunction with the uniform freestream, the camber line becomes a streamline of the flow while at the same time satisfying the Kutta condition. The strength of such a vortex sheet is obtained from the fundamental equation of thin airfoil theory:

$$\frac{1}{2\pi}\int_0^c \frac{\gamma(\xi)\,d\xi}{x-\xi} = V_\infty\left(\alpha - \frac{dz}{dx}\right) \tag{4.18}$$

Results of thin airfoil theory:

Symmetric airfoil

1. $c_l = 2\pi\alpha$.
2. Lift slope $= dc_l/d\alpha = 2\pi$.
3. The center of pressure and the aerodynamic center are both at the quarter-chord point.
4. $c_{m,c/4} = c_{m,\text{ac}} = 0$.

Cambered airfoil

1. $$c_l = 2\pi\left[\alpha + \frac{1}{\pi}\int_0^\pi \frac{dz}{dx}(\cos\theta_0 - 1)\,d\theta_0\right] \tag{4.57}$$
2. Lift slope $= dc_l/d\alpha = 2\pi$.
3. The aerodynamic center is at the quarter-chord point.
4. The center of pressure varies with the lift coefficient.

The vortex panel method is an important numerical technique for the solution of the inviscid, incompressible flow over bodies of arbitrary shape, thickness, and angle of attack. For panels of constant strength, the governing equations are

$$V_\infty \cos \beta_i - \sum_{j=1}^{n} \frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_i} \, ds_j = 0 \qquad (i = 1, 2, \ldots, n)$$

and

$$\gamma_i = -\gamma_{i-1}$$

which is one way of expressing the Kutta condition for the panels immediately above and below the trailing edge.

4.17 PROBLEMS

4.1 Consider the data for the NACA 2412 airfoil given in Figure 4.10. Calculate the lift and moment about the quarter chord (per unit span) for this airfoil when the angle of attack is 4° and the freestream is at standard sea level conditions with a velocity of 50 ft/s. The chord of the airfoil is 2 ft.

4.2 Consider an NACA 2412 airfoil with a 2-m chord in an airstream with a velocity of 50 m/s at standard sea level conditions. If the lift per unit span is 1353 N, what is the angle of attack?

4.3 Starting with the definition of circulation, derive Kelvin's circulation theorem, Equation (4.11).

4.4 Starting with Equation (4.35), derive Equation (4.36).

4.5 Consider a thin, symmetric airfoil at 1.5° angle of attack. From the results of thin airfoil theory, calculate the lift coefficient and the moment coefficient about the leading edge.

4.6 The NACA 4412 airfoil has a mean camber line given by

$$\frac{z}{c} = \begin{cases} 0.25 \left[0.8 \dfrac{x}{c} - \left(\dfrac{x}{c} \right)^2 \right] & \text{for } 0 \le \dfrac{x}{c} \le 0.4 \\ 0.111 \left[0.2 + 0.8 \dfrac{x}{c} - \left(\dfrac{x}{c} \right)^2 \right] & \text{for } 0.4 \le \dfrac{x}{c} \le 1 \end{cases}$$

Using thin airfoil theory, calculate
(*a*) $\alpha_{L=0}$ (*b*) c_l when $\alpha = 3^\circ$

4.7 For the airfoil given in Problem 4.6, calculate $c_{m,c/4}$ and x_{cp}/c when $\alpha = 3^\circ$.

4.8 Compare the results of Problems 4.6 and 4.7 with experimental data for the NACA 4412 airfoil, and note the percentage difference between theory and experiment. (*Hint:* A good source of experimental airfoil data is Reference 11.)

4.9 Starting with Equations (4.35) and (4.43), derive Equation (4.62).

4.10 For the NACA 2412 airfoil, the lift coefficient and moment coefficient about the quarter-chord at -6° angle of attack are -0.39 and -0.045, respectively. At 4° angle of attack, these coefficients are 0.65 and -0.037, respectively. Calculate the location of the aerodynamic center.

4.11 Consider again the NACA 2412 airfoil discussed in Problem 4.10. The airfoil is flying at a velocity of 60 m/s at a standard altitude of 3 km (see Appendix D). The chord length of the airfoil is 2 m. Calculate the lift per unit span when the angle of attack is 4°.

4.12 For the airfoil in Problem 4.11, calculate the value of the circulation around the airfoil.

4.13 In Section 3.15 we studied the case of the lifting flow over a circular cylinder. In real life, a rotating cylinder in a flow will produce lift; such real flow fields are shown in the photographs in Figures 3.34(*b*) and (*c*). Here, the viscous shear stress acting between the flow and the surface of the cylinder drags the flow around in the direction of rotation of the cylinder. For a cylinder of radius R rotating with an angular velocity ω in an otherwise stationary fluid, the viscous flow solution for the velocity field obtained from the Navier-Stokes equations (Chapter 15) is

$$V_\theta = \frac{R^2\omega}{r}$$

where V_θ is the tangential velocity along the circular streamlines and r is the radial distance from the center of the cylinder. (See Schlichting, *Boundary-Layer Theory,* 6th ed., McGraw-Hill, 1968, page 81.) Note that V_θ varies inversely with r and is of the same form as the inviscid flow velocity for a point vortex given by Equation (3.105). If the rotating cylinder has a radius of 1 m and is flying at the same velocity and altitude as the airfoil in Problem 4.11, what must its angular velocity be to produce the same lift as the airfoil in Problem 4.11? (*Note:* You can check your results with the experimental data for lift on rotating cylinders in Hoerner, *Fluid-Dynamic Lift,* published by the author, 1975, p. 21–4, Fig. 5.)

4.14 The question is often asked: Can an airfoil fly upside-down? To answer this, make the following calculation. Consider a positively cambered airfoil with a zero-lift angle of -3°. The lift slope is 0.1 per degree. (*a*) Calculate the lift coefficient at an angle of attack of 5°. (*b*) Now imagine the same airfoil turned upside-down, but at the same 5° angle of attack as part (*a*). Calculate its lift coefficient. (*c*) At what angle of attack must the upside-down airfoil be set to generate the same lift as that when it is right-side-up at a 5° angle of attack?

4.15 The airfoil section of the wing of the British Spitfire of World War II fame (see Figure 5.19) is an NACA 2213 at the wing root, tapering to an NACA 2205 at the wing tip. The root chord is 8.33 ft. The measured profile drag coefficient of the NACA 2213 airfoil is 0.006 at a Reynolds number of

9×10^6. Consider the Spitfire cruising at an altitude of 18,000 ft. (a) At what velocity is it flying for the root chord Reynolds number to be 9×10^6? (b) At this velocity and altitude, assuming completely turbulent flow, estimate the skin-friction drag coefficient for the NACA 2213 airfoil, and compare this with the total profile drag coefficient. Calculate the percentage of the profile drag coefficient that is due to pressure drag. *Note:* Assume that μ varies as the square root of temperature, as first discussed in Section 1.8.

4.16 For the conditions given in Problem 4.15, a more reasonable calculation of the skin friction coefficient would be to assume an initially laminar boundary layer starting at the leading edge, and then transitioning to a turbulent boundary layer at some point downstream. Calculate the skin-friction coefficient for the Spitfire's airfoil described in Problem 4.15, but this time assuming a critical Reynolds number of 10^6 for transition.

CHAPTER 5

Incompressible Flow over Finite Wings

The one who has most carefully watched the soaring birds of prey sees man with wings and the faculty of using them.

James Means, Editor of
the Aeronautical Annual, 1895

PREVIEW BOX

The Beechcraft Baron 58 twin-engine business aircraft is shown in three-view in Figure 5.1. The wing on this airplane has a 15 percent thick NACA 23015 airfoil at the root, tapering to a 10 percent thickness at the tip. We studied airfoil properties in Chapter 4, ostensibly to be able to predict the lift and drag characteristics of an airplane wing utilizing a given airfoil shape. Wind tunnel data for the NACA 23015 airfoil is given in Figure 5.2 in the standard format used by the NACA as described in Reference 11. The airfoil lift coefficient and the moment coefficient about the quarter-chord point are given as a function of airfoil section angle of attack in Figure 5.2*a*. The drag coefficient and the moment coefficient about the aerodynamic center are given as a function of the lift coefficient in Figure 5.2*b*. The airfoil shape is shown to scale at the top of Figure 5.2*b*. Results for three different values of the Reynolds number based on chord length are shown, as indicated by the code near the bottom of Figure 5.2*b*. The data labeled "standard roughness" applies to a special case where the model surface was covered with a sandpaper-like grit; we are not concerned with this special case.

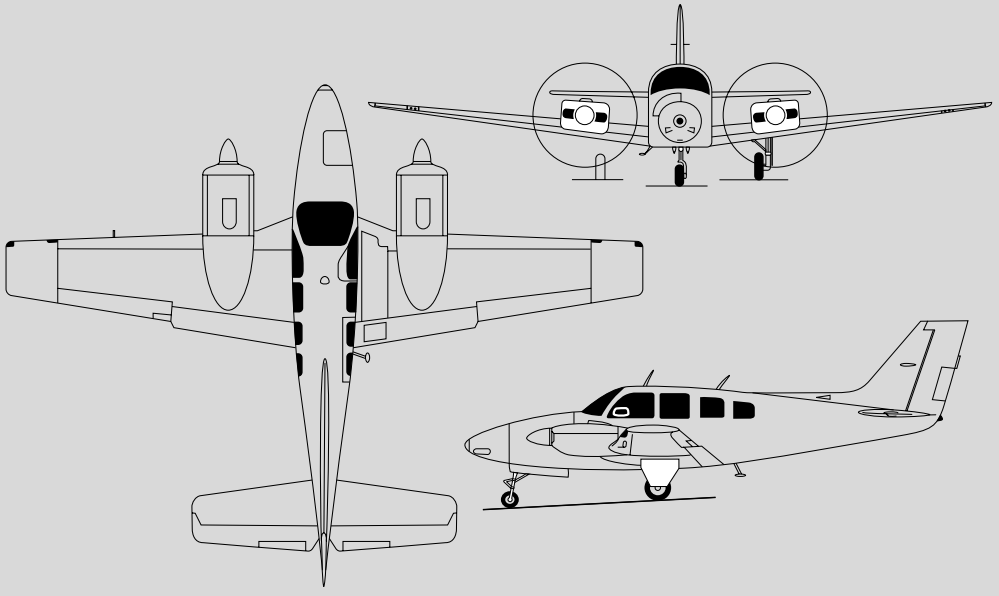


Figure 5.1 Beechcraft Baron 58, four- to six-seat general aviation airplane.

Consider the Beechcraft Baron 58 cruising such that the wing is at a four-degree angle of attack. What are the lift and drag coefficients of the wing? From Figure 5.2*a*, for the airfoil at $\alpha = 4^\circ$, we have $c_l = 0.54$. Using the drag coefficient data at the highest Reynolds number shown in Figure 5.2*b*, for $c_l = 0.54$ the corresponding drag coefficient $c_d = 0.0068$. Using capital letters for the aerodynamic coefficients of a wing, in contrast to the lower case letters for an airfoil, we pose the question: Are C_L and C_D for the wing the same as those for the airfoil? That is,

$$C_L = 0.54 \text{ (?)} \qquad C_D = 0.0068 \text{ (?)}$$

The answer is a resounding NO! Not even close! Surprised? How can this be? Why are the aerodynamic coefficients of the wing *not* the same as those for the airfoil shape from which the wing is made? Surely, the aerodynamic properties of the airfoil must have something to do with the aerodynamic properties of the finite wing. Or else our studies of airfoils in Chapter 4 would be a big waste of time.

The single, most important purpose of this chapter is to solve this mystery. This chapter is focused on the aerodynamic properties of real, finite wings. The solution involves some interesting and (yes) some rather intense mathematics combined with some knowledge of the physical properties of the flow over a finite wing. This is very important stuff. This chapter gets at the very core of the aerodynamics of a real airplane flying at speeds appropriate to the assumption of incompressible flow. Read on with vigor, and let us solve this mystery.

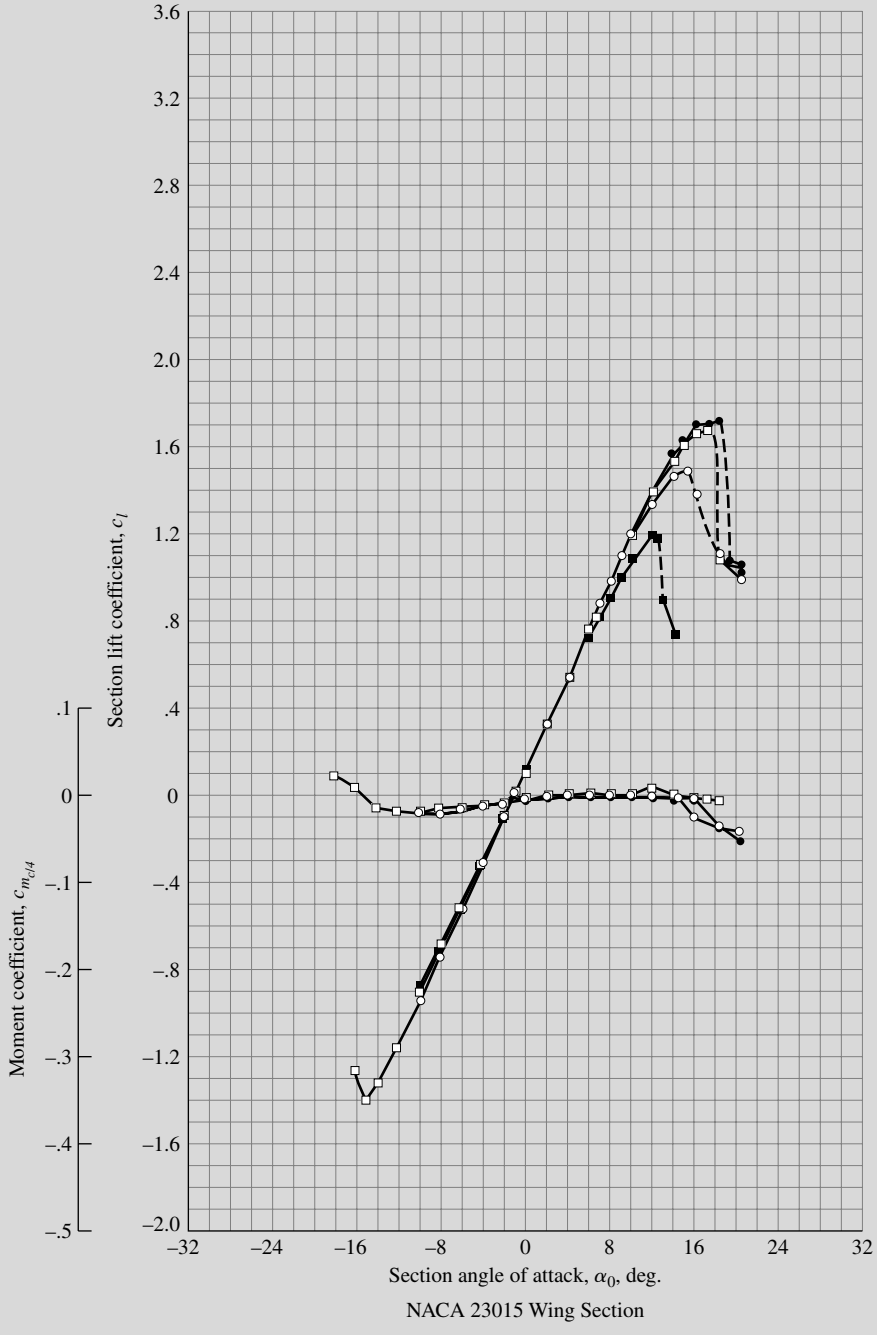


Figure 5.2*a* Lift coefficient and moment coefficient about the quarter-chord for the NACA 23015 airfoil.

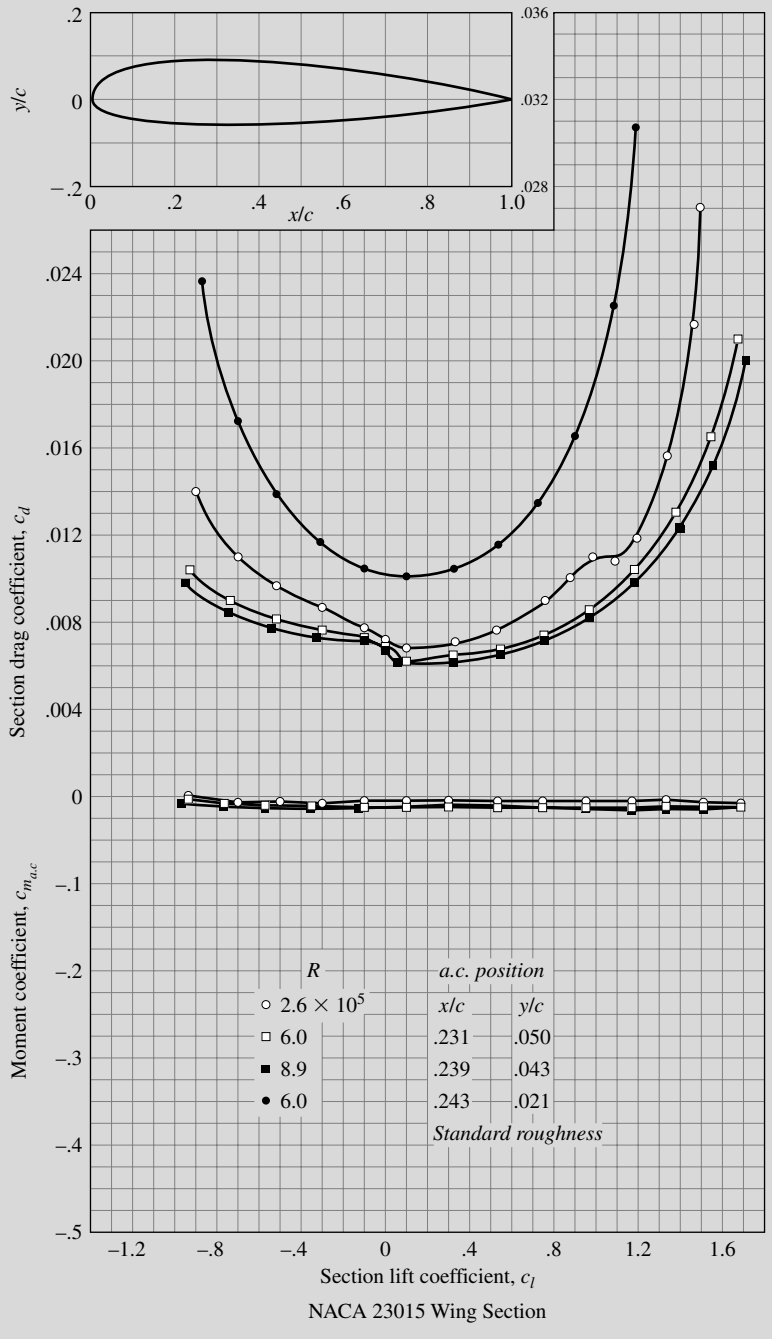


Figure 5.2*b* Drag coefficient and moment coefficient about the aerodynamic center for the NACA 23015 airfoil.

5.1 INTRODUCTION: DOWNWASH AND INDUCED DRAG

In Chapter 4 we discussed the properties of airfoils, which are the same as the properties of a wing of infinite span; indeed, airfoil data are frequently denoted as "infinite wing" data. However, all real airplanes have wings of finite span, and the purpose of the present chapter is to apply our knowledge of airfoil properties to the analysis of such finite wings. This is the second step in Prandtl's philosophy of wing theory, as described in Section 4.1. You should review Section 4.1 before proceeding further.

Question: Why are the aerodynamic characteristics of a finite wing any different from the properties of its airfoil sections? Indeed, an airfoil is simply a section of a wing, and at first thought, you might expect the wing to behave exactly the same as the airfoil. However, as studied in Chapter 4, the flow over an airfoil is two-dimensional. In contrast, a finite wing is a three-dimensional body, and consequently the flow over the finite wing is three-dimensional; that is, there is a component of flow in the spanwise direction. To see this more clearly, examine Figure 5.3, which gives the top and front views of a finite wing. The physical mechanism for generating lift on the wing is the existence of a high pressure on the bottom surface and a low pressure on the top surface. The net imbalance of the pressure distribution creates the lift, as discussed in Section 1.5. However, as

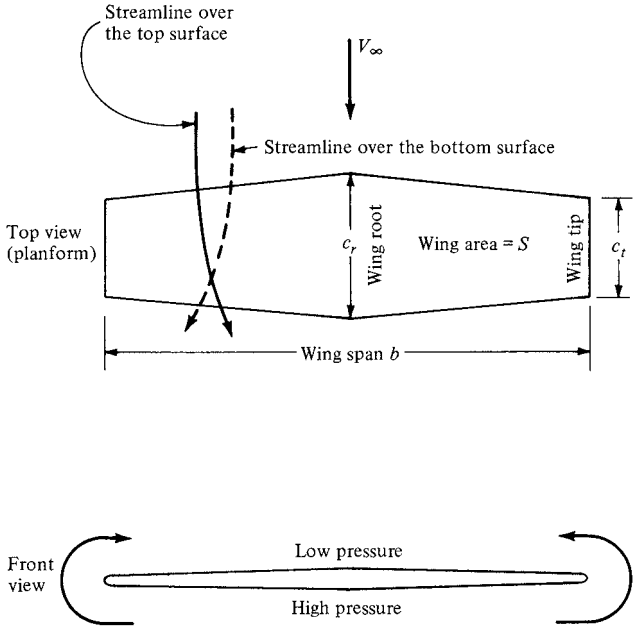


Figure 5.3 Finite wing. In this figure, the curvature of the streamlines over the top and bottom of the wing is exaggerated for clarity.

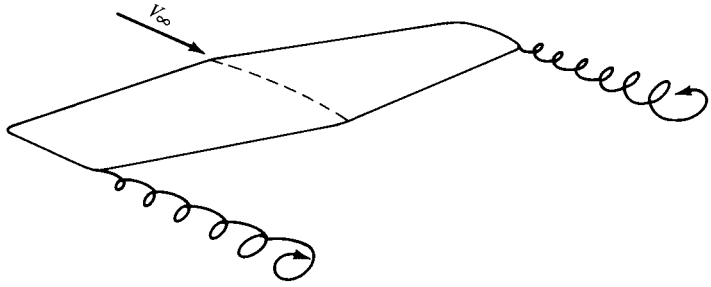


Figure 5.4 Schematic of wing-tip vortices.

a by-product of this pressure imbalance, the flow near the wing tips tends to curl around the tips, being forced from the high-pressure region just underneath the tips to the low-pressure region on top. This flow around the wing tips is shown in the front view of the wing in Figure 5.3. As a result, on the top surface of the wing, there is generally a spanwise component of flow from the tip toward the wing root, causing the streamlines over the top surface to bend toward the root, as sketched on the top view shown in Figure 5.3. Similarly, on the bottom surface of the wing, there is generally a spanwise component of flow from the root toward the tip, causing the streamlines over the bottom surface to bend toward the tip. Clearly, the flow over the finite wing is three-dimensional, and therefore you would expect the overall aerodynamic properties of such a wing to differ from those of its airfoil sections.

The tendency for the flow to "leak" around the wing tips has another important effect on the aerodynamics of the wing. This flow establishes a circulatory motion that trails downstream of the wing; that is, a trailing *vortex* is created at each wing tip. These wing-tip vortices are sketched in Figure 5.4 and are illustrated in Figure 5.5. The tip vortices are essentially weak "tornadoes" that trail downstream of the finite wing. (For large airplanes such as a Boeing 747, these tip vortices can be powerful enough to cause light airplanes following too closely to go out of control. Such accidents have occurred, and this is one reason for large spacings between aircraft landing or taking off consecutively at airports.) These wing-tip vortices downstream of the wing induce a small downward component of air velocity in the neighborhood of the wing itself. This can be seen by inspecting Figure 5.5; the two vortices tend to drag the surrounding air around with them, and this secondary movement induces a small velocity component in the downward direction at the wing. This downward component is called *downwash,* denoted by the symbol w. In turn, the downwash combines with the freestream velocity V_∞ to produce a *local* relative wind which is canted downward in the vicinity of each airfoil section of the wing, as sketched in Figure 5.6.

Examine Figure 5.6 closely. The angle between the chord line and the direction of V_∞ is the angle of attack α, as defined in Section 1.5 and as used throughout our discussion of airfoil theory in Chapter 4. We now more precisely define α as the *geometric* angle of attack. In Figure 5.6, the local relative wind is inclined

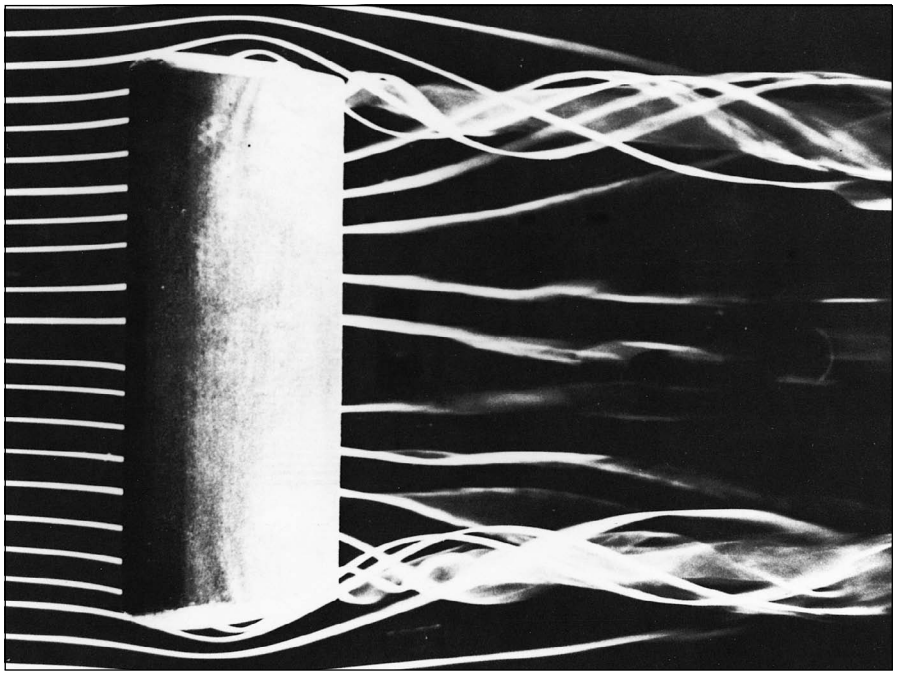


Figure 5.5 Wing-tip vortices from a rectangular wing. The wing is in a smoke tunnel, where individual streamtubes are made visible by means of smoke filaments. (*Source:* Head, M. R., in *Flow Visualization II,* W. Merzkirch (Ed.), Hemisphere Publishing Co., New York, 1982, pp. 399–403. Also available in Van Dyke, Milton, *An Album of Fluid Motion,* The Parabolic Press, Stanford, CA, 1982.)

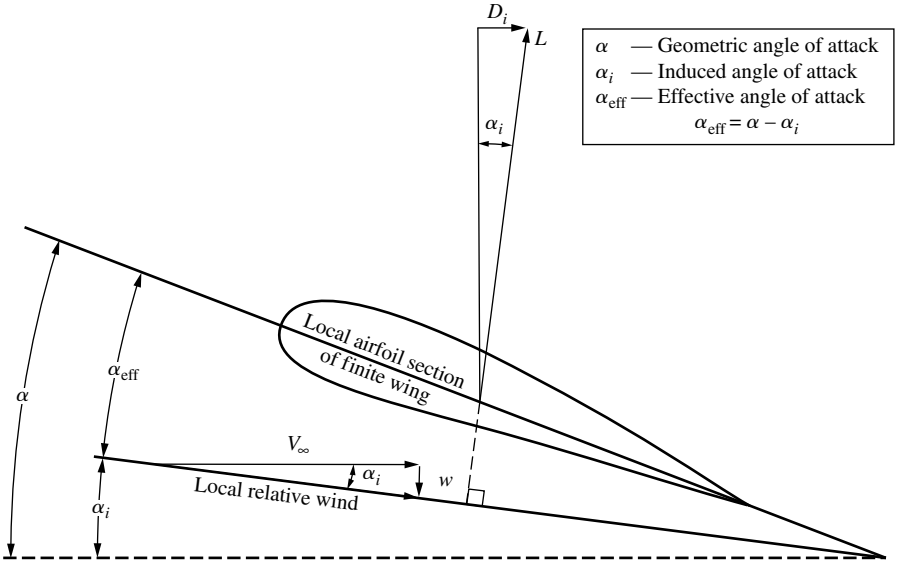


Figure 5.6 Effect of downwash on the local flow over a local airfoil section of a finite wing.

below the direction of V_∞ by the angle α_i, called the *induced* angle of attack. The presence of downwash, and its effect on inclining the local relative wind in the downward direction, has two important effects on the local airfoil section, as follows:

1. The angle of attack actually seen by the local airfoil section is the angle between the chord line and the local relative wind. This angle is given by α_{eff} in Figure 5.4 and is defined as the *effective* angle of attack. Hence, although the wing is at a geometric angle of attack α, the local airfoil section is seeing a smaller angle, namely, the effective angle of attack α_{eff}. From Figure 5.6,

$$\alpha_{\text{eff}} = \alpha - \alpha_i \tag{5.1}$$

2. The local lift vector is aligned perpendicular to the local relative wind, and hence is inclined behind the vertical by the angle α_i, as shown in Figure 5.6. Consequently, there is a component of the local lift vector in the direction of V_∞; that is, there is a *drag* created by the presence of downwash. This drag is defined as *induced drag,* denoted by D_i in Figure 5.6.

Hence, we see that the presence of downwash over a finite wing reduces the angle of attack that each section effectively sees, and moreover, it creates a component of drag—the induced drag D_i. Keep in mind that we are still dealing with an inviscid, incompressible flow, where there is no skin friction or flow separation. For such a flow, there is a *finite* drag—the induced drag—on a finite wing. D'Alembert's paradox does *not* occur for a finite wing.

The tilting backward of the lift vector shown in Figure 5.6 is one way of visualizing the physical generation of induced drag. Two alternate ways are as follows:

1. The three-dimensional flow induced by the wing-tip vortices shown in Figures 5.4 and 5.5 simply alters the pressure distribution on the finite wing in such a fashion that a net pressure imbalance exists in the direction of V_∞ (i.e., drag is created). In this sense, induced drag is a type of "pressure drag."
2. The wing-tip vortices contain a large amount of translational and rotational kinetic energy. This energy has to come from somewhere; indeed, it is ultimately provided by the aircraft engine, which is the only source of power associated with the airplane. Since the energy of the vortices serves no useful purpose, this power is essentially lost. In effect, the extra power provided by the engine that goes into the vortices is the extra power required from the engine to overcome the induced drag.

Clearly, from the discussion in this section, the characteristics of a finite wing are *not* identical to the characteristics of its airfoil sections. Therefore, let us proceed to develop a theory that will enable us to analyze the aerodynamic properties of finite wings. In the process, we follow the road map given in Figure 5.7—keep in touch with this road map as we progress through the present chapter.

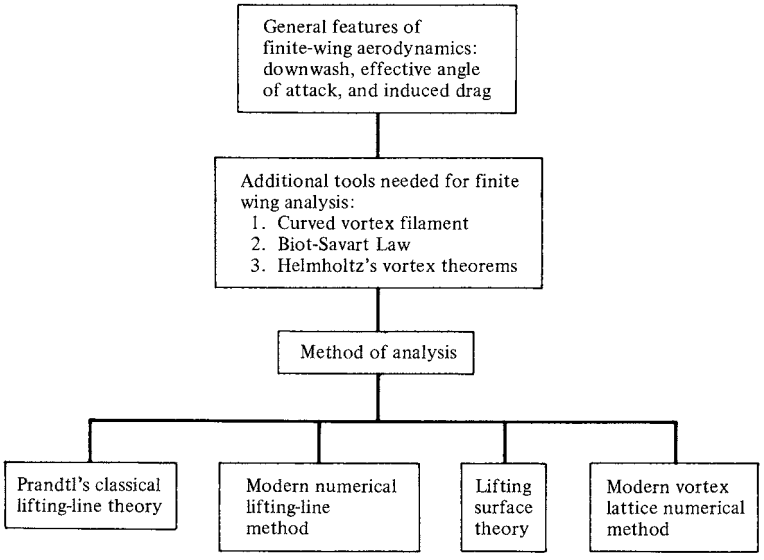


Figure 5.7 Road map for Chapter 5.

In this chapter, we note a difference in nomenclature. For the two-dimensional bodies considered in the previous chapters, the lift, drag, and moments per unit span have been denoted with primes, for example, L', D', and M', and the corresponding lift, drag, and moment coefficients have been denoted by lowercase letters, for example, c_l, c_d, and c_m. In contrast, the lift, drag, and moments on a complete three-dimensional body such as a finite wing are given without primes, for example, L, D, and M, and the corresponding lift, drag, and moment coefficients are given by capital letters, for example, C_L, C_D, and C_M. This distinction has already been mentioned in Section 1.5.

Finally, we note that the total drag on a subsonic finite wing in real life is the sum of the induced drag D_i, the skin friction drag D_f, and the pressure drag D_p due to flow separation. The latter two contributions are due to viscous effects (see Section 4.12 and Chapters 15 to 20). The sum of these two viscous-dominated drag contributions is called profile drag, as discussed in Section 4.3. The profile drag coefficient c_d for an NACA 2412 airfoil was given in Figure 4.11. At moderate angle of attack, the profile drag coefficient for a finite wing is essentially the same as for its airfoil sections. Hence, defining the profile drag coefficient as

$$c_d = \frac{D_f + D_p}{q_\infty S} \tag{5.2}$$

and the induced drag coefficient as

$$C_{D,i} = \frac{D_i}{q_\infty S} \tag{5.3}$$

the total drag coefficient for the finite wing C_D is given by

$$C_D = c_d + C_{D,i} \tag{5.4}$$

In Equation (5.4), the value of c_d is usually obtained from airfoil data, such as given in Figures 4.11 and 5.2*b*. The value of $C_{D,i}$ can be obtained from finite-wing theory as presented in this chapter. Indeed, one of the central objectives of the present chapter is to obtain an expression for induced drag and to study its variation with certain design characteristics of the finite wing. (See Chapter 5 of Reference 2 for an additional discussion of the characteristics of finite wings.)

5.2 THE VORTEX FILAMENT, THE BIOT-SAVART LAW, AND HELMHOLTZ'S THEOREMS

To establish a rational aerodynamic theory for a finite wing, we need to introduce a few additional aerodynamic tools. To begin with, we expand the concept of a vortex filament first introduced in Section 4.4. In Section 4.4, we discussed a *straight* vortex filament extending to $\pm\infty$. (Review the first paragraph of Section 4.4 before proceeding further.)

In general, a vortex filament can be *curved,* as shown in Figure 5.8. Here, only a portion of the filament is illustrated. The filament induces a flow field in the surrounding space. If the circulation is taken about any path enclosing the filament, a constant value Γ is obtained. Hence, the strength of the vortex filament is defined as Γ. Consider a directed segment of the filament **dl**, as shown in Figure 5.8. The radius vector from **dl** to an arbitrary point P in space is **r**. The segment **dl** induces a velocity at P equal to

$$\boxed{\mathbf{dV} = \frac{\Gamma}{4\pi} \frac{\mathbf{dl} \times \mathbf{r}}{|\mathbf{r}|^3}} \tag{5.5}$$

Equation (5.5) is called the *Biot-Savart law* and is one of the most fundamental relations in the theory of inviscid, incompressible flow. Its derivation is given in more advanced books (see, e.g., Reference 9). Here, we must accept it without

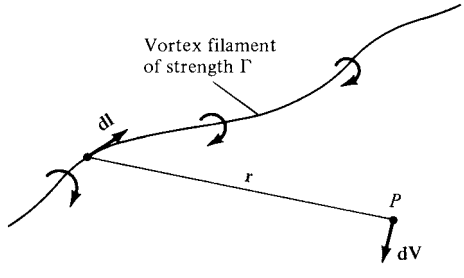


Figure 5.8 Vortex filament and illustration of the Biot-Savart law.

proof. However, you might feel more comfortable if we draw an analogy with electromagnetic theory. If the vortex filament in Figure 5.8 were instead visualized as a wire carrying an electrical current I, then the magnetic field strength **dB** induced at point P by a segment of the wire **dl** with the current moving in the direction of **dl** is

$$\mathbf{dB} = \frac{\mu I}{4\pi} \frac{\mathbf{dl} \times \mathbf{r}}{|\mathbf{r}|^3} \tag{5.6}$$

where μ is the permeability of the medium surrounding the wire. Equation (5.6) is identical in form to Equation (5.5). Indeed, the Biot-Savart law is a general result of potential theory, and potential theory describes electromagnetic fields as well as inviscid, incompressible flows. In fact, our use of the word "induced" in describing velocities generated by the presence of vortices, sources, etc. is a carry-over from the study of electromagnetic fields induced by electrical currents. When developing their finite-wing theory during the period 1911–1918, Prandtl and his colleagues even carried the electrical terminology over to the generation of drag, hence the term "induced" drag.

Return again to our picture of the vortex filament in Figure 5.8. Keep in mind that this single vortex filament and the associated Biot-Savart law [Equation (5.5)] are simply conceptual aerodynamic tools to be used for synthesizing more complex flows of an inviscid, incompressible fluid. They are, for all practical purposes, a solution of the governing equation for inviscid, incompressible flow—Laplace's equation (see Section 3.7)—and, by themselves, are not of particular value. However, when a number of vortex filaments are used in conjunction with a uniform freestream, it is possible to synthesize a flow which has a practical application. The flow over a finite wing is one such example, as we will soon see.

Let us apply the Biot-Savart law to a straight vortex filament of infinite length, as sketched in Figure 5.9. The strength of the filament is Γ. The velocity induced at point P by the directed segment of the vortex filament dl is given by

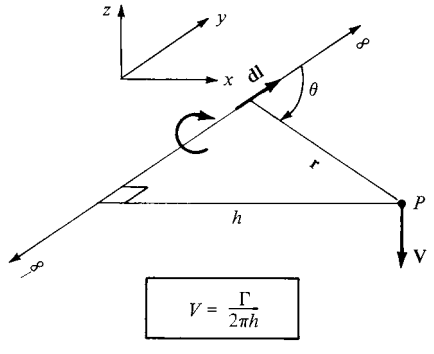


Figure 5.9 Velocity induced at point P by an infinite, straight vortex filament.

Equation (5.5). Hence, the velocity induced at P by the entire vortex filament is

$$\mathbf{V} = \int_{-\infty}^{\infty} \frac{\Gamma}{4\pi} \frac{\mathbf{dl} \times \mathbf{r}}{|\mathbf{r}|^3} \tag{5.7}$$

From the definition of the vector cross product (see Section 2.2), the direction of $\mathbf{V}$ is downward in Figure 5.9. The magnitude of the velocity, $V = |\mathbf{V}|$, is given by

$$V = \frac{\Gamma}{4\pi} \int_{-\infty}^{\infty} \frac{\sin\theta}{r^2}\, dl \tag{5.8}$$

In Figure 5.9, let h be the perpendicular distance from point P to the vortex filament. Then, from the geometry shown in Figure 5.9,

$$r = \frac{h}{\sin\theta} \tag{5.9a}$$

$$l = \frac{h}{\tan\theta} \tag{5.9b}$$

$$dl = -\frac{h}{\sin^2\theta}\, d\theta \tag{5.9c}$$

Substituting Equations (5.9a to c) in Equation (5.8), we have

$$V = \frac{\Gamma}{4\pi} \int_{-\infty}^{\infty} \frac{\sin\theta}{r^2} dl = -\frac{\Gamma}{4\pi h} \int_{\pi}^{0} \sin\theta\, d\theta$$

or

$$V = \frac{\Gamma}{2\pi h} \tag{5.10}$$

Thus, the velocity induced at a given point P by an infinite, straight vortex filament at a perpendicular distance h from P is simply $\Gamma/2\pi h$, which is precisely the result given by Equation (3.105) for a point vortex in two-dimensional flow. [Note that the minus sign in Equation (3.105) does not appear in Equation (5.10); this is because V in Equation (5.10) is simply the absolute magnitude of V, and hence it is positive by definition.]

Consider the *semi*-infinite vortex filament shown in Figure 5.10. The filament extends from point A to ∞. Point A can be considered a boundary of the flow. Let P be a point in the plane through A perpendicular to the filament. Then, by

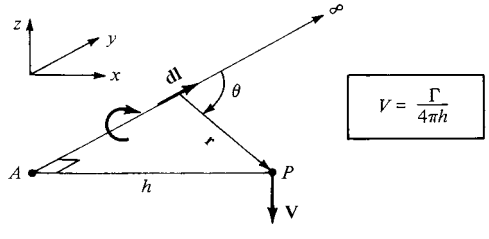


Figure 5.10 Velocity induced at point P by a semi-infinite straight vortex filament.

an integration similar to that above (try it yourself), the velocity induced at P by the semi-infinite vortex filament is

$$V = \frac{\Gamma}{4\pi h} \tag{5.11}$$

We use Equation (5.11) in the next section.

The great German mathematician, physicist, and physician Hermann von Helmholtz (1821–1894) was the first to make use of the vortex filament concept in the analysis of inviscid, incompressible flow. In the process, he established several basic principles of vortex behavior which have become known as Helmholtz's vortex theorems:

1. The strength of a vortex filament is constant along its length.
2. A vortex filament cannot end in a fluid; it must extend to the boundaries of the fluid (which can be $\pm\infty$) or form a closed path.

We make use of these theorems in the following sections.

Finally, let us introduce the concept of *lift distribution* along the span of a finite wing. Consider a given spanwise location y_1, where the local chord is c, the local geometric angle of attack is α, and the airfoil section is a given shape. The lift per unit span at this location is $L'(y_1)$. Now consider another location y_2 along the span, where c, α, and the airfoil shape may be different. (Most finite wings have a variable chord, with the exception of a simple rectangular wing. Also, many wings are geometrically twisted so that α is different at different spanwise locations—so-called geometric twist. If the tip is at a lower α than the root, the wing is said to have *washout;* if the tip is at a higher α than the root, the wing has *washin.* In addition, the wings on a number of modern airplanes have different airfoil sections along the span, with different values of $\alpha_{L=0}$; this is called *aerodynamic twist.*) Consequently, the lift per unit span at this different location, $L'(y_2)$, will, in general, be different from $L'(y_1)$. Therefore, there is a distribution of lift per unit span along the wing, that is, $L' = L'(y)$, as sketched in Figure 5.11. In turn, the circulation is also a function of y, $\Gamma(y) = L'(y)/\rho_\infty V_\infty$. Note from Figure 5.11 that the lift distribution goes to zero at the tips; that is because there is a pressure equalization from the bottom to the top of the wing precisely at $y = -b/2$ and $b/2$, and hence no lift is created at these points. The

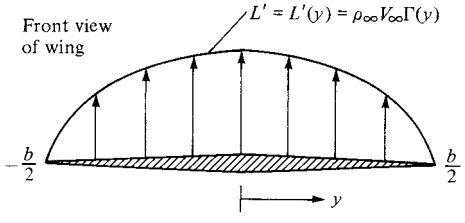


Figure 5.11 Sketch of the lift distribution along the span of a wing.

calculation of the lift distribution $L(y)$ [or the circulation distriution $\Gamma(y)$] is one of the central problems of finite-wing theory. It is addressed in the following sections.

In summary, we wish to calculate the induced drag, the total lift, and the lift distribution for a finite wing. This is the purpose of the remainder of this chapter.

5.3 PRANDTL'S CLASSICAL LIFTING-LINE THEORY

The first practical theory for predicting the aerodynamic properties of a finite wing was developed by Ludwig Prandtl and his colleagues at Göttingen, Germany, during the period 1911–1918, spanning World War I. The utility of Prandtl's theory is so great that it is still in use today for preliminary calculations of finite-wing characteristics. The purpose of this section is to describe Prandtl's theory and to lay the groundwork for the modern numerical methods described in subsequent sections.

Prandtl reasoned as follows. A vortex filament of strength Γ that is somehow bound to a fixed location in a flow—a so-called bound vortex—will experience a force $L' = \rho_\infty V_\infty \Gamma$ from the Kutta-Joukowski theorem. This bound vortex is in contrast to a *free vortex,* which moves with the same fluid elements throughout a flow. Therefore, let us replace a finite wing of span b with a bound vortex, extending from $y = -b/2$ to $y = b/2$, as sketched in Figure 5.12. However, due to Helmholtz's theorem, a vortex filament cannot end in the fluid. Therefore, assume the vortex filament continues as two free vortices trailing downstream from the wing tips to infinity, as also shown in Figure 5.12. This vortex (the bound plus the two free) is in the shape of a horseshoe, and therefore is called a *horseshoe vortex.*

A single horseshoe vortex is shown in Figure 5.13. Consider the downwash w induced along the bound vortex from $-b/2$ to $b/2$ by the horseshoe vortex. Examining Figure 5.13, we see that the bound vortex induces no velocity along itself; however, the two trailing vortices both contribute to the induced velocity

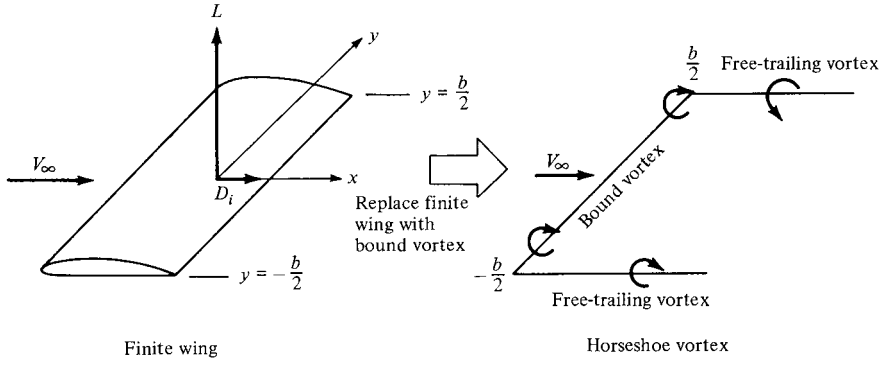


Figure 5.12 Replacement of the finite wing with a bound vortex.

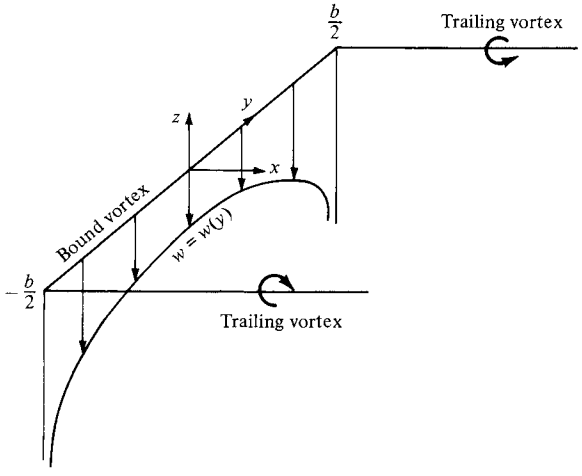


Figure 5.13 Downwash distribution along the y axis for a single horseshoe vortex.

along the bound vortex, and both contributions are in the downward direction. Consistent with the xyz coordinate system in Figure 5.13, such a downward velocity is negative; that is, w (which is in the z direction) is a negative value when directed downward and a positive value when directed upward. If the origin is taken at the center of the bound vortex, then the velocity at any point y along the bound vortex induced by the trailing semi-infinite vortices is, from Equation (5.11),

$$w(y) = -\frac{\Gamma}{4\pi(b/2 + y)} - \frac{\Gamma}{4\pi(b/2 - y)} \tag{5.12}$$

In Equation (5.12), the first term on the right-hand side is the contribution from the left trailing vortex (trailing from $-b/2$), and the second term is the contribution from the right trailing vortex (trailing from $b/2$). Equation (5.12) reduces to

$$w(y) = -\frac{\Gamma}{4\pi} \frac{b}{(b/2)^2 - y^2} \tag{5.13}$$

This variation of $w(y)$ is sketched in Figure 5.13. Note that w approaches $-\infty$ as y approaches $-b/2$ or $b/2$.

The downwash distribution due to the single horseshoe vortex shown in Figure 5.13 does not realistically simulate that of a finite wing; the downwash approaching an infinite value at the tips is especially disconcerting. During the early evolution of finite-wing theory, this problem perplexed Prandtl and his colleagues. After several years of effort, a resolution of this problem was obtained which, in hindsight, was simple and straightforward. Instead of representing the wing by a single horseshoe vortex, let us superimpose a large number of horseshoe vortices, each with a different length of the bound vortex, but with all the bound vortices

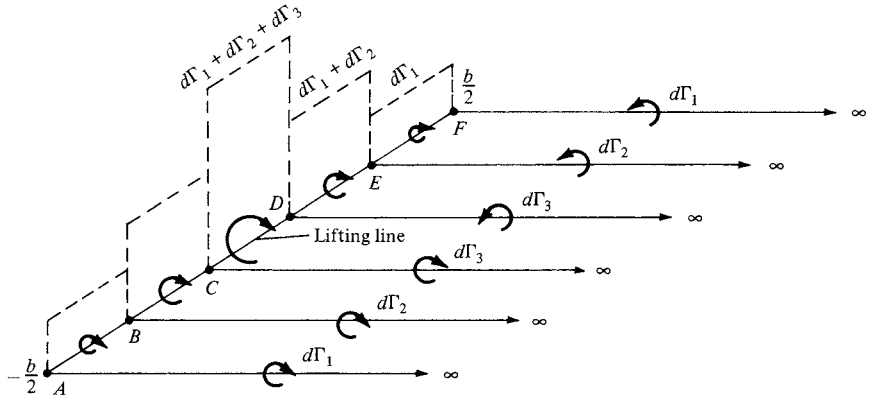


Figure 5.14 Superposition of a finite number of horseshoe vortices along the lifting line.

coincident along a single line, called the *lifting line*. This concept is illustrated in Figure 5.14, where only three horseshoe vortices are shown for the sake of clarity. In Figure 5.14, a horseshoe vortex of strength $d\Gamma_1$ is shown, where the bound vortex spans the entire wing from $-b/2$ to $b/2$ (from point A to point F). Superimposed on this is a second horseshoe vortex of strength $d\Gamma_2$, where its bound vortex spans only part of the wing, from point B to point E. Finally, superimposed on this is a third horseshoe vortex of strength $d\Gamma_3$, where its bound vortex spans only the part of the wing from point C to point D. As a result, the circulation varies along the line of bound vortices—the lifting line defined above. Along AB and EF, where only one vortex is present, the circulation is $d\Gamma_1$. However, along BC and DE, where two vortices are superimposed, the circulation is the sum of their strengths $d\Gamma_1 + d\Gamma_2$. Along CD, three vortices are superimposed, and hence the circulation is $d\Gamma_1 + d\Gamma_2 + d\Gamma_3$. This variation of Γ along the lifting line is denoted by the vertical bars in Figure 5.14. Also, note from Figure 5.14 that we now have a series of trailing vortices distributed over the span, rather than just two vortices trailing downstream of the tips as shown in Figure 5.13. The series of trailing vortices in Figure 5.14 represents pairs of vortices, each pair associated with a given horseshoe vortex. Note that the strength of each trailing vortex is equal to the *change in circulation* along the lifting line.

Let us extrapolate Figure 5.14 to the case where an *infinite number* of horseshoe vortices are superimposed along the lifting line, each with a vanishingly small strength $d\Gamma$. This case is illustrated in Figure 5.15. Note that the vertical bars in Figure 5.14 have now become a continuous distribution of $\Gamma(y)$ along the lifting line in Figure 5.15. The value of the circulation at the origin is Γ_0. Also, note that the finite number of trailing vortices in Figure 5.14 have become a *continuous vortex sheet* trailing downstream of the lifting line in Figure 5.15. This vortex sheet is parallel to the direction of V_∞. The total strength of the sheet integrated across the span of the wing is zero, because it consists of pairs of trailing vortices of equal strength but in opposite directions.

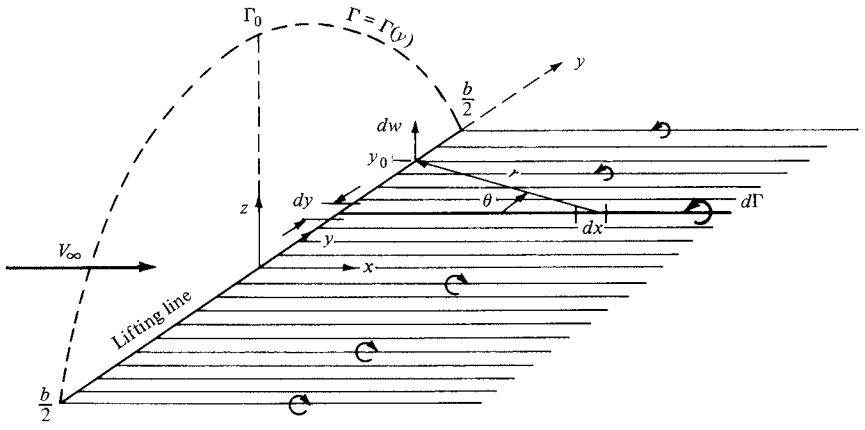


Figure 5.15 Superposition of an infinite number of horseshoe vortices along the lifting line.

Let us single out an infinitesimally small segment of the lifting line dy located at the coordinate y as shown in Figure 5.15. The circulation at y is $\Gamma(y)$, and the change in circulation over the segment dy is $d\Gamma = (d\Gamma/dy)\,dy$. In turn, the strength of the trailing vortex at y must equal the change in circulation $d\Gamma$ along the lifting line; this is simply an extrapolation of our result obtained for the strength of the finite trailing vortices in Figure 5.14. Consider more closely the trailing vortex of strength $d\Gamma$ that intersects the lifting line at coordinate y, as shown in Figure 5.15. Also consider the arbitrary location y_0 along the lifting line. Any segment of the trailing vortex dx will induce a velocity at y_0 with a magnitude and direction given by the Biot-Savart law, Equation (5.5). In turn, the velocity dw at y_0 induced by the entire semi-infinite trailing vortex located at y is given by Equation (5.11), which in terms of the picture given in Figure 5.15 yields

$$dw = -\frac{(d\Gamma/dy)\,dy}{4\pi(y_0 - y)} \tag{5.14}$$

The minus sign in Equation (5.14) is needed for consistency with the picture shown in Figure 5.15; for the trailing vortex shown, the direction of dw at y_0 is upward and hence is a positive value, whereas Γ is decreasing in the y direction, making $d\Gamma/dy$ a negative quantity. The minus sign in Equation (5.14) makes the positive dw consistent with the negative $d\Gamma/dy$.

The total velocity w induced at y_0 by the entire trailing vortex sheet is the summation of Equation (5.14) over all the vortex filaments, that is, the integral of Equation (5.14) from $-b/2$ to $b/2$:

$$\boxed{w(y_0) = -\frac{1}{4\pi}\int_{-b/2}^{b/2} \frac{(d\Gamma/dy)\,dy}{y_0 - y}} \tag{5.15}$$

Equation (5.15) is important in that it gives the value of the downwash at y_0 due to all the trailing vortices. (Keep in mind that although we label w as downwash, w is treated as positive in the upward direction in order to be consistent with the normal convention in an xyz rectangular coordinate system.)

Pause for a moment and assess the status of our discussion so far. We have replaced the finite wing with the model of a lifting line along which the circulation $\Gamma(y)$ varies continuously, as shown in Figure 5.15. In turn, we have obtained an expression for the downwash along the lifting line, given by Equation (5.15). However, our central problem still remains to be solved; that is, we want to *calculate* $\Gamma(y)$ for a given finite wing, along with its corresponding total lift and induced drag. Therefore, we must press on.

Return to Figure 5.6, which shows the local airfoil section of a finite wing. Assume this section is located at the arbitrary spanwise station y_0. From Figure 5.6, the induced angle of attack α_i is given by

$$\alpha_i(y_0) = \tan^{-1}\left(\frac{-w(y_0)}{V_\infty}\right) \tag{5.16}$$

[Note in Figure 5.6 that w is downward, and hence is a negative quantity. Since α_i in Figure 5.6 is positive, the negative sign in Equation (5.16) is necessary for consistency.] Generally, w is much smaller than V_∞, and hence α_i is a small angle, on the order of a few degrees at most. For small angles, Equation (5.16) yields

$$\alpha_i(y_0) = -\frac{w(y_0)}{V_\infty} \tag{5.17}$$

Substituting Equation (5.15) into (5.17), we obtain

$$\boxed{\alpha_i(y_0) = \frac{1}{4\pi V_\infty}\int_{-b/2}^{b/2}\frac{(d\Gamma/dy)\,dy}{y_0 - y}} \tag{5.18}$$

that is, an expression for the induced angle of attack in terms of the circulation distribution $\Gamma(y)$ along the wing.

Consider again the *effective* angle of attack α_{eff}, as shown in Figure 5.6. As explained in Section 5.1, α_{eff} is the angle of attack actually seen by the local airfoil section. Since the downwash varies across the span, then α_{eff} is also variable; $\alpha_{\text{eff}} = \alpha_{\text{eff}}(y_0)$. The lift coefficient for the airfoil section located at $y = y_0$ is

$$c_l = a_0[\alpha_{\text{eff}}(y_0) - \alpha_{L=0}] = 2\pi[\alpha_{\text{eff}}(y_0) - \alpha_{L=0}] \tag{5.19}$$

In Equation (5.19), the local section lift slope a_0 has been replaced by the thin airfoil theoretical value of $2\pi\,(\text{rad}^{-1})$. Also, for a wing with aerodynamic twist, the angle of zero lift $\alpha_{L=0}$ in Equation (5.19) varies with y_0. If there is no aerodynamic twist, $\alpha_{L=0}$ is constant across the span. In any event, $\alpha_{L=0}$ is a known property of the local airfoil sections. From the definition of lift coefficient and from the Kutta-Joukowski theorem, we have, for the local airfoil section located at y_0,

$$L' = \tfrac{1}{2}\rho_\infty V_\infty^2 c(y_0) c_l = \rho_\infty V_\infty \Gamma(y_0) \tag{5.20}$$

From Equation (5.20), we obtain

$$c_l = \frac{2\Gamma(y_0)}{V_\infty c(y_0)} \tag{5.21}$$

Substituting Equation (5.21) into (5.19) and solving for α_{eff}, we have

$$\alpha_{\text{eff}} = \frac{\Gamma(y_0)}{\pi V_\infty c(y_0)} + \alpha_{L=0} \tag{5.22}$$

The above results come into focus if we refer to Equation (5.1):

$$\alpha_{\text{eff}} = \alpha - \alpha_i \tag{5.1}$$

Substituting Equations (5.18) and (5.22) into (5.1), we obtain

$$\boxed{\alpha(y_0) = \frac{\Gamma(y_0)}{\pi V_\infty c(y_0)} + \alpha_{L=0}(y_0) + \frac{1}{4\pi V_\infty} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy)\, dy}{y_0 - y}} \tag{5.23}$$

the *fundamental equation of Prandtl's lifting-line theory;* it simply states that the geometric angle of attack is equal to the sum of the effective angle plus the induced angle of attack. In Equation (5.23), α_{eff} is expressed in terms of Γ, and α_i is expressed in terms of an integral containing $d\Gamma/dy$. Hence, Equation (5.23) is an integro-differential equation, in which the only unknown is Γ; all the other quantities, α, c, V_∞, and $\alpha_{L=0}$, are known for a finite wing of given design at a given geometric angle of attack in a freestream with given velocity. Thus, a solution of Equation (5.23) yields $\Gamma = \Gamma(y_0)$, where y_0 ranges along the span from $-b/2$ to $b/2$.

The solution $\Gamma = \Gamma(y_0)$ obtained from Equation (5.23) gives us the three main aerodynamic characteristics of a finite wing, as follows:

1. The lift distribution is obtained from the Kutta-Joukowski theorem:

$$L'(y_0) = \rho_\infty V_\infty \Gamma(y_0) \tag{5.24}$$

2. The total lift is obtained by integrating Equation (5.24) over the span:

$$L = \int_{-b/2}^{b/2} L'(y)\, dy$$

or

$$L = \rho_\infty V_\infty \int_{-b/2}^{b/2} \Gamma(y)\, dy \tag{5.25}$$

(Note that we have dropped the subscript on y, for simplicity.) The lift coefficient follows immediately from Equation (5.25):

$$C_L = \frac{L}{q_\infty S} = \frac{2}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y)\, dy \tag{5.26}$$

3. The induced drag is obtained by inspection of Figure 5.4. The induced drag per unit span is

$$D_i' = L_i' \sin \alpha_i$$

Since α_i is small, this relation becomes

$$D_i' = L_i' \alpha_i \tag{5.27}$$

The total induced drag is obtained by integrating Equation (5.27) over the span:

$$D_i = \int_{-b/2}^{b/2} L'(y) \alpha_i(y)\, dy \tag{5.28}$$

or

$$D_i = \rho_\infty V_\infty \int_{-b/2}^{b/2} \Gamma(y) \alpha_i(y)\, dy \tag{5.29}$$

In turn, the induced drag coefficient is

$$C_{D,i} = \frac{D_i}{q_\infty S} = \frac{2}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y) \alpha_i(y)\, dy \tag{5.30}$$

In Equations (5.27) to (5.30), $\alpha_i(y)$ is obtained from Equation (5.18). Therefore, in Prandtl's lifting-line theory the solution of Equation (5.23) for $\Gamma(y)$ is clearly the key to obtaining the aerodynamic characteristics of a finite wing. Before discussing the general solution of this equation, let us consider a special case, as outlined below.

5.3.1 Elliptical Lift Distribution

Consider a circulation distribution given by

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} \tag{5.31}$$

In Equation (5.31), note the following:

1. Γ_0 is the circulation at the origin, as shown in Figure 5.15.

2. The circulation varies elliptically with distance y along the span; hence, it is designated as an *elliptical circulation distribution*. Since $L'(y) = \rho_\infty V_\infty \Gamma(y)$, we also have

$$L'(y) = \rho_\infty V_\infty \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2}$$

Hence, we are dealing with an *elliptical lift distribution*.

3. $\Gamma(b/2) = \Gamma(-b/2) = 0$. Thus, the circulation, hence lift, properly goes to zero at the wing tips, as shown in Figure 5.15. We have not obtained Equation (5.31) as a direct solution of Equation (5.23); rather, we are simply stipulating a lift distribution that is elliptic. We now ask the question, 'What are the aerodynamic properties of a finite wing with such an elliptic lift distribution?'

First, let us calculate the downwash. Differentiating Equation (5.31), we obtain

$$\frac{d\Gamma}{dy} = -\frac{4\Gamma_0}{b^2}\frac{y}{(1-4y^2/b^2)^{1/2}} \tag{5.32}$$

Substituting Equation (5.32) into (5.15), we have

$$w(y_0) = \frac{\Gamma_0}{\pi b^2}\int_{-b/2}^{b/2}\frac{y}{(1-4y^2/b^2)^{1/2}(y_0-y)}\,dy \tag{5.33}$$

The integral can be evaluated easily by making the substitution

$$y = \frac{b}{2}\cos\theta \quad dy = -\frac{b}{2}\sin\theta\,d\theta$$

Hence, Equation (5.33) becomes

$$w(\theta_0) = -\frac{\Gamma_0}{2\pi b}\int_{\pi}^{0}\frac{\cos\theta}{\cos\theta_0-\cos\theta}\,d\theta$$

or

$$w(\theta_0) = -\frac{\Gamma_0}{2\pi b}\int_{0}^{\pi}\frac{\cos\theta}{\cos\theta-\cos\theta_0}\,d\theta \tag{5.34}$$

The integral in Equation (5.34) is the standard form given by Equation (4.26) for $n = 1$. Hence, Equation (5.34) becomes

$$\boxed{w(\theta_0) = -\frac{\Gamma_0}{2b}} \tag{5.35}$$

which states the interesting and important result that the *downwash is constant over the span for an elliptical lift distribution*. In turn, from Equation (5.17), we obtain, for the induced angle of attack,

$$\alpha_i = -\frac{w}{V_\infty} = \frac{\Gamma_0}{2bV_\infty} \tag{5.36}$$

For an elliptic lift distribution, the induced angle of attack is also constant along the span. Note from Equations (5.35) and (5.36) that both the downwash and induced angle of attack go to zero as the wing span becomes infinite—which is consistent with our previous discussions on airfoil theory.

A more useful expression for α_i can be obtained as follows. Substituting Equation (5.31) into (5.25), we have

$$L = \rho_\infty V_\infty \Gamma_0 \int_{-b/2}^{b/2}\left(1-\frac{4y^2}{b^2}\right)^{1/2} dy \tag{5.37}$$

Again, using the transformation $y = (b/2)\cos\theta$, Equation (5.37) readily integrates to

$$L = \rho_\infty V_\infty \Gamma_0 \frac{b}{2} \int_0^\pi \sin^2\theta \, d\theta = \rho_\infty V_\infty \Gamma_0 \frac{b}{4}\pi \tag{5.38}$$

Solving Equation (5.38) for Γ_0, we have

$$\Gamma_0 = \frac{4L}{\rho_\infty V_\infty b\pi} \tag{5.39}$$

However, $L = \frac{1}{2}\rho_\infty V_\infty^2 S C_L$. Hence, Equation (5.39) becomes

$$\Gamma_0 = \frac{2V_\infty S C_L}{b\pi} \tag{5.40}$$

Substituting Equation (5.40) into (5.36), we obtain

$$\alpha_i = \frac{2V_\infty S C_L}{b\pi} \frac{1}{2bV_\infty}$$

or

$$\alpha_i = \frac{SC_L}{\pi b^2} \tag{5.41}$$

An important geometric property of a finite wing is the *aspect ratio,* denoted by AR and defined as

$$\text{AR} \equiv \frac{b^2}{S}$$

Hence, Equation (5.41) becomes

$$\boxed{\alpha_i = \frac{C_L}{\pi \text{AR}}} \tag{5.42}$$

Equation (5.42) is a useful expression for the induced angle of attack, as shown below.

The induced drag coefficient is obtained from Equation (5.30), noting that α_i is constant:

$$C_{D,i} = \frac{2\alpha_i}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y)\, dy = \frac{2\alpha_i \Gamma_0}{V_\infty S} \frac{b}{2} \int_0^\pi \sin^2\theta \, d\theta - \frac{\pi \alpha_i \Gamma_0 b}{2V_\infty S} \tag{5.42a}$$

Substituting Equations (5.40) and (5.42) into (5.42*a*), we obtain

$$C_{D,i} = \frac{\pi b}{2V_\infty S} \left(\frac{C_L}{\pi \text{AR}} \right) \frac{2V_\infty S C_L}{b\pi}$$

or

$$\boxed{C_{D,i} = \frac{C_L^2}{\pi \text{AR}}} \tag{5.43}$$

Equation (5.43) is an important result. It states that the induced drag coefficient is directly proportional to the square of the lift coefficient. The dependence of induced drag on the lift is not surprising, for the following reason. In Section 5.1, we saw that induced drag is a consequence of the presence of the wing-tip vortices, which in turn are produced by the difference in pressure between the lower and upper wing surfaces. The lift is produced by this same pressure difference. Hence, induced drag is intimately related to the production of lift on a finite wing; indeed, induced drag is frequently called the *drag due to lift*. Equation (5.43) dramatically illustrates this point. Clearly, an airplane cannot generate lift for free; the induced drag is the price for the generation of lift. The power required from an aircraft engine to overcome the induced drag is simply the power required to generate the lift of the aircraft. Also, note that because $C_{D,i} \propto C_L^2$, the induced drag coefficient increases rapidly as C_L increases and becomes a substantial part of the total drag coefficient when C_L is high (e.g., when the airplane is flying slowly such as on takeoff or landing). Even at relatively high cruising speeds, induced drag is typically 25 percent of the total drag.

Another important aspect of induced drag is evident in Equation (5.43); that is, $C_{D,i}$ is *inversely proportional to aspect ratio*. Hence, to reduce the induced drag, we want a finite wing with the highest possible aspect ratio. Wings with high and low aspect ratios are sketched in Figure 5.16. Unfortunately, the design of very high aspect ratio wings with sufficient structural strength is difficult. Therefore, the aspect ratio of a conventional aircraft is a compromise between conflicting aerodynamic and structural requirements. It is interesting to note that the aspect ratio of the 1903 Wright Flyer was 6 and that today the aspect ratios of conventional subsonic aircraft range typically from 6 to 8. (Exceptions are the Lockheed U-2 high-altitude reconnaissance aircraft with AR = 14.3 and sailplanes with aspect ratios as high as 51. For example, the Schempp-Hirth

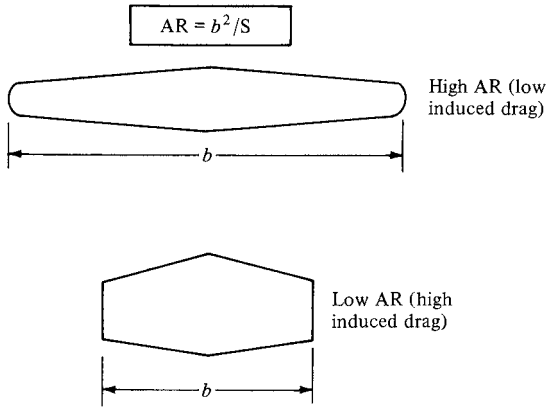


Figure 5.16 Schematic of high- and low-aspect-ratio wings.

Nimbus 4 sailplane, designed in 1994 with over 100 built by 2004, has an aspect ratio of 39. The ETA sailplane, designed in 2000 with 6 built by 2004, has an aspect ratio of 51.3.)

Another property of the elliptical lift distribution is as follows. Consider a wing with no geometric twist (i.e., α is constant along the span) and no aerodynamic twist (i.e., $\alpha_{L=0}$ is constant along the span). From Equation (5.42), we have seen that α_i is constant along the span. Hence, $\alpha_{\text{eff}} = \alpha - \alpha_i$ is also constant along the span. Since the local section lift coefficient c_l is given by

$$c_l = a_0(\alpha_{\text{eff}} - \alpha_{L=0})$$

then assuming that a_0 is the same for each section ($a_0 = 2\pi$ from thin airfoil theory), c_l must be constant along the span. The lift per unit span is given by

$$L'(y) = q_\infty c c_l \tag{5.44}$$

Solving Equation (5.44) for the chord, we have

$$c(y) = \frac{L'(y)}{q_\infty c_l} \tag{5.45}$$

In Equation (5.45), q_∞ and c_l are constant along the span. However, $L'(y)$ varies elliptically along the span. Thus, Equation (5.45) dictates that for such an elliptic lift distribution, the *chord must vary elliptically along the span;* that is, for the conditions given above, the *wing planform is elliptical*.

The related characteristics—the elliptic lift distribution, the elliptic planform, and the constant downwash—are sketched in Figure 5.17. Although an elliptical lift distribution may appear to be a restricted, isolated case, in reality it gives a reasonable approximation for the induced drag coefficient for an arbitrary finite wing. The form of $C_{D,i}$ given by Equation (5.43) is only slightly modified for

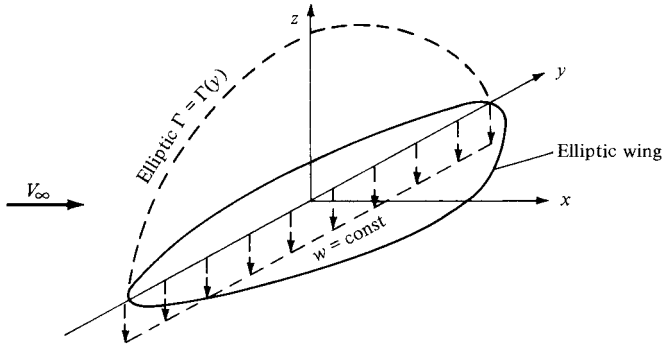


Figure 5.17 Illustration of the related quantities: an elliptic lift distribution, elliptic planform, and constant downwash.

the general case. Let us now consider the case of a finite wing with a general lift distribution.

5.3.2 General Lift Distribution

Consider the transformation

$$y = -\frac{b}{2}\cos\theta \tag{5.46}$$

where the coordinate in the spanwise direction is now given by θ, with $0 \leq \theta \leq \pi$. In terms of θ, the elliptic lift distribution given by Equation (5.31) is written as

$$\Gamma(\theta) = \Gamma_0 \sin\theta \tag{5.47}$$

Equation (5.47) hints that a Fourier sine series would be an appropriate expression for the general circulation distribution along an arbitrary finite wing. Hence, assume for the general case that

$$\Gamma(\theta) = 2bV_\infty \sum_1^N A_n \sin n\theta \tag{5.48}$$

where as many terms N in the series can be taken as we desire for accuracy. The coefficients A_n (where $n = 1, \ldots, N$) in Equation (5.48) are unknowns; however, they must satisfy the fundamental equation of Prandtl's lifting-line theory; that is, the A_n's must satisfy Equation (5.23). Differentiating Equation (5.48), we obtain

$$\frac{d\Gamma}{dy} = \frac{d\Gamma}{d\theta}\frac{d\theta}{dy} = 2bV_\infty \sum_1^N nA_n \cos n\theta \frac{d\theta}{dy} \tag{5.49}$$

Substituting Equations (5.48) and (5.49) into (5.23), we obtain

$$\alpha(\theta_0) = \frac{2b}{\pi c(\theta_0)} \sum_1^N A_n \sin n\theta_0 + \alpha_{L=0}(\theta_0) + \frac{1}{\pi}\int_0^\pi \frac{\sum_1^N nA_n \cos n\theta}{\cos\theta - \cos\theta_0} d\theta \tag{5.50}$$

The integral in Equation (5.50) is the standard form given by Equation (4.26). Hence, Equation (5.50) becomes

$$\boxed{\alpha(\theta_0) = \frac{2b}{\pi c(\theta_0)} \sum_1^N A_n \sin n\theta_0 + \alpha_{L=0}(\theta_0) + \sum_1^N nA_n \frac{\sin n\theta_0}{\sin\theta_0}} \tag{5.51}$$

Examine Equation (5.51) closely. It is evaluated at a given spanwise location; hence, θ_0 is specified. In turn, b, $c(\theta_0)$, and $\alpha_{L=0}(\theta_0)$ are known quantities from the geometry and airfoil section of the finite wing. The only unknowns in Equation (5.51) are the A_n's. Hence, written at a given spanwise location (a specified θ_0), Equation (5.51) is one algebraic equation with N unknowns, $A_1, A_2, \ldots, A_n$. However, let us choose N different spanwise stations, and let us evaluate Equation (5.51) at each of these N stations. We then obtain a system of N independent

algebraic equations with N unknowns, namely, $A_1, A_2, \ldots, A_N$. In this fashion, actual numerical values are obtained for the A_n's—numerical values that ensure that the general circulation distribution given by Equation (5.48) satisfies the fundamental equation of finite-wing theory, Equation (5.23).

Now that $\Gamma(\theta)$ is known via Equation (5.48), the lift coefficient for the finite wing follows immediately from the substitution of Equation (5.48) into (5.26):

$$C_L = \frac{2}{V_\infty S}\int_{-b/2}^{b/2}\Gamma(y)\,dy = \frac{2b^2}{S}\sum_1^N A_n\int_0^\pi \sin n\theta\sin\theta\,d\theta \tag{5.52}$$

In Equation (5.52), the integral is

$$\int_0^\pi \sin n\theta\sin\theta\,d\theta = \begin{cases}\pi/2 & \text{for } n=1\\ 0 & \text{for } n\neq 1\end{cases}$$

Hence, Equation (5.52) becomes

$$C_L = A_1\pi\frac{b^2}{S} = A_1\pi\,\mathrm{AR} \tag{5.53}$$

Note that C_L depends only on the leading coefficient of the Fourier series expansion. (However, although C_L depends on A_1 only, we must solve for all the A_n's simultaneously in order to obtain A_1.)

The induced drag coefficient is obtained from the substitution of Equation (5.48) into Equation (5.30) as follows:

$$C_{D,i} = \frac{2}{V_\infty S}\int_{-b/2}^{b/2}\Gamma(y)\alpha_i(y)\,dy \tag{5.54}$$

$$= \frac{2b^2}{S}\int_0^\pi\left(\sum_1^N A_n\sin n\theta\right)\alpha_i(\theta)\sin\theta\,d\theta$$

The induced angle of attack $\alpha_i(\theta)$ in Equation (5.54) is obtained from the substitution of Equations (5.46) and (5.49) into (5.18), which yields

$$\alpha_i(y_0) = \frac{1}{4\pi V_\infty}\int_{-b/2}^{b/2}\frac{(d\Gamma/dy)\,dy}{y_0 - y}$$

$$= \frac{1}{\pi}\sum_1^N nA_n\int_0^\pi\frac{\cos n\theta}{\cos\theta - \cos\theta_0}d\theta \tag{5.55}$$

The integral in Equation (5.55) is the standard form given by Equation (4.26). Hence, Equation (5.55) becomes

$$\alpha_i(\theta_0) = \sum_1^N nA_n\frac{\sin n\theta_0}{\sin\theta_0} \tag{5.56}$$

In Equation (5.56), θ_0 is simply a dummy variable which ranges from 0 to π across the span of the wing; it can therefore be replaced by θ, and Equation (5.56)

can be written as

$$\alpha_i(\theta) = \sum_1^N nA_n \frac{\sin n\theta}{\sin \theta} \tag{5.57}$$

Substituting Equation (5.57) into (5.54), we have

$$C_{D,i} = \frac{2b^2}{S} \int_0^\pi \left(\sum_1^N A_n \sin n\theta \right) \left(\sum_1^N nA_n \sin n\theta \right) d\theta \tag{5.58}$$

Examine Equation (5.58) closely; it involves the product of two summations. Also, note that, from the standard integral,

$$\int_0^\pi \sin m\theta \sin k\theta = \begin{cases} 0 & \text{for } m \neq k \\ \pi/2 & \text{for } m = k \end{cases} \tag{5.59}$$

Hence, in Equation (5.58), the mixed product terms involving unequal subscripts (such as A_1A_2, A_2A_4) are, from Equation (5.59), equal to zero. Hence, Equation (5.58) becomes

$$\begin{aligned} C_{D,i} &= \frac{2b^2}{S} \left(\sum_1^N nA_n^2 \right) \frac{\pi}{2} = \pi \text{AR} \sum_1^N nA_n^2 \\ &= \pi \text{AR} \left(A_1^2 + \sum_2^N nA_n^2 \right) \\ &= \pi \text{AR}\, A_1^2 \left[1 + \sum_2^N n \left(\frac{A_n}{A_1} \right)^2 \right] \end{aligned} \tag{5.60}$$

Substituting Equation (5.53) for C_L into Equation (5.60), we obtain

$$\boxed{C_{D,i} = \frac{C_L^2}{\pi \text{AR}} (1 + \delta)} \tag{5.61}$$

where $\delta = \sum_2^N n(A_n/A_1)^2$. Note that $\delta \geq 0$; hence, the factor $1 + \delta$ in Equation (5.61) is either greater than 1 or at least equal to 1. Let us define a span efficiency factor, e, as $e = (1 + \delta)^{-1}$. Then Equation (5.61) can be written as

$$\boxed{C_{D,i} = \frac{C_L^2}{\pi e \text{AR}}} \tag{5.62}$$

where $e \leq 1$. Comparing Equations (5.61) and (5.62) for the general lift distribution with Equation (5.43) for the elliptical lift distribution, note that $\delta = 0$ and $e = 1$ for the elliptical lift distribution. Hence, the lift distribution which yields minimum induced drag is the *elliptical lift distribution*. This is why we have a practical interest in the elliptical lift distribution.

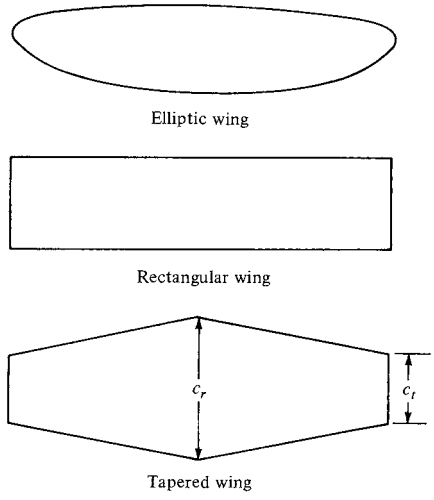


Figure 5.18 Various planforms for straight wings.

Recall that for a wing with no aerodynamic twist and no geometric twist, an elliptical lift distribution is generated by a wing with an elliptical planform, as sketched at the top of Figure 5.18. Several aircraft have been designed in the past with elliptical wings; the most famous, perhaps, being the British Spitfire from World War II, shown in Figure 5.19. However, elliptic planforms are more expensive to manufacture than, say, a simple rectangular wing as sketched in the middle of Figure 5.18. On the other hand, a rectangular wing generates a lift distribution far from optimum. A compromise is the tapered wing shown at the bottom of Figure 5.18. The tapered wing can be designed with a taper ratio, that is, tip chord/root chord $\equiv c_t/c_r$, such that the lift distribution closely approximates the elliptic case. The variation of δ as a function of taper ratio for wings of different aspect ratio is illustrated in Figure 5.20. Such calculations of δ were first performed by the famous English aerodynamicist, Hermann Glauert and published in Reference 18 in the year 1926. Glauert used only four terms in the series expansion given in Equation (5.60). The results shown in Figure 5.20 are based on the recent computer calculations carried out by B.W. McCormick at Penn State University using the equivalent of 50 terms in the series expansion. Note from Figure 5.20 that a tapered wing can be designed with an induced drag coefficient reasonably close to the minimum value. In addition, tapered wings with straight leading and trailing edges are considerably easier to manufacture than elliptic planforms. Therefore, most conventional aircraft employ tapered rather than elliptical wing planforms.

5.3.3 Effect of Aspect Ratio

Returning to Equations (5.61) and (5.62), note that the induced drag coefficient for a finite wing with a general lift distribution is inversely proportional to the

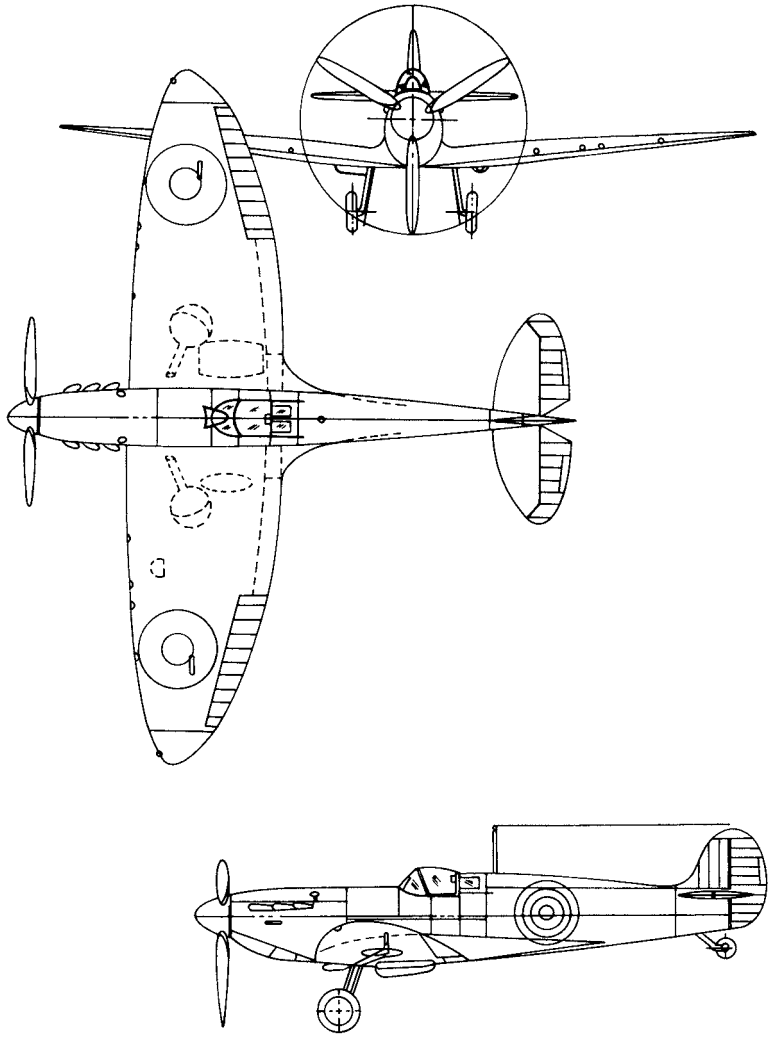


Figure 5.19 Three views of the Supermarine Spitfire, a famous British World War II fighter.

aspect ratio, as was discussed earlier in conjunction with the case of the elliptic lift distribution. Note that AR, which typically varies from 6 to 22 for standard subsonic airplanes and sailplanes, has a much stronger effect on $C_{D,i}$ than the value of δ, which from Figure 5.20 varies only by about 10 percent over the practical range of taper ratio. Hence, the primary design factor for minimizing induced drag is not the closeness to an elliptical lift distribution, but rather, the ability to make the aspect ratio as large as possible. The determination that $C_{D,i}$ is inversely proportional to AR was one of the great victories of Prandtl's lifting-line theory. In 1915, Prandtl verified this result with a series of classic experiments wherein the lift and drag of seven rectangular wings with different aspect ratios

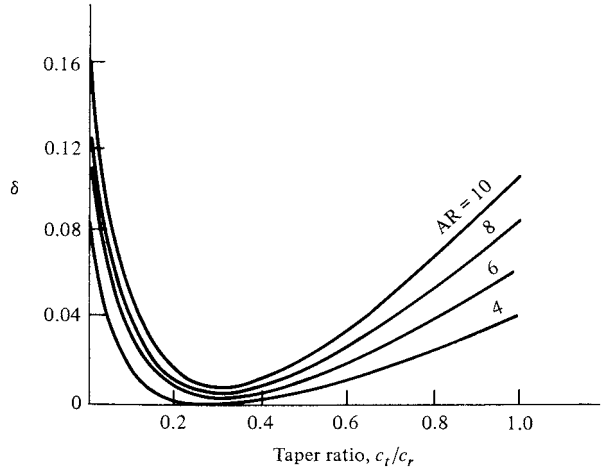


Figure 5.20 Induced drag factor δ as a function of taper ratio. Computer calculations using the equivalent of 50 terms in the series expansion, Eq. (5.60). (*Data Source:* McCormick, B. W., *Aerodynamics, Aeronautics, and Flight Mechanics,* John Wiley & Sons, New York, 1979.)

were measured. The data are given in Figure 5.21. Recall from Equation (5.4), that the total drag of a finite wing is given by

$$C_D = c_d + \frac{C_L^2}{\pi e \mathrm{AR}} \tag{5.63}$$

The parabolic variation of C_D with C_L as expressed in Equation (5.63) is reflected in the data of Figure 5.21. This is a plot of lift coefficient versus drag coefficient, and is called a *drag polar*. Equation (5.63) is similarly the equation of a drag polar. If we consider two wings with different aspect ratios AR_1 and AR_2, Equation (5.63) gives the drag coefficients $C_{D,1}$ and $C_{D,2}$ for the two wings as

$$C_{D,1} = c_d + \frac{C_L^2}{\pi e \mathrm{AR}_1} \tag{5.64a}$$

and

$$C_{D,2} = c_d + \frac{C_L^2}{\pi e \mathrm{AR}_2} \tag{5.64b}$$

Assume that the wings are at the same C_L. Also, since the airfoil section is the same for both wings, c_d is essentially the same. Moreover, the variation of e between the wings is only a few percent and can be ignored. Hence, subtracting Equation (5.64*b*) from (5.64*a*), we obtain

$$C_{D,1} = C_{D,2} + \frac{C_L^2}{\pi e}\left(\frac{1}{\mathrm{AR}_1} - \frac{1}{\mathrm{AR}_2}\right) \tag{5.65}$$

Equation (5.65) can be used to scale the data of a wing with aspect ratio AR_2 to correspond to the case of another aspect ratio AR_1. For example, Prandtl scaled

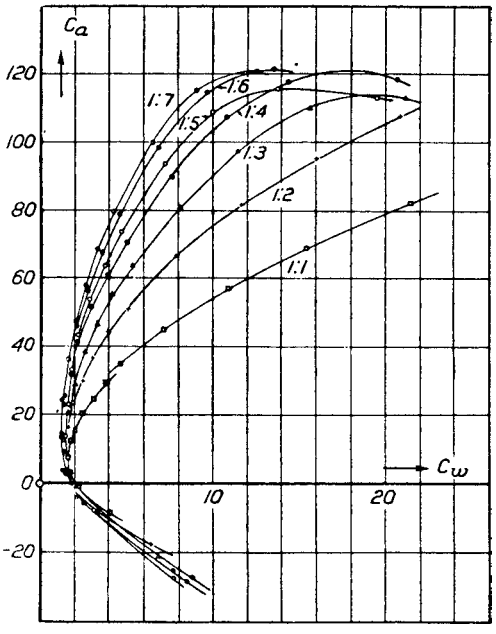


Figure 5.21 Prandtl's classic rectangular wing data for seven different aspect ratios from 1 to 7; variation of lift coefficient versus drag coefficient. For historical interest, we reproduce here Prandtl's actual graphs. Note that, in his nomenclature, C_a = lift coefficient and C_w = drag coefficient. Also, the numbers on both the ordinate and abscissa are 100 times the actual values of the coefficients. (*Source:* Prandtl, L., *Applications of Modern Hydrodynamics to Aeronautics,* NACA Report No. 116, 1921.)

the data of Figure 5.21 to correspond to a wing with an aspect ratio of 5. For this case, Equation (5.65) becomes

$$C_{D,1} = C_{D,2} + \frac{C_L^2}{\pi e}\left(\frac{1}{5} - \frac{1}{\text{AR}_2}\right) \tag{5.66}$$

Inserting the respective values of $C_{D,2}$ and AR_2 from Figure 5.21 into Equation (5.66), Prandtl found that the resulting data for $C_{D,1}$, versus C_L collapsed to essentially the same curve, as shown in Figure 5.22. Hence, the inverse dependence of $C_{D,i}$ on AR was substantially verified as early as 1915.

There are two primary differences between airfoil and finite-wing properties. We have discussed one difference, namely, a finite wing generates induced drag. However, a second major difference appears in the lift slope. In Figure 4.9, the lift slope for an airfoil was defined as $a_0 \equiv dc_l/d\alpha$. Let us denote the lift slope for a

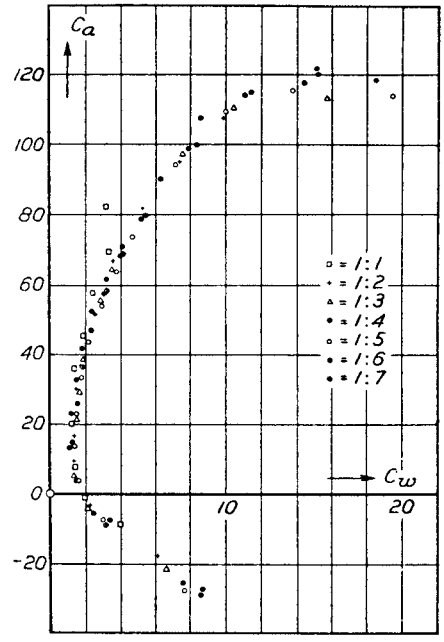


Figure 5.22 Data of Figure 5.21 scaled by Prandtl to an aspect ratio of 5.

finite wing as $a \equiv dC_L/d\alpha$. When the lift slope of a finite wing is compared with that of its airfoil section, we find that $a < a_0$. To see this more clearly, return to Figure 5.6, which illustrates the influence of downwash on the flow over a local airfoil section of a finite wing. Note that although the geometric angle of attack of the finite wing is α, the airfoil section effectively senses a smaller angle of attack, namely, α_{eff}, where $\alpha_{\text{eff}} = \alpha - \alpha_i$. For the time being, consider an elliptic wing with no twist; hence, α_i and α_{eff} are both constant along the span. Moreover, c_l is also constant along the span, and therefore $C_L = c_l$. Assume that we plot C_L for the finite wing versus α_{eff}, as shown at the top of Figure 5.23. Because we are using α_{eff} the lift slope corresponds to that for an infinite wing a_0. However, in real life, our naked eyes cannot see α_{eff}; instead, what we actually observe is a finite wing with a certain angle between the chord line and the relative wind; that is, in practice, we always observe the geometric angle of attack α. Hence, C_L for a finite wing is generally given as a function of α, as sketched at the bottom of Figure 5.23. Since $\alpha > \alpha_{\text{eff}}$, the bottom abscissa is stretched, and hence the bottom lift curve is less inclined; it has a slope equal to a, and Figure 5.23 clearly shows that $a < a_0$. The effect of a finite wing is to *reduce* the lift slope. Also, recall that at zero lift, there are no induced effects; that is, $\alpha_i = C_{D,i} = 0$. Thus, when $C_L = 0$, $\alpha = \alpha_{\text{eff}}$. As a result, $\alpha_{L=0}$ is the same for the finite and the infinite wings, as shown in Figure 5.23.

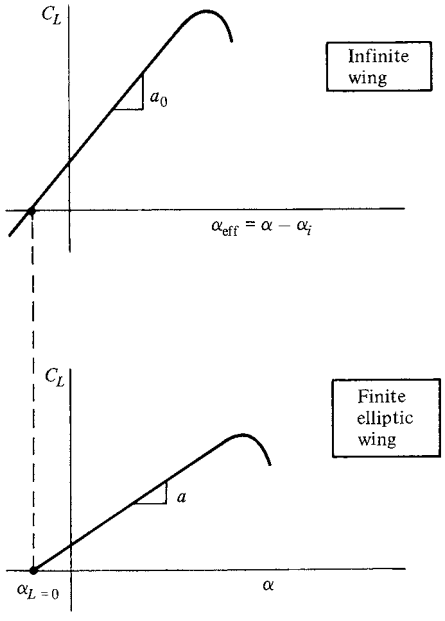


Figure 5.23 Lift curves for an infinite wing versus a finite elliptic wing.

The values of a_0 and a are related as follows. From the top of Figure 5.23,

$$\frac{dC_L}{d(\alpha - \alpha_i)} = a_0$$

Integrating, we find

$$C_L = a_0(\alpha - \alpha_i) + \text{const} \tag{5.67}$$

Substituting Equation (5.42) into (5.67), we obtain

$$C_L = a_0\left(\alpha - \frac{C_L}{\pi \text{AR}}\right) + \text{const} \tag{5.68}$$

Differentiating Equation (5.68) with respect to α, and solving for $dC_L/d\alpha$, we obtain

$$\frac{dC_L}{d\alpha} = a = \frac{a_0}{1 + a_0/\pi \text{AR}} \tag{5.69}$$

Equation (5.69) gives the desired relation between a_0 and a for an elliptic finite wing. For a finite wing of general planform, Equation (5.69) is slightly modified, as given below:

$$a = \frac{a_0}{1 + (a_0/\pi \text{AR})(1 + \tau)} \tag{5.70}$$

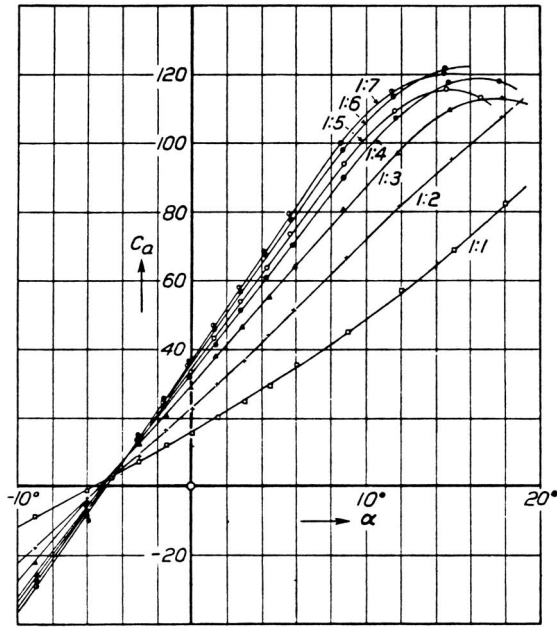


Figure 5.24 Prandtl's classic rectangular wing data. Variation of lift coefficient with angle of attack for seven different aspect ratios from 1 to 7. Nomenclature and scale are the same as given in Figure 5.21.

In Equation (5.70), τ is a function of the Fourier coefficients A_n. Values of τ were first calculated by Glauert in the early 1920s and were published in Reference 18, which should be consulted for more details. Values of τ typically range between 0.05 and 0.25.

Of most importance in Equations (5.69) and (5.70) is the aspect-ratio variation. Note that for low-AR wings, a substantial difference can exist between a_0 and a. However, as AR $\rightarrow \infty$, $a \rightarrow a_0$. The effect of aspect ratio on the lift curve is dramatically shown in Figure 5.24, which gives classic data obtained on rectangular wings by Prandtl in 1915. Note the reduction in $dC_L/d\alpha$ as AR is reduced. Moreover, using the equations obtained above, Prandtl scaled the data in Figure 5.24 to correspond to an aspect ratio of 5; his results collapsed to essentially the same curve, as shown in Figure 5.25. In this manner, the aspect-ratio variation given in Equations (5.69) and (5.70) was confirmed as early as the year 1915.

5.3.4 Physical Significance

Consider again the basic model underlying Prandtl's lifting-line theory. Return to Figure 5.15 and study it carefully. An infinite number of infinitesimally weak horseshoe vortices are superimposed in such a fashion as to generate a lifting line which spans the wing, along with a vortex sheet which trails downstream. This

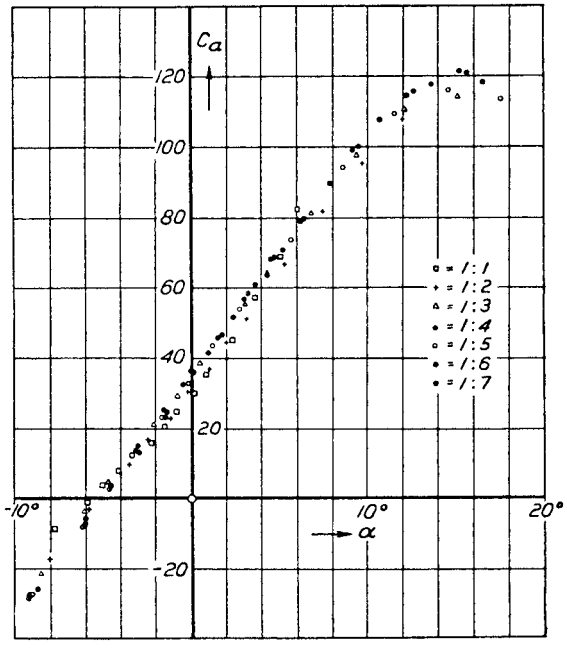


Figure 5.25 Data of Figure 5.24 scaled by Prandtl to an aspect ratio of 5.

trailing-vortex sheet is the instrument that induces downwash at the lifting line. At first thought, you might consider this model to be somewhat abstract—a mathematical convenience that somehow produces surprisingly useful results. However, to the contrary, the model shown in Figure 5.15 has real physical significance. To see this more clearly, return to Figure 5.3. Note that in the three-dimensional flow over a finite wing, the streamlines leaving the trailing edge from the top and bottom surfaces are in different directions; that is, there is a discontinuity in the tangential velocity at the trailing edge. We know from Chapter 4 that a discontinuous change in tangential velocity is theoretically allowed across a vortex sheet. In real life, such discontinuities do not exist; rather, the different velocities at the trailing edge generate a thin region of large velocity gradients—a thin region of shear flow with very large vorticity. Hence, a sheet of vorticity actually trails downstream from the trailing edge of a finite wing. This sheet tends to roll up at the edges and helps to form the wing-tip vortices sketched in Figure 5.4. Thus, Prandtl's lifting-line model with its trailing-vortex sheet is physically consistent with the actual flow downstream of a finite wing.

EXAMPLE 5.1

Consider a finite wing with an aspect ratio of 8 and a taper ratio of 0.8. The airfoil section is thin and symmetric. Calculate the lift and induced drag coefficients for the wing when it is at an angle of attack of 5°. Assume that $\delta = \tau$.

■ Solution

From Figure 5.20, $\delta = 0.055$. Hence, from the stated assumption, τ also equals 0.055. From Equation (5.70), assuming $a_0 = 2\pi$ from thin airfoil theory,

$$a = \frac{a_0}{1 + a_0/\pi \text{AR}(1+\tau)} = \frac{2\pi}{1 + 2\pi(1.055)/8\pi} = 4.97 \text{ rad}^{-1}$$

$$= 0.0867 \text{ degree}^{-1}$$

Since the airfoil is symmetric, $\alpha_{L=0} = 0^\circ$. Thus,

$$C_L = a\alpha = (0.0867 \text{ degree}^{-1}(5^\circ) = \boxed{0.4335}$$

From Equation (5.61),

$$C_{D,i} = \frac{C_L^2}{\pi \text{AR}}(1+\delta) = \frac{(0.4335)^2(1+0.055)}{8\pi} = \boxed{0.00789}$$

EXAMPLE 5.2

Consider a rectangular wing with an aspect ratio of 6, an induced drag factor $\delta = 0.055$, and a zero-lift angle of attack of -2°. At an angle of attack of 3.4°, the induced drag coefficient for this wing is 0.01. Calculate the induced drag coefficient for a similar wing (a rectangular wing with the same airfoil section) at the same angle of attack, but with an aspect ratio of 10. Assume that the induced factors for drag and the lift slope, δ and τ, respectively, are equal to each other (i.e., $\delta = \tau$). Also, for AR = 10, $\delta = 0.105$.

■ Solution

We must recall that although the angle of attack is the same for the two cases compared here (AR = 6 and 10), the value of C_L is different because of the aspect-ratio effect on the lift slope. First, let us calculate C_L for the wing with aspect ratio 6. From Equation (5.61),

$$C_L^2 = \frac{\pi \text{AR} C_{D,i}}{1+\delta} = \frac{\pi(6)(0.01)}{1+0.055} = 0.1787$$

Hence,

$$C_L = 0.423$$

The lift slope of this wing is therefore

$$\frac{dC_L}{d\alpha} = \frac{0.423}{3.4^\circ - (-2^\circ)} = 0.078/\text{degree} = 4.485/\text{rad}$$

The lift slope for the airfoil (the infinite wing) can be obtained from Equation (5.70):

$$\frac{dC_L}{d\alpha} = a = \frac{a_0}{1 + (a_0/\pi \text{AR})(1+\tau)}$$

$$4.485 = \frac{a_0}{1 + [(1.055)a_0/\pi(6)]} = \frac{a_0}{1 + 0.056a_0}$$

Solving for a_0, we find that this yields $a_0 = 5.989$/rad. Since the second wing (with AR = 10) has the same airfoil section, then a_0 is the same. The lift slope of the second

wing is given by

$$a = \frac{a_0}{1 + (a_0/\pi \text{AR})(1+\tau)} = \frac{5.989}{1 + [(5.989)(1.105)/\pi(10)]} = 4.95/\text{rad}$$
$$= 0.086/\text{degree}$$

The lift coefficient for the second wing is therefore

$$C_L = a(\alpha - \alpha_{L=0}) = 0.086[3.4^\circ - (-2^\circ)] = 0.464$$

In turn, the induced drag coefficient is

$$C_{D,i} = \frac{C_L^2}{\pi \text{AR}}(1+\delta) = \frac{(0.464)^2(1.105)}{\pi(10)} = \boxed{0.0076}$$

Note: This problem would have been more straightforward if the lift coefficients had been stipulated to be the same between the two wings rather than the angle of attack. Then Equation (5.61) would have yielded the induced drag coefficient directly. A purpose of this example is to reinforce the rationale behind Equation (5.65), which readily allows the scaling of drag coefficients from one aspect ratio to another, as long as the *lift coefficient is the same*. This allows the scaled drag-coefficient data to be plotted versus C_L (not the angle of attack) as in Figure 5.22. However, in the present example where the angle of attack is the same between both cases, the effect of aspect ratio on the lift slope must be explicitly considered, as we have done above.

EXAMPLE 5.3

Consider the twin-jet executive transport discussed in Example 1.6. In addition to the information given in Example 1.6, for this airplane the zero-lift angle of attack is -2°, the lift slope of the airfoil section is 0.1 per degree, the lift efficiency factor $\tau = 0.04$, and the wing aspect ratio is 7.96. At the cruising condition treated in Example 1.6, calculate the angle of attack of the airplane.

■ Solution

The lift slope of the airfoil section in radians is

$$a_0 = 0.1 \text{ per degree} = 0.1(57.3) = 5.73 \text{ rad}$$

From Equation (5.70) repeated below

$$a = \frac{a_0}{1 + (a_0/\pi \text{AR})(1+\tau)}$$

we have

$$a = \frac{5.73}{1 + \left(\dfrac{5.73}{7.96\pi}\right)(1 + 0.04)} = 4.627 \text{ per rad}$$

or

$$a = \frac{4.627}{57.3} = 0.0808 \text{ per degree}$$

From Example 1.6, the airplane is cruising at a lift coefficient equal to 0.21. Since

$$C_L = a(\alpha - \alpha_{L=0})$$

we have

$$\alpha = \frac{C_L}{a} + \alpha_{L=0} = \frac{0.21}{0.0808} + (-2) = \boxed{0.6^\circ}$$

EXAMPLE 5.4

In the Preview Box for this chapter, we considered the Beechcraft Baron 58 (Figure 5.1) flying such that the wing is at a 4-degree angle of attack. The wing of this airplane has an NACA 23015 airfoil at the root, tapering to a 23010 airfoil at the tip. The data for the NACA 23015 airfoil is given in Figure 5.2. In the Preview Box, we teased you by reading from Figure 5.2 the airfoil lift and drag coefficients at $\alpha = 4^\circ$, namely, $c_l = 0.54$ and $c_d = 0.0068$, and posed the question: Are the lift and drag coefficients of the wing the same values, that is, $C_L = 0.54$ (?) and $C_D = 0.0068$ (?) The answer given in the Preview Box was a resounding NO! We now know why. Moreover, we now know how to calculate C_L and C_D for the wing. Let us proceed to do just that. Consider the wing of the Beechcraft Baron 58 at a 4-degree angle of attack. The wing has an aspect ratio of 7.61 and a taper ratio of 0.45. Calculate C_L and C_D for the wing.

■ Solution

From Figure 5.2*a*, the zero-lift angle of attack of the airfoil, which is the same for the finite wing, is

$$\alpha_{L=0} = -1^\circ$$

The airfoil lift slope is also obtained from Figure 5.2*a*. Since the lift curve is linear below the stall, we arbitrarily pick two points on this curve: $\alpha = 7^\circ$ where $c_l = 0.9$, and $\alpha = -1^\circ$ where $c_l = 0$. Thus

$$a_0 = \frac{0.9 - 0}{7 - (-1)} = \frac{0.9}{8} = 0.113 \text{ per degree}$$

The lift slope in radians is:

$$a_0 = 0.113(57.3) = 6.47 \text{ per rad}$$

From Figure 5.20, for AR = 7.61 and taper ratio = 0.45

$$\delta = 0.01$$

Hence,

$$e = \frac{1}{1 + \delta} = \frac{1}{1 + 0.01} = 0.99$$

From Equation (5.70), assuming $\tau = \delta$,

$$a = \frac{a_0}{1 + \left(\dfrac{a_0}{\pi \text{AR}}\right)(1 + \tau)} (a \text{ and } a_0 \text{ are per rad})$$

where
$$\frac{a_0}{\pi \text{AR}} = \frac{6.47}{\pi(7.61)} = 0.271$$
$$(1 + \tau) = 1 + 0.01 = 1.01$$

we have
$$a = \frac{6.47}{1 + (0.271)(1.01)} = 5.08 \text{ per rad}$$

Converting back to degrees:
$$a = \frac{5.08}{57.3} = 0.0887 \text{ per degree}$$

For the linear lift curve for the finite wing
$$C_L = a(\alpha - \alpha_{L=0})$$

For $\alpha = 4^\circ$, we have
$$C_L = 0.0887[4 - (-1)] = 0.0887(5)$$
$$C_L = \boxed{0.443}$$

The drag coefficient is given by Equation (5.63);
$$C_D = c_d + \frac{C_L^2}{\pi e \text{AR}}$$

Here, c_d is the section drag coefficient given in Figure 5.2*b*. Note that in Figure 5.2*b*, c_d is plotted versus the section lift coefficient c_l. To accurately read c_d from Figure 5.2*b*, we need to know the value of c_l actually sensed by the airfoil section on the finite wing, that is, the value of the airfoil c_l for the airfoil at its effective angle of attack, α_{eff}. To estimate α_{eff}, we will assume an elliptical lift distribution over the wing. We know this is not quite correct, but with a value of $\delta = 0.01$, it is not very far off. From Equation (5.42) for an elliptical lift distribution, the induced angle of attack is
$$\alpha_i = \frac{C_L}{\pi \text{AR}} = \frac{(0.443)}{\pi(7.61)} = 0.0185 \text{ rad}$$

In degrees
$$\alpha_i = (0.0185)(57.3) = 1.06^\circ$$

From Figure 5.6,
$$\alpha_{\text{eff}} = \alpha - \alpha_i = 4^\circ - 1.06^\circ = 2.94^\circ \approx 3^\circ$$

The lift coefficient sensed by the airfoil is then
$$c_l = a_0(\alpha_{\text{eff}} - \alpha_{L=0})$$
$$= 0.113[3 - (-1)] = 0.113(4) = 0.452$$

(Note how close this section lift coefficient is to the overall lift coefficient of the wing of 0.443.) From Figure 5.2*b*, taking the data at the highest Reynolds number shown, for $c_l = 0.452$, we have
$$c_d = 0.0065$$

Returning to Equation (5.63),

$$C_D = c_d + \frac{C_L^2}{\pi e \mathrm{AR}}$$

$$= 0.0065 + \frac{(0.443)^2}{\pi(0.99)(7.61)}$$

$$= 0.0065 + 0.0083 = \boxed{0.0148}$$

So finally, with the results of this worked example, we see why the answer given in the Preview Box was a resounding NO! The lift coefficient for the finite wing is 0.443 compared to the airfoil value of 0.54 given in the Preview Box; the finite wing value is 18 percent lower than the airfoil value—a substantial difference. The drag coefficient for the finite wing is 0.0148 compared to the airfoil value of 0.0068; the finite wing value is more than a factor of two larger—a dramatic difference. These differences are the reason for the studies covered in this chapter.

DESIGN BOX

In airplane design, the aspect ratio is a much more important consideration than wing planform shape from the point of view of reducing the induced drag coefficient. Although the elliptical planform, as sketched at the top of Figure 5.18, leads to the optimum lift distribution to minimize induced drag, the tapered wing, as sketched at the bottom of Figure 5.18, can yield a near-optimum lift distribution, with induced drag coefficients only a few percent higher than the elliptical wing. Because a tapered wing with its straight leading and trailing edges is much cheaper and easier to manufacture, the design choice for wing planform is almost always a tapered wing and hardly ever an elliptical wing.

Why then did the Supermarine Spitfire, shown in Figure 5.19, have such a beautiful elliptical wing planform? The answer has nothing to do with aerodynamics. In 1935, the Supermarine Company was responding to the British Air Ministry specification F.37/34 for a new fighter aircraft. Designer Reginald Mitchell had originally laid out on the drawing board an aircraft with tapered wings. However, Mitchell was also coping with the Air Ministry requirement that the airplane be armed with eight 0.303 caliber Browning machine guns. Mounting four of these guns in each wing far enough outboard to be outside of the propeller disk, Mitchell had a problem—the outboard sections of the tapered wing did not have enough chord length to accommodate the guns. His solution was an elliptical planform, which provided sufficient chord length far enough out on the span to allow the guns to fit. The result was the beautiful elliptic planform shown in Figure 5.19. The enhanced aerodynamic efficiency of this wing was only a by-product of a practical design solution. Predictably, the elliptic wings were difficult to produce, and this contributed to production delays in the critical months before the beginning of World War II. A second enhanced aerodynamic by-product was afforded by the large chord lengths along most of the elliptic wing. This allowed Mitchell to choose a thinner airfoil section, with 13 percent thickness ratio at the wing root and 7 percent at the wing tip, and still maintain sufficient absolute thickness for internal structural design. Because of this, the Spitfire had a larger critical Mach number (to be discussed in Section 11.6) and could reach the unusually high freestream Mach number of 0.92 in a dive.

For aerodynamic efficiency at subsonic speeds, the airplane designer would love to have very large aspect ratio wings—wings that would look like a long slat out of a common venetian blind. However, existing airplanes do not fly around with venetian blind slats for wings. The reason is that the structural design of such wings poses a compromise. The larger the aspect ratio, the larger are the bending movements at the wing root caused by the lift distribution reaching farther away from the root. Such wings require heavier internal structure. Hence, as the aspect ratio of a wing increases, so does the structural weight of the wing. As a result of this compromise between aerodynamics and structures, typical aspect ratios for conventional subsonic airplanes are on the order of 6 to 8.

However, examine the three-view of the Lockheed U-2 high altitude reconnaissance aircraft shown in Figure 5.26. This airplane has the unusually high aspect ratio of 14.3. Why? The answer is keyed to its mission. The U-2 was essentially a *point design;* it was to cruise at the exceptionally high altitude of 70,000 ft or higher in order to not be reached by interceptor aircraft or ground-to-air-missiles during overflights of the Soviet Union in the 1950s. To achieve this mission, the need for incorporating a very high aspect ratio wing was paramount, for the following reason. In steady, level flight, where the airplane lift L must equal its weight W,

$$L = W = q_\infty S C_L = \tfrac{1}{2}\rho_\infty V_\infty^2 S C_L \qquad (5.71)$$

As the airplane flies higher, ρ_∞ decreases and hence from Equation (5.71) C_L must be increased in order to keep the lift equal to the weight. As its high-altitude cruise design point, the U-2 flies at a high value of C_L, just on the verge of stalling. (This is in stark contrast to the normal cruise conditions of conventional airplanes at conventional altitudes, where the cruise lift coefficient is relatively small.) At the high value of C_L for the U-2 at cruising altitude, its induced drag coefficient [which from Equation (5.62) varies as C_L^2] would be unacceptably high if a conventional aspect ratio were used. Hence, the Lockheed design group (at the Lockheed Skunk Works) had to opt for as high an aspect ratio as possible to keep the induced drag coefficient within reasonable bounds. The wing design shown in Figure 5.26 was the result.

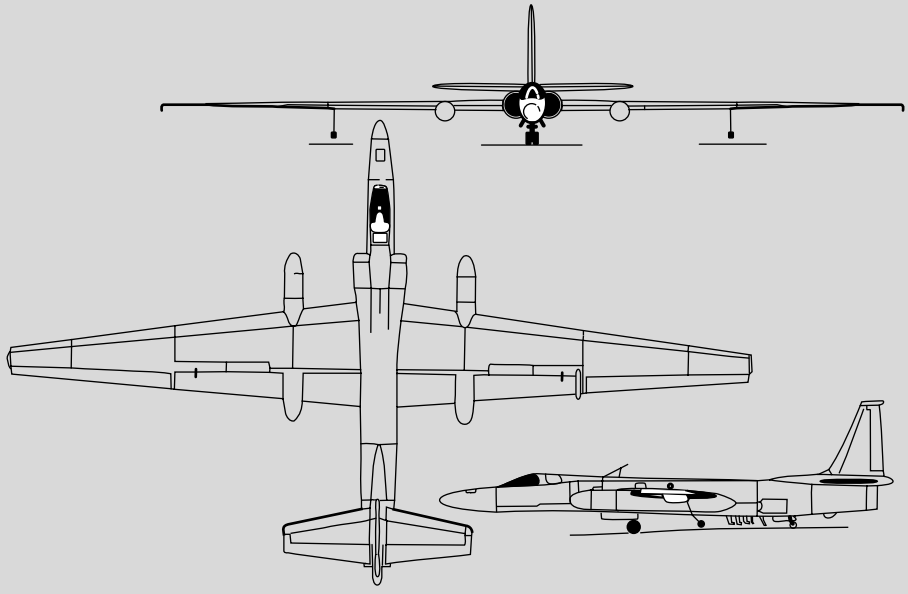


Figure 5.26 Three-view of the Lockheed U-2 high-altitude reconnaissance airplane.

We made an observation about induced drag D_i itself, in contrast to the induced drag coefficient $C_{D,i}$. We have emphasized, based on Equation (5.62), that $C_{D,i}$ can be reduced by increasing the aspect ratio. For an airplane in steady, level flight, however, the induced drag force itself is governed by another design parameter, rather than the aspect ratio per se, as follows. From Equation (5.62), we have

$$D_i = q_\infty S C_{D,i} = q_\infty S \frac{C_L^2}{\pi e \text{AR}} \tag{5.72}$$

For steady, level flight, from Equation (5.71), we have

$$C_L^2 = \left(\frac{L}{q_\infty S}\right)^2 = \left(\frac{W}{q_\infty S}\right)^2 \tag{5.73}$$

Substituting Equation (5.73) into (5.72), we have

$$D_i = q_\infty S \left(\frac{W}{q_\infty S}\right)^2 \frac{1}{\pi e \text{AR}}$$

$$= \frac{1}{\pi e} q_\infty S \left(\frac{W}{q_\infty S}\right)^2 \left(\frac{S}{b^2}\right)$$

or

$$\boxed{D_i = \frac{1}{\pi e q_\infty} \left(\frac{W}{b}\right)^2} \tag{5.74}$$

This is a rather revealing result! The induced drag in steady, level flight—the force itself—does not depend explicitly on the aspect ratio, but rather on another design parameter W/b called the *span loading:*

$$\boxed{\text{Span loading} \equiv \frac{W}{b}}$$

From Equation (5.74), we see that the induced drag for a given weight airplane, can be reduced simply by increasing the wingspan b. In so doing, the wing tip vortices (the physical source of induced drag) are simply moved farther away, hence lessening their effect on the rest of the wing and in turn reducing the induced drag. This makes good intuitive sense.

However, in the preliminary design of an airplane, the wing area S is usually dictated by the landing or take-off speed, which is only about 10 to 20 percent above V_{stall}. This is seen by Equation (1.47) repeated below

$$V_{\text{stall}} = \sqrt{\frac{2W}{\rho_\infty S C_{L,\max}}} \tag{1.47}$$

For a specified V_{stall} at sea level, and given $C_{L,\max}$ for the airplane, Equation (1.47) determines the necessary wing area for a given weight airplane. Therefore, reflecting on Equation (5.74) when we say that D_i can be reduced for a given weight airplane simply by increasing the wing span, since S is usually fixed for the given weight, then clearly the aspect ratio b^2/S increases as we increase b. So when we use Equation (5.74) to say that D_i can be reduced by increasing the span for a given weight airplane, this also has the connotation of increasing the aspect ratio. However, it is instructional to note that D_i depends explicitly on the design parameter, W/b, and not the aspect ratio; this is the message in Equation (5.74).

Question: How much of the total drag of an airplane is induced drag? A generic answer to this question is shown in the bar chart in Figure 5.27. Here, the induced drag (shaded portion) relative to parasite drag (white portion) is shown for typical cruise and takeoff conditions for a generic subsonic jet transport. The parasite drag is the sum of the drag due to skin friction and pressure drag due to flow separation associated with the complete airplane, including the wing.

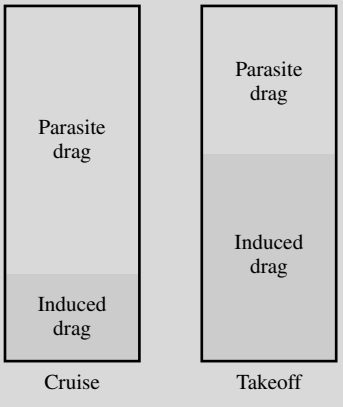


Figure 5.27 Comparison of relative amounts of induced and parasite drag for cruise (high speed) and takeoff (low speed).

The total drag of the airplane is the sum of the parasite drag and the induced drag. Figure 5.27 indicates that induced drag is about 25 percent of the total drag at cruise, but can be 60 percent or more of the total drag at takeoff (where the airplane is flying at high C_L).

For more details on the drag characteristics of airplanes, and their associated impact on airplane design, see the author's book *Aircraft Performance and Design,* McGraw-Hill, Boston, 1999.

5.4 A NUMERICAL NONLINEAR LIFTING-LINE METHOD

The classical Prandtl lifting-line theory described in Section 5.3 assumes a linear variation of c_l versus α_{eff}. This is clearly seen in Equation (5.19). However, as the angle of attack approaches and exceeds the stall angle, the lift curve becomes nonlinear, as shown in Figure 4.9. This high-angle-of-attack regime is of interest to modern aerodynamicists. For example, when an airplane is in a spin, the angle of attack can range from 40 to 90°; an understanding of high-angle-of-attack aerodynamics is essential to the prevention of such spins. In addition, modern fighter airplanes achieve optimum maneuverability by pulling high angles of attack at subsonic speeds. Therefore, there are practical reasons for extending Prandtl's classical theory to account for a nonlinear lift curve. One simple extension is described in this section.

The classical theory developed in Section 5.4 is essentially closed form; that is, the results are analytical equations as opposed to a purely numerical solution. Of course, in the end, the Fourier coefficients A_n for a given wing must come from a solution of a system of simultaneous linear algebraic equations. Until the advent of the modern digital computer, these coefficients were calculated by hand. Today, they are readily solved on a computer using standard matrix methods. However, the elements of the lifting-line theory lend themselves to a straightforward purely numerical solution which allows the treatment of nonlinear effects. Moreover, this numerical solution emphasizes the fundamental aspects of lifting-line theory. For these reasons, such a numerical solution is outlined in this section.

Consider the most general case of a finite wing of given planform and geometric twist, with different airfoil sections at different spanwise stations. Assume that we have experimental data for the lift curves of the airfoil sections, including the nonlinear regime (i.e., assume we have the conditions of Figure 4.9 for all the given airfoil sections). A numerical iterative solution for the finite-wing properties can be obtained as follows:

1. Divide the wing into a number of spanwise stations, as shown in Figure 5.28. Here $k + 1$ stations are shown, with n designating any specific station.
2. For the given wing at a given α, *assume* the lift distribution along the span; that is, assume values for Γ at all the stations $\Gamma_1, \Gamma_2, \ldots, \Gamma_n, \ldots, \Gamma_{k+1}$. An elliptical lift distribution is satisfactory for such an assumed distribution.

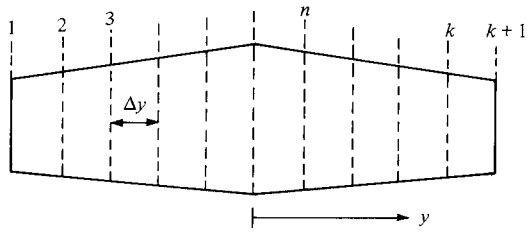


Figure 5.28 Stations along the span for a numerical solution.

3. With this assumed variation of Γ, calculate the induced angle of attack α_i from Equation (5.18) at each of the stations:

$$\alpha_i(y_n) = \frac{1}{4\pi V_\infty} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy)\, dy}{y_n - y} \tag{5.75}$$

The integral is evaluated numerically. If Simpson's rule is used, Equation (5.75) becomes

$$\alpha_i(y_n) = \frac{1}{4\pi V_\infty} \frac{\Delta y}{3} \sum_{j=2,4,6}^{k} \frac{(d\Gamma/dy)_{j-1}}{(y_n - y_{j-1})} + 4\frac{(d\Gamma/dy)_j}{y_n - y_j} + \frac{(d\Gamma/dy)_{j+1}}{y_n - y_{j+1}} \tag{5.76}$$

where Δy is the distance between stations. In Equation (5.76), when $y_n = y_{j-1}$, y_j, or y_{j+1}, a singularity occurs (a denominator goes to zero). When this singularity occurs, it can be avoided by replacing the given term by its average value based on the two adjacent sections.

4. Using α_i from step 3, obtain the effective angle of attack α_{eff} at each station from

$$\alpha_{\text{eff}}(y_n) = \alpha - \alpha_i(y_n)$$

5. With the distribution of α_{eff} calculated from step 4, obtain the section lift coefficient $(c_l)_n$ at each station. These values are read from the known lift curve for the airfoil.

6. From $(c_l)_n$ obtained in step 5, a *new* circulation distribution is calculated from the Kutta-Joukowski theorem and the definition of lift coefficient:

$$L'(y_n) = \rho_\infty V_\infty \Gamma(y_n) = \tfrac{1}{2}\rho_\infty V_\infty^2 c_n (c_l)_n$$

Hence,

$$\Gamma(y_n) = \tfrac{1}{2} V_\infty c_n (c_l)_n$$

where c_n is the local section chord. Keep in mind that in all the above steps, n ranges from 1 to $k + 1$.

7. The new distribution of Γ obtained in step 6 is compared with the values that were initially fed into step 3. If the results from step 6 do not agree with the input to step 3, then a new input is generated. If the previous input to step 3 is designated as Γ_{old} and the result of step 6 is designated as Γ_{new},

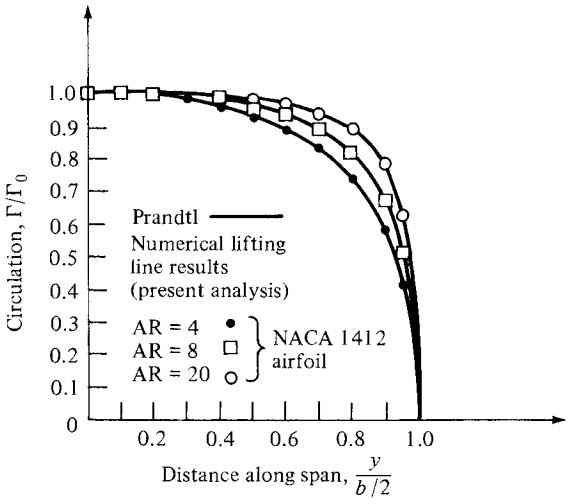


Figure 5.29 Lift distribution for a rectangular wing; comparison between Prandtl's classical theory and the numerical lifting-line method of Reference 20.

then the new input to step 3 is determined from

$$\Gamma_{\text{input}} = \Gamma_{\text{old}} + D(\Gamma_{\text{new}} - \Gamma_{\text{old}})$$

where D is a damping factor for the iterations. Experience has found that the iterative procedure requires heavy damping, with typical values of D on the order of 0.05.

8. Steps 3 to 7 are repeated a sufficient number of cycles until Γ_{new} and Γ_{old} agree at each spanwise station to within acceptable accuracy. If this accuracy is stipulated to be within 0.01 percent for a stretch of five previous iterations, then a minimum of 50 and sometimes as many as 150 iterations may be required for convergence.

9. From the converged $\Gamma(y)$, the lift and induced drag coefficients are obtained from Equations (5.26) and (5.30), respectively. The integrations in these equations can again be carried out by Simpson's rule.

The procedure outlined above generally works smoothly and quickly on a high-speed digital computer. Typical results are shown in Figure 5.29, which shows the circulation distributions for rectangular wings with three different aspect ratios. The solid lines are from the classical calculations of Prandtl (Section 5.3), and the symbols are from the numerical method described above. Excellent agreement is obtained, thus verifying the integrity and accuracy of the numerical method. Also, Figure 5.29 should be studied as an example of typical circulation distributions over general finite wings, with Γ reasonably high over the center section of the wing but rapidly dropping to zero at the tips.

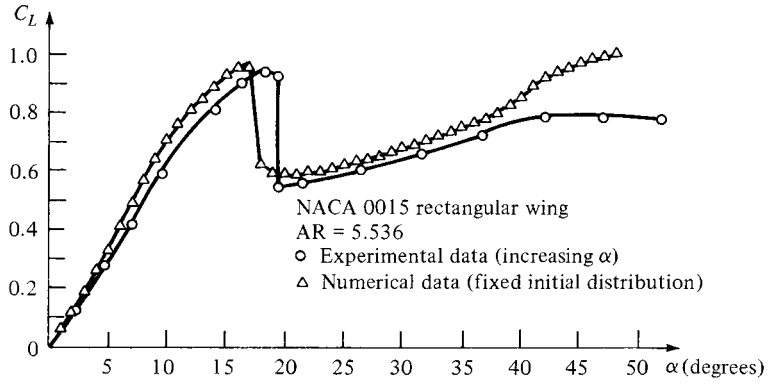


Figure 5.30 Lift coefficient versus angle of attack; comparison between experimental and numerical results.

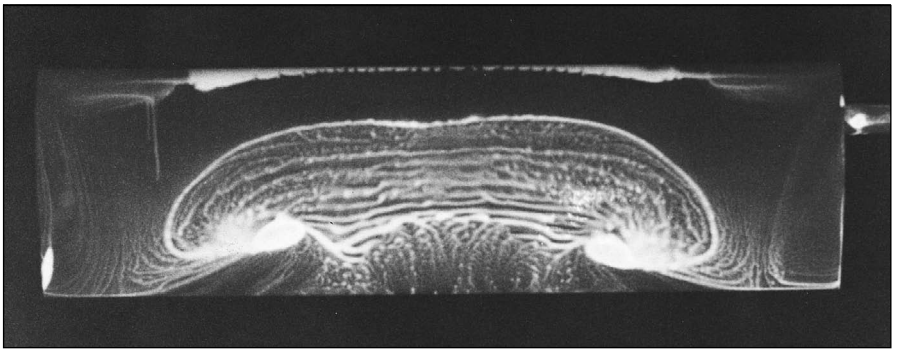


Figure 5.31 Surface oil flow pattern on a stalled, finite rectangular wing with a Clark Y-14 airfoil section. AR = 3.5, $\alpha = 22.8°$, Re = 245,000 (based on chord length). This pattern was established by coating the wing surface with pigmented mineral oil and inserting the model in a low-speed subsonic wind tunnel. In the photograph shown, flow is from top to bottom. Note the highly three-dimensional flow pattern. *(Courtesy of Allen E. Winkelmann, University of Maryland.)*

An example of the use of the numerical method for the nonlinear regime is shown in Figure 5.30. Here, C_L versus α is given for a rectangular wing up to an angle of attack of 50°—well beyond stall. The numerical results are compared with existing experimental data obtained at the University of Maryland (Reference 19). The numerical lifting-line solution at high angle of attack agrees with the experiment to within 20 percent, and much closer for many cases. Therefore, such solutions give reasonable preliminary engineering results for the high-angle-of-attack poststall region. However, it is wise not to stretch the applicability of lifting-line theory too far. At high angles of attack, the flow is highly three-dimensional. This is clearly seen in the surface oil pattern on a rectangular wing at high angle of attack shown in Figure 5.31. At high α, there is a strong

spanwise flow, in combination with mushroom-shaped flow separation regions. Clearly, the basic assumptions of lifting-line theory, classical or numerical, cannot properly account for such three-dimensional flows.

For more details and results on the numerical lifting-line method, please see Reference 20.

5.5 THE LIFTING-SURFACE THEORY AND THE VORTEX LATTICE NUMERICAL METHOD

Prandtl's classical lifting-line theory (Section 5.3) gives reasonable results for straight wings at moderate to high aspect ratio. However, for low-aspect-ratio straight wings, swept wings, and delta wings, classical lifting-line theory is inappropriate. For such planforms, sketched in Figure 5.32, a more sophisticated model must be used. The purpose of this section is to introduce such a model and to discuss its numerical implementation. However, it is beyond the scope of this book to elaborate on the details of such higher-order models; rather, only the flavor is given here. You are encouraged to pursue this subject by reading the literature and by taking more advanced studies in aerodynamics.

Return to Figure 5.15. Here, a simple lifting line spans the wing, with its associated trailing vortices. The circulation Γ varies with y along the lifting line. Let us extend this model by placing a *series* of lifting lines on the plane of the wing, at different chordwise stations; that is, consider a large number of lifting lines all parallel to the y axis, located at different values of x, as shown in Figure 5.33. In the limit of an infinite number of lines of infinitesimal strength, we obtain a vortex sheet, where the vortex lines run parallel to the y axis. The strength of this sheet (per unit length in the x direction) is denoted by γ, where γ varies in the y direction, analogous to the variation of Γ for the single lifting line in Figure 5.15. Moreover, each lifting line will have, in general, a different overall strength, so that γ varies with x also. Hence, $\gamma = \gamma(x, y)$ as shown in Figure 5.33. In addition, recall that each lifting line has a system of trailing vortices; hence, the series of lifting lines is crossed by a series of superimposed trailing vortices parallel to the x axis. In the limit of an infinite number of infinitesimally weak vortices, these trailing vortices form another vortex sheet of strength δ (per unit length in the

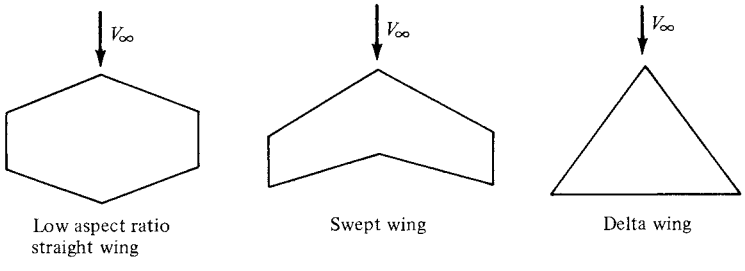


Figure 5.32 Types of wing planforms for which classical lifting-line theory is not appropriate.

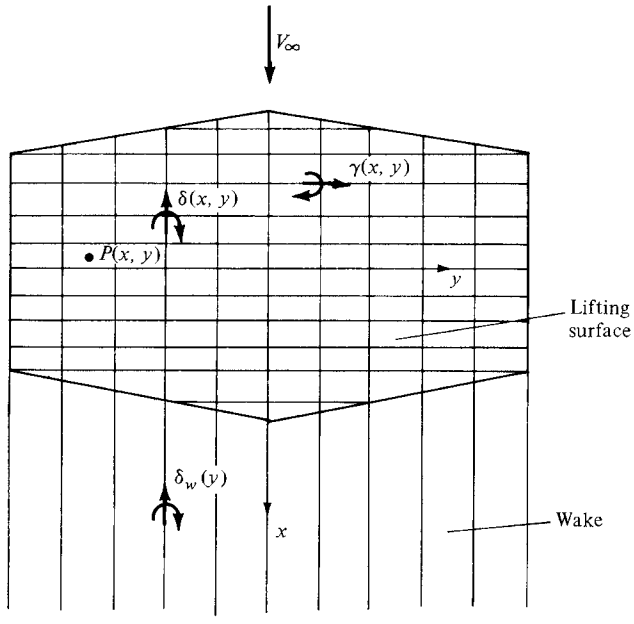


Figure 5.33 Schematic of a lifting surface.

y direction). [Note that this δ is different from the δ used in Equation (5.61); the use of the same symbol in both cases is standard, and there should be no confusion since the meanings and context are completely different.] To see this more clearly, consider a single line parallel to the x axis. As we move along this line from the leading edge to the trailing edge, we pick up an additional superimposed trailing vortex each time we cross a lifting line. Hence, δ must vary with x. Moreover, the trailing vortices are simply parts of the horseshoe vortex systems, the leading edges of which make up the various lifting lines. Since the circulation about each lifting line varies in the y direction, the strengths of different trailing vortices will, in general, be different. Hence, δ also varies in the y direction, that is, $\delta = \delta(x, y)$, as shown in Figure 5.33. The two vortex sheets—the one with vortex lines running parallel to y with strength γ (per unit length in the x direction) and the other with vortex lines running parallel to x with strength δ (per unit length in the y direction)—result in a *lifting surface* distributed over the entire planform of the wing, as shown in Figure 5.33. At any given point on the surface, the strength of the lifting surface is given by both γ and δ, which are functions of x and y. We denote $\gamma = \gamma(x, y)$ as the spanwise vortex strength distribution and $\delta = \delta(x, y)$ as the chordwise vortex strength distribution.

Note that downstream of the trailing edge we have no spanwise vortex lines, only trailing vortices. Hence, the wake consists of only chordwise vortices. The strength of this wake vortex sheet is given by δ_w (per unit length in the y direction). Since in the wake the trailing vortices do not cross any vortex lines, the strength

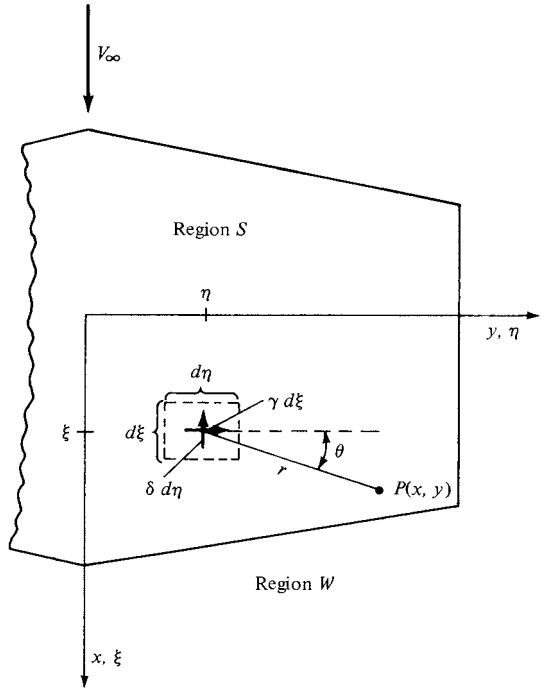


Figure 5.34 Velocity induced at point P by an infinitesimal segment of the lifting surface. The velocity is perpendicular to the plane of the paper.

of any given trailing vortex is constant with x. Hence, δ_w depends only on y and, throughout the wake, $\delta_w(y)$ is equal to its value at the trailing edge.

Now that we have defined the lifting surface, of what use is it? Consider point P located at (x, y) on the wing, as shown in Figure 5.33. The lifting surface and the wake vortex sheet both induce a normal component of velocity at point P. Denote this normal velocity by $w(x, y)$. We want the wing planform to be a stream surface of the flow; that is, we want the sum of the induced $w(x, y)$ and the normal component of the freestream velocity to be zero at point P and for all points on the wing—this is the flow-tangency condition on the wing surface. (Keep in mind that we are treating the wing as a flat surface in this discussion.) The central theme of lifting-surface theory is to find $\gamma(x, y)$ and $\delta(x, y)$ such that the flow-tangency condition is satisfied at all points on the wing. [Recall that in the wake, $\delta_w(y)$ is fixed by the trailing-edge values of $\delta(x, y)$; hence, $\delta_w(y)$ is not, strictly speaking, one of the unknown dependent variables.]

Let us obtain an expression for the induced normal velocity $w(x, y)$ in terms of γ, δ, and δ_w. Consider the sketch given in Figure 5.34, which shows a portion of the planview of a finite wing. Consider the point given by the coordinates (ξ, η). At this point, the spanwise vortex strength is $\gamma(\xi, \eta)$. Consider a thin ribbon, or

filament, of the spanwise vortex sheet of incremental length $d\xi$ in the x direction. Hence, the strength of this filament is $\gamma\, d\xi$, and the filament stretches in the y (or η) direction. Also, consider point P located at (x, y) and removed a distance r from the point (ξ, η). From the Biot-Savart law, Equation (5.5), the incremental velocity induced at P by a segment $d\eta$ of this vortex filament of strength $\gamma\, d\xi$ is

$$|\mathbf{dV}| = \left| \frac{\Gamma}{4\pi} \frac{\mathbf{dl} \times \mathbf{x}}{|\mathbf{r}|^3} \right| = \frac{\gamma\, d\xi}{4\pi} \frac{(d\eta) r \sin\theta}{r^3} \tag{5.77}$$

Examining Figure 5.34, and following the right-hand rule for the strength γ, note that $|\mathbf{dV}|$ is induced downward, into the plane of the wing (i.e., in the negative z direction). Following the usual sign convention that w is positive in the upward direction (i.e., in the positive z direction), we denote the contribution of Equation (5.77) to the induced velocity w as $(dw)_\gamma = -|\mathbf{dV}|$. Also, note that $\sin\theta = (x - \xi)/r$. Hence, Equation (5.77) becomes

$$(dw)_\gamma = -\frac{\gamma}{4\pi} \frac{(x - \xi)\, d\xi\, d\eta}{r^3} \tag{5.78}$$

Considering the contribution of the elemental chordwise vortex of strength $\delta\, d\eta$ to the induced velocity at P, we find by an analogous argument that

$$(dw)_\delta = -\frac{\delta}{4\pi} \frac{(y - \eta)\, d\xi\, d\eta}{r^3} \tag{5.79}$$

To obtain the velocity induced at P by the entire lifting surface, Equations (5.78) and (5.79) must be integrated over the wing planform, designated as region S in Figure 5.34. Moreover, the velocity induced at P by the complete wake is given by an equation analogous to Equation (5.79), but with δ_w instead of δ, and integrated over the wake, designated as region W in Figure 5.34. Noting that

$$r = \sqrt{(x - \xi)^2 + (y - \eta)^2}$$

the normal velocity induced at P by both the lifting surface and the wake is

$$\boxed{\begin{aligned} w(x, y) = &-\frac{1}{4\pi} \iint_S \frac{(x - \xi)\gamma(\xi, \eta) + (y - \eta)\delta(\xi, \eta)}{[(x - \xi)^2 + (y - \eta)^2]^{3/2}}\, d\xi\, d\eta \\ &-\frac{1}{4\pi} \iint_W \frac{(y - \eta)\delta_w(\xi, \eta)}{[(x - \xi)^2 + (y - \eta)^2]^{3/2}}\, d\xi\, d\eta \end{aligned}} \tag{5.80}$$

The central problem of lifting-surface theory is to solve Equation (5.80) for $\gamma(\xi, \eta)$ and $\delta(\xi, \eta)$ such that the sum of $w(x, y)$ and the normal component of the freestream is zero, that is, such that the flow is tangent to the planform surface S. The details of various lifting-surface solutions are beyond the scope of this book; rather, our purpose here was simply to present the flavor of the basic model.

The advent of the high-speed digital computer has made possible the implementation of numerical solutions based on the lifting-surface concept. These solutions are similar to the panel solutions for two-dimensional flow discussed in Chapters 3 and 4 in that the wing planform is divided into a number of panels, or

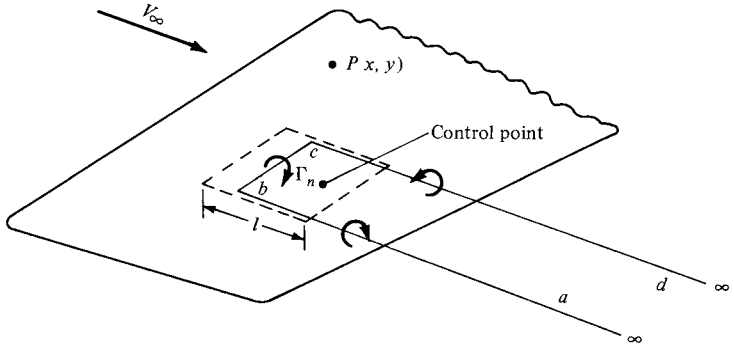


Figure 5.35 Schematic of a single horseshoe vortex, which is part of a vortex system on the wing.

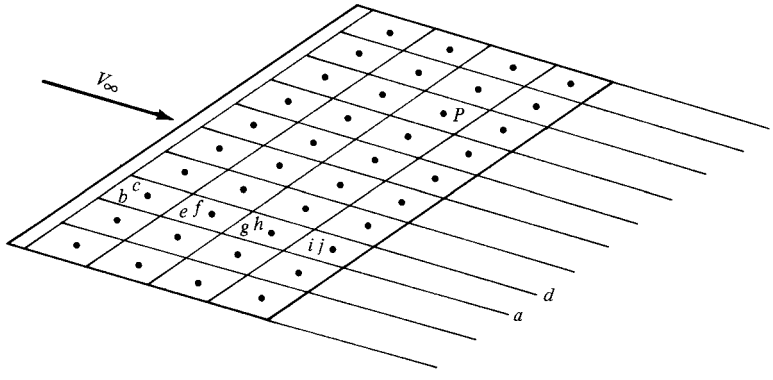


Figure 5.36 Vortex lattice system on a finite wing.

elements. On each panel, either constant or prescribed variations of both γ and δ can be made. Control points on the panels can be chosen, where the net normal flow velocity is zero. The evaluation of equations like Equation (5.80) at these control points results in a system of simultaneous algebraic equations that can be solved for the values of the γ's and δ's on all the panels.

A related but somewhat simpler approach is to superimpose a finite number of horseshoe vortices of different strengths Γ_n on the wing surface. For example, consider Figure 5.35, which shows part of a finite wing. The dashed lines define a panel on the wing planform, where l is the length of the panel in the flow direction. The panel is a trapezoid; it does not have to be a square, or even a rectangle. A horseshoe vortex *abcd* of strength Γ_n is placed on the panel such that the segment *bc* is a distance $l/4$ from the front of the panel. A control point is placed on the centerline of the panel at a distance $\frac{3}{4}l$ from the front. The velocity induced at an arbitrary point P only by the single horseshoe vortex can be calculated from the Biot-Savart law by treating each of the vortex filaments *ab*, *bc*, and *cd* separately. Now consider the entire wing covered by a finite number of panels, as sketched in Figure 5.36. A series of horseshoe vortices is now superimposed. For example,

on one panel at the leading edge, we have the horseshoe vortex *abcd*. On the panel behind it, we have the horseshoe vortex *aefd*. On the next panel, we have *aghd*, and on the next, *aijd*, etc. The entire wing is covered by this lattice of horseshoe vortices, each of different unknown strength Γ_n. At any control point P, the normal velocity induced by *all* the horseshoe vortices can be obtained from the Biot-Savart law. When the flow-tangency condition is applied at all the control points, a system of simultaneous algebraic equations results which can be solved for the unknown Γ_n's. This numerical approach is called the *vortex lattice method* and is in wide use today for the analysis of finite-wing properties. Once again, only the flavor of the method is given above; you are encouraged to read the volumes of literature that now exist on various versions of the vortex lattice method. In particular, Reference 13 has an excellent introductory discussion on the vortex lattice method, including a worked example that clearly illustrates the salient points of the technique.

DESIGN BOX

The lift slope for a high-aspect-ratio straight wing with an elliptical lift distribution is predicted by Prandtl's lifting-line theory and is given by Equation (5.69), repeated below:

$$a = \frac{a_0}{1 + a_0/\pi \text{AR}}$$

high-aspect-ratio straight wings (5.69)

This relation, and others like it, is useful for the conceptual design process, where simple formulas, albeit approximate, can lead to fast, back-of-the-envelope calculations. However, Equation (5.69), like all results from simple lifting-line theory, is valid only for high-aspect-ratio straight wings (AR $>$ 4, as a rule of thumb).

The German aerodynamicist H. B. Helmbold in 1942 modified Equation (5.69) to obtain the following form applicable to low-aspect-ratio straight wings:

$$a = \frac{a_0}{\sqrt{1 + (a_0/\pi \text{AR})^2} + a_0/(\pi \text{AR})}$$

low-aspect-ratio straight wing (5.81)

Equation (5.81) is remarkably accurate for wings with AR $<$ 4. This is demonstrated in Figure 5.37, which gives experimental data for the lift slope for rectangular wings as a function of AR from 0.5 to 6; these data are compared with the predictions from Prandtl's lifting-line theory, Equation (5.69), and Helmbold's equation, Equation (5.81). Note from Figure 5.37 that Helmbold's equation gives excellent agreement with the data for AR $<$ 4, and that Equation (5.69) is preferable for AR $>$ 6.

For swept wings, Kuchemann (Reference 70) suggests the following modification to Helmbold's equation:

$$a = \frac{a_0 \cos \Lambda}{\sqrt{1 + [(a_0 \cos \Lambda)/(\pi \text{AR})]^2} + (a_0 \cos \Lambda/(\pi \text{AR}))}$$

swept wing (5.82)

where Λ is the sweep angle of the wing, referenced to the half-chord line, as shown in Figure 5.38.

Keep in mind that Equations (5.69), (5.81), and (5.82) apply to *incompressible flow*. Compressibility corrections to account for high Mach number effects will be discussed in Chapter 10.

Also, the equations above are simple formulas intended to provide quick and easy "back of the envelope" calculations for conceptual design purposes. In contrast, today the power of computational fluid

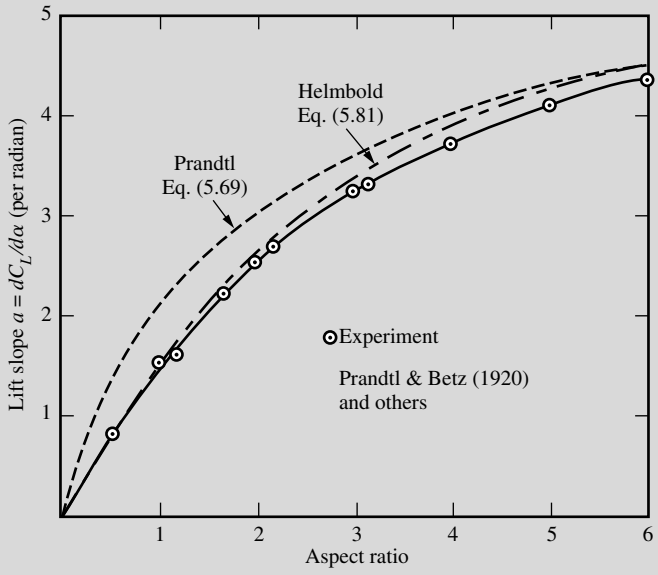


Figure 5.37 Lift slope versus aspect ratio for straight wings in low-speed flow.

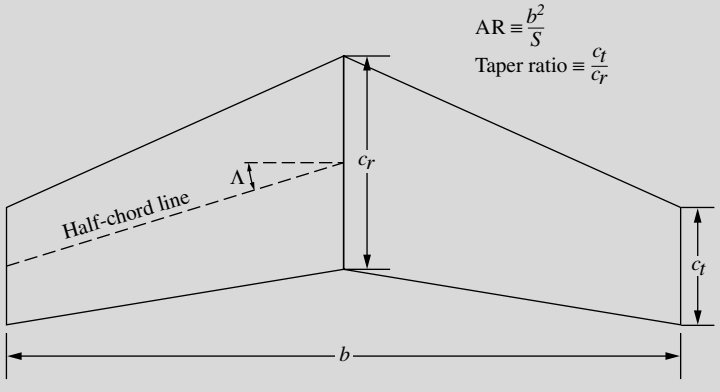


Figure 5.38 Geometry of a swept wing.

dynamics (CFD) allows the detailed calculation of the flow field over a finite wing of any general shape (any aspect ratio, wing sweep, taper ratio, etc.). Moreover, the combination of CFD with modern optimization procedures allows the optimized design of the complete wing, including not only the optimum planform shape, but also the optimum variation of airfoil shape along the span. Because this modern, optimized wing design is usually applied to high-speed airplanes where the flow must be treated as compressible, we defer any discussion of this modern design process until Part 3 of this book.

5.6 APPLIED AERODYNAMICS: THE DELTA WING

In Part 3 of this book, we will see that supersonic flow is dramatically different from subsonic flow in virtually all respects—the mathematics and physics of these two flow regimes are totally different. Such differences impact the design philosophy of aircraft for supersonic flight in comparison to aircraft for subsonic flight. In particular, supersonic airplanes usually have highly swept wings (the reasons for this are discussed in Part 3). A special case of swept wings is those aircraft with a triangular planform—called delta wings. A comparison of the planform of a conventional swept wing was shown in Figure 5.32. Two classic examples of aircraft with delta wings are the Convair F-102A, the first operational jet airplane in the United States to be designed with a delta wing, shown in Figure 5.39*a*, and the space shuttle, basically a hypersonic airplane, shown in Figure 5.39*b*. In reality, the planform of the space shuttle is more correctly denoted as a double-delta shape. Indeed, there are several variants of the basic delta wing used on modern aircraft; these are shown in Figure 5.40. Delta wings are used on many different types of high-speed airplanes around the world; hence, the delta planform is an important aerodynamic configuration.

Question: Since delta-winged aircraft are high-speed vehicles, why are we discussing this topic in the present chapter, which deals with the low-speed, incompressible flow over finite wings? The obvious answer is that all high-speed aircraft fly at low speeds for takeoff and landing; moreover, in most cases, these aircraft spend the vast majority of their flight time at subsonic speeds, using their supersonic capability for short "supersonic dashes," depending on their mission. Several exceptions are, of course, the Concorde supersonic transport which cruised at supersonic speeds across oceans, and the space shuttle, which is hypersonic for most of its reentry into the earth's atmosphere. However, a vast majority of delta-winged aircraft spend a great deal of their flight time at subsonic speeds. For this reason, the low-speed aerodynamic characteristics of delta wings are of great importance; this is accentuated by the rather different and unique aerodynamic aspects associated with such delta wings. Therefore, the low-speed aerodynamics of delta wings has been a subject of much serious study over the past years, going back as far as the early work on delta wings by Alexander Lippisch in Germany during the 1930s. This is the answer to our question posed above—in the context of our discussion on finite wings, we must give the delta wing some special attention.

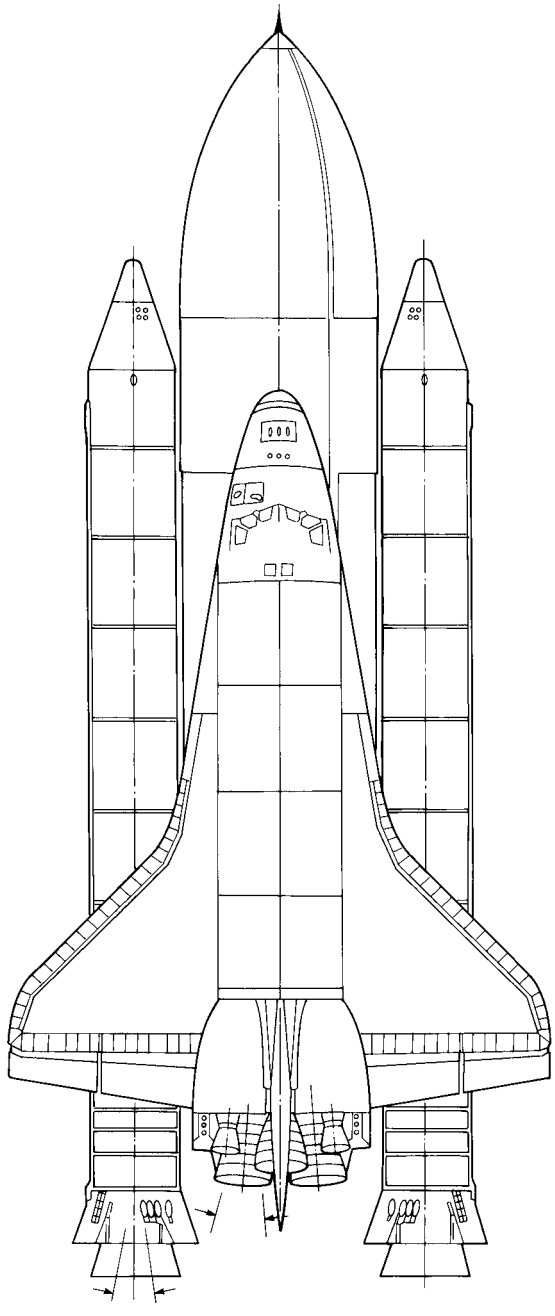
The subsonic flow pattern over the top of a delta wing at angle of attack is sketched in Figure 5.41. The dominant aspect of this flow are the two vortex patterns that occur in the vicinity of the highly swept leading edges. These vortex patterns are created by the following mechanism. The pressure on the bottom surface of the wing at the angle of attack is higher than the pressure on the top surface. Thus, the flow on the bottom surface in the vicinity of the leading edge tries to curl around the leading edge from the bottom to the top. If the leading edge is sharp, the flow will separate along its entire length. (We have already mentioned several times that when low-speed, subsonic flow passes over a sharp convex corner, inviscid flow theory predicts an infinite velocity at the corner, and



(*a*)

Figure 5.39 Some delta-winged vehicles. (*a*) The Convair F-102A. *(Courtesy of the U.S. Air Force.)*

that nature copes with this situation by having the flow separate at the corner. The leading edge of a delta wing is such a case.) This separated flow curls into a primary vortex which exists above the wing just inboard of each leading edge, as sketched in Figure 5.41. The stream surface which has separated at the leading edge (the primary separation line S_1 in Figure 5.41) loops above the wing and then



(*b*)

Figure 5.39 (continued) Some delta-winged vehicles. (*b*) The space shuttle. *(Courtesy of NASA.)*

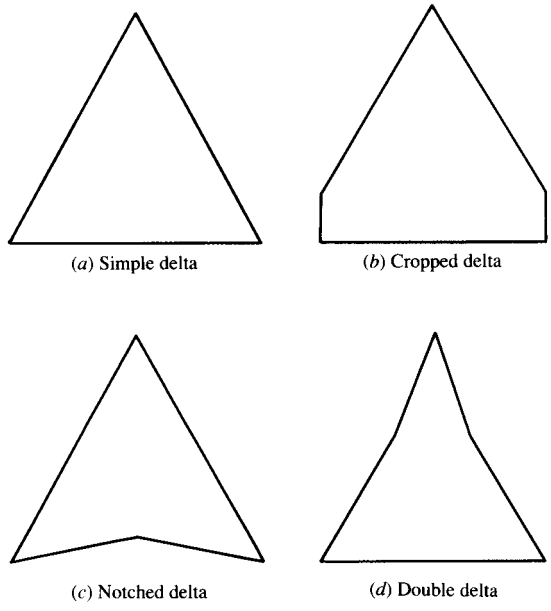


Figure 5.40 Four versions of a delta-wing planform.
(From Loftin, Reference 48.)

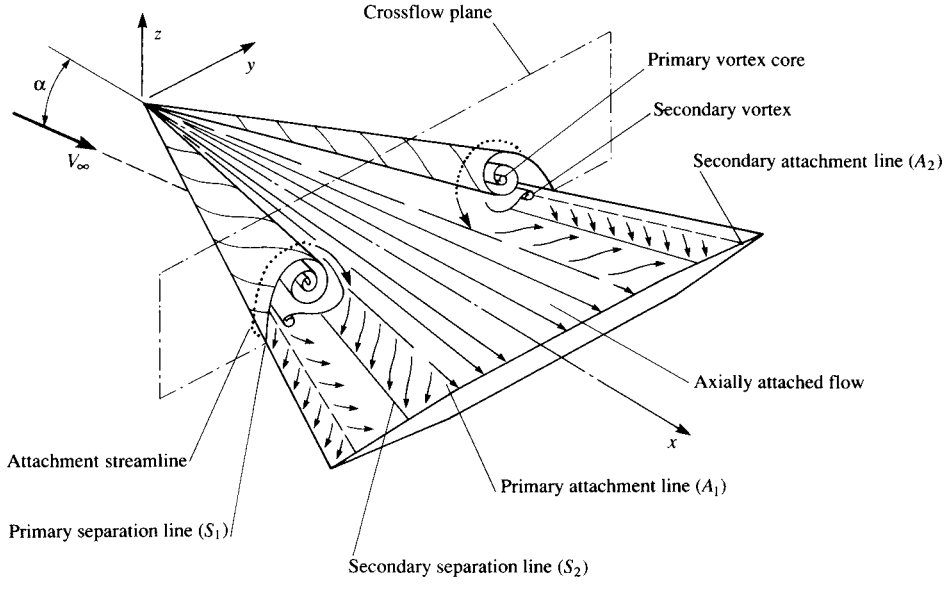


Figure 5.41 Schematic of the subsonic flow field over the top of a delta wing at angle of attack.
(Courtesy of John Stollery, Cranfield Institute of Technology, England.)

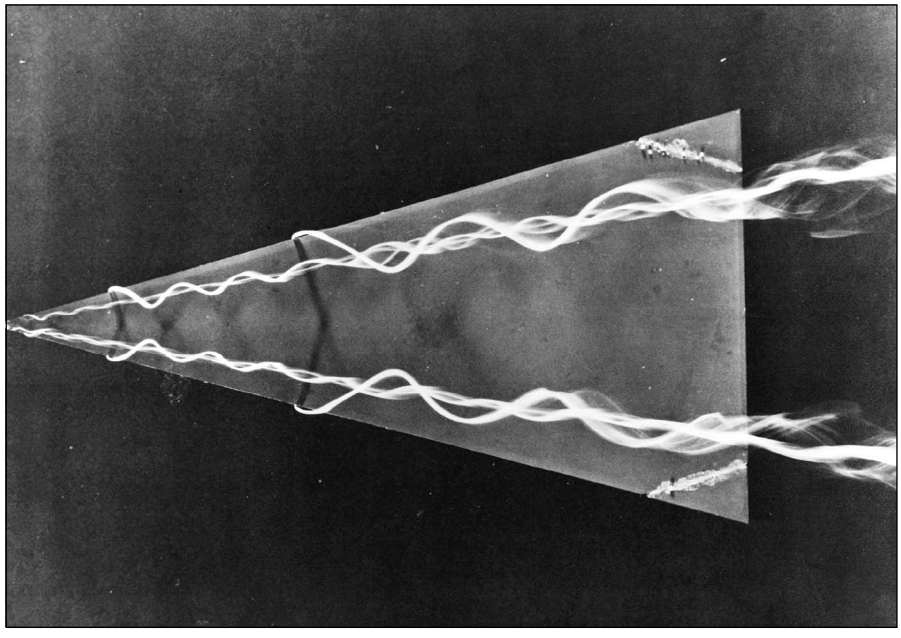


Figure 5.42 Leading-edge vortices over the top surface of a delta wing at angle of attack. The vortices are made visible by dye streaks in water flow. *(Courtesy of H. Werle, ONERA, France. Also in Van Dyke, Milton,* An Album of Fluid Motion, *The Parabolic Press, Stanford, CA, 1982.)*

reattaches along the primary attachment line (line A_1 in Figure 5.41). The primary vortex is contained within this loop. A secondary vortex is formed underneath the primary vortex, with its own separation line, denoted by S_2 in Figure 5.41, and its own reattachment line A_2. Notice that the surface streamlines flow *away* from the attachment lines A_1 and A_2 on both sides of these lines, whereas the surface streamlines tend to flow *toward* the separation lines S_1 and S_2 and then simply lift off the surface along these lines. Inboard of the leading-edge vortices, the surface streamlines are attached, and flow downstream virtually is undisturbed along a series of straight-line rays emanating from the vertex of the triangular shape. A graphic illustration of the leading-edge vortices is shown in both Figures 5.42 and 5.43. In Figure 5.42, we see a highly swept delta wing mounted in a water tunnel. Filaments of colored dye are introduced at two locations along each leading edge. This photograph, taken from an angle looking down on the top of the wing, clearly shows the entrainment of the colored dye in the vortices. Figure 5.43 is a photograph of the vortex pattern in the crossflow plane (the crossflow plane is shown in Figure 5.41). From the photographs in Figures 5.42 and 5.43, we clearly see that the leading-edge vortex is real and is positioned above and somewhat inboard of the leading edge itself.

The leading-edge vortices are strong and stable. Being a source of high energy, relatively high-vorticity flow, the local static pressure in the vicinity of the vortices

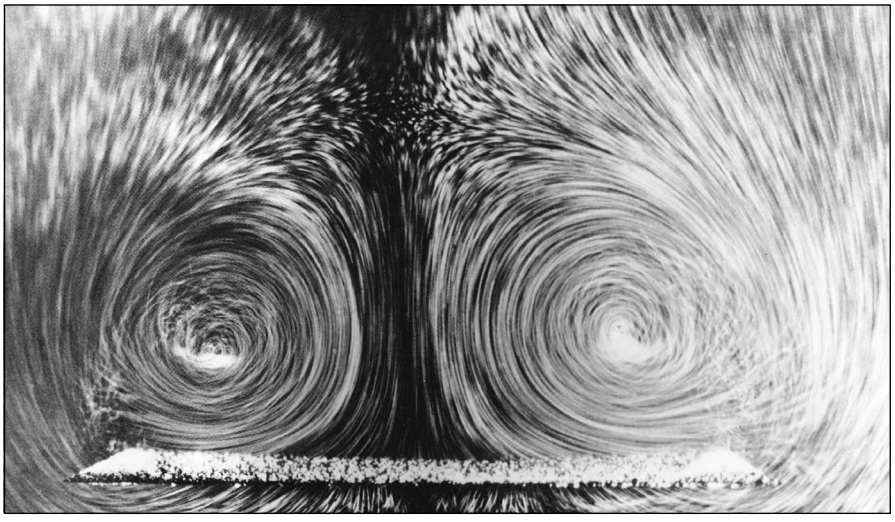


Figure 5.43 The flow field in the crossflow plane above a delta wing at angle of attack, showing the two primary leading-edge vortices. The vortices are made visible by small air bubbles in water. *(Courtesy of H. Werle, ONERA, France. Also in Van Dyke, Milton,* An Album of Fluid Motion, *The Parabolic Press, Stanford, CA, 1982.)*

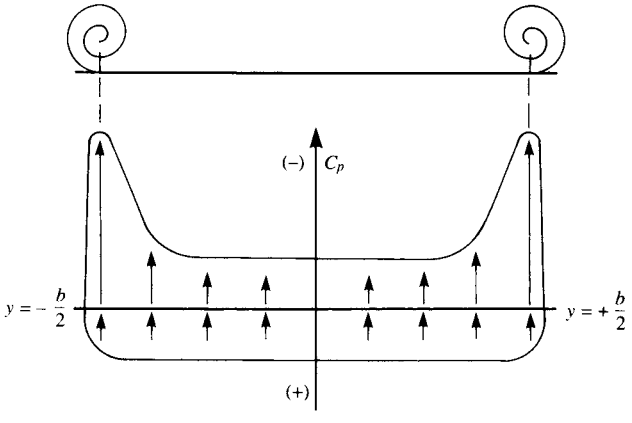


Figure 5.44 Schematic of the spanwise pressure coefficient distribution across a delta wing. *(Data Courtesy of John Stollery, Cranfield Institute of Technology, England.)*

is small. Hence, the surface pressure on the top surface of the delta wing is reduced near the leading edge and is higher and reasonably constant over the middle of the wing. The qualitative variation of the pressure coefficient in the spanwise direction (the y direction as shown in Figure 5.41) is sketched in Figure 5.44. The spanwise variation of pressure over the bottom surface is essentially constant

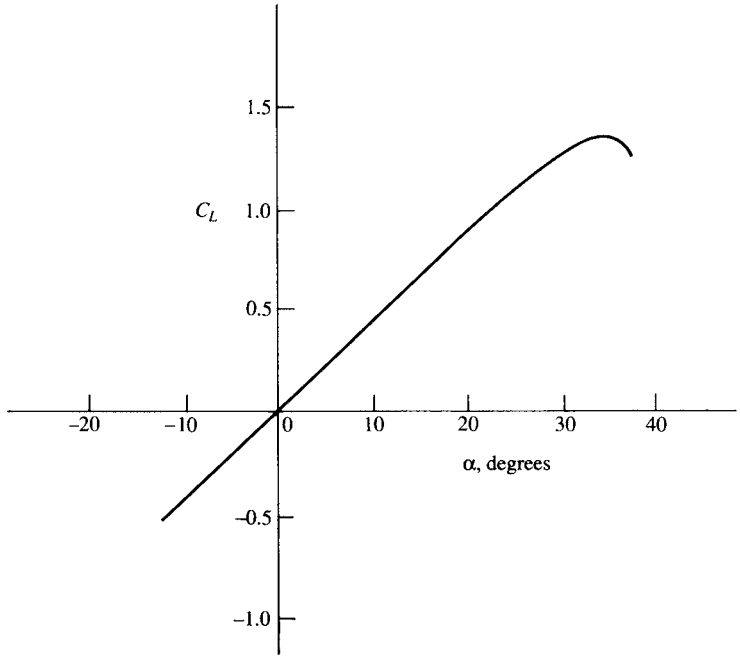


Figure 5.45 Variation of lift coefficient for a flat delta wing with angle of attack. *(Data Courtesy of John Stollery, Cranfield Institute of Technology, England.)*

and higher than the freestream pressure (a positive C_p). Over the top surface, the spanwise variation in the midsection of the wing is essentially constant and lower than the freestream pressure (a negative C_p). However, near the leading edges the static pressure drops considerably (the values of C_p become more negative). The leading-edge vortices are literally creating a strong "suction" on the top surface near the leading edges. In Figure 5.44, vertical arrows are shown to indicate further the effect on the spanwise lift distribution; the upward direction of these arrows as well as their relative length show the local contribution of each section of the wing to the normal force distribution. The suction effect of the leading-edge vortices is clearly shown by these arrows.

The suction effect of the leading-edge vortices enhances the lift; for this reason, the lift coefficient curve for a delta wing exhibits an increase in C_L for values of α at which conventional wing planforms would be stalled. A typical variation of C_L with α for a 60° delta wing is shown in Figure 5.45. Note the following characteristics:

1. The lift slope is small, on the order of 0.05/degree.
2. However, the lift continues to increase to large values of α; in Figure 5.45, the stalling angle of attack is on the order of 35°. The net result is a reasonable value of $C_{L,\max}$, on the order of 1.3.

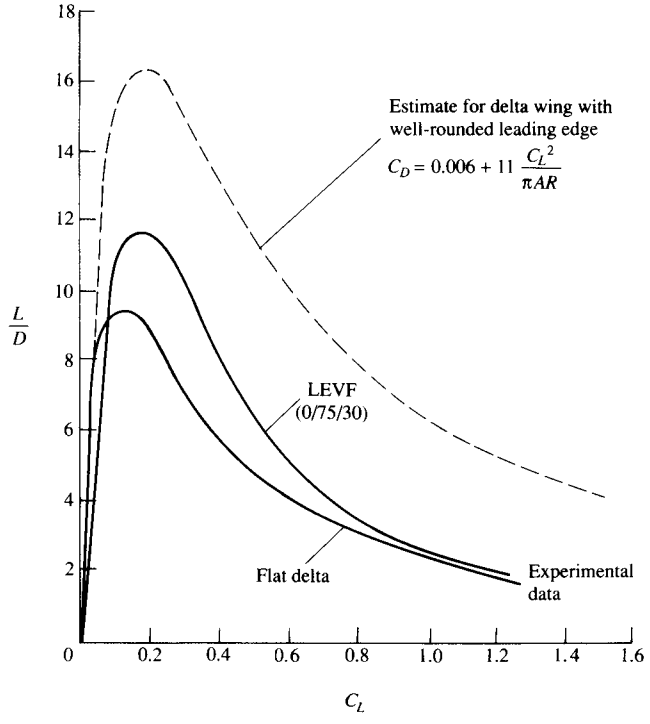


Figure 5.46 The effect of leading-edge shape on the lift-to-drag ratio for a delta wing of aspect ratio 2.31. The two solid curves apply to a sharp leading edge, and the dashed curve applies to a rounded leading edge. LEVF denotes a wing with a leading-edge vortex flap. *(Data Courtesy of John Stollery, Cranfield Institute of Technology, England.)*

The next time you have an opportunity to watch a delta-winged airplane take off or land, say, for example, the televised landing of the space shuttle, note the large angle of attack of the vehicle. Moreover, you will understand why the angle of attack is large—because the lift slope is small, and hence the angle of attack must be large enough to generate the high values of C_L required for low-speed flight.

The suction effect of the leading-edge vortices, in acting to increase the normal force, consequently, increases the drag at the same time it increases the lift. Hence, the aerodynamic effect of these vortices is not necessarily advantageous. In fact, the lift-to-drag ratio L/D for a delta planform is not so high as conventional wings. The typical variation of L/D with C_L for a delta wing is shown in Figure 5.46, the results for the sharp leading edge, 60° delta wing are given by the lower curve. Note that the maximum value of L/D for this case is about 9.3—not a particularly exciting value for a low-speed aircraft.

There are two other phenomena that are reflected by the data in Figure 5.46. The first is the effect of greatly rounding the leading edges of the delta wing. In our

previous discussions, we have treated the case of a sharp leading edge; such sharp edges cause the flow to separate at the leading edge, forming the leading-edge vortices. On the other hand, if the leading-edge radius is large, the flow separation will be minimized, or possibly will not occur. In turn, the drag penalty discussed above will not be present, and hence the L/D ratio will increase. The dashed curve in Figure 5.46 is the case for a 60° delta wing with well-rounded leading edges. Note that $(L/D)_{max}$ for this case is about 16.5, almost a factor of 2 higher than the sharp leading-edge case. However, keep in mind that these are results for subsonic speeds. There is a major design compromise reflected in these results. At the beginning of this section, we mentioned that the delta-wing planform with sharp leading edges is advantageous for supersonic flight—its highly swept shape in combination with sharp leading edges has a low supersonic drag. However, at supersonic speeds this advantage will be negated if the leading edges are rounded to any great extent. We will find in our study of supersonic flow in Part 3 that a blunt-nosed body creates very large values of wave drag. Therefore, leading edges with large radii are not appropriate for supersonic aircraft; indeed, it is desirable to have as sharp a leading edge as is practically possible for supersonic airplanes. A singular exception is the design of the space shuttle. The leading-edge radius of the space shuttle is large; this is due to three features that combine to make such blunt leading edges advantageous for the shuttle. First, the shuttle must slow down early during reentry into the earth's atmosphere to avoid massive aerodynamic heating (aspects of aerodynamic heating are discussed in Part 4). Therefore, in order to obtain this deceleration, a high drag is desirable for the space shuttle; indeed, the maximum L/D ratio of the space shuttle during reentry is about 2. A large leading-edge radius, with its attendant high drag, is therefore advantageous. Second, as we will see in Part 4, the rate of aerodynamic heating to the leading edge itself—a region of high heating—is inversely proportional to the square root of the leading-edge radius. Hence, the larger the radius, the smaller will be the heating rate to the leading edge. Third, as already explained above, a highly rounded leading edge is certainly advantageous to the shuttle's subsonic aerodynamic characteristics. Hence, a well-rounded leading edge is an important design feature for the space shuttle on all accounts. However, we must be reminded that this is not the case for more conventional supersonic aircraft, which demand very sharp leading edges. For these aircraft, a delta wing with a sharp leading edge has relatively poor subsonic performance.

This leads to the second of the phenomena reflected in Figure 5.46. The middle curve in Figure 5.46 is labeled LEVF, which denotes the case for a leading-edge vortex flap. This pertains to a mechanical configuration where the leading edges can be deflected downward through a variable angle, analogous to the deflection of a conventional trailing-edge flap. The spanwise pressure-coefficient distribution for this case is sketched in Figure 5.47; note that the direction of the suction due to the leading-edge vortice is now modified in comparison to the case with no leading-edge flap shown earlier in Figure 5.44. Also, returning to Figure 5.41, you can visualize what the wing geometry would look like with the leading edge

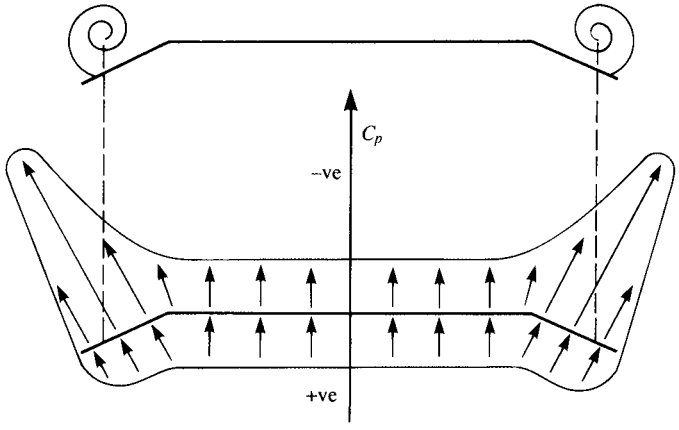


Figure 5.47 A schematic of the spanwise pressure coefficient distribution over the top of a delta wing as modified by leading-edge vortex flaps. *(Data Courtesy of John Stollery, Cranfield Institute of Technology, England.)*

drooped down; a front view of the downward deflected flap would actually show some projected frontal area. Since the pressure is low over this frontal area, the net drag can decrease. This phenomenon is illustrated by the middle curve in Figure 5.46, which shows a generally higher L/D for the leading-edge vortex flap in comparison to the case with no flap (the flat delta wing).

Finally, we note something drastic that occurs in the flow over the top surface of a delta wing when it is at a high enough angle of attack. The primary vortices shown in Figures 5.41 and 5.42 begin to fall apart somewhere along the length of the vortex; this is called *vortex breakdown,* illustrated in Figure 5.48. Compare this photograph with that shown in Figure 5.42 for well-behaved vortices at lower angle of attack. In Figure 5.48 the two leading-edge vortices show vortex breakdown at a location about two-thirds along their length over the top of the wing. This photograph is particularly interesting because it shows two types of vortex breakdown. The vortex at the top of the photograph exhibits a spiral-type of vortex breakdown, where the breakdown occurs progressively along the core and causes the core to twist in various directions. The vortex at the bottom of the photograph exhibits a bubble-type of vortex breakdown, where the vortex suddenly bursts, forming a large bubble of chaotic flow. The spiral type of vortex breakdown is more common. When vortex breakdown occurs, the lift and pitching moment of the delta wing decrease, the flow becomes unsteady, and buffeting of the wing occurs.

The progressive development of vortex breakdown is shown by the computational fluid dynamic (CFD) results in Figure 5.49 for a delta wing with a 60-degree sweep angle. At $\alpha = 5^\circ$ (Figure 5.49*a*), the vortex core is well-behaved. At $\alpha = 15^\circ$ (Figure 5.49*b*), vortex breakdown is starting. At $\alpha = 40^\circ$ (Figure 5.49*c*), the flow over the top of the delta wing is completely separated,

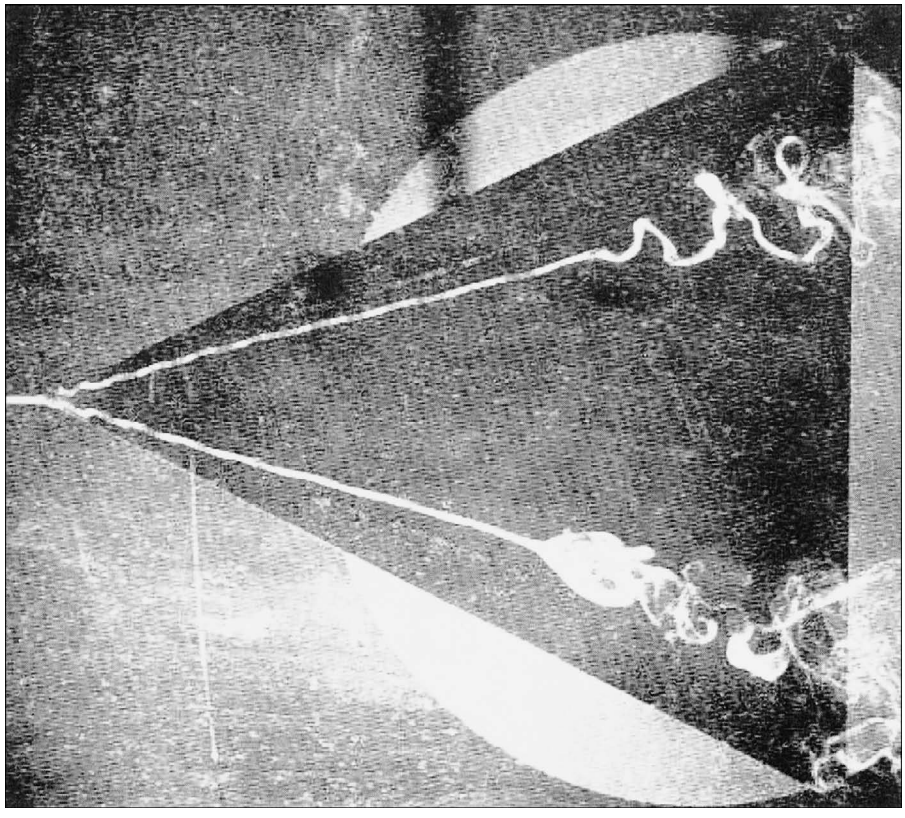


Figure 5.48 Vortex breakdown over a delta wing. (*Source:* N. C. Lamboune, and D. W. Bryer, "The Bursting of Leading Edge Vortices: Some Observation and Discussion of the Phenomenon," Aeronautical Research Council R&M 3282, 1962.)

and the wing is stalled. The results in Figure 5.49 are interesting for another reason as well. In a footnote to Section 4.4 we noted that inviscid flow calculations sometimes predict the location and nature of flow separation. Here we see another such case. The vortex breakdown and separated flow shown in Figure 5.49 are calculated from a CFD solution of the Euler equations (i.e., an inviscid flow calculation). It appears that friction does not play a critical role in vortex formation and breakdown.

(For more information on vortex bursting over delta wings, see the recent survey by I. Gursul, "Recent Developments in Delta Wing Aerodynamics," *The Aeronautical Journal,* vol. 108, number 1087, September 2004, pp. 437–452.)

In summary, the delta wing is a common planform for supersonic aircraft. In this section, we have examined the low-speed aerodynamic characteristics of such wings and have found that these characteristics are in some ways quite different from a conventional planform.

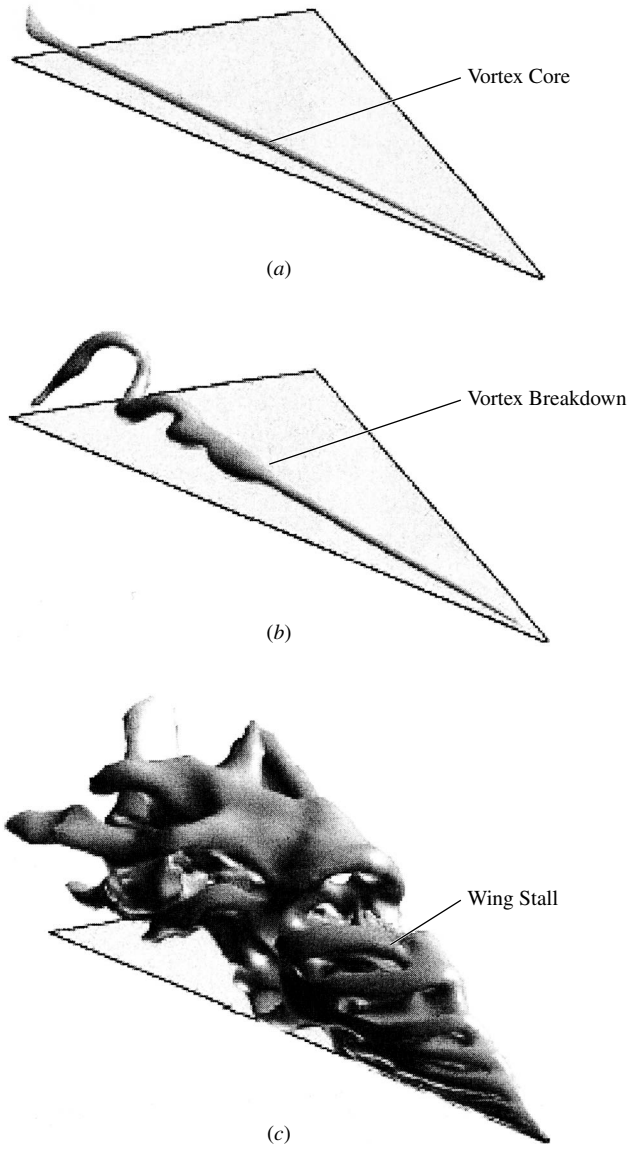


Figure 5.49 Vortex behavior and breakdown progression with increasing angles of attack of (*a*) 5°, (*b*) 15°, (*c*) 40°. (*Source:* R. E. Gordnier and M. R. Visbal, "Computation of the Aeroelastic Response of a Flexible Delta Wing at High Angles-of-Attack," AIAA Paper 2003-1728, 2003.)

5.7 HISTORICAL NOTE: LANCHESTER AND PRANDTL—THE EARLY DEVELOPMENT OF FINITE-WING THEORY

On June 27, 1866, in a paper entitled "Aerial Locomotion" given to the Aeronautical Society of Great Britain, the Englishman Francis Wenham expressed for the first time in history the effect of aspect ratio on finite-wing aerodynamics. He theorized (correctly) that most of the lift of a wing occurs from the portion near the leading edge, and hence a long, narrow wing would be most efficient. He suggested stacking a number of long thin wings above each other to generate the required lift, and he built two full-size gliders in 1858, both with five wings each, to demonstrate (successfully) his ideas. (Wenham is also known for designing and building the first wind tunnel in history, at Greenwich, England, in 1871.)

However, the true understanding of finite-wing aerodynamics, as well as ideas for the theoretical analysis of finite wings, did not come until 1907. In that year, Frederick W. Lanchester published his now famous book entitled *Aerodynamics*. We have met Lanchester before—in Section 4.15 concerning his role in the development of the circulation theory of lift. Here, we examine his contributions to finite-wing theory.

In Lanchester's *Aerodynamics,* we find the first mention of vortices that trail downstream of the wing tips. Figure 5.50 is one of Lanchester's own drawings from his 1907 book, showing the "vortex trunk" which forms at the wing tip. Moreover, he knew that a vortex filament could not end in space (see Section 5.2), and he theorized that the vortex filaments that constituted the two wing-tip vortices must cross the wing along its span—the first concept of bound vortices in the spanwise direction. Hence, the essence of the horseshoe vortex concept originated with Lanchester. In his own words:

> Thus the author regards the two trailed vortices as a definite proof of the existence of a cyclic component of equal strength in the motion surrounding the airfoil itself.

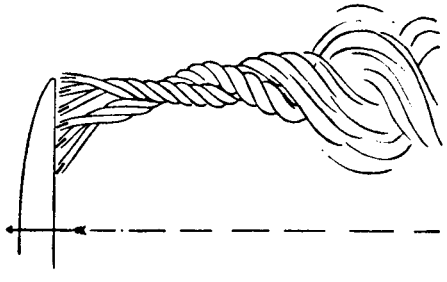


Figure 5.50 A figure from Lanchester's *Aerodynamics,* 1907; this is his own drawing of the wing-tip vortex on a finite wing.

Considering the foresight and originality of Lanchester's thinking, let us pause for a moment and look at the man himself. Lanchester was born on October 23, 1868, in Lewisham, England. The son of an architect, Lanchester became interested in engineering at an early age. (He was told by his family that his mind was made up at the age of 4.) He studied engineering and mining during the years 1886–1889 at the Royal College of Science in South Kensington, London, but never officially graduated. He was a quick-minded and innovative thinker and became a designer at the Forward Gas Engine Company in 1889, specializing in internal combustion engines. He rose to the post of assistant works manager. In the early 1890s, Lanchester became very interested in aeronautics, and along with his development of high-speed engines, he also carried out numerous aerodynamics experiments. It was during this period that he formulated his ideas on both the circulation theory of lift and the finite-wing vortex concepts. A serious paper written by Lanchester first for the Royal Society, and then for the Physical Society, was turned down for publication—something Lanchester never forgot. Finally, his aeronautical concepts were published in his two books *Aerodynamics* and *Aerodonetics* in 1907 and 1908, respectively. To his detriment, Lanchester had a style of writing and a means of explanation that were not easy to follow and his works were not immediately seized upon by other researchers. Lanchester's bitter feelings about the public's receipt of his papers and books are graphically seen in his letter to the Daniel Guggenheim Medal Fund decades later. In a letter dated June 6, 1931, Lanchester writes:

> So far as aeronautical science is concerned, I cannot say that I experienced anything but discouragement; in the early days my theoretical work (backed by a certain amount of experimental verification), mainly concerning the vortex theory of sustentation and the screw propeller, was refused by the two leading scientific societies in this country, and I was seriously warned that my profession as an engineer would suffer if I dabbled in a subject that was merely a dream of madmen! When I published my two volumes in 1907 and 1908 they were well received on the whole, but this was mainly due to the success of the brothers Wright, and the general interest aroused on the subject.

In 1899, he formed the Lanchester Motor Company, Limited, and sold automobiles of his own design. He married in 1919, but had no children. Lanchester maintained his interest in automobiles and related mechanical devices until his death on March 8, 1946, at the age of 77.

In 1908, Lanchester visited Göttingen, Germany, and fully discussed his wing theory with Ludwig Prandtl and his student Theodore von Karman. Prandtl spoke no English, Lanchester spoke no German, and in light of Lanchester's unclear way of explaining his ideas, there appeared to be little chance of understanding between the two parties. However, shortly after, Prandtl began to develop his own wing theory, using a bound vortex along the span and assuming that the vortex trails downstream from both wing tips. The first mention of Prandtl's work on finite-wing theory was made in a paper by O. Foppl in 1911, discussing some

of Foppl's experimental work on finite wings. Commenting on his results, Foppl says:

> They agree very closely with the theoretical investigation by Professor Prandtl on the current around an airplane with a finite span wing. Already Lanchester in his work, "Aerodynamics" (translated into German by C. and A. Runge), indicated that to the two extremities of an airplane wing are attached two vortex ropes (Wirbelzopfe) which make possible the transition from the flow around the airplane, which occurs nearly according to Kutta's theory, to the flow of the undisturbed fluid at both sides. These two vortex ropes continue the vortex which, according to Kutta's theory, takes place on the lamina.
>
> We are led to admit this owing to the Helmholtz theorem that vortices cannot end in the fluid. At any rate these two vortex ropes have been made visible in the Göttingen Institute by emitting an ammonia cloud into the air. Prandtl's theory is constructed on the consideration of this current in reality existing.

In the same year, Prandtl expressed his own first published words on the subject. In a paper given at a meeting of the Representatives of Aeronautical Science in Göttingen in November 1911, entitled "Results and Purposes of the Model Experimental Institute of Göttingen," Prandtl states:

> Another theoretical research relates to the conditions of the current which is formed by the air behind an airplane. The lift generated by the airplane is, on account of the principle of action and reaction, necessarily connected with a descending current behind the airplane. Now it seemed very useful to investigate this descending current in all its details. It appears that the descending current is formed by a pair of vortices, the vortex filaments of which start from the airplane wing tips. The distance of the two vortices is equal to the span of the airplane, their strength is equal to the circulation of the current around the airplane and the current in the vicinity of the airplane is fully given by the superposition of the uniform current with that of a vortex consisting of three rectilinear sections.

In discussing the results of his theory, Prandtl goes on to state in the same paper:

> The same theory supplies, taking into account the variations of the current on the airplane which came from the lateral vortices, a relationship showing the dependence of the airplane lift on the aspect ratio; in particular it gives the possibility of extrapolating the results thus obtained experimentally to the airplane of infinite span wing. From the maximum aspect ratios measured by us (1:9 to that of $1{:}\infty$) the lifts increase further in marked degree—by some 30 or 40 percent. I would add here a remarkable result of this extrapolation, which is, that the results of Kutta's theory of the infinite wing, at least so far as we are dealing with small cambers and small angles of incidence, have been confirmed by these experimental results.
>
> Starting from this line of thought we can attack the problem of calculating the surface of an airplane so that lift is distributed along its span in a determined manner, previously fixed. The experimental trial of these calculations has not yet been made, but it will be in the near future.

It is clear from the above comments that Prandtl was definitely following the model proposed earlier by Lanchester. Moreover, the major concern of the finite-wing theory was first in the calculation of lift—no mention is made of induced drag.

It is interesting to note that Prandtl's theory first began with a single horseshoe vortex, such as sketched in Figure 5.13. The results were not entirely satisfactory. During the period 1911–1918, Prandtl and his colleagues expanded and refined his finite-wing theory, which evolved to the concept of a lifting line consisting of an infinite number of horseshoe vortices, as sketched in Figure 5.15. In 1918, the term "induced drag" was coined by Max Munk, a colleague of Prandtl at Göttingen. Much of Prandtl's development of finite-wing theory was classified secret by the German government during World War I. Finally, his lifting-line theory was released to the outside world, and his ideas were published in English in a special NACA report written by Prandtl and published in 1922, entitled "Applications of Modern Hydrodynamics to Aeronautics" (NACA TR 116). Hence, the theory we have outlined in Section 5.3 was well-established more than 80 years ago.

One of Prandtl's strengths was the ability to base his thinking on sound ideas, and to apply intuition that resulted in relatively straightforward theories that most engineers could understand and appreciate. This is in contrast to the difficult writings of Lanchester. As a result, the lifting theory for finite wings has come down through the years identified as *Prandtl's lifting-line theory,* although we have seen that Lanchester was the first to propose the basic model on which lifting-line theory is built.

In light of Lanchester's 1908 visit with Prandtl and Prandtl's subsequent development of the lifting-line theory, there has been some discussion over the years that Prandtl basically stole Lanchester's ideas. However, this is clearly not the case. We have seen in the above quotes that Prandtl's group at Göttingen was giving full credit to Lanchester as early as 1911. Moreover, Lanchester never gave the world a clear and practical theory with which results could be readily obtained—Prandtl did. Therefore, in this book we have continued the tradition of identifying the lifting-line theory with Prandtl's name. On the other hand, for very good reasons, in England and various places in western Europe, the theory is labeled the *Lanchester-Prandtl theory*.

To help put the propriety in perspective, Lanchester was awarded the Daniel Guggenheim Medal in 1936 (Prandtl had received this award some years earlier). In the medal citation, we find the following words:

> Lanchester was the foremost person to propound the now famous theory of flight based on the Vortex theory, so brilliantly followed up by Prandtl and others. He first put forward his theory in a paper read before the Birmingham Natural History and Philosophical Society on 19th June, 1894. In a second paper in 1897, in his two books published in 1907 and 1908, and in his paper read before the Institution of Automobile Engineers in 1916, he further developed this doctrine.

Perhaps the best final words on Lanchester are contained in this excerpt from his obituary found in the British periodical *Flight* in March 1946:

> And now Lanchester has passed from our ken but not from our thoughts. It is to be hoped that the nation which neglected him during much of his lifetime will at any rate perpetuate his work by a memorial worthy of the "Grand Old Man" of aerodynamics.

5.8 HISTORICAL NOTE: PRANDTL—THE MAN

The modern science of aerodynamics rests on a strong fundamental foundation, a large percentage of which was established in one place by one man—at the University of Göttingen by Ludwig Prandtl. Prandtl never received a Nobel Prize, although his contributions to aerodynamics and fluid mechanics are felt by many to be of that caliber. Throughout this book, you will encounter his name in conjunction with major advances in aerodynamics: thin airfoil theory in Chapter 4, finite-wing theory in Chapter 5, supersonic shock- and expansion-wave theory in Chapter 9, compressibility corrections in Chapter 11, and what may be his most important contribution, namely, the boundary-layer concept in Chapter 17. Who was this man who has had such a major impact on fluid dynamics? Let us take a closer look.

Ludwig Prandtl was born on February 4, 1874, in Freising, Bavaria. His father was Alexander Prandtl, a professor of surveying and engineering at the agricultural college at Weihenstephan, near Freising. Although three children were born into the Prandtl family, two died at birth, and Ludwig grew up as an only child. His mother, the former Magdalene Ostermann, had a protracted illness, and partly as a result of this, Prandtl became very close to his father. At an early age, Prandtl became interested in his father's books on physics, machinery, and instruments. Much of Prandtl's remarkable ability to go intuitively to the heart of a physical problem can be traced to his environment at home as a child, where his father, a great lover of nature, induced Ludwig to observe natural phenomena and to reflect on them.

In 1894, Prandtl began his formal scientific studies at the Technische Hochschule in Munich, where his principal teacher was the well-known mechanics professor, August Foppl. Six years later, he graduated from the University of Munich with a Ph.D., with Foppl as his advisor. However, by this time Prandtl was alone, his father having died in 1896 and his mother in 1898.

By 1900, Prandtl had not done any work or shown any interest in fluid mechanics. Indeed, his Ph.D. thesis at Munich was in solid mechanics, dealing with unstable elastic equilibrium in which bending and distortion acted together. (It is not generally recognized by people in fluid dynamics that Prandtl continued his interest and research in solid mechanics through most of his life—this work is eclipsed, however, by his major contributions to the study of fluid flow.) However, soon after graduation from Munich, Prandtl had his first major encounter with fluid mechanics. Joining the Nuremburg works of the Maschinenfabrick Augsburg as an engineer, Prandtl worked in an office designing mechanical equipment for the new factory. He was made responsible for redesigning an apparatus for removing machine shavings by suction. Finding no reliable information in the scientific literature about the fluid mechanics of suction, Prandtl arranged his own experiments to answer a few fundamental questions about the flow. The result of this work was his new design for shavings cleaners. The apparatus was modified with pipes of improved shape and size, and carried out satisfactory operation at one-third its original power consumption. Prandtl's contributions in fluid mechanics had begun.

One year later, in 1901, he became Professor of Mechanics in the Mathematical Engineering Department at the Technische Hochschule in Hanover. (Please note that in Germany a "technical high school" is equivalent to a technical university in the United States.) It was at Hanover that Prandtl enhanced and continued his new-found interest in fluid mechanics. He also developed his boundary-layer theory and became interested in supersonic flow through nozzles at Hanover. In 1904, Prandtl delivered his famous paper on the concept of the boundary layer to the Third Congress on Mathematicians at Heidelberg. Entitled "Über Flussigkeitsbewegung bei sehr kleiner Reibung," Prandtl's Heidelberg paper established the basis for most modern calculations of skin friction, heat transfer, and flow separation (see Chapters 15 to 20). From that time on, the star of Prandtl was to rise meteorically. Later that year, he moved to the prestigious University of Göttingen to become Director of the Institute for Technical Physics, later to be renamed Applied Mechanics. Prandtl spent the remainder of his life at Göttingen, building his laboratory into the world's greatest aerodynamic research center of the 1904–1930 time period.

At Göttingen, during 1905–1908 Prandtl carried out numerous experiments on supersonic flow through nozzles and developed oblique shock- and expansion-wave theory (see Chapter 9). He took the first photographs of the supersonic flow through nozzles, using a special schlieren optical system (see chapter 4 of Reference 21). From 1910 to 1920, he devoted most of his efforts to low-speed aerodynamics, principally airfoil and wing theory, developing the famous lifting-line theory for finite wings (see Section 5.3). Prandtl returned to high-speed flows in the 1920s, during which he contributed to the evolution of the famous Prandtl-Glauert compressibility correction (see Sections 11.4 and 11.11).

By the 1930s, Prandtl was recognized worldwide as the "elder statesman" of fluid dynamics. Although he continued to do research in various areas, including structural mechanics and meteorology, his "Nobel Prize-level" contributions to fluid dynamics had all been made. Prandtl remained at Göttingen throughout the turmoil of World War II, engrossed in his work and seemingly insulated from the intense political and physical disruptions brought about by Nazi Germany. In fact, the German Air Ministry provided Prandtl's laboratory with new equipment and financial support. Prandtl's attitude at the end of the war is reflected in his comments to a U.S. Army interrogation team that swept through Göttingen in 1945; he complained about bomb damage to the roof of his house, and he asked how the Americans planned to support his current and future research. Prandtl was 70 at the time and was still going strong. However, the fate of Prandtl's laboratory at this time is summed up in the words of Irmgard Flugge-Lotz and Wilhelm Flugge, colleagues of Prandtl, who wrote 28 years later in the *Annual Review of Fluid Mechanics* (Vol. 5, 1973):

> World War II swept over all of us. At its end some of the research equipment was dismantled, and most of the research staff was scattered with the winds. Many are now in this country (the United States) and in England, some have returned. The seeds sown by Prandtl have sprouted in many places, and there are now many "second growth" Göttingers who do not even know that they are.

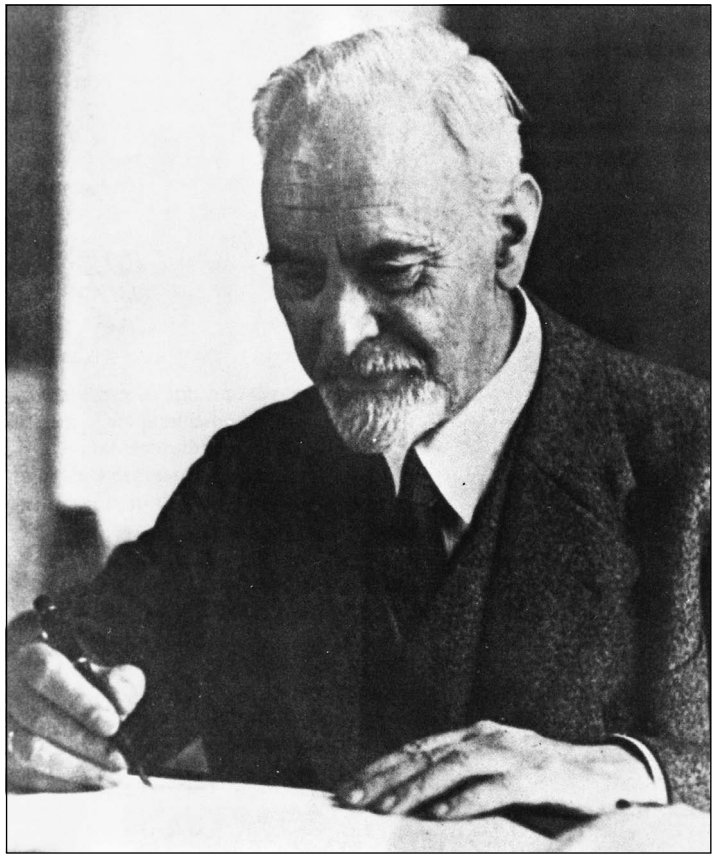


Figure 5.51 Ludwig Prandtl (1875–1953). *(Courtesy of the John Anderson collection)*

What type of person was Prandtl? By all accounts he was a gracious man, studious, likable, friendly, and totally focused on those things that interested him. He enjoyed music and was an accomplished pianist. Figure 5.51 shows a rather introspective man busily at work. One of Prandtl's most famous students, Theodor von Kármán, wrote in his autobiography *The Wind and Beyond* (Little, Brown and Company, Boston, 1967) that Prandtl bordered on being naive. A favorite story along these lines is that, in 1909, Prandtl decided that he should be married, but he did not know quite what to do. He finally wrote to Mrs. Foppl, the wife of his respected teacher, asking permission to marry one of her two daughters. Prandtl and Foppl's daughters were acquainted, but nothing more than that. Moreover, Prandtl did not stipulate which daughter. The Foppl's made a family decision that Prandtl should marry the elder daughter, Gertrude. This story from von Karman has been disputed in recent years by a Prandtl relative. Nevertheless, the marriage took place, leading to a happy relationship. The Prandtl's had two daughters, born in 1914 and 1917.

Prandtl was considered a tedious lecturer because he could hardly make a statement without qualifying it. However, he attracted excellent students who later went on to distinguish themselves in fluid mechanics—such as Jakob Ackeret in Zurich, Switzerland, Adolf Busemann in Germany, and Theodor von Kármán at Aachen, Germany, and later at Cal Tech in the United States.

Prandtl died in 1953. He was clearly the father of modern aerodynamics—a monumental figure in fluid dynamics. His impact will be felt for centuries to come.

5.9 SUMMARY

Return to the chapter road map in Figure 5.7, and review the straightforward path we have taken during the development of finite-wing theory. Make certain that you feel comfortable with the flow of ideas before proceeding further.

A brief summary of the important results of this chapter follows:

The wing-tip vortices from a finite wing induce a downwash which reduces the angle of attack effectively seen by a local airfoil section:

$$\alpha_{\text{eff}} = \alpha - \alpha_i \tag{5.1}$$

In turn, the presence of downwash results in a component of drag defined as induced drag D_i.

Vortex sheets and vortex filaments are useful in modeling the aerodynamics of finite wings. The velocity induced by a directed segment **dl** of a vortex filament is given by the Biot-Savart law:

$$\mathbf{dV} = \frac{\Gamma}{4\pi} \frac{\mathbf{dl} \times \mathbf{r}}{|\mathbf{r}|^3} \tag{5.5}$$

In Prandtl's classical lifting-line theory, the finite wing is replaced by a single spanwise lifting line along which the circulation $\Gamma(y)$ varies. A system of vortices trails downstream from the lifting line, which induces a downwash at the lifting line. The circulation distribution is determined from the fundamental equation

$$\alpha(y_0) = \frac{\Gamma(y_0)}{\pi V_\infty c(y_0)} + \alpha_{L=0}(y_0) + \frac{1}{4\pi V_\infty} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy)\, dy}{y_0 - y} \tag{5.23}$$

Results from classical lifting-line theory:

Elliptic wing:

Downwash is constant:

$$w = -\frac{\Gamma_0}{2b} \tag{5.35}$$

$$\alpha_i = \frac{C_L}{\pi \text{AR}} \tag{5.42}$$

$$C_{D,i} = \frac{C_L^2}{\pi \text{AR}} \tag{5.43}$$

$$a = \frac{a_0}{1 + a_0/\pi \text{AR}} \tag{5.69}$$

General wing:

$$C_{D,i} = \frac{C_L^2}{\pi \text{AR}}(1 + \delta) = \frac{C_L^2}{\pi e \text{AR}} \tag{5.61) and (5.62}$$

$$a = \frac{a_0}{1 + (a_0/\pi \text{AR})(1 + \tau)} \tag{5.70}$$

For low-aspect-ratio wings, swept wings, and delta wings, lifting-surface theory must be used. In modern aerodynamics, such lifting-surface theory is implemented by the vortex panel or the vortex lattice techniques.

5.10 PROBLEMS

5.1 Consider a vortex filament of strength Γ in the shape of a closed circular loop of radius R. Obtain an expression for the velocity induced at the center of the loop in terms of Γ and R.

5.2 Consider the same vortex filament as in Problem 5.1. Consider also a straight line through the center of the loop, perpendicular to the plane of the loop. Let A be the distance along this line, measured from the plane of the loop. Obtain an expression for the velocity at distance A on the line, as induced by the vortex filament.

5.3 The measured lift slope for the NACA 23012 airfoil is 0.1080 degree^{-1}, and $\alpha_{L=0} = -1.3^\circ$. Consider a finite wing using this airfoil, with AR = 8 and taper ratio = 0.8. Assume that $\delta = \tau$. Calculate the lift and induced drag coefficients for this wing at a geometric angle of attack = 7°.

5.4 The Piper Cherokee (a light, single-engine general aviation aircraft) has a wing area of 170 ft^2 and a wing span of 32 ft. Its maximum gross weight is 2450 lb. The wing uses an NACA 65-415 airfoil, which has a lift slope of 0.1033 $degree^{-1}$ and $\alpha_{L=0} = -3^\circ$. Assume $\tau = 0.12$. If the airplane is cruising at 120 mi/h at standard sea level at its maximum gross weight and is in straight-and-level flight, calculate the geometric angle of attack of the wing.

5.5 Consider the airplane and flight conditions given in Problem 5.4. The span efficiency factor e for the complete airplane is generally much less than that for the finite wing alone. Assume $e = 0.64$. Calculate the induced drag for the airplane in Problem 5.4.

5.6 Consider a finite wing with an aspect ratio of 6. Assume an elliptical lift distribution. The lift slope for the airfoil section is 0.1/degree. Calculate and compare the lift slopes for (a) a straight wing, and (b) a swept wing, with a half-chord line sweep of 45 degrees.

5.7 Repeat Problem 5.6, except for a lower aspect ratio of 3. From a comparison of the results from these two problems, draw some conclusions about the effect of wing sweep on the lift slope, and how the magnitude of this effect is affected by aspect ratio.

5.8 In Problem 1.19 we noted that the Wright brothers, in the design of their 1900 and 1901 gliders, used aerodynamic data from the Lilienthal table given in Figure 1.65. They chose a design angle of attack of 3 degrees, corresponding to a design lift coefficient of 0.546. When they tested their gliders at Kill Devil Hills near Kitty Hawk , North Carolina, in 1900 and 1901, however, they measured only one-third the amount of lift they had originally calculated on the basis of the Lilienthal table. This led the Wrights to question the validity of Lilienthal's data, and this cast a pall on the Lilienthal table that has persisted to the present time. However, in Reference 62 this author shows that the Lilienthal data are reasonably valid, and that the Wrights misinterpreted the data in the Lilienthal table in three respects (see pages 209–216 of Reference 62). One of these respects was the difference in aspect ratio. The Wrights' 1900 glider had rectangular wings with an aspect ratio of 3.5, whereas the data in the Lilienthal table were taken with a wing with an ogival planform tapering to a point at the tip and with an aspect ratio of 6.48. The Wrights seemed not to appreciate the aerodynamic importance of aspect ratio at the time, and even if they had, there was no existing theory that would have allowed them to correct the Lilienthal data for their design. (Prandtl's lifting line theory appeared 18 years later.) Given just the difference in aspect ratio between the Wrights' glider and the test model used by Lilienthal, what value of lift coefficient should the Wrights have used instead of the value of 0.546 they took straight from the table? (*Note:* There are two other misinterpretations by the Wrights that resulted in their calculation of lift being too high; see Reference 62 for details.)

5.9 Consider the Supermarine Spitfire shown in Figure 5.19. The first version of the Spitfire was the Mk I, which first flew in 1936. Its maximum velocity is 362 mi/h at an altitude of 18,500 ft. Its weight is 5820 lb, wing area is 242 ft^2, and wing span is 36.1 ft. It is powered by a supercharged Merlin engine, which produced 1050 horsepower at 18,500 ft. (a) Calculate the induced drag coefficient of the Spitfire at the flight condition of V_{max} at 18,500 ft. (b) What percentage of the total drag coefficient is the induced drag coefficient? *Note:* To calculate the total drag, we note that in steady, level flight of the airplane, $T = D$, where T is the thrust from the propeller. In turn, the thrust is related to the power by the basic mechanical relation $TV_\infty = P$, where P is the power supplied by the propeller-engine combination. Because of aerodynamic losses experienced by the propeller, P is less than the shaft power provided by the engine by a ratio, η, defined as the propeller efficiency. That is, if HP is the shaft horsepower provided by the engine, and since 550 ft · lb/s equals 1 horsepower, then the power provided by the engine-propeller combination in foot-pounds per second is $P = 550\eta$HP. See Chapter 6 of Reference 2 for more details. For this problem, assume the propeller efficiency for the Spitfire is 0.9.

5.10 If the elliptical wing of the Spitfire in Problem 5.9 were replaced by a tapered wing with a taper ratio of 0.4, everything else remaining the same, calculate the induced drag coefficient. Compare this value with that obtained in Problem 5.9. What can you conclude about the relative effect of planform shape change on the drag of the airplane at high speeds?

5.11 Consider the Spitfire in Problem 5.9 on its landing approach at sea level with a landing velocity of 70 mi/h. Calculate the induced drag coefficient for this low-speed case. Compare your result with the high-speed case in Problem 5.9. From this, what can you conclude about the relative importance of the induced drag coefficient at low speeds compared to that at high speeds?

CHAPTER 6

Three-Dimensional Incompressible Flow

Treat nature in terms of the cylinder, the sphere, the cone, all in perspective.
Paul Cézanne, 1890

PREVIEW BOX

We go three-dimensional in this chapter. For such a huge and complex subject, this chapter is mercifully short. It has only three objectives. This first is to see what happens when the circular cylinder studied in Chapter 3 morphs into a sphere—how do we modify the theory to account for the three-dimensional flow over a sphere, and how are the results changed from those for a circular cylinder? The second is to demonstrate a general phenomenon in aerodynamics known as the *three-dimensional relieving effect.* The third is to briefly examine the aerodynamic flow over a complete three-dimensional flight vehicle. Achieving these objectives is important; they will further open your mind to the wonders of aerodynamics. Read on, and enjoy.

6.1 INTRODUCTION

To this point in our aerodynamic discussions, we have been working mainly in a two-dimensional world; the flows over the bodies treated in Chapter 3 and the airfoils in Chapter 4 involved only two dimensions in a single plane—so-called planar flows. In Chapter 5, the analyses of a finite wing were carried out in the plane of the wing, in spite of the fact that the detailed flow over a finite wing is truly three-dimensional. The relative simplicity of dealing with two dimensions (i.e., having only two independent variables) is self-evident and is the reason why a large bulk of aerodynamic theory deals with two-dimensional flows. Fortunately,

the two-dimensional analyses go a long way toward understanding many practical flows, but they also have distinct limitations.

The real world of aerodynamic applications is three-dimensional. However, because of the addition of one more independent variable, the analyses generally become more complex. The accurate calculation of three-dimensional flow fields has been, and still is, one of the most active areas of aerodynamic research.

The purpose of this book is to present the fundamentals of aerodynamics. Therefore, it is important to recognize the predominance of three-dimensional flows, although it is beyond our scope to go into detail. Therefore, the purpose of this chapter is to introduce some very basic considerations of three-dimensional incompressible flow. This chapter is short; we do not even need a road map to guide us through it. Its function is simply to open the door to the analysis of three-dimensional flow.

The governing fluid flow equations have already been developed in three dimensions in Chapters 2 and 3. In particular, if the flow is irrotational, Equation (2.154) states that

$$\mathbf{V} = \nabla\phi \tag{2.154}$$

where, if the flow is also incompressible, the velocity potential is given by Laplace's equation:

$$\nabla^2\phi = 0 \tag{3.40}$$

Solutions of Equation (3.40) for flow over a body must satisfy the flow-tangency boundary condition on the body, that is,

$$\mathbf{V} \cdot \mathbf{n} = 0 \tag{3.48a}$$

where $\mathbf{n}$ is a unit vector normal to the body surface. In all of the above equations, ϕ is, in general, a function of three-dimensional space; for example, in spherical coordinates $\phi = \phi(r, \theta, \Phi)$. Let us use these equations to treat some elementary three-dimensional incompressible flows.

6.2 THREE-DIMENSIONAL SOURCE

Return to Laplace's equation written in spherical coordinates, as given by Equation (3.43). Consider the velocity potential given by

$$\phi = -\frac{C}{r} \tag{6.1}$$

where C is a constant and r is the radial coordinate from the origin. Equation (6.1) satisfies Equation (3.43), and hence it describes a physically possible incompressible, irrotational three-dimensional flow. Combining Equation (6.1) with the definition of the gradient in spherical coordinates, Equation (2.18), we obtain

$$\mathbf{V} = \nabla\phi = \frac{C}{r^2}\mathbf{e}_r \tag{6.2}$$

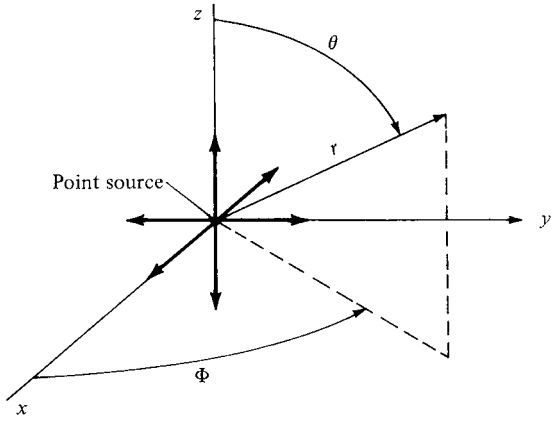


Figure 6.1 Three-dimensional (point) source.

In terms of the velocity components, we have

$$V_r = \frac{C}{r^2} \tag{6.3a}$$

$$V_\theta = 0 \tag{6.3b}$$

$$V_\Phi = 0 \tag{6.3c}$$

Clearly, Equation (6.2), or Equations (6.3*a* to *c*), describes a flow with straight streamlines emanating from the origin, as sketched in Figure 6.1. Moreover, from Equation (6.2) or (6.3*a*), the velocity varies inversely as the square of the distance from the origin. Such a flow is defined as a *three-dimensional source*. Sometimes it is called simply a *point source,* in contrast to the two-dimensional line source discussed in Section 3.10.

To evaluate the constant C in Equation (6.3*a*), consider a sphere of radius r and surface S centered at the origin. From Equation (2.46), the mass flow across the surface of this sphere is

$$\text{Mass flow} = \oiint_S \rho \mathbf{V} \cdot \mathbf{dS}$$

Hence, the *volume* flow, denoted by λ, is

$$\lambda = \oiint_S \mathbf{V} \cdot \mathbf{dS} \tag{6.4}$$

On the surface of the sphere, the velocity is a constant value equal to $V_r = C/r^2$ and is normal to the surface. Hence, Equation (6.4) becomes

$$\lambda = \frac{C}{r^2} 4\pi r^2 = 4\pi C$$

Hence,

$$C = \frac{\lambda}{4\pi} \tag{6.5}$$

Substituting Equation (6.5) into (6.3*a*), we find

$$\boxed{V_r = \frac{\lambda}{4\pi r^2}} \tag{6.6}$$

Compare Equation (6.6) with its counterpart for a two-dimensional source given by Equation (3.62). Note that the three-dimensional effect is to cause an inverse *r-squared* variation and that the quantity 4π appears rather than 2π. Also, substituting Equation (6.5) into (6.1), we obtain, for a point source,

$$\boxed{\phi = -\frac{\lambda}{4\pi r}} \tag{6.7}$$

In the above equations, λ is defined as the *strength* of the source. When λ is a negative quantity, we have a point sink.

6.3 THREE-DIMENSIONAL DOUBLET

Consider a sink and source of equal but opposite strength located at points O and A, as sketched in Figure 6.2. The distance between the source and sink is l. Consider an arbitrary point P located a distance r from the sink and a distance r_1 from the source. From Equation (6.7), the velocity potential at P is

$$\phi = -\frac{\lambda}{4\pi}\left(\frac{1}{r_1} - \frac{1}{r}\right)$$

or

$$\phi = -\frac{\lambda}{4\pi}\frac{r - r_1}{r r_1} \tag{6.8}$$

Let the source approach the sink as their strengths become infinite; that is, let

$$l \to 0 \qquad \text{as} \qquad \lambda \to \infty$$

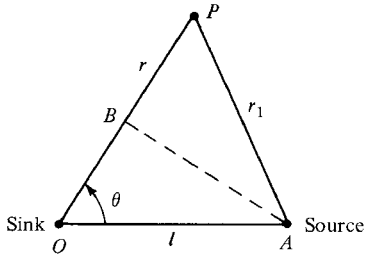


Figure 6.2 Source-sink pair. In the limit as $l \to 0$, a three-dimensional doublet is obtained.

In the limit, as $l \to 0$, $r - r_1 \to OB = l\cos\theta$, and $rr_1 \to r^2$. Thus, in the limit, Equation (6.8) becomes

$$\phi = -\lim_{\substack{l\to 0\\ \lambda\to\infty}} \frac{\lambda}{4\pi}\frac{r-r_1}{rr_1} = -\frac{\lambda}{4\pi}\frac{l\cos\theta}{r^2}$$

or

$$\boxed{\phi = -\frac{\mu}{4\pi}\frac{\cos\theta}{r^2}} \tag{6.9}$$

where $\mu = \Lambda l$. The flow field produced by Equation (6.9) is a *three-dimensional doublet;* μ is defined as the strength of the doublet. Compare Equation (6.9) with its two-dimensional counterpart given in Equation (3.88). Note that the three-dimensional effects lead to an inverse r-squared variation and introduce a factor 4π, versus 2π for the two-dimensional case.

From Equations (2.18) and (6.9), we find

$$\mathbf{V} = \nabla\phi = \frac{\mu}{2\pi}\frac{\cos\theta}{r^3}\mathbf{e}_r + \frac{\mu}{4\pi}\frac{\sin\theta}{r^3}\mathbf{e}_\theta + 0\mathbf{e}_\Phi \tag{6.10}$$

The streamlines of this velocity field are sketched in Figure 6.3. Shown are the streamlines in the zr plane; they are the same in all the zr planes (i.e., for all values of Φ). Hence, the flow induced by the three-dimensional doublet is a series of stream surfaces generated by revolving the streamlines in Figure 6.3 about the z axis. Compare these streamlines with the two-dimensional case illustrated in Figure 3.18; they are qualitatively similar but quantitatively different.

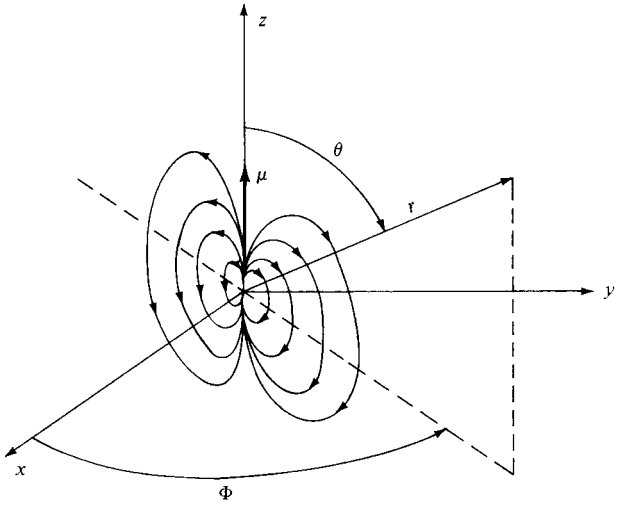


Figure 6.3 Sketch of the streamlines in the zr plane (Φ = constant plane) for a three-dimensional doublet.

Note that the flow in Figure 6.3 is independent of Φ; indeed, Equation (6.10) clearly shows that the velocity field depends only on r and θ. Such a flow is defined as *axisymmetric flow*. Once again, we have a flow with two independent variables. For this reason, axisymmetric flow is sometimes labeled "two-dimensional" flow. However, it is quite different from the two-dimensional planar flows discussed earlier. In reality, axisymmetric flow is a degenerate three-dimensional flow, and it is somewhat misleading to refer to it as "two-dimensional." Mathematically, it has only two independent variables, but it exhibits some of the same physical characteristics as general three-dimensional flows, such as the three-dimensional relieving effect to be discussed later.

6.4 FLOW OVER A SPHERE

Consider again the flow induced by the three-dimensional doublet illustrated in Figure 6.3. Superimpose on this flow a uniform velocity field of magnitude V_∞ in the negative z direction. Since we are more comfortable visualizing a freestream that moves horizontally, say, from left to right, let us flip the coordinate system in Figure 6.3 on its side. The picture shown in Figure 6.4 results.

Examining Figure 6.4, the spherical coordinates of the freestream are

$$V_r = -V_\infty \cos\theta \tag{6.11a}$$

$$V_\theta = V_\infty \sin\theta \tag{6.11b}$$

$$V_\Phi = 0 \tag{6.11c}$$

Adding V_r, V_θ, and V_Φ for the free stream, Equations (6.11a to c), to the representative components for the doublet given in Equation (6.10), we obtain, for the

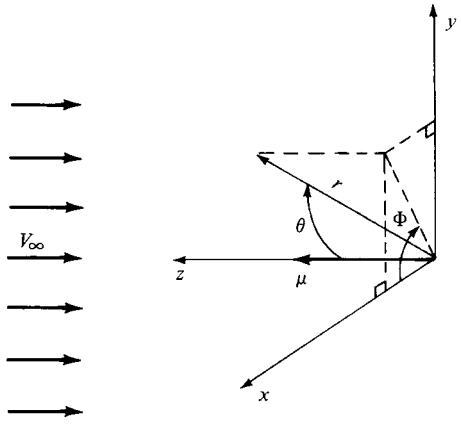


Figure 6.4 The superposition of a uniform flow and a three-dimensional doublet.

combined flow,

$$V_r = -V_\infty \cos\theta + \frac{\mu}{2\pi}\frac{\cos\theta}{r^3} = -\left(V_\infty - \frac{\mu}{2\pi r^3}\right)\cos\theta \tag{6.12}$$

$$V_\theta = V_\infty \sin\theta + \frac{\mu}{4\pi}\frac{\sin\theta}{r^3} = \left(V_\infty + \frac{\mu}{4\pi r^3}\right)\sin\theta \tag{6.13}$$

$$V_\Phi = 0 \tag{6.14}$$

To find the stagnation points in the flow, set $V_r = V_\theta = 0$ in Equations (6.12) and (6.13). From Equation (6.13), $V_\theta = 0$ gives $\sin\theta = 0$; hence, the stagnation points are located at $\theta = 0$ and π. From Equation (6.12), with $V_r = 0$, we obtain

$$V_\infty - \frac{\mu}{2\pi R^3} = 0 \tag{6.15}$$

where $r = R$ is the radial coordinate of the stagnation points. Solving Equation (6.15) for R, we obtain

$$R = \left(\frac{\mu}{2\pi V_\infty}\right)^{1/3} \tag{6.16}$$

Hence, there are two stagnation points, both on the z axis, with (r, θ) coordinates

$$\left[\left(\frac{\mu}{2\pi V_\infty}\right)^{1/3}, 0\right] \quad \text{and} \quad \left[\left(\frac{\mu}{2\pi V_\infty}\right)^{1/3}, \pi\right]$$

Insert the value of $r = R$ from Equation (6.16) into the expression for V_r given by Equation (6.12). We obtain

$$\begin{aligned} V_r &= -\left(V_\infty - \frac{\mu}{2\pi R^3}\right)\cos\theta = -\left[V_\infty - \frac{\mu}{2\pi}\left(\frac{2\pi V_\infty}{\mu}\right)\right]\cos\theta \\ &= -(V_\infty - V_\infty)\cos\theta = 0 \end{aligned}$$

Thus, $V_r = 0$ when $r = R$ for *all values* of θ and Φ. This is precisely the flow-tangency condition for flow over a sphere of radius R. Hence, the velocity field given by Equations (6.12) to (6.14) is the *incompressible flow over a sphere of radius R*. This flow is shown in Figure 6.5; it is qualitatively similar to the flow over the cylinder shown in Figure 3.19, but quantitatively the two flows are different.

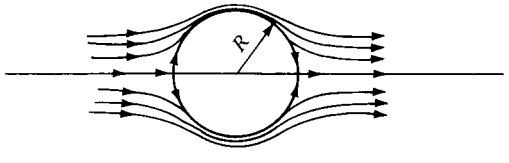


Figure 6.5 Schematic of the incompressible flow over a sphere.

On the surface of the sphere, where $r = R$, the tangential velocity is obtained from Equation (6.13) as follows:

$$V_\theta = \left(V_\infty + \frac{\mu}{4\pi R^3} \right) \sin\theta \tag{6.17}$$

From Equation (6.16),

$$\mu = 2\pi R^3 V_\infty \tag{6.18}$$

Substituting Equation (6.18) into (6.17), we have

$$V_\theta = \left(V_\infty + \frac{1}{4\pi} \frac{2\pi R^3 V_\infty}{R^3} \right) \sin\theta$$

or

$$\boxed{V_\theta = \tfrac{3}{2} V_\infty \sin\theta} \tag{6.19}$$

The maximum velocity occurs at the top and bottom points of the sphere, and its magnitude is $\frac{3}{2}V_\infty$. Compare these results with the two-dimensional circular cylinder case given by Equation (3.100). For the two-dimensional flow, the maximum velocity is $2V_\infty$. Hence, for the same V_∞, the maximum surface velocity on a sphere is *less* than that for a cylinder. The flow over a sphere is somewhat "relieved" in comparison with the flow over a cylinder. The flow over a sphere has an extra dimension in which to move out of the way of the solid body; the flow can move sideways as well as up and down. In contrast, the flow over a cylinder is more constrained; it can only move up and down. Hence, the maximum velocity on a sphere is less than that on a cylinder. This is an example of the *three-dimensional relieving effect,* which is a general phenomenon for all types of three-dimensional flows.

The pressure distribution on the surface of the sphere is given by Equations (3.38) and (6.19) as follows:

$$C_p = 1 - \left(\frac{V}{V_\infty} \right)^2 = 1 - \left(\frac{3}{2} \sin\theta \right)^2$$

or

$$\boxed{C_p = 1 - \tfrac{9}{4} \sin^2\theta} \tag{6.20}$$

Compare Equation (6.20) with the analogous result for a circular cylinder given by Equation (3.101). Note that the absolute magnitude of the pressure coefficient on a sphere is less than that for a cylinder—again, an example of the three-dimensional relieving effect. The pressure distributions over a sphere and a cylinder are compared in Figure 6.6, which dramatically illustrates the three-dimensional relieving effect.

6.4.1 Comment on the Three-Dimensional Relieving Effect

There is a good physical reason for the three-dimensional relieving effect. First, visualize the two-dimensional flow over a circular cylinder. In order to move out

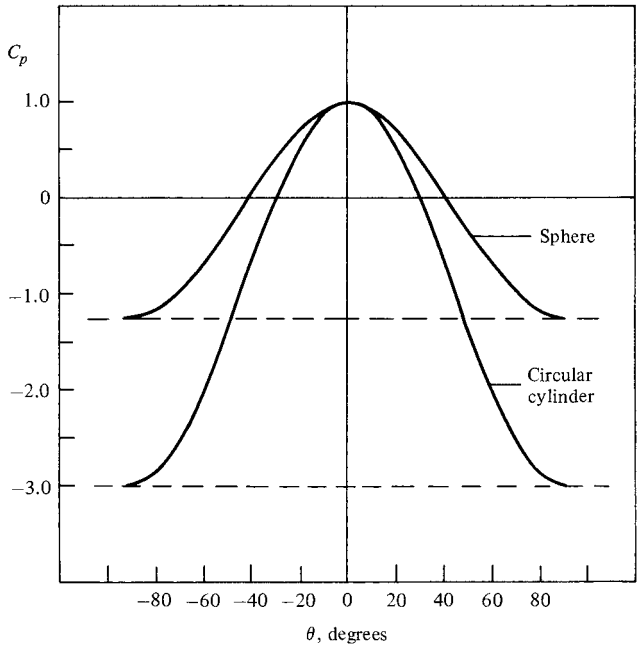


Figure 6.6 The pressure distribution over the surface of a sphere and a cylinder. Illustration of the three-dimensional relieving effect.

of the way of the cylinder, the flow has only two ways to go: riding up-and-over and down-and-under the cylinder. In contrast, visualize the three-dimensional flow over a sphere. In addition to moving up-and-over and down-and-under the sphere, the flow can now move sideways, to the left and right over the sphere. This sidewise movement relieves the previous constraint on the flow; the flow does not have to speed up so much to get out of the way of the sphere, and therefore the pressure in the flow does not have to change so much. The flow is "less stressed"; it moves around the sphere in a more relaxed fashion—it is "relieved," and consequently the changes in velocity and pressure are smaller.

6.5 GENERAL THREE-DIMENSIONAL FLOWS: PANEL TECHNIQUES

In modern aerodynamic applications, three-dimensional, inviscid, incompressible flows are almost always calculated by means of numerical panel techniques. The philosophy of the two-dimensional panel methods discussed in previous chapters is readily extended to three dimensions. The details are beyond the scope of this book—indeed, there are dozens of different variations, and the resulting computer programs are frequently long and sophisticated. However, the general idea behind all such panel programs is to cover the three-dimensional body with panels over

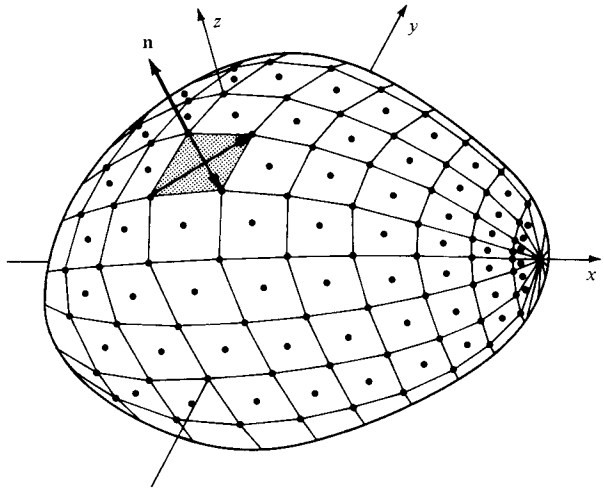


Figure 6.7 Distribution of three-dimensional source panels over a general nonlifting body *(Reference 14)*.

which there is an unknown distribution of singularities (such as point sources, doublets, or vortices). Such paneling is illustrated in Figure 6.7. These unknowns are solved through a system of simultaneous linear algebraic equations generated by calculating the induced velocity at control points on the panels and applying the flow-tangency condition. For a nonlifting body such as illustrated in Figure 6.7, a distribution of source panels is sufficient. However, for a lifting body, both source and vortex panels (or their equivalent) are necessary. A striking example of the extent to which panel methods are now used for three-dimensional lifting bodies is shown in Figure 6.8, which illustrates the paneling used for calculations made by the Boeing Company of the potential flow over a Boeing 747–space shuttle piggyback combination. Such applications are very impressive; moreover, they have become an industry standard and are today used routinely as part of the airplane design process by the major aircraft companies.

Examining Figures 6.7 and 6.8, one aspect stands out, namely, the geometric complexity of distributing panels over the three-dimensional bodies. How do you get the computer to "see" the precise shape of the body? How do you distribute the panels over the body; that is, do you put more at the wing leading edges and less on the fuselage, etc.? How many panels do you use? These are all nontrivial questions. It is not unusual for an aerodynamicist to spend weeks or even a few months determining the best geometric distribution of panels over a complex body.

We end this section on the following note. From the time they were introduced in the 1960s, panel techniques have revolutionized the calculation of three-dimensional potential flows. However, no matter how complex the application of these methods may be, the techniques are still based on the fundamentals

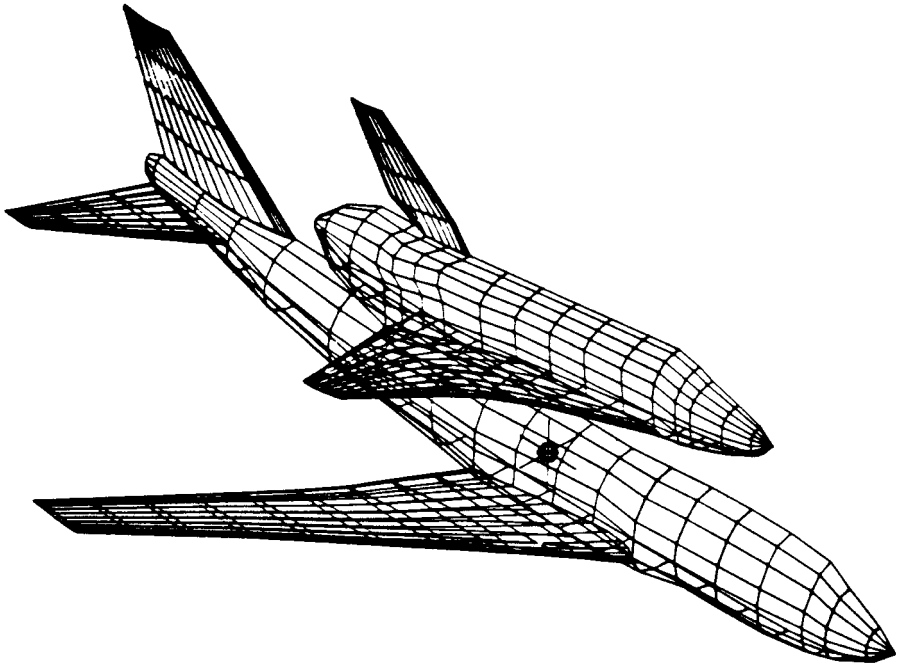


Figure 6.8 Panel distribution for the analysis of the Boeing 747 carrying the space shuttle orbiter.

we have discussed in this and all the preceding chapters. You are encouraged to pursue these matters further by reading the literature, particularly as it appears in such journals as the *Journal of Aircraft* and the *AIAA Journal*.

6.6 APPLIED AERODYNAMICS: THE FLOW OVER A SPHERE—THE REAL CASE

The present section is a complement to Section 3.18, in which the real flow over a circular cylinder was discussed. Since the present chapter deals with three-dimensional flows, it is fitting at this stage to discuss the three-dimensional analog of the circular cylinder, namely, the sphere. The qualitative features of the real flow over a sphere are similar to those discussed for a cylinder in Section 3.18—the phenomenon of flow separation, the variation of drag coefficient with a Reynolds number, the precipitous drop in drag coefficient when the flow transits from laminar to turbulent ahead of the separation point at the critical Reynolds number, and the general structure of the wake. These items are similar for both cases. However, because of the three-dimensional relieving effect, the flow over a sphere is *quantitatively* different from that for a cylinder. These differences are the subject of the present section.

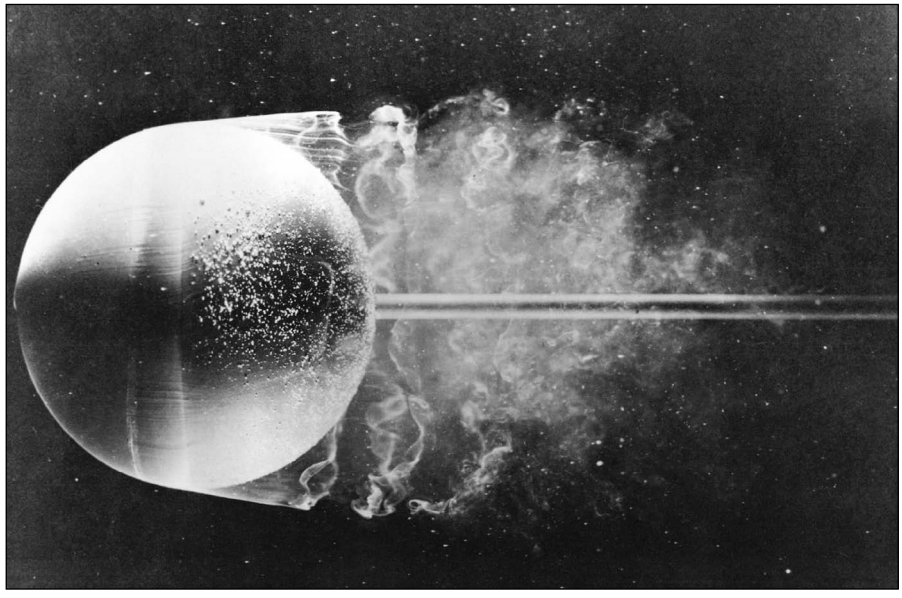


Figure 6.9 Laminar flow case: Instantaneous flow past a sphere in water. Re = 15,000. Flow is made visible by dye in the water. *(Courtesy of H. Werle, ONERA, France. Also in Van Dyke, Milton,* An Album of Fluid Motion, *The Parabolic Press, Stanford, CA, 1982.)*

The laminar flow over a sphere is shown in Figure 6.9. Here, the Reynolds number is 15,000, certainly low enough to maintain laminar flow over the spherical surface. However, in response to the adverse pressure gradient on the back surface of the sphere predicted by inviscid, incompressible flow theory (see Section 6.4 and Figure 6.6), the laminar flow readily separates from the surface. Indeed, in Figure 6.9, separation is clearly seen on the *forward* surface, slightly ahead of the vertical equator of the sphere. Thus, a large, fat wake trails downstream of the sphere, with a consequent large pressure drag on the body (analogous to that discussed in Section 3.18 for a cylinder.) In contrast, the turbulent flow case is shown in Figure 6.10. Here, the Reynolds number is 30,000, still a low number normally conducive to laminar flow. However, in this case, turbulent flow is induced artificially by the presence of a wire loop in a vertical plane on the forward face. (Trip wires are frequently used in experimental aerodynamics to induce transition to turbulent flow; this is in order to study such turbulent flows under conditions where they would not naturally exist.) Because the flow is turbulent, separation takes place much farther over the back surface, resulting in a thinner wake, as can be seen by comparing Figures 6.9 and 6.10. Consequently, the pressure drag is less for the turbulent case.

The variation of drag coefficient C_D with the Reynolds number for a sphere, is shown in Figure 6.11. Compare this figure with Figure 3.44 for a circular cylinder;

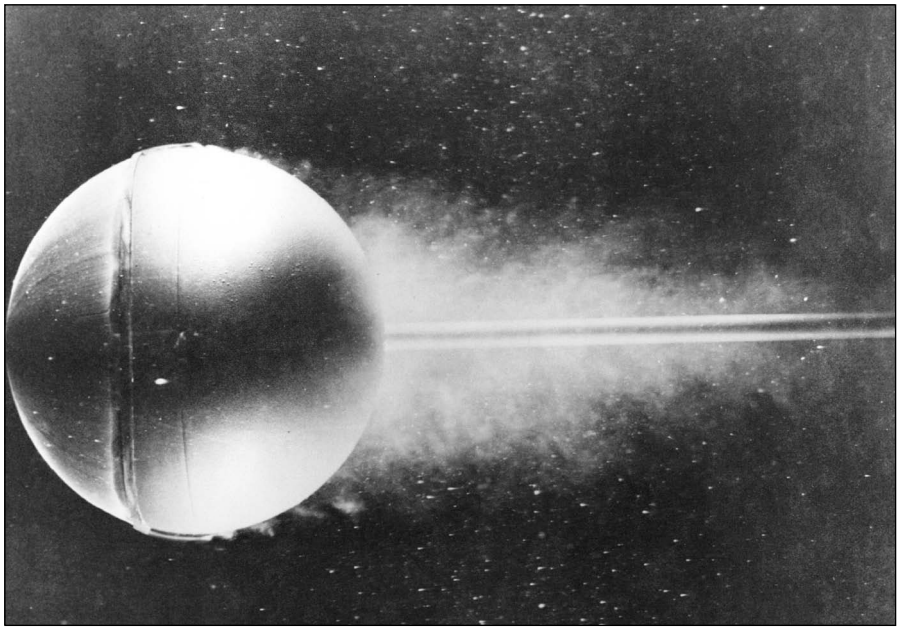


Figure 6.10 Turbulent flow case: Instantaneous flow past a sphere in water. Re = 30,000. The turbulent flow is forced by a trip wire hoop ahead of the equator, causing the laminar flow to become turbulent suddenly. The flow is made visible by air bubbles in water. *(Courtesy of H. Werle, ONERA, France. Also in Van Dyke, Milton,* An Album of Fluid Motion, *The Parabolic Press, Stanford, CA, 1982.)*

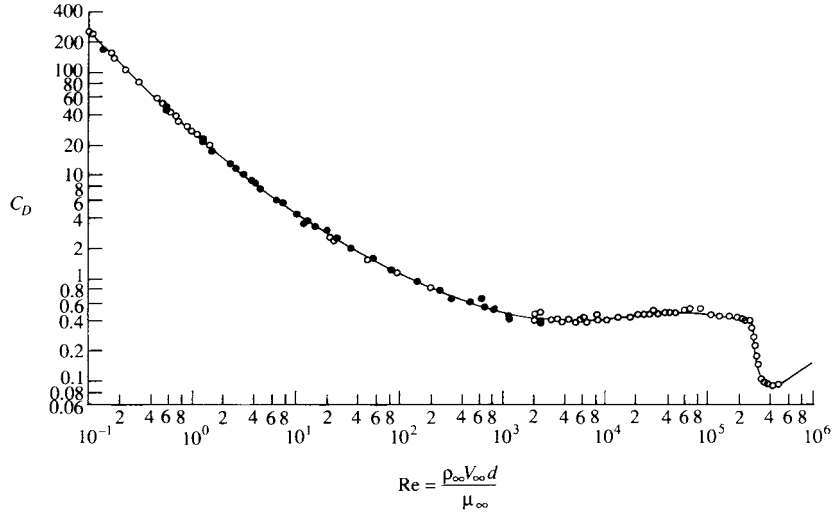


Figure 6.11 Variation of drag coefficient with Reynolds number for a sphere. *(Data taken from Schlichting, Reference 42.)*

the C_D variations are qualitatively similar, both with a precipitous decrease in C_D near a critical Reynolds number of 300,000, coinciding with natural transition from laminar to turbulent flow. However, *quantitatively* the two curves are quite different. In the Reynolds number range most appropriate to practical problems, that is, for Re $>$ 1000, the values of C_D for the sphere are considerably smaller than those for a cylinder—a classic example of the three-dimensional relieving effect. Reflecting on Figure 3.44 for the cylinder, note that the value of C_D for Re slightly less than the critical value is about 1 and drops to 0.3 for Re slightly above the critical value. In contrast, for the sphere as shown in Figure 6.11, C_D is about 0.4 in the Reynolds number range below the critical value and drops to about 0.1 for Reynolds numbers above the critical value. These variations in C_D for both the cylinder and sphere are classic results in aerodynamics; you should keep the actual C_D values in mind for future reference and comparisons.

As a final point in regard to both Figures 3.44 and 6.11, the value of the critical Reynolds number at which transition to turbulent flow takes place upstream of the separation point is not a fixed, universal number. Quite the contrary, transition is influenced by many factors, as will be discussed in Part 4. Among these is the amount of turbulence in the freestream; the higher the freestream turbulence, the more readily transition takes place. In turn, the higher the freestream turbulence, the lower is the value of the critical Reynolds number. Because of this trend, calibrated spheres are used in wind-tunnel testing actually to assess the degree of freestream turbulence in the test section, simply by measuring the value of the critical Reynolds number on the sphere.

6.7 APPLIED AERODYNAMICS: AIRPLANE LIFT AND DRAG

A three-dimensional object of primary interest to aerospace engineers is a whole airplane such as shown in Figure 6.8, not just the finite wing discussed in Chapter 5. In this section we expand our horizons to consider lift and drag of a complete airplane configuration.

We emphasized in Section 1.5 that the aerodynamic force on any body moving through the air is due only to two basic sources, the pressure and shear stress distributions exerted over the body surface. Lift is primarily created by the pressure distribution; shear stress has only a minor effect on lift. We have used this fact, beginning in Chapter 3 through to the present chapter, where the assumption of inviscid flow has given us reasonable predictions of the lift on cylinders with circulation, airfoils, and finite wings. Drag, on the other hand, is created by both the pressure and shear stress distributions, and analyses based on just inviscid flow are not sufficient for the prediction of drag.

6.7.1 Airplane Lift

We normally think of wings as the primary component producing the lift of an airplane in flight, and quite rightly so. However, even a pencil at an angle of

attack will generate lift, albeit small. Hence, lift is produced by the fuselage of an airplane as well as the wing. The mating of a wing with a fuselage is called a *wing-body combination*. The lift of a wing-body combination is *not* obtained by simply adding the lift of the wing alone to the lift of the body alone. Rather, as soon as the wing and body are mated, the flow field over the body modifies the flow field over the wing, and visa versa—this is called the *wing-body interaction*.

There is no accurate analytical equation that can predict the lift of a wing-body combination, properly taking into account the nature of the wing-body aerodynamic interaction. Either the configuration must be tested in a wind tunnel, or a computational fluid dynamic calculation must be made. We cannot even say in advance whether the combined lift will be greater or smaller than the sum of the two parts. For subsonic speeds, however, data obtained using different fuselage thicknesses, d, mounted on wings with different spans, b, show that the total lift for a wing-body combination is essentially constant for d/b ranging from 0 (wing only) to 6 (which would be an inordinately fat fuselage, with a short, stubby wing). Hence, the lift of the wing-body combination can be treated as simply the lift on the complete wing by itself, including that portion of the wing that is masked by the fuselage. This is illustrated in Figure 6.12. See Chapter 2 of Reference 69 for more details.

Of course, other components of the airplane such as a horizontal tail, canard surfaces, and wing strakes can contribute to the lift, either in a positive or negative sense. Once again we emphasize that reasonably accurate predictions of lift on a complete airplane can come only from wind tunnel tests, detailed computational fluid dynamic calculations (such as the panel calculations illustrated by Figure 6.8), and, of course, from actual flight tests of the airplane.

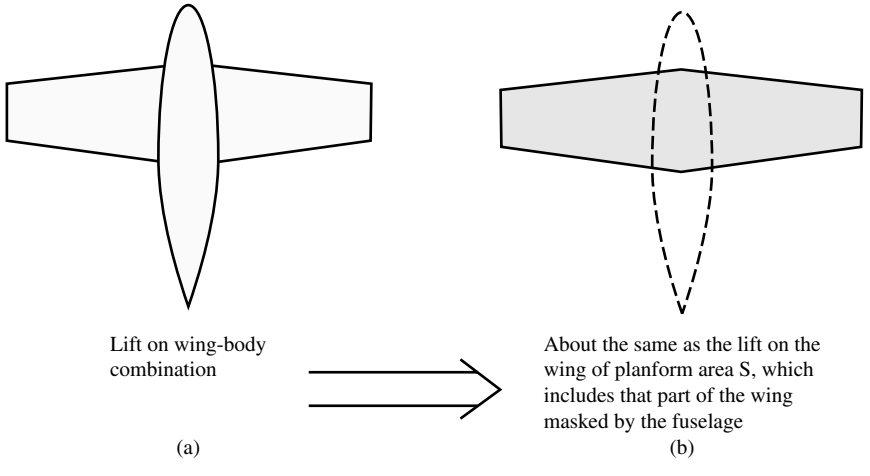


Figure 6.12 Lift on a wing-body combination.

6.7.2 Airplane Drag

When you watch an airplane flying overhead, or when you ride in an airplane, it is almost intuitive that your first aerodynamic thought is about lift. You are witnessing a machine that, in straight and level flight, is producing enough aerodynamic lift to equal the weight of the machine. This keeps it in the air—a vital concern. But this is only part of the role of airplane aerodynamics. It is equally important to produce this lift as *efficiently as possible*, that is, with as little drag as possible. The ratio of lift to drag, *L*/*D*, is a good measure of aerodynamic efficiency. A barn door will produce lift at angle of attack, but it also produces a lot of drag at the same time—the *L*/*D* for a barn door is terrible. For such reasons, minimizing drag has been one of the strongest drivers in the historical development of applied aerodynamics. And to minimize drag, we first have to provide methods for its estimation.

As in the case of lift, the drag of an airplane can *not* be obtained as the simple sum of the drag on each component. For example, for a wing-body combination, the drag is usually higher than the sum of the separate drag forces on the wing and the body, giving rise to an extra drag component called interference drag. For a more detailed discussion of airplane drag prediction, see Reference 69. The subject of drag prediction is so complex that whole books have been written about it; one classic is the book by Hoerner, Reference 116.

In this section we will limit our discussion to the simple extension of Equation (5.63) for application to the whole airplane. Equation (5.63), copied below, applies to a finite wing.

$$C_D = c_d + \frac{C_L^2}{\pi e \mathrm{AR}} \tag{5.63}$$

In Equation (5.63), C_D is the total drag coefficient for a finite wing, c_d is the profile drag coefficient caused by skin friction and pressure drag due to flow separation, and $C_L^2/\pi e\mathrm{AR}$ is the induced drag coefficient with the span efficiency factor e defined by Equation (5.62). For the whole airplane, Equation (5.63) is rewritten as

$$C_D = C_{D,e} + \frac{C_L^2}{\pi e \mathrm{AR}} \tag{6.21}$$

where C_D is the total drag coefficient for the airplane and $C_{D,e}$ is defined as the *parasite drag coefficient*, which contains not only the profile drag of the wing [c_d in Equation (5.63)] but also the friction and pressure drag of the tail surfaces, fuselage, engine nacelles, landing gear, and any other component of the airplane that is exposed to the airflow. Because of changes in the flow field around the airplane—especially changes in the amount of separated flow over parts of the airplane—as the angle of attack is varied, $C_{D,e}$ will change with angle of attack. Because the lift coefficient, C_L, is a specific function of angle of attack, we can consider that $C_{D,e}$ is a function of C_L. A reasonable approximation for this

function is

$$C_{D,e} = C_{D,o} + rC_L^2 \tag{6.22}$$

where r is an empirically determined constant. Since at zero lift, $C_L = 0$, then Equation (6.22) defines $C_{D,o}$ as the parasite drag coefficient at zero lift, or more commonly, the *zero-lift drag coefficient*. With Equation (6.22), we can write Equation (6.21) as

$$C_D = C_{D,o} + \left(r + \frac{1}{\pi e \text{AR}}\right) C_L^2 \tag{6.23}$$

In Equations (6.21) and (6.23), e is the familiar span efficiency factor, which takes into account the nonelliptical lift distribution on wings of general shape (see Section 5.3.2). Let us now *redefine* e so that it also includes the effect of the variation of parasite drag with lift; that is, let us write Equation (6.23) in the form

$$\boxed{C_D = C_{D,o} + \frac{C_L^2}{\pi e \text{AR}}} \tag{6.24}$$

where $C_{D,o}$ is the parasite drag coefficient at *zero lift* (or simply the *zero-lift drag coefficient* for the airplane) and the term $C_L^2/(\pi e \text{AR})$ is the *drag coefficient due to lift* including both induced drag and the contribution to parasite drag due to lift.

In Equation (6.24), the redefined e is called the *Oswald efficiency factor* (named after W. Bailey Oswald, who first established this terminology in NACA Report No. 408 in 1932). The use of the symbol e for the Oswald efficiency factor has become standard in the literature, and that is why we continue this standard here. To avoid confusion, keep in mind that e introduced for a finite wing in Section 5.3.2 and used in Equation (6.21) is the span efficiency factor for a finite wing, and the e used in Equation (6.24) is the Oswald efficiency factor for a complete airplane. These are two different numbers; the Oswald efficiency factor for different airplanes typically varies between 0.7 and 0.85 whereas the span efficiency factor typically varies between 0.9 and at most 1.0 and is a function of wing aspect ratio and taper ratio as demonstrated in Figure 5.20. Daniel Raymer in Reference 117 gives the following empirical expression for the Oswald efficiency factor for straight-wing aircraft, based on data obtained from actual airplanes:

$$e = 1.78\left(1 - 0.045\ \text{AR}^{0.68}\right) - 0.64 \tag{6.25}$$

Raymer notes that Equation (6.25) should be used for conventional aspect ratios for normal airplanes, and not for the very large aspect ratios (on the order of 25 or higher) associated with sailplanes.

Equation (6.24) conveys all the information you need to calculate the drag of a complete airplane, but to use it you have to know the zero-lift drag coefficient and the Oswald efficiency factor. Equation (6.24) is called the *drag polar* for the airplane, representing the variation of C_D with C_L. It is the cornerstone for conceptual airplane design and for predictions of the performance of a given aircraft (see Reference 69 for more details).

EXAMPLE 6.1

Return again to the photograph of the Seversky P-35 shown in Figure 3.2. This airplane has a wing planform area of 220 ft^2 and a wingspan of 36 ft. Also, examine again the drag breakdown for the Seversky XP-41 given in Figure 1.58. In Example 1.12 we assumed that the drag breakdown for the XP-41, being an airplane very similar to the P-35, applied to the P-35 as well. We do the same here. Using the data given in Figure 1.58, calculate the zero-lift drag coefficient for the P-35.

■ Solution

For the drag breakdown shown in Figure 1.58, condition 18 is that for the complete airplane configuration. For condition 18, the total drag coefficient is given as $C_D = 0.0275$ when the aircraft is at the particular angle of attack where $C_L = 0.15$. That is, we know simultaneous values of C_D and C_L that can be used in Equation (6.24). In that equation,

$$\text{AR} = \frac{b^2}{s} = \frac{(36)^2}{220} = 5.89$$

and the Oswald efficiency factor from Equation (6.25) is

$$\begin{aligned} e &= 1.78\,(1 - 0.045\text{AR}^{0.68}) - 0.64 \\ &= 1.78\,[1 - 0.045(5.89)^{0.68}] - 0.64 \\ &= 1.78\,[1 - 0.045(3.339)] - 0.64 \\ &= 0.873 \end{aligned}$$

Thus, Equation (6.24) gives for the zero-lift drag coefficient

$$\begin{aligned} C_{D,o} &= C_D - \frac{C_L^2}{\pi e\ \text{AR}} \\ &= 0.0275 - \frac{(0.15)^2}{\pi(0.873)(5.89)} \end{aligned}$$

or,

$$\boxed{C_{D,o} = 0.026}$$

The late Larry Loftin, in his excellent book *Quest for Performance: The Evolution of Modern Aircraft* (Reference 48), tabulated the zero-lift drag coefficient extracted from flight performance data for a large number of historic airplanes from the twentieth century. His tabulated value for the Seversky P-35 is $C_{D,o} = 0.0251$. Note that the value of $C_{D,o} = 0.026$ calculated in this example agrees within 3.6 percent.

Airplane Lift-to-Drag Ratio The dimensional analysis discussed in Section 1.7 proves that C_L, C_D, and hence the lift-to-drag ratio C_L/C_D, at a given Mach number and Reynolds number depend only on the shape of the body and

the angle of attack. This is reinforced by the sketches shown in Figure 6.13. For a given airplane shape, Figure 6.13*a* gives the variation of C_L with the airplane angle of attack,α; Figure 6.13*b* gives the variation of C_D with α; and Figure 6.13*c* gives the lift-to-drag ratio C_L/C_D as a function of α. These are *aerodynamic properties* associated with a given airplane. Within reasonable Mach number and Reynolds number ranges, we can simply talk about *the* lift coefficient, drag coefficient, and lift-to-drag ratio as specific values at any specific angle of attack. Indeed, in

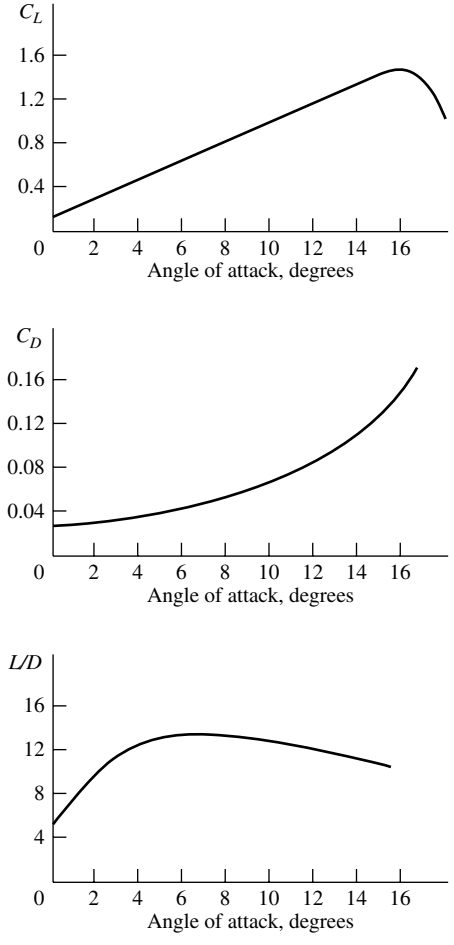


Figure 6.13 Typical variations of lift and drag coefficients and lift-to-drag ratio for a generic small propeller-driven general aviation airplane (based on calculations from Chapter 6 of Reference 2).

Example 6.1 we calculated *the* zero-lift drag coefficient for the Seversky P-35. We know that, strictly speaking, $C_{D,o}$ will depend on Mach number and Reynolds number, but for the normal flight regime of the low-speed subsonic aircraft germane to our present discussion, the variation of the airplane aerodynamic coefficients with Mach number and Reynolds number is considered small. Hence, for example, we can meaningfully talk about *the* $C_{D,o}$ for the airplane.

Examining Figure 6.13*c*, note that C_L/C_D first increases as α increases, reaches a maximum value at a certain value of α, and then subsequently decreases as α increases further. The maximum lift-to-drag ratio, $(L/D)_{\max} = (C_L/C_D)_{\max}$, is a direct measure of the aerodynamic efficiency of the airplane, and therefore its value is of great importance in airplane design and in the prediction of airplane performance (see, for example, References 2 and 69). Since $(L/D)_{\max}$ is an aerodynamic property of the airplane, we should be able to calculate its value from other known aerodynamic properties. Let us see.

$$\frac{C_L}{C_D} = \frac{C_L}{C_{D,o} + C_L^2/(\pi e \text{AR})} \tag{6.26}$$

For maximum C_L/C_D, differentiate Equation (6.26) with respect to C_L and set the result equal to 0:

$$\frac{d(C_L/C_D)}{dC_L} = \frac{C_{D,o} + \dfrac{C_L^2}{\pi e \text{AR}} - C_L[2C_L/(\pi e \text{AR})]}{[C_{D,o} + C_L^2/(\pi e \text{AR})]^2} = 0$$

Thus,

$$C_{D,o} + \frac{C_L^2}{\pi e \text{AR}} - \frac{2C_L^2}{\pi e \text{AR}} = 0$$

or

$$C_{D,o} = \frac{C_L^2}{\pi e \text{AR}} \tag{6.27}$$

Equation (6.27) is an interesting intermediate result. It states that when the airplane is flying at the specific angle of attack where the lift-to-drag ratio is maximum, the zero-lift drag and the drag due to lift are precisely equal. Solving Equation (6.27) for C_L, we have

$$C_L = \sqrt{\pi e \text{AR}\ C_{D,o}} \tag{6.28}$$

Equation (6.28) gives the value of C_L when the airplane is flying at $(L/D)_{\max}$. Return to Equation (6.25), which gives C_L/C_D as a function of C_L. By substituting the value of C_L from Equation (6.28), which pertains just to the maximum value of L/D, into Equation (6.25), we obtain for the maximum lift-to-drag ratio

$$\left(\frac{C_L}{C_D}\right)_{\max} = \frac{(\pi e \text{AR}\ C_{D,o})^{1/2}}{C_{D,o} + \dfrac{\pi e \text{AR} C_{D,o}}{\pi e \text{AR}}}$$

or,

$$\boxed{\left(\frac{C_L}{C_D}\right)_{\max} = \frac{(\pi e \text{AR}\ C_{D,o})^{1/2}}{2C_{D,o}}} \tag{6.29}$$

Equation (6.29) is *powerful*. It tells us that the maximum value of lift-to-drag ratio for a given airplane depends *only* on the zero-lift drag coefficient $C_{D,o}$, the Oswald efficiency factor e, and the wing aspect ratio. So our earlier supposition that $(L/D)_{\max}$, being an aerodynamic property of the given airplane, should depend only on other aerodynamic properties, is correct, the other aerodynamic properties being simply $C_{D,o}$ and e.

EXAMPLE 6.2

Using the information obtained in Example 6.1, calculate the maximum lift-to-drag ratio for the Seversky P-35.

■ Solution

From Example 6.1, we have

$$C_{D,o} = 0.026$$
$$e = 0.873$$
$$\text{AR} = 5.89$$

From Equation (6.29), we have

$$\left(\frac{C_L}{C_D}\right)_{\max} = \frac{(\pi e \text{AR}\ C_{D,o})^{1/2}}{2C_{D,o}}$$
$$= \frac{[\pi(0.873)(5.89)(0.026)]^{1/2}}{2(0.026)}$$
$$\left(\frac{C_L}{C_D}\right)_{\max} = \boxed{12.46}$$

The value for $(L/D)_{\max}$ for the P-35 as tabulated by Loftin in Reference 84 is $(L/D)_{\max} = 11.8$, which is within 5 percent of the value calculated here.

6.7.3 Application of Computational Fluid Dynamics for the Calculation of Lift and Drag

The role of computational fluid dynamics (CFD) for the numerical solution of the continuity, momentum, and energy equations is discussed in Section 2.17.2. Numerical solutions of the purely inviscid flow equations are labeled "Euler solutions"; the CFD results discussed in Chapter 13 are examples of such Euler solutions. Numerical solutions of the general viscous flow equations are labeled "Navier-Stokes solutions"; examples of such Navier-Stokes solutions are given in Chapter 20.

These numerical solutions of the continuity, momentum, and energy equations give the variation of the flowfield properties, (p, T, V, etc) as a function of space and time throughout the flow. This includes, of course, the pressure at the body surface. The shear stress at the surface is obtained from Equation (1.59), repeated below

$$\tau_w = \mu \left(\frac{dV}{dy} \right)_{y=0} \tag{1.59}$$

where the velocity gradient at the wall, $(dV/dy)_{y=0}$, is obtained from the CFD solution of the flow velocity at gridpoints adjacent to the wall using one-sided differences (see Section 2.17.2). Finally, by numerically integrating the pressure and shear stress distributions over the surface, the lift and drag of the airplane can be obtained (see Section 1.5). This is how CFD results can be used to give lift and drag on a body.

Some very recent CFD results for the flowfield over a complete airplane are described in seven coordinated papers in the *Journal of Aircraft*, Vol. 46, No. 2, March–April 2009. These papers report CFD results obtained by different investigators using different computer programs and algorithms for the flowfield

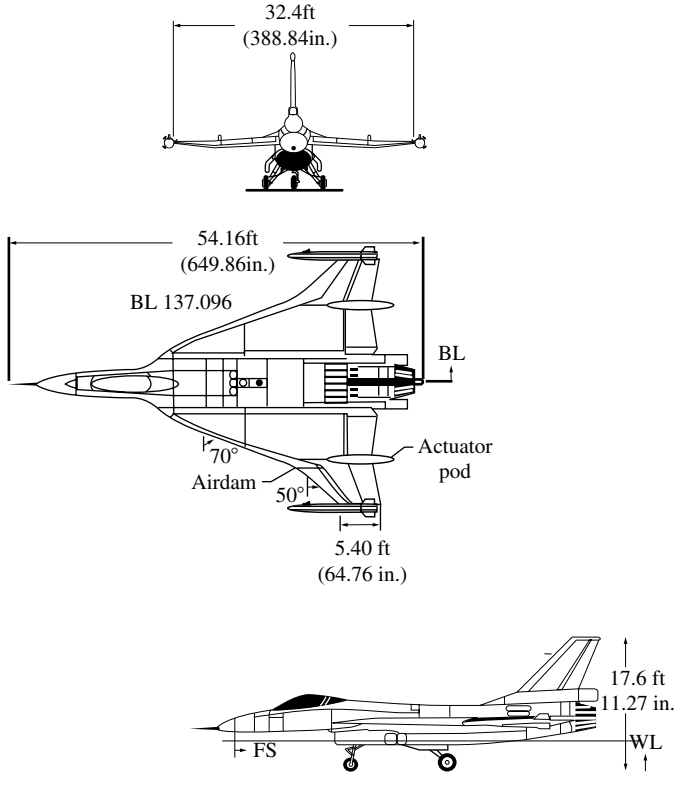


Figure 6.14 Three-view of the F-16XL.

over the F-16XL cranked-wing configuration shown in Figure 6.14. As part of the Cranked-Arrow Wing Aerodynamics Project (CAWAP) organized by NASA and administered through the AIAA Applied Aerodynamics Technical Committee, various investigators were invited to make CFD flowfield calculations over the F-16XL at various flight conditions. The purpose is to compare the results in order to assess the state of the art of CFD calculations of flowfields around complete airplane configurations, particularly with the airplane at relatively high angle of attack with large regions of separated flow. A summary of the comparisons and conclusions are given in Reference 118. Although the main thrust of this project was to evaluate and compare calculations of detailed flowfield structure and surface pressure distributions, some comparisons of lift and drag coefficients were made. A representative comparison is given in Table 6.1, where C_L and C_D obtained from seven different investigations are tabulated. The results apply to the F-16XL flying at $M_\infty = 0.36$, angle of attack $\alpha = 11.85^\circ$, sideslip angle $= 0.612^\circ$, and Reynolds number $= 46.8 \times 10^6$. In Table 6.1, the different investigators are simply labeled by number; the actual sources and the particular CFD codes are identified in Reference 118.

Note that the discrepancy between the lowest and highest number obtained is 26 percent for C_L and 42 percent for C_D. However, if the results from Investigator 3 are not counted, the discrepancies are 6.7 percent for C_L and 21.5 percent for C_D.

The three-dimensional flowfield associated with the values of C_L and C_D in Table 6.1 is complex; it contains primary and secondary vortices much like those shown for flow over a delta wing in Figure 5.41. The airplane is at both an angle of attack and angle of sideslip, and the resulting flowfield is highly three-dimensional with embedded vortices and large regions of flow separation. This is a severe test for any CFD code, and indeed is the reason why this case has been chosen here to illustrate the use of CFD for the calculation of lift and drag of a complete airplane. The test cases used in the Cranked-Arrow Wing Aerodynamics Project represent perhaps the upper limit of complexity, and therefore the discrepancies between the results of the different CFD codes may represent an upper bound — a kind of worst-case scenario.

With that caveat in mind, note that the discrepancies in the calculation of C_L are remarkably small, but that the results for C_D vary considerably. The accurate calculation of drag for most practical aerodynamic vehicles has been a challenge

Table 6.1

Investigator No.	C_L	C_D
1	0.43846	0.13289
2	0.44693	0.13469
3	0.37006	0.11084
4	0.43851	0.15788
5	0.46798	0.13648
6	0.44190	0.16158
7	0.44590	0.14265

and a problem for centuries, going back to the early nineteenth century flying machine inventors (see References 2, 62, and 115, for example). Amazingly, in our modern world of high technology and advanced CFD techniques, accurate drag prediction remains a problem, although improvements are gradually being made. Accurate CFD predictions of drag are compromised by at least the following:

1. The calculation of skin friction drag requires the accurate calculation of shear stress, which requires an accurate calculation of the velocity gradient at the surface [see for example Equation (1.59)], which requires a very fine, closely spaced computational grid adjacent to the wall to obtain very accurate values of the flow velocity at the first several gridpoints above the wall. The velocity gradient at the surface is then obtained from these velocities by using one-sided differencing.

2. The boundary layers on any practical-sized vehicle are turbulent, and any CFD calculation of this flow must include this effect. Turbulence remains one of the few unsolved problems in classical physics, so its effect must be *modeled* in aerodynamic calculations. Most CFD calculations of turbulent flows use the Reynolds averaged Navier-Stokes equations (RANS), discussed in Part 4 of this book, and must incorporate some type of turbulence model. There are literally dozens of different turbulence models in existence, each one depending in one way or another on empirical data. Turbulence models by themselves introduce a great deal of uncertainty in the calculation of drag. The seven different CFD calculations noted in Table 6.1 all used different turbulence models.

3. The calculation of locations on a body where the flow separates is also uncertain. For CFD, the calculation of separated flows can only be made with Navier-Stokes solutions; only in a few (but interesting) instances can a solution of the Euler equations yield a semblance of flow separation. The nature and location of flow separation is different for laminar and turbulent flows (see for example the discussion of the real flow over a sphere in Section 6.6). The uncertainty in the calculation of separated flows, which is in part related to the uncertainty in turbulence modeling discussed earlier, is another reason for the discrepancies in drag coefficient as tabulated in Table 6.1. The high angle of attack flows associated with these test cases for the F-16XL have large regions of complex flow separation.

Considering these uncertainties, the discrepancy in the calculations of C_D listed in Table 6.1, all told, are not bad. Further advances in algorithms and modeling will inevitably lead to even better results. Furthermore, because C_L was accurately calculated and the details of the flowfield itself were accurately captured by the CFD calculations, the investigators participating in the Cranked-Arrow Wing Aerodynamics Project International (AWAPI) were moved to conclude (Reference 118):

> Although differences were observed in the comparison of results from 10 different CFD solvers with measurements, these solvers all functioned robustly on an actual aircraft at flight conditions, with sufficient agreement among them to conclude

that the overall objectives of the CAWAPI endeavor have been achieved. In particular, the status of CFD as a tool for understanding flight-test observations has been confirmed.

This is also an appropriate conclusion to end our discussion of airplane lift and drag.

6.8 SUMMARY

For a three-dimensional (point) source,

$$V_r = \frac{\lambda}{4\pi r^2} \tag{6.6}$$

and

$$\phi = -\frac{\lambda}{4\pi r} \tag{6.7}$$

For a three-dimensional doublet,

$$\phi = -\frac{\mu}{4\pi}\frac{\cos\theta}{r^2} \tag{6.9}$$

and

$$\mathbf{V} = \frac{\mu}{2\pi}\frac{\cos\theta}{r^3}\mathbf{e}_r + \frac{\mu}{4\pi}\frac{\sin\theta}{r^3}\mathbf{e}_\theta \tag{6.10}$$

The flow over a sphere is generated by superimposing a three-dimensional doublet and a uniform flow. The resulting surface velocity and pressure distributions are given by

$$V_\theta = \tfrac{3}{2}V_\infty \sin\theta \tag{6.19}$$

and

$$C_p = 1 - \tfrac{9}{4}\sin^2\theta \tag{6.20}$$

In comparison with flow over a cylinder, the surface velocity and magnitude of the pressure coefficient are smaller for the sphere—an example of the three-dimensional relieving effect.

In modern aerodynamic applications, inviscid, incompressible flows over complex three-dimensional bodies are usually computed via three-dimensional panel techniques.

6.9 PROBLEMS

6.1 Prove that three-dimensional source flow is irrotational.

6.2 Prove that three-dimensional source flow is a physically possible incompressible flow.

6.3 A sphere and a circular cylinder (with its axis perpendicular to the flow) are mounted in the same freestream. A pressure tap exists at the top of the sphere, and this is connected via a tube to one side of a manometer. The other side of the manometer is connected to a pressure tap on the surface of the cylinder. This tap is located on the cylindrical surface such that no deflection of the manometer fluid takes place. Calculate the location of this tap.

PART 3

Inviscid, Compressible Flow

In Part 3, we deal with high-speed flows—subsonic, supersonic, and hypersonic. In such flows, the density is a variable—this is compressible flow. ■

CHAPTER 7

Compressible Flow: Some Preliminary Aspects

With the realization of aeroplane and missile speeds equal to or even surpassing many times the speed of sound, thermodynamics has entered the scene and will never again leave our considerations.

Jakob Ackeret, 1962

PREVIEW BOX

With this chapter we move into the wild and exciting world of high-speed flow. To jump into this world, however, we need some preparation. This chapter prepares us to deal with the high-speed subsonic, transonic, supersonic, and hypersonic flows that are discussed in subsequent chapters. In the present chapter, we once again add to our inventory of fundamental principles and relations—those necessary to understand and predict high-speed flow. View this chapter as a continuation of Chapter 2 on some fundamental principles and equations, but a continuation with a flair laced by high energy (figuratively and literally).

Energy? What do we mean? Consider that you are in an automobile traveling at 65 miles per hour along a highway. If you stick your hand out the window (not recommended by the way, for safety reasons) you will sense a certain amount of energy in the airflow, and your hand will be forced back to some extent. Now imagine that you are traveling at 650 mi/h and you stick your hand out the window (figuratively). You can just imagine what tremendous energy you would sense, and what disaster would happen to your hand. The point is that high-speed flow is *high-energy flow*. The science of energy is *thermodynamics*. So this chapter is, for the most part, a discussion of thermodynamics—but only to the extent necessary for our subsequent applications in high-speed flow.

High-speed flow is also *compressible flow*. With this chapter, we can no longer assume that the density is constant. Rather, the density is a variable, which introduces some neat physics and analytical challenges for our study of compressible flow. Indeed, the world we are entering, beginning with this chapter, is quite different than the incompressible world we dealt with in Chapters 3 to 6. This is like a fresh start. So get on board, and let's start.

7.1 INTRODUCTION

On September 30, 1935, the leading aerodynamicists from all corners of the world converged on Rome, Italy. Some of them arrived in airplanes which, in those days, lumbered along at speeds of 130 mi/h. Ironically, these people were gathering to discuss airplane aerodynamics not at 130 mi/h but rather at the unbelievable speeds of 500 mi/h and faster. By invitation only, such aerodynamic giants as Theodore von Karman and Eastman Jacobs from the United States, Ludwig Prandtl and Adolf Busemann from Germany, Jakob Ackeret from Switzerland, G. I. Taylor from England, Arturo Crocco and Enrico Pistolesi from Italy, and others assembled for the fifth Volta Conference, which had as its topic "High Velocities in Aviation." Although the jet engine had not yet been developed, these men were convinced that the future of aviation was "faster and higher." At that time, some aeronautical engineers felt that airplanes would never fly faster than the speed of sound—the myth of the "sound barrier" was propagating through the ranks of aviation. However, the people who attended the fifth Volta Conference knew better. For 6 days, inside an impressive Renaissance building that served as the city hall during the Holy Roman Empire, these individuals presented papers that discussed flight at high subsonic, supersonic, and even hypersonic speeds. Among these presentations was the first public revelation of the concept of a swept wing for high-speed flight; Adolf Busemann, who originated the concept, discussed the technical reasons why swept wings would have less drag at high speeds than conventional straight wings. (One year later, the swept-wing concept was classified by the German Luftwaffe as a military secret. The Germans went on to produce a large bulk of swept-wing research during World War II, resulting in the design of the first operational jet airplane—the Me 262—which had a moderate degree of sweep.) Many of the discussions at the Volta Conference centered on the effects of "compressibility" at high subsonic speeds, that is, the effects of variable density, because this was clearly going to be the first problem to be encountered by future high-speed airplanes. For example, Eastman Jacobs presented wind-tunnel test results for compressibility effects on standard NACA four- and five-digit airfoils at high subsonic speeds and noted extraordinarily large increases in drag beyond certain freestream Mach numbers. In regard to supersonic flows, Ludwig Prandtl presented a series of photographs showing shock waves inside nozzles and on various bodies—with some of the photographs dating as far back as 1907, when Prandtl started serious work in supersonic aerodynamics. (Clearly, Ludwig Prandtl was busy with much more than just the development of his incompressible airfoil and finite-wing theory discussed in Chapters 4 and 5.) Jakob Ackeret gave a paper on the design of supersonic wind tunnels, which, under his direction, were being established in Italy, Switzerland, and Germany. There were also presentations on propulsion techniques for high-speed flight, including rockets and ramjets. The atmosphere surrounding the participants in the Volta Conference was exciting and heady; the conference launched the world aerodynamic community into the area of high-speed subsonic and supersonic flight—an area which today is as commonplace as the 130-mi/h flight speeds of 1935. Indeed, the purpose of the next eight chapters of this book is to present the fundamentals of such high-speed flight.

In contrast to the low-speed, incompressible flows discussed in Chapters 3 to 6, the pivotal aspect of high-speed flow is that the density is a variable. Such flows are called *compressible flows* and are the subject of Chapters 7 to 14. Return to Figure 1.45, which gives a block diagram categorizing types of aerodynamic flows. In Chapters 7 to 14, we discuss flows which fall into blocks D and F; that is, we will deal with *inviscid compressible* flow. In the process, we touch all the flow regimes itemized in blocks G through J. These flow regimes are illustrated in Figure 1.44; study Figures 1.44 and 1.45 carefully, and review the surrounding discussion in Section 1.10 before proceeding further.

In addition to variable density, another pivotal aspect of high-speed compressible flow is *energy*. A high-speed flow is a high-energy flow. For example, consider the flow of air at standard sea level conditions moving at twice the speed of sound. The internal energy of 1 kg of this air is 2.07×10^5 J, whereas the kinetic energy is larger, namely, 2.31×10^5 J. When the flow velocity is decreased, some of this kinetic energy is lost and reappears as an increase in internal energy, hence increasing the temperature of the gas. Therefore, in a high-speed flow, energy transformations and temperature changes are important considerations. Such considerations come under the science of *thermodynamics*. For this reason, thermodynamics is a vital ingredient in the study of compressible flow. One purpose of the present chapter is to review briefly the particular aspects of thermodynamics which are essential to our subsequent discussions of compressible flow.

The road map for this chapter is given in Figure 7.1. As our discussion proceeds, refer to this road map in order to provide an orientation for our ideas.

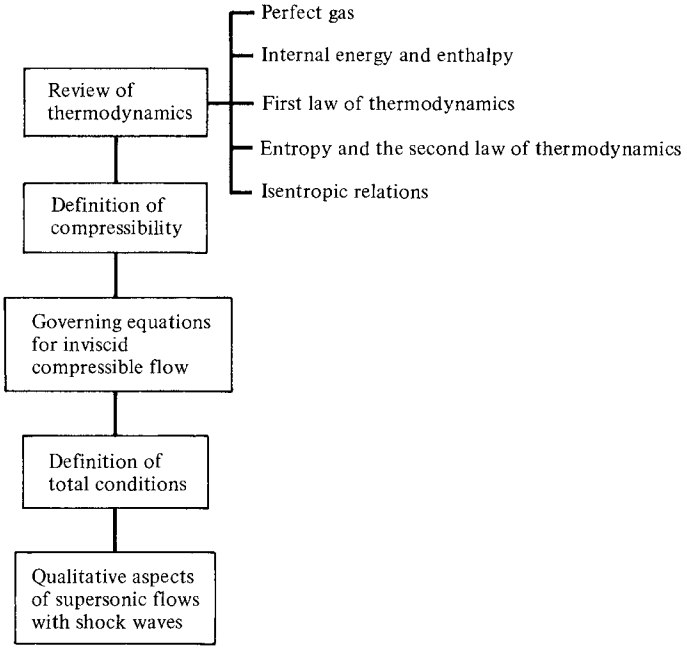


Figure 7.1 Road map for Chapter 7.

7.2 A BRIEF REVIEW OF THERMODYNAMICS

The importance of thermodynamics in the analysis and understanding of compressible flow was underscored in Section 7.1. Hence, the purpose of the present section is to review those aspects of thermodynamics that are important to compressible flows. This is in no way intended to be an exhaustive discussion of thermodynamics; rather, it is a review of only those fundamental ideas and equations that will be of direct use in subsequent chapters. If you have studied thermodynamics, this review should serve as a ready reminder of some important relations. If you are not familiar with thermodynamics, this section is somewhat self-contained so as to give you a feeling for the fundamental ideas and equations that we use frequently in subsequent chapters.

7.2.1 Perfect Gas

As described in Section 1.2, a gas is a collection of particles (molecules, atoms, ions, electrons, etc.) which are in more or less random motion. Due to the electronic structure of these particles, a force field pervades the space around them. The force field due to one particle reaches out and interacts with neighboring particles, and vice versa. Hence, these fields are called *intermolecular forces*. However, if the particles of the gas are far enough apart, the influence of the intermolecular forces is small and can be neglected. A gas in which the intermolecular forces are neglected is defined as a *perfect gas*. For a perfect gas, p, ρ, and T are related through the following *equation of state:*

$$\boxed{p = \rho RT} \tag{7.1}$$

where R is the specific gas constant, which is a different value for different gases. For air at standard conditions, $R = 287$ J/(kg · K) $= 1716$ (ft · lb)/(slug · °R).

At the temperatures and pressures characteristic of many compressible flow applications, the gas particles are, on the average, more than 10 molecular diameters apart; this is far enough to justify the assumption of a perfect gas. Therefore, throughout the remainder of this book, we use the equation of state in the form of Equation (7.1), or its counterpart,

$$\boxed{pv = RT} \tag{7.2}$$

where v is the specific volume, that is, the volume per unit mass; $v = 1/\rho$. (*Please note:* Starting with this chapter, we use the symbol v to denote both specific volume and the y component of velocity. This usage is standard, and in all cases it should be obvious and cause no confusion.)

7.2.2 Internal Energy and Enthalpy

Consider an individual molecule of a gas, say, an O_2 molecule in air. This molecule is moving through space in a random fashion, occasionally colliding with a neighboring molecule. Because of its velocity through space, the molecule has

translational kinetic energy. In addition, the molecule is made up of individual atoms which we can visualize as connected to each other along various axes; for example, we can visualize the O_2 molecule as a "dumbbell" shape, with an O atom at each end of a connecting axis. In addition to its translational motion, such a molecule can execute a rotational motion in space; the kinetic energy of this rotation contributes to the net energy of the molecule. Also, the atoms of a given molecule can vibrate back and forth along and across the molecular axis, thus contributing a potential and kinetic energy of vibration to the molecule. Finally, the motion of the electrons around each of the nuclei of the molecule contributes an "electronic" energy to the molecule. Hence, the energy of a given molecule is the sum of its translational, rotational, vibrational, and electronic energies.

Now consider a finite volume of gas consisting of a large number of molecules. The sum of the energies of all the molecules in this volume is defined as the *internal energy* of the gas. The internal energy per unit mass of gas is defined as the specific internal energy, denoted by e. A related quantity is the specific enthalpy, denoted by h and defined as

$$h = e + pv \tag{7.3}$$

For a perfect gas, both e and h are functions of temperature only:

$$e = e(T) \tag{7.4a}$$

$$h = h(T) \tag{7.4b}$$

Let de and dh represent differentials of e and h, respectively. Then, for a perfect gas,

$$de = c_v\, dT \tag{7.5a}$$

$$dh = c_p\, dT \tag{7.5b}$$

where c_v and c_p are the specific heats at constant volume and constant pressure, respectively. In Equations (7.5*a* and *b*), c_v and c_p can themselves be functions of T. However, for moderate temperatures (for air, for $T < 1000$ K), the specific heats are reasonably constant. A perfect gas where c_v and c_p are constants is defined as a *calorically perfect gas,* for which Equations (7.5*a* and *b*) becomes

$$\boxed{e = c_v T} \tag{7.6a}$$

$$\boxed{h = c_p T} \tag{7.6b}$$

For a large number of practical compressible flow problems, the temperatures are moderate. For this reason, in this book we always treat the gas as calorically perfect; that is, we assume that the specific heats are constant. For a discussion of compressible flow problems where the specific heats are not constant (such as the high-temperature chemically reacting flow over a high-speed atmospheric entry vehicle, that is, the space shuttle), see Reference 21.

Note that e and h in Equations (7.3) through (7.6) are thermodynamic state variables—they depend only on the state of the gas and are independent of any

process. Although c_v and c_p appear in these equations, there is no restriction to just a constant volume or a constant pressure process. Rather, Equations (7.5*a* and *b*) and (7.6*a* and *b*) are relations for thermodynamic state variables, namely, e and h as functions of T, and have nothing to do with the process that may be taking place.

For a specific gas, c_p and c_v are related through the equation

$$c_p - c_v = R \tag{7.7}$$

Dividing Equation (7.7) by c_p, we obtain

$$1 - \frac{c_v}{c_p} = \frac{R}{c_p} \tag{7.8}$$

Define $\gamma \equiv c_p/c_v$. For air at standard conditions, $\gamma = 1.4$. Then Equation (7.8) becomes

$$1 - \frac{1}{\gamma} = \frac{R}{c_p}$$

or

$$\boxed{c_p = \frac{\gamma R}{\gamma - 1}} \tag{7.9}$$

Similarly, dividing Equation (7.7) by c_v, we obtain

$$\boxed{c_v = \frac{R}{\gamma - 1}} \tag{7.10}$$

Equations (7.9) and (7.10) are particularly useful in our subsequent discussion of compressible flow.

EXAMPLE 7.1

Consider a room with a rectangular floor that is 5 m by 7 m, and a 3.3 m high ceiling. The air pressure and temperature in the room are 1 atm and 25°C, respectively. Calculate the internal energy and the enthalpy of the air in the room.

■ Solution

We first need to calculate the mass of air in the room. From Equation (7.1)

$$\rho = \frac{p}{RT}$$

where each quantity must be expressed in consistent SI units. Since 1 atm $= 1.01 \times 10^5$ N/m^2 and 0°C is 273 K, we have

$$p = 1.01 \times 10^5 \text{ N/m}^2 \quad \text{and} \quad T = 273 + 25 = 298 \text{ K}$$

Hence,

$$\rho = \frac{p}{RT} = \frac{1.01 \times 10^5}{(287)(298)} = 1.181 \text{ kg/m}^3$$

The volume of the room is $(5)(7)(3.3) = 115.5\ \text{m}^3$. The mass of air in the room is therefore

$$M = (1.181)(115.5) = 136.4\ \text{kg}$$

From Equation (7.6*a*), the internal energy per unit mass is

$$e = c_v T$$

where $$c_v = \frac{R}{\gamma - 1} = \frac{(287)}{1.4 - 1} = \frac{287}{0.4} = 717.5 \quad \text{joule/(kg} \cdot \text{K)}$$

Thus, $$e = c_v T = (717.5)(298) = 2.138 \times 10^5\ \text{joule/kg}$$

The internal energy in the room, E, is then

$$E = Me = (136.4)(2.138 \times 10^5) = \boxed{2.92 \times 10^7\ \text{joule}}$$

From Equation (7.6*b*), the enthalpy per unit mass is

$$h = c_p T$$

where $$c_p = \frac{\gamma R}{\gamma - 1} = \frac{(1.4)(287)}{0.4} = 1004.5 \quad \text{joule/(kg} \cdot \text{K)}$$

Thus, $$h = c_p T = (1004.5)(298) = 2.993 \times 10^5 \quad \text{joule/kg}$$

The enthalpy in the room, H, is then

$$H = Mh = (136.4)(2.993 \times 10^5) = \boxed{4.08 \times 10^7\ \text{joule}}$$

A check on two answers can be made knowing that

$$\frac{h}{e} = \frac{c_p T}{c_v T} = \frac{c_p}{c_v} = \gamma = 1.4$$

From the answers,

$$\frac{H}{E} = \frac{4.08 \times 10^7}{2.92 \times 10^7} = 1.4 \qquad \text{It checks.}$$

This simple example is intended to reinforce two basic points.

1. Consistent units must be used when making a calculation using basic equations from physics, such as Equations (7.1), (7.6*a*), and (7.6*b*). Here we used SI units, because the given information was given in terms of meters and degrees Celsius, which is readily put in terms of the consistent unit of temperature, degrees Kelvin.
2. The internal energy and enthalpy per unit mass were calculated directly from the temperature of the gas, using Equations (7.6*a* and *b*). There was no need to consider a "constant volume process" or a "constant pressure process"; there is no process at all to consider here. Internal energy and enthalpy per unit mass, e and h, are simply

state variables, depending only on the thermodynamic state of the system. Even though Equation (7.6*a*) contains c_v, a constant volume process is not relevant here. Similarly, even though Equation (7.6*b*) contains c_p, a constant pressure process is not relevant here.

EXAMPLE 7.2

One type of supersonic wind tunnel is a blow-down tunnel, where air is stored in a high-pressure reservoir, and then, upon the opening of a valve, exhausted through the tunnel into a vacuum tank or simply into the open atmosphere at the downstream end of the tunnel. Supersonic wind tunnels are discussed in Chapter 10. For this example, we consider just the high-pressure reservoir as a storage tank that is being charged with air by a high-pressure pump. As air is being pumped into the constant-volume reservoir, the air pressure inside the reservoir increases. The pump continues to charge the reservoir until the desired pressure is achieved.

Consider a reservoir with an internal volume of 30 m^3. As air is pumped into the reservoir, the air pressure inside the reservoir continually increases with time. Consider the instant during the charging process when the reservoir pressure is 10 atm. Assume the air temperature inside the reservoir is held constant at 300 K by means of a heat exchanger. Air is pumped into the reservoir at the rate of 1 kg/s. Calculate the time rate of increase of pressure in the reservoir at this instant.

■ Solution

Let M be the total mass of air inside the reservoir at any instant. Since air is being pumped into the reservoir at the rate of 1 kg/s, then the total mass of air is increasing at the rate of $dM/dt = 1$ kg/s. The density of the air at any instant is

$$\rho = \frac{M}{V} \tag{E.7.1}$$

where V is the total volume of the reservoir and is constant; $V = 30$ m^3. Since V is constant, from Equation (E.7.1)

$$\frac{d\rho}{dt} = \frac{1}{V}\frac{dM}{dt} = \frac{1 \text{ kg/s}}{30 \text{ m}^3} = 0.0333$$

Differentiating Equation (7.1) with respect to time, and recalling that R and T are constant,

$$\frac{dp}{dt} = RT\frac{d\rho}{dt} \tag{E.7.2}$$

$$\frac{dp}{dt} = (287)(300)(0.0333) = \boxed{2867.13 \ \frac{\text{N}}{\text{m}^2\text{s}}}$$

EXAMPLE 7.3

In Example 7.2, if the pumping rate of 1 kg/s were maintained constant throughout the charging process, how long will it take to increase the reservoir pressure from 10 to 20 atm?

■ **Solution**

If the pumping rate is held constant at 1 kg/s, then the time rate of change of density, $d\rho/dt = 0.0333$ kg/m^{3}s will remain constant, and consequently the time rate of change of pressure, $dp/dt = 2867.13$ N/m^{2}s, will remain constant. Hence, the time required to increase the reservoir pressure from 10 atm to 20 atm is

$$\frac{(20 \text{ atm} - 10 \text{ atm})(1.01 \times 10^5)}{2867.13} = 352.27 \text{ s}$$

or, in minutes,

$$\frac{352.27}{60} = \boxed{5.87 \text{ min}}$$

7.2.3 First Law of Thermodynamics

Consider a fixed mass of gas, which we define as the *system*. (For simplicity, assume a unit mass, for example, 1 kg or 1 slug.) The region outside the system is called the *surroundings*. The interface between the system and its surroundings is called the *boundary,* as shown in Figure 7.2. Assume that the system is stationary. Let δq be an incremental amount of heat added to the system across the boundary, as sketched in Figure 7.2. Examples of the source of δq are radiation from the surroundings that is absorbed by the mass in the system and thermal conduction due to temperature gradients across the boundary. Also, let δw denote the work done on the system by the surroundings (say, by a displacement of the boundary, squeezing the volume of the system to a smaller value). As discussed earlier, due to the molecular motion of the gas, the system has an internal energy e. The heat added and work done on the system cause a change in energy, and since the system is stationary, this change in energy is simply de:

$$\boxed{\delta q + \delta w = de} \tag{7.11}$$

This is the *first law of thermodynamics:* It is an empirical result confirmed by experience. In Equation (7.11), e is a state variable. Hence, de is an exact

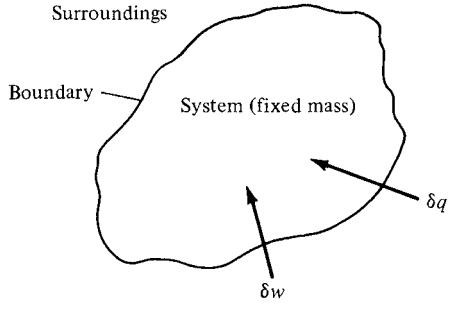


Figure 7.2 Thermodynamic system.

differential, and its value depends only on the initial and final states of the system. In contrast, δq and δw depend on the process in going from the initial to the final states.

For a given de, there are in general an infinite number of different ways (processes) by which heat can be added and work done on the system. We are primarily concerned with three types of processes:

1. *Adiabatic process.* One in which no heat is added to or taken away from the system
2. *Reversible process.* One in which no dissipative phenomena occur, that is, where the effects of viscosity, thermal conductivity, and mass diffusion are absent
3. *Isentropic process.* One that is both adiabatic and reversible

For a reversible process, it can be easily shown that $\delta w = -p\,dv$, where dv is an incremental change in the volume due to a displacement of the boundary of the system. Thus, Equation (7.11) becomes

$$\delta q - p\,dv = de \tag{7.12}$$

7.2.4 Entropy and the Second Law of Thermodynamics

Consider a block of ice in contact with a red-hot plate of steel. Experience tells us that the ice will warm up (and probably melt) and the steel plate will cool down. However, Equation (7.11) does not necessarily say this will happen. Indeed, the first law allows that the ice may get cooler and the steel plate hotter—just as long as energy is conserved during the process. Obviously, in real life this does not happen; instead, nature imposes another condition on the process, a condition that tells us *which direction* a process will take. To ascertain the proper direction of a process, let us define a new state variable, the entropy, as follows:

$$ds = \frac{\delta q_{\text{rev}}}{T} \tag{7.13}$$

where s is the entropy of the system, δq_{rev} is an incremental amount of heat added reversibly to the system, and T is the system temperature. Do not be confused by the above definition. It defines a change in entropy in terms of a reversible addition of heat δq_{rev}. However, entropy is a state variable, and it can be used in conjunction with any type of process, reversible or irreversible. The quantity δq_{rev} in Equation (7.13) is just an artifice; an effective value of δq_{rev} can always be assigned to relate the initial and end points of an irreversible process, where the actual amount of heat added is δq. Indeed, an alternative and probably more lucid relation is

$$\boxed{ds = \frac{\delta q}{T} + ds_{\text{irrev}}} \tag{7.14}$$

In Equation (7.14), δq is the actual amount of heat added to the system during an actual irreversible process, and ds_{irrev} is the generation of entropy due to

the irreversible, dissipative phenomena of viscosity, thermal conductivity, and mass diffusion occurring *within* the system. These dissipative phenomena always increase the entropy:

$$\boxed{ds_{\text{irrev}} \geq 0} \tag{7.15}$$

In Equation (7.15), the equals sign denotes a reversible process, where by definition no dissipative phenomena occur within the system. Combining Equations (7.14) and (7.15), we have

$$\boxed{ds \geq \frac{\delta q}{T}} \tag{7.16}$$

Furthermore, if the process is adiabatic, $\delta q = 0$, and Equation (7.16) becomes

$$\boxed{ds \geq 0} \tag{7.17}$$

Equations (7.16) and (7.17) are forms of the *second law of thermodynamics*. The second law tells us in what direction a process will take place. A process will proceed in a direction such that the entropy of the system plus that of its surroundings always increases or, at best, stays the same. In our example of the ice in contact with hot steel, consider the system to be both the ice and steel plate combined. The simultaneous heating of the ice and cooling of the plate yield a net increase in entropy for the system. On the other hand, the impossible situation of the ice getting cooler and the plate hotter would yield a net decrease in entropy, a situation forbidden by the second law. In summary, the concept of entropy in combination with the second law allows us to predict the *direction* that nature takes.

The practical calculation of entropy is carried out as follows. In Equation (7.12), assume that heat is added reversibly; then the definition of entropy, Equation (7.13), substituted in Equation (7.12) yields

$$T\,ds - p\,dv = de$$

or

$$\boxed{T\,ds = de + p\,dv} \tag{7.18}$$

From the definition of enthalpy, Equation (7.3), we have

$$dh = de + p\,dv + v\,dp \tag{7.19}$$

Combining Equations (7.18) and (7.19), we obtain

$$\boxed{T\,ds = dh - v\,dp} \tag{7.20}$$

Equations (7.18) and (7.20) are important; they are essentially alternate forms of the first law expressed in terms of entropy. For a perfect gas, recall Equations (7.5*a* and *b*), namely, $de = c_v\,dT$ and $dh = c_p\,dT$. Substituting these relations into

Equations (7.18) and (7.20), we obtain

$$ds = c_v \frac{dT}{T} + \frac{p\,dv}{T} \tag{7.21}$$

and

$$ds = c_p \frac{dT}{T} - \frac{v\,dp}{T} \tag{7.22}$$

Working with Equation (7.22), substitute the equation of state $pv = RT$, or $v/T = R/p$, into the last term:

$$ds = c_p \frac{dT}{T} - R\frac{dp}{p} \tag{7.23}$$

Consider a thermodynamic process with initial and end states denoted by 1 and 2, respectively. Equation (7.23), integrated between states 1 and 2, becomes

$$s_2 - s_1 = \int_{T_1}^{T_2} c_p \frac{dT}{T} - \int_{p_1}^{p_2} R\frac{dp}{p} \tag{7.24}$$

For a calorically perfect gas, both R and c_p are constants; hence, Equation (7.24) becomes

$$\boxed{s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}} \tag{7.25}$$

In a similar fashion, Equation (7.21) leads to

$$\boxed{s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}} \tag{7.26}$$

Equations (7.25) and (7.26) are practical expressions for the calculation of the entropy change of a calorically perfect gas between two states. Note from these equations that s is a function of two thermodynamic variables, for example, $s = s(p, T)$, $s = s(v, T)$.

7.2.5 Isentropic Relations

We have defined an isentropic process as one which is both adiabatic and reversible. Consider Equation (7.14). For an adiabatic process, $\delta q = 0$. Also, for a reversible process, $ds_{\text{irrev}} = 0$. Thus, for an adiabatic, reversible process, Equation (7.14) yields $ds = 0$, or entropy is constant; hence, the word "isentropic." For such an isentropic process, Equation (7.25) is written as

$$0 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}$$

$$\ln \frac{p_2}{p_1} = \frac{c_p}{R} \ln \frac{T_2}{T_1}$$

or

$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{c_p/R} \tag{7.27}$$

However, from Equation (7.9),

$$\frac{c_p}{R} = \frac{\gamma}{\gamma - 1}$$

and hence Equation (7.27) is written as

$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)} \tag{7.28}$$

In a similar fashion, Equation (7.26) written for an isentropic process gives

$$0 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}$$

$$\ln \frac{v_2}{v_1} = -\frac{c_v}{R} \ln \frac{T_2}{T_1}$$

$$\frac{v_2}{v_1} = \left(\frac{T_2}{T_1}\right)^{-c_v/R} \tag{7.29}$$

From Equation (7.10),

$$\frac{c_v}{R} = \frac{1}{\gamma - 1}$$

and hence Equation (7.29) is written as

$$\frac{v_2}{v_1} = \left(\frac{T_2}{T_1}\right)^{-1/(\gamma-1)} \tag{7.30}$$

Since $\rho_2/\rho_1 = v_1/v_2$, Equation (7.30) becomes

$$\frac{\rho_2}{\rho_1} = \left(\frac{T_2}{T_1}\right)^{1/(\gamma-1)} \tag{7.31}$$

Combining Equations (7.28) and (7.31), we can summarize the isentropic relations as

$$\boxed{\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma} = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)}} \tag{7.32}$$

Equation (7.32) is very important; it relates pressure, density, and temperature for an isentropic process. We use this equation frequently, so make certain to brand it on your mind. Also, keep in mind the source of Equation (7.32); it stems from the first law and the definition of entropy. Therefore, Equation (7.32) is basically an energy relation for an isentropic process.

Why is Equation (7.32) so important? Why is it frequently used? Why are we so interested in an isentropic process when it seems so restrictive—requiring both adiabatic and reversible conditions? The answers rest on the fact that a large number of practical compressible flow problems can be assumed to be isentropic—contrary to what you might initially think. For example, consider the flow over an airfoil or through a rocket engine. In the regions adjacent to the airfoil surface

and the rocket nozzle walls, a boundary layer is formed wherein the dissipative mechanisms of viscosity, thermal conduction, and diffusion are strong. Hence, the entropy increases within these boundary layers. However, consider the fluid elements moving outside the boundary layer. Here, the dissipative effects of viscosity, etc., are very small and can be neglected. Moreover, no heat is being transferred to or from the fluid element (i.e., we are not heating the fluid element with a Bunsen burner or cooling it in a refrigerator); thus, the flow outside the boundary layer is adiabatic. Consequently, the fluid elements outside the boundary layer are experiencing an adiabatic reversible process—namely, isentropic flow. In the vast majority of practical applications, the viscous boundary layer adjacent to the surface is thin compared with the entire flow field, and hence large regions of the flow can be assumed isentropic. This is why a study of isentropic flow is directly applicable to many types of practical compressible flow problems. In turn, Equation (7.32) is a powerful relation for such flows, valid for a calorically perfect gas.

This ends our brief review of thermodynamics. Its purpose has been to give a quick summary of ideas and equations that will be employed throughout our subsequent discussions of compressible flow. Indeed, this author knows of no practical problem dealing with compressible flow that can be solved without invoking some aspect of thermodynamics—it is that important. For a more thorough discussion of the power and beauty of thermodynamics, see any good thermodynamics text, such as References 22 to 24.

EXAMPLE 7.4

Consider a Boeing 747 flying at a standard altitude of 36,000 ft. The pressure at a point on the wing is 400 lb/ft^2. Assuming isentropic flow over the wing, calculate the temperature at this point.

■ Solution

From Appendix E, at a standard altitude of 36,000 ft, $p_\infty = 476$ lb/ft^2 and $T_\infty = 391$ °R. From Equation (7.32),

$$\frac{p}{p_\infty} = \left(\frac{T}{T_\infty}\right)^{\gamma/(\gamma-1)}$$

or

$$T = T_\infty \left(\frac{p}{p_\infty}\right)^{(\gamma-1)/\gamma} = 391 \left(\frac{400}{476}\right)^{0.4/1.4} = \boxed{372\ ^\circ\text{R}}$$

EXAMPLE 7.5

Consider the gas in the reservoir of the supersonic wind tunnel discussed in Examples 7.2 and 7.3. The pressure and temperature of the air in the reservoir are 20 atm and 300 K respectively. The air in the reservoir expands through the wind tunnel duct. At a certain location in the duct, the pressure is 1 atm. Calculate the air temperature at this location if: (a) the expansion is isentropic and (b) the expansion is nonisentropic with an entropy increase through the duct to this location of 320 J/(kg · K).

■ Solution

(a) From Equation (7.32),

$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)}$$

or,

$$T_2 = T_1 \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}} = 300 \left(\frac{1}{20}\right)^{\frac{0.4}{1.4}} = 300\,(0.05)^{0.2857}$$

$$= 300\,(0.4249) = \boxed{127.5 \text{ K}}$$

(b) From Equation (7.25),

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}$$

Using Equation (7.9) to obtain the value of cp,

$$c_p = \frac{\gamma R}{\gamma - 1} = \frac{(1.4)(287)}{0.4} = 1004.5 \ \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

we have from Equation (7.25),

$$320 = 1004.5 \ln \left(\frac{T_2}{300}\right) - (287) \ln \left(\frac{1}{20}\right)$$

$$= 1004.5 \ln \left(\frac{T_2}{300}\right) - (-859.78)$$

Thus,

$$\ln \left(\frac{T_2}{300}\right) = \frac{320 - 859.78}{1004.5} = -0.5374$$

$$\frac{T_2}{300} = e^{-0.5374} = 0.5843$$

$$T_2 = (0.5843)(300) = \boxed{175.3 \text{ K}}$$

Comment: Comparing the results from parts (a) and (b), note that the entropy increase results in a higher temperature at the point in the expansion where $p = 1$ atm compared to that for the isentropic expansion. This makes sense. From Equation (7.25) we see that entropy is a function of both temperature and pressure, increasing with an increase in temperature and decreasing with an increase in pressure. In this example, the final pressure for both cases (a) and (b) is the same, but the entropy for case (b) is higher. Thus, from Equation (7.25), we see that the final temperature for case (b) must be higher than that for case (a). On a more qualitative basis, the physical mechanisms that could produce the change in entropy would be viscous dissipation (friction), the presence of shock waves in the duct, or heat addition from the surroundings through the walls of the duct. Intuitively, all these irreversible mechanisms would result in a higher gas temperature than the isentropic expansion that, by definition, assumes an adiabatic and reversible (no friction) expansion.

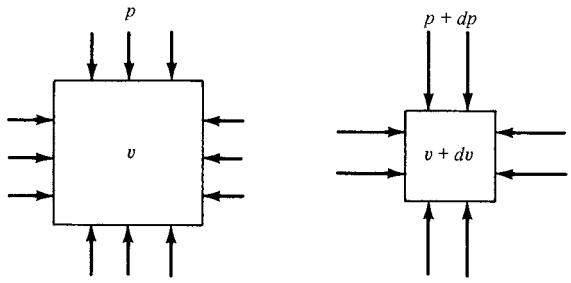


Figure 7.3 Definition of compressibility.

7.3 DEFINITION OF COMPRESSIBILITY

All real substances are compressible to some greater or lesser extent; that is, when you squeeze or press on them, their density will change. This is particularly true of gases, much less so for liquids, and virtually unnoticeable for solids. The amount by which a substance can be compressed is given by a specific property of the substance called the *compressibility,* defined below.

Consider a small element of fluid of volume v, as sketched in Figure 7.3. The pressure exerted on the sides of the element is p. Assume the pressure is now increased by an infinitesimal amount dp. The volume of the element will change by a corresponding amount dv; here, the volume will decrease; hence, dv shown in Figure 7.3 is a negative quantity. By definition, the compressibility τ of the fluid is

$$\tau = -\frac{1}{v}\frac{dv}{dp} \tag{7.33}$$

Physically, the compressibility is the fractional change in volume of the fluid element per unit change in pressure. However, Equation (7.33) is not precise enough. We know from experience that when a gas is compressed (say, in a bicycle pump), its temperature tends to increase, depending on the amount of heat transferred into or out of the gas through the boundaries of the system. If the temperature of the fluid element in Figure 7.3 is held constant (due to some heat transfer mechanism), then τ is identified as the *isothermal compressibility* τ_T, defined from Equation (7.33) as

$$\tau_T = -\frac{1}{v}\left(\frac{\partial v}{\partial p}\right)_T \tag{7.34}$$

On the other hand, if no heat is added to or taken away from the fluid element, and if friction is ignored, the compression of the fluid element takes place isentropically, and τ is identified as the isentropic compressibility τ_s, defined from Equation (7.33) as

$$\tau_s = -\frac{1}{v}\left(\frac{\partial v}{\partial p}\right)_s \tag{7.35}$$

where the subscript s denotes that the partial derivative is taken at constant entropy. Both τ_T and τ_s are precise thermodynamic properties of the fluid; their values for

different gases and liquids can be obtained from various handbooks of physical properties. In general, the compressibility of gases is several orders of magnitude larger than that of liquids.

The role of the compressibility τ in determining the properties of a fluid in motion is seen as follows. Define v as the specific volume (i.e., the volume per unit mass). Hence, $v = 1/\rho$. Substituting this definition into Equation (7.33), we obtain

$$\tau = \frac{1}{\rho}\frac{d\rho}{dp} \tag{7.36}$$

Thus, whenever the fluid experiences a change in pressure dp, the corresponding change in density $d\rho$ from Equation (7.36) is

$$d\rho = \rho\tau\, dp \tag{7.37}$$

Consider a fluid flow, say, for example, the flow over an airfoil. If the fluid is a *liquid,* where the compressibility τ is very small, then for a given pressure change dp from one point to another in the flow, Equation (7.37) states that $d\rho$ will be negligibly small. In turn, we can reasonably assume that ρ is constant and that the flow of a liquid is incompressible. On the other hand, if the fluid is a *gas,* where the compressibility τ is large, then for a given pressure change dp from one point to another in the flow, Equation (7.37) states that $d\rho$ can be large. Thus, ρ is *not* constant, and in general, the flow of a gas is a *compressible flow*. The exception to this is the *low-speed flow* of a gas; in such flows, the actual magnitude of the pressure changes throughout the flow field is small compared with the pressure itself. Thus, for a low-speed flow, dp in Equation (7.37) is small, and even though τ is large, the value of $d\rho$ can be dominated by the small dp. In such cases, ρ can be assumed to be constant, hence allowing us to analyze low-speed gas flows as incompressible flows (such as discussed in Chapters 3 to 6).

Later, we demonstrate that the most convenient index to gage whether a gas flow can be considered incompressible, or whether it must be treated as compressible, is the Mach number M, defined in Chapter 1 as the ratio of local flow velocity V to the local speed of sound a:

$$M \equiv \frac{V}{a} \tag{7.38}$$

We show that, when $M > 0.3$, the flow should be considered compressible. Also, we show that the speed of sound in a gas is related to the isentropic compressibility τ_s, given by Equation (7.35).

7.4 GOVERNING EQUATIONS FOR INVISCID, COMPRESSIBLE FLOW

In Chapters 3 to 6, we studied inviscid, incompressible flow; recall that the primary dependent variables for such flows are p and $\mathbf{V}$, and hence we need only two basic equations, namely, the continuity and momentum equations, to solve for these two unknowns. Indeed, the basic equations are combined to obtain Laplace's

equation and Bernoulli's equation, which are the primary tools used for the applications discussed in Chapters 3 to 6. Note that both ρ and T are assumed to be constant throughout such inviscid, incompressible flows. As a result, no additional governing equations are required; in particular, there is no need for the energy equation or energy concepts in general. Basically, incompressible flow obeys purely mechanical laws and does not require thermodynamic considerations.

In contrast, for compressible flow, ρ is variable and becomes an unknown. Hence, we need an additional governing equation—the energy equation—which in turn introduces internal energy e as an unknown. Since e is related to temperature, then T also becomes an important variable. Therefore, the primary dependent variables for the study of compressible flow are p, $\mathbf{V}$, ρ, e, and T; to solve for these five variables, we need five governing equations. Let us examine this situation further.

To begin with, the flow of a compressible fluid is governed by the basic equations derived in Chapter 2. At this point in our discussion, it is most important for you to be familiar with these equations as well as their derivation. Therefore, before proceeding further, return to Chapter 2 and carefully review the basic ideas and relations contained therein. This is a serious study tip, and if you follow it, the material in our next seven chapters will flow much easier for you. In particular, review the integral and differential forms of the continuity equation (Section 2.4), the momentum equation (Section 2.5), and the energy equation (Section 2.7); indeed, pay particular attention to the energy equation because this is an important aspect which sets compressible flow apart from incompressible flow.

For convenience, some of the more important forms of the governing equations for an inviscid, compressible flow from Chapter 2 are repeated below:

Continuity: From Equation (2.48),

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \, d\mathcal{V} + \oiint_{S} \rho \mathbf{V} \cdot \mathbf{dS} = 0 \tag{7.39}$$

From Equation (2.52),

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{V} = 0 \tag{7.40}$$

Momentum: From Equation (2.64),

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \mathbf{V} \, d\mathcal{V} + \oiint_{S} (\rho \mathbf{V} \cdot \mathbf{dS})\mathbf{V} = -\oiint_{S} p \, \mathbf{dS} + \iiint_{\mathcal{V}} \rho \mathbf{f} \, d\mathcal{V} \tag{7.41}$$

From Equations (2.113*a* to *c*),

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \rho f_x \tag{7.42a}$$

$$\rho \frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \rho f_y \tag{7.42b}$$

$$\rho \frac{Dw}{Dt} = -\frac{\partial p}{\partial z} + \rho f_z \tag{7.42c}$$

Energy: From Equation (2.95),

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho\left(e+\frac{V^2}{2}\right)d\mathcal{V}+\oint\!\!\!\oint_{S} \rho\left(e+\frac{V^2}{2}\right)\mathbf{V}\cdot\mathbf{dS}$$
$$=\iiint_{\mathcal{V}} \dot{q}\rho\, d\mathcal{V}-\oint\!\!\!\oint_{S} p\mathbf{V}\cdot\mathbf{dS}+\iiint_{\mathcal{V}} \rho(\mathbf{f}\cdot\mathbf{V})\,d\mathcal{V} \qquad (7.43)$$

From Equation (2.114),

$$\rho\frac{D(e+V^2/2)}{Dt}=\rho\dot{q}-\nabla\cdot p\mathbf{V}+\rho(\mathbf{f}\cdot\mathbf{V}) \qquad (7.44)$$

The above continuity, momentum, and energy equations are three equations in terms of the five unknowns p, $\mathbf{V}$, ρ, T, and e. Assuming a calorically perfect gas, the additional two equations needed to complete the system are obtained from Section 7.2:

Equation of state: $$p=\rho RT \qquad (7.1)$$

Internal energy: $$e=c_v T \qquad (7.6a)$$

In regard to the basic equations for compressible flow, please note that Bernoulli's equation as derived in Section 3.2 and given by Equation (3.13) does *not* hold for compressible flow; it clearly contains the assumption of constant density, and hence is invalid for compressible flow. This warning is necessary because experience shows that a certain number of students of aerodynamics, apparently attracted by the simplicity of Bernoulli's equation, attempt to use it for all situations, compressible as well as incompressible. Do not do it! Always remember that Bernoulli's equation in the form of Equation (3.13) holds for incompressible flow only, and we must dismiss it from our thinking when dealing with compressible flow.

As a final note, we use both the integral and differential forms of the above equations in our subsequent discussions. Make certain that you feel comfortable with these equations before proceeding further.

7.5 DEFINITION OF TOTAL (STAGNATION) CONDITIONS

At the beginning of Section 3.4, the concept of static pressure p was discussed in some detail. Static pressure is a measure of the purely random motion of molecules in a gas; it is the pressure you feel when you ride along with the gas at the local flow velocity. In contrast, the total (or stagnation) pressure was defined in Section 3.4 as the pressure existing at a point (or points) in the flow where $\mathbf{V}=0$. Let us now define the concept of total conditions more precisely.

Consider a fluid element passing through a given point in a flow where the local pressure, temperature, density, Mach number, and velocity are p, T, ρ, M, and $\mathbf{V}$, respectively. Here, p, T, and ρ are static quantities (i.e., static pressure, static

temperature, and static density, respectively); they are the pressure, temperature, and density you feel when you ride along with the gas at the local flow velocity. Now imagine that you grab hold of the fluid element and *adiabatically* slow it down to zero velocity. Clearly, you would expect (correctly) that the values of p, T, and ρ would change as the fluid element is brought to rest. In particular, the value of the temperature of the fluid element after it has been brought to rest adiabatically is defined as the *total temperature,* denoted by T_0. The corresponding value of enthalpy is defined as the *total enthalpy* h_0, where $h_0 = c_p T_0$ for a calorically perfect gas. Keep in mind that we do not *actually* have to bring the flow to rest in real life in order to talk about the total temperature or total enthalpy; rather, they are *defined quantities* that would exist at a point in a flow *if* (in our imagination) the fluid element passing through that point were brought to rest adiabatically. Therefore, at a given point in a flow, where the static temperature and enthalpy are T and h, respectively, we can also assign a value of total temperature T_0 and a value of total enthalpy h_0 defined as above.

The energy equation, Equation (7.44), provides some important information about total enthalpy and hence total temperature, as follows. Assume that the flow is adiabatic ($\dot{q} = 0$), and that body forces are negligible ($\mathbf{f} = 0$). For such a flow, Equation (7.44) becomes

$$\rho \frac{D(e + V^2/2)}{Dt} = -\nabla \cdot p\mathbf{V} \tag{7.45}$$

Expand the right-hand side of Equation (7.45) using the following vector identity:

$$\nabla \cdot p\mathbf{V} \equiv p\nabla \cdot \mathbf{V} + \mathbf{V} \cdot \nabla p \tag{7.46}$$

Also, note that the substantial derivative defined in Section 2.9 follows the normal laws of differentiation; for example,

$$\rho \frac{D(p/\rho)}{Dt} = \rho \frac{\rho Dp/Dt - pD\rho/Dt}{\rho^2} = \frac{Dp}{Dt} - \frac{p}{\rho}\frac{D\rho}{Dt} \tag{7.47}$$

Recall the form of the continuity equation given by Equation (2.108):

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{V} = 0 \tag{2.108}$$

Substituting Equation (2.108) into (7.47), we obtain

$$\rho \frac{D(p/\rho)}{Dt} = \frac{Dp}{Dt} + p\nabla \cdot \mathbf{V} = \frac{\partial p}{\partial t} + \mathbf{V} \cdot \nabla p + p\nabla \cdot \mathbf{V} \tag{7.48}$$

Substituting Equation (7.46) into (7.45), and adding Equation (7.48) to the result, we obtain

$$\rho \frac{D}{Dt}\left(e + \frac{p}{\rho} + \frac{V^2}{2}\right) = -p\nabla \cdot \mathbf{V} - \mathbf{V} \cdot \nabla p + \frac{\partial p}{\partial t} + \mathbf{V} \cdot \nabla p + p\nabla \cdot \mathbf{V} \tag{7.49}$$

Note that

$$e + \frac{p}{\rho} = e + pv \equiv h \tag{7.50}$$

Substituting Equation (7.50) into (7.49), and noting that some of the terms on the right-hand side of Equation (7.49) cancel each other, we have

$$\rho \frac{D(h + V^2/2)}{Dt} = \frac{\partial p}{\partial t} \tag{7.51}$$

If the flow is steady, $\partial p/\partial t = 0$, and Equation (7.51) becomes

$$\rho \frac{D(h + V^2/2)}{Dt} = 0 \tag{7.52}$$

From the definition of the substantial derivative given in Section 2.9, Equation (7.52) states that the time rate of change of $h + V^2/2$ following a moving fluid element is zero; that is,

$$\boxed{h + \frac{V^2}{2} = \text{const}} \tag{7.53}$$

along a streamline. Recall that the assumptions which led to Equation (7.53) are that the flow is steady, adiabatic, and inviscid. In particular, since Equation (7.53) holds for an adiabatic flow, it can be used to elaborate on our previous definition of total enthalpy. Since h_0 is defined as that enthalpy that would exist at a point if the fluid element were brought to rest adiabatically, we find from Equation (7.53) with $V = 0$ and hence $h = h_0$ that the value of the constant in Equation (7.53) is h_0. Hence, Equation (7.53) can be written as

$$\boxed{h + \frac{V^2}{2} = h_0} \tag{7.54}$$

Equation (7.54) is important; it states that at any point in a flow, the total enthalpy is given by the sum of the static enthalpy plus the kinetic energy, all per unit mass. Whenever we have the combination $h + V^2/2$ in any subsequent equations, it can be identically replaced by h_0. For example, Equation (7.52), which was derived for a steady, adiabatic, inviscid flow, states that

$$\rho \frac{Dh_0}{Dt} = 0$$

that is, the total enthalpy is constant along a streamline. Moreover, if all the streamlines of the flow originate from a common uniform freestream (as is usually the case), then h_0 is the same for each streamline. Consequently, we have for such a steady, adiabatic flow that

$$\boxed{h_0 = \text{const}} \tag{7.55}$$

throughout the *entire* flow, and h_0 is equal to its freestream value. Equation (7.55), although simple in form, is a powerful tool. For steady, inviscid, adiabatic flow, Equation (7.55) is a statement of the energy equation, and hence it can be used in *place of* the more complex partial differential equation given by Equation (7.52). This is a great simplification, as we will see in subsequent discussions.

For a calorically perfect gas, $h_0 = c_p T_0$. Thus, the above results also state that the total temperature is constant throughout the steady, inviscid, adiabatic flow of a calorically perfect gas; that is,

$$\boxed{T_0 = \text{const}} \tag{7.56}$$

For such a flow, Equation (7.56) can be used as a form of the governing energy equation.

Keep in mind that the above discussion marbled two trains of thought: On the one hand, we dealt with the general concept of an adiabatic flow field [which led to Equations (7.51) to (7.53)], and on the other hand, we dealt with the definition of total enthalpy [which led to Equation (7.54)]. These two trains of thought are really separate and should not be confused. Consider, for example, a general *nonadiabatic* flow, such as a viscous boundary layer with heat transfer. A generic nonadiabatic flow is sketched in Figure 7.4*a*. Clearly, Equations (7.51) to (7.53) do not hold for such a flow. However, Equation (7.54) holds locally at each point in the flow, because the assumption of an adiabatic flow contained in Equation (7.54) is made through the *definition* of h_0 and has nothing to do with the general overall flow field. For example, consider two different points, 1 and 2, in the general flow, as shown in Figure 7.4*a*. At point 1, the local static enthalpy and velocity are h_1 and V_1, respectively. Hence, the local total enthalpy at point 1 is $h_{0,1} = h_1 + V_1^2/2$. At point 2, the local static enthalpy and velocity are h_2 and V_2, respectively. Hence, the local total enthalpy at point 2 is $h_{0,2} = h_2 + V_2^2/2$. If the flow between points 1 and 2 is nonadiabatic, then $h_{0,1} \neq h_{0,2}$. Only for the special case where the flow is adiabatic between the two points would $h_{0,1} = h_{0,2}$. This case is illustrated in Figure 7.4*b*. Of course, this is the special case treated by Equations (7.55) and (7.56).

Return to the beginning of this section, where we considered a fluid element passing through a point in a flow where the local properties are p, T, ρ, M, and $\mathbf{V}$. Once again, imagine that you grab hold of the fluid element and slow it down to zero velocity, but this time, let us slow it down both adiabatically *and* reversibly. That is, let us slow the fluid element down to zero velocity *isentropically*. When the fluid element is brought to rest isentropically, the resulting pressure and density are defined as the *total pressure* p_0 and *total density* ρ_0. (Since an isentropic process is also adiabatic, the resulting temperature is the same total temperature T_0 as discussed earlier.) As before, keep in mind that we do not have to actually bring the flow to rest in real life in order to talk about total pressure and total density; rather, they are *defined* quantities that would exist at a point in a flow *if* (in our imagination) the fluid element passing through that point were brought to rest isentropically. Therefore, at a given point in a flow, where the static pressure and static density are p and ρ, respectively, we can also assign a value of total pressure p_0, and total density ρ_0 defined as above.

The definition of p_0 and ρ_0 deals with an isentropic compression to zero velocity. Keep in mind that the isentropic assumption is involved with the definition only. The concept of total pressure and density can be applied throughout any general *nonisentropic* flow. For example, consider two different points, 1 and

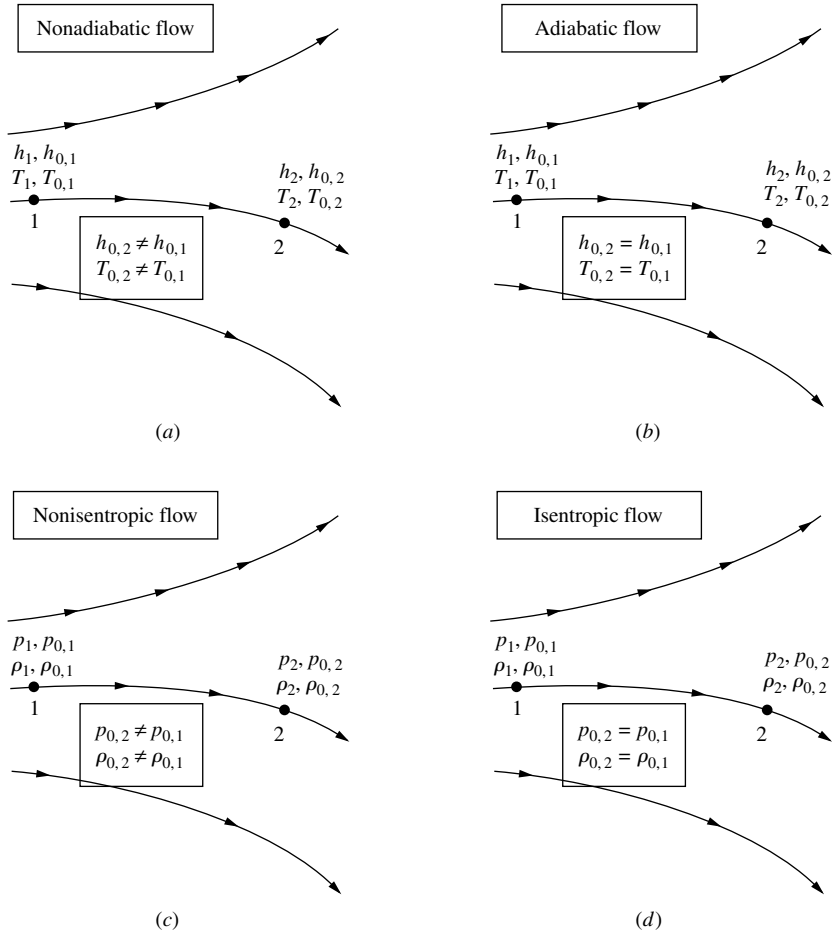


Figure 7.4 Comparisons between (*a*) nonadiabatic, (*b*) adiabatic, (*c*) nonisentropic, and (*d*) isentropic flows.

2, in a general flow field, as sketched in Figure 7.4*c*. At point 1, the local static pressure and static density are p_1 and ρ_1, respectively; also the local total pressure and total density are $p_{0,1}$ and $\rho_{0,1}$, respectively, defined as above. Similarly, at point 2, the local static pressure and static density are p_2 and ρ_2, respectively, and the local total pressure and total density are $p_{0,2}$ and $\rho_{0,2}$, respectively. If the flow is nonisentropic between points 1 and 2, then $p_{0,1} \neq p_{0,2}$ and $\rho_{0,1} \neq \rho_{0,2}$, as shown in Figure 7.4*c*. On the other hand, if the flow is isentropic between points 1 and 2, then $p_{0,1} = p_{0,2}$ and $\rho_{0,1} = \rho_{0,2}$, as shown in Figure 7.4*d*. Indeed, if the general flow field is isentropic throughout, then both p_0 and ρ_0 are constant values throughout the flow.

As a corollary to the above considerations, we need another defined temperature, denoted by T^*, and defined as follows. Consider a point in a subsonic flow where the local static temperature is T. At this point, imagine that the fluid

element is speeded up to *sonic velocity, adiabatically*. The temperature it would have at such sonic conditions is denoted as T^*. Similarly, consider a point in a supersonic flow, where the local static temperature is T. At this point, imagine that the fluid element is slowed down to sonic velocity, adiabatically. Again, the temperature it would have at such sonic conditions is denoted as T^*. The quantity T^* is simply a defined quantity at a given point in a flow, in exactly the same vein as T_0, p_0, and ρ_0 are defined quantities. Also, $a^* = \sqrt{\gamma RT^*}$.

EXAMPLE 7.6

At a point in an airflow the pressure, temperature, and velocity are 1 atm, 320 K, and 1000 m/s. Calculate the total temperature and total pressure at this point.

■ Solution

From Equation (7.54)

$$h + \frac{V^2}{2} = h_0$$

Since

$$h = c_p T$$

and

$$c_p = \frac{\gamma R}{\gamma - 1},$$

we have

$$c_p T + \frac{V^2}{2} = c_p T_0$$

$$T_0 = T + \frac{V^2}{2c_p} = T + \left(\frac{\gamma - 1}{2\gamma R}\right) V^2$$

$$T_0 = 320 + \left[\frac{0.4}{2(1.4)(287)}\right](1000)^2 = 320 + 497.8$$

$$T_0 = \boxed{817.8 \text{ K}}$$

By definition, the total pressure is the pressure that would exist if the flow at the point were slowed isentropically to zero velocity. Hence, we can use the isentropic relations in Equation (7.32) to relate total to static conditions. That is, from Equation (7.32),

$$\frac{p_0}{p} = \left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}}$$

Hence,

$$p_0 = p\left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}} = (1 \text{ atm})\left(\frac{817.8}{320}\right)^{\frac{1.4}{0.4}}$$

$$P_0 = \boxed{26.7 \text{ atm}}$$

Note: In the above calculation of total pressure, we continued to use the *nonconsistent* unit of atmospheres. This is okay because Equation (7.32) contains a *ratio* of pressures, and therefore it would just be extra work to convert 1 atm to 1.01×10^5 N/m^2, and then carry out the calculation. The result is the same.

EXAMPLE 7.7

An airplane is flying at a standard altitude of 10,000 ft. A Pitot tube mounted at the nose measures a pressure of 2220 lb/ft^2. The airplane is flying at a high subsonic speed, faster than 300 mph. From our comments in Section 3.1, the flow should be considered *compressible*. Calculate the velocity of the airplane.

■ Solution

From our discussion in Section 3.4, the pressure measured by a Pitot tube immersed in an incompressible flow is the total pressure. For the same *physical reasons* discussed in Section 3.4, a Pitot tube also measures the total pressure in a high-speed subsonic compressible flow. (This is further discussed in Section 8.7.1 on the measurement of velocity in a subsonic compressible flow.) *Caution*: Because we are dealing with a compressible flow in this example, we can *not* use Bernoulli's equation to calculate the velocity.

The flow in front of the Pitot tube is compressed isentropically to zero velocity at the mouth of the tube, hence the pressure at the mouth is the total pressure, p_0. From Equation (7.32), we can write:

$$\frac{p_0}{p_\infty} = \left(\frac{T_0}{T_\infty}\right)^{\gamma/(\gamma-1)} \tag{E.7.3}$$

where p_0 and T_0 are the total pressure and temperature, respectively, at the mouth of the Pitot tube, and p_∞ and T_∞ are the freestream static pressure and static temperature, respectively. Solving Equation (E7.3) above for T_0, we get

$$T_0 = T_\infty \left(\frac{p_0}{p_\infty}\right)^{(\gamma-1)/\gamma} \tag{E.7.4}$$

From Appendix E, the pressure and temperature at a standard altitude of 10,000 ft are 1455.6 lb/ft^2 and 483.04 °R respectively. These are the values of p_∞ and T_∞ in Equation (E7.4). Thus, from Equation (E7.4),

$$T_0 = (483.04)\left(\frac{2220}{1455.6}\right)^{0.4/1.4} = 544.9\ ^\circ\text{R}$$

From the energy equation, Equation (7.54), written in terms of temperature, we have

$$c_p\, T + \frac{V^2}{2} = c_p\, T_0 \tag{E.7.5}$$

In Equation (E7.5), both T and V are the freestream values, hence we have

$$c_p\, T_\infty + \frac{V_\infty^2}{2} = c_p\, T_0 \tag{E.7.6}$$

Also,

$$c_p = \frac{\gamma R}{\gamma - 1} = \frac{(1.4)(1716)}{0.4} = 6006\ \frac{\text{ft} \cdot \text{lb}}{\text{slug} \cdot {}^\circ\text{R}}$$

Solving Equation (E7.6) for V_∞, we have

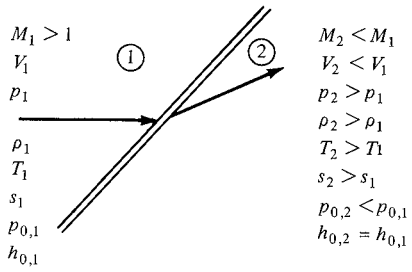
$$\begin{aligned} V_\infty &= [2\, c_p\, (T_0 - T_\infty)]^{1/2} \\ &= [2\,(6006)(544.9 - 483.04]^{1/2} \\ &= \boxed{862 \text{ ft/s}} \end{aligned}$$

Note: From this example, we see that the total pressure measured by a Pitot tube in a subsonic compressible flow is a measure of the flow velocity, but we need also the value of the flow static temperature in order to calculate the velocity. In Section 8.7, we show more fundamentally that the *ratio* of Pitot pressure to flow static pressure in a compressible flow, subsonic or supersonic, is a *direct measure* of the *Mach number*, not the velocity. But more on this later.

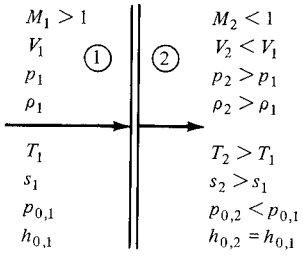
7.6 SOME ASPECTS OF SUPERSONIC FLOW: SHOCK WAVES

Return to the different regimes of flow sketched in Figure 1.44. Note that subsonic compressible flow is qualitatively (but not quantitatively) the same as incompressible flow; Figure 1.44*a* shows a subsonic flow with a smoothly varying streamline pattern, where the flow far ahead of the body is forewarned about the presence of the body and begins to adjust accordingly. In contrast, supersonic flow is quite different, as sketched in Figure 1.44*d* and *e*. Here, the flow is dominated by shock waves, and the flow upstream of the body does not know about the presence of the body until it encounters the leading-edge shock wave. In fact, any flow with a supersonic region, such as those sketched in Figure 1.44*b* to *e*, is subject to shock waves. Thus, an essential ingredient of a study of supersonic flow is the calculation of the shape and strength of shock waves. This is the main thrust of Chapters 8 and 9.

A shock wave is an extremely thin region, typically on the order of 10^{-5} cm, across which the flow properties can change drastically. The shock wave is usually at an oblique angle to the flow, such as sketched in Figure 7.5*a*; however, there are many cases where we are interested in a shock wave normal to the flow, as sketched in Figure 7.5*b*. Normal shock waves are discussed at length in Chapter 8, whereas oblique shocks are considered in Chapter 9. In both cases, the shock wave is an almost explosive compression process, where the pressure increases almost discontinuously across the wave. Examine Figure 7.5 closely. In region 1 ahead of the shock, the Mach number, flow velocity, pressure, density, temperature, entropy, total pressure, and total enthalpy are denoted by M_1, V_1, p_1, ρ_1, T_1, s_1, $p_{0,1}$, and $h_{0,1}$, respectively. The analogous quantities in region 2 behind the shock are M_2, V_2, p_2, ρ_2, T_2, s_2, $p_{0,2}$, and $h_{0,2}$, respectively. The qualitative changes across the wave are noted in Figure 7.5. The pressure, density, temperature, and entropy increase across the shock, whereas the total pressure, Mach number, and velocity decrease. Physically, the flow across a shock wave is adiabatic (we are not heating the gas with a laser beam or cooling it in a refrigerator, for example). Therefore, recalling the discussion in Section 7.5, the total enthalpy is constant across the wave. In both the oblique shock and normal shock cases, the flow ahead of the shock wave must be supersonic (i.e., $M_1 > 1$). Behind the oblique shock, the flow usually remains supersonic (i.e., $M_2 > 1$), but at a reduced Mach number (i.e., $M_2 < M_1$). However, as discussed in Chapter 9, there are special cases where the oblique shock is strong enough to decelerate the downstream flow to a subsonic Mach number; hence, $M_2 < 1$ can occur behind an oblique shock. For the normal



(*a*) Oblique shock wave

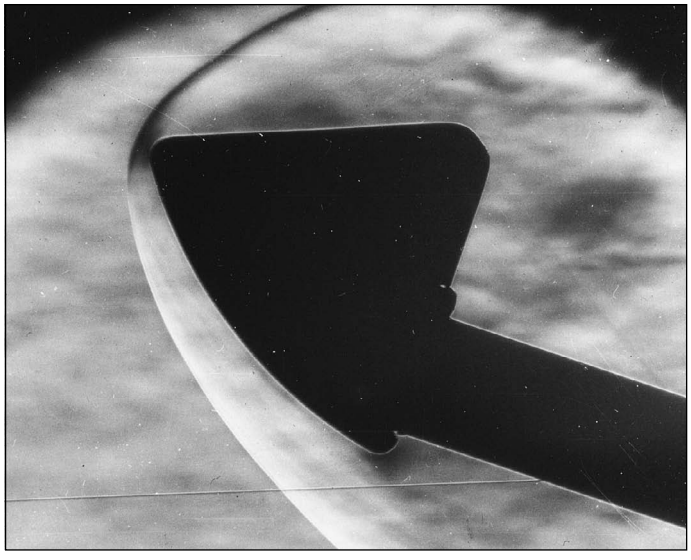


(*b*) Normal shock wave

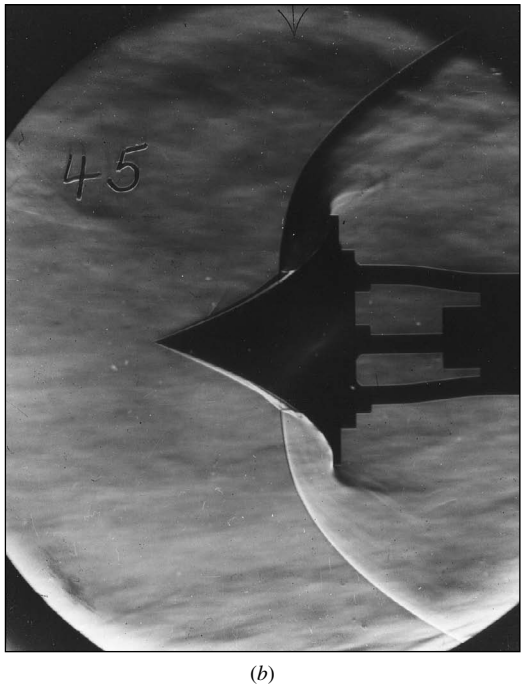
Figure 7.5 Qualitative pictures of flow through oblique and normal shock waves.

shock, as sketched in Figure 7.5*b*, the downstream flow is always subsonic (i.e., $M_2 < 1$). Study the qualitative variations illustrated in Figure 7.5 closely. They are important, and you should have them in mind for our subsequent discussions. One of the primary purposes of Chapters 8 and 9 is to develop a shock-wave theory that allows the quantitative evaluation of these variations. We prove that pressure increases across the shock, that the upstream Mach number must be supersonic, etc. Moreover, we obtain equations that allow the direct calculation of changes across the shock.

Several photographs of shock waves are shown in Figure 7.6. Since air is transparent, we cannot usually see shock waves with the naked eye. However, because the density changes across the shock wave, light rays propagating through the flow will be refracted across the shock. Special optical systems, such as shadowgraphs, schlieren, and interferometers, take advantage of this refraction and allow the visual imaging of shock waves on a screen or a photographic negative. For details of the design and characteristics of these optical systems, see References 25 and 26. (Under certain conditions, you can see the refracted light from a shock wave with your naked eye. Recall from Figure 1.43*b* that a shock wave can form in the locally supersonic region on the top surface of an airfoil if the freestream subsonic Mach number is high enough. The next time you are flying in a jet transport, and the sun is directly overhead, look out the window along the span of the wing. If you are lucky, you will see the shock wave dancing back and forth over the top of the wing.)

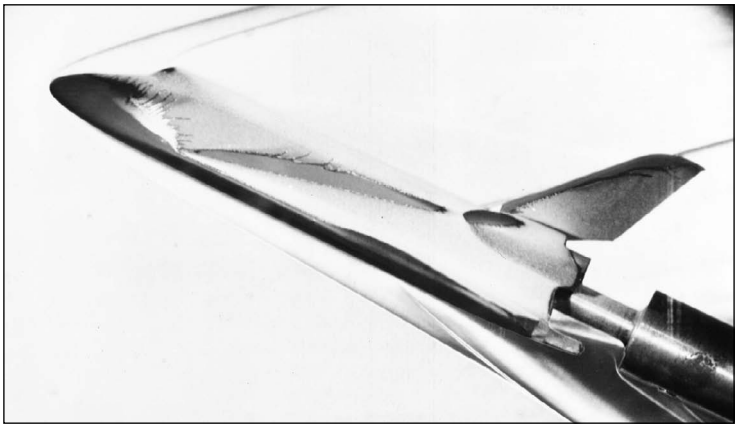


(*a*)

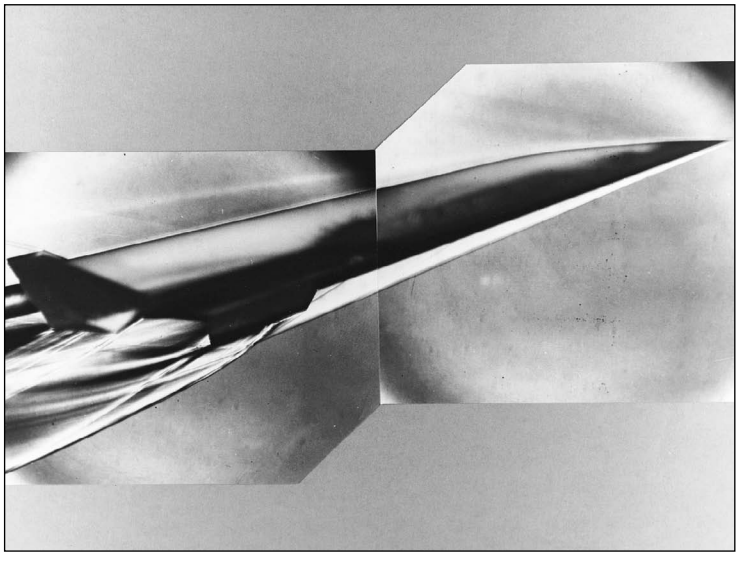


(*b*)

Figure 7.6 (*a*) Schlieren photographs illustrating shock waves on various bodies. Apollo Command Module wind-tunnel model at Mach 8. (*b*) "Tension shell" body once proposed for a Martian entry vehicle at Mach 8. *(Courtesy of the NASA Langley Research Center.)*



(*c*)



(*d*)

Figure 7.6 (continued) (*c*) Space Shuttle Orbiter model at Mach 6. This photo also shows regions of high aerodynamic heating on the model surface by means of the visible phase-change paint pattern. (*d*) A conceptual hypersonic aircraft at Mach 6. *(Courtesy of the NASA Langley Research Center.)*

In summary, compressible flows introduce some very exciting physical phenomena into our aerodynamic studies. Moreover, as the flow changes from subsonic to supersonic, the complete nature of the flow changes, not the least of which is the occurrence of shock waves. The purpose of the next seven chapters is to describe and analyze these flows.

7.7 SUMMARY

As usual, examine the road map for this chapter (Figure 7.1), and make certain that you feel comfortable with the material represented by this road map before continuing further.

Some of the highlights of this chapter are summarized below:

Thermodynamic relations:

Equation of state:
$$p = \rho RT \tag{7.1}$$

For a calorically perfect gas,
$$e = c_v T \qquad \text{and} \qquad h = c_p T \tag{7.6a and b}$$
$$c_p = \frac{\gamma R}{\gamma - 1} \tag{7.9}$$
$$c_v = \frac{R}{\gamma - 1} \tag{7.10}$$

Forms of the first law:
$$\delta q + \delta w = de \tag{7.11}$$
$$T\,ds = de + p\,dv \tag{7.18}$$
$$T\,ds = dh - v\,dp \tag{7.20}$$

Definition of entropy:
$$ds = \frac{\delta q_{\text{rev}}}{T} \tag{7.13}$$

Also
$$ds = \frac{\delta q}{T} + ds_{\text{irrev}} \tag{7.14}$$

The second law:
$$ds \geq \frac{\delta q}{T} \tag{7.16}$$

or, for an adiabatic process,
$$ds \geq 0 \tag{7.17}$$

Entropy changes can be calculated from (for a calorically perfect gas)
$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \tag{7.25}$$

and
$$s_2 - s_1 = c_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1} \tag{7.26}$$

For an isentropic flow,

$$\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma} = \left(\frac{T_2}{T_1}\right)^{\gamma/(\gamma-1)} \tag{7.32}$$

General definition of compressibility:

$$\tau = -\frac{1}{v}\frac{dv}{dp} \tag{7.33}$$

For an isothermal process,

$$\tau_T = -\frac{1}{v}\left(\frac{\partial v}{\partial p}\right)_T \tag{7.34}$$

For an isentropic process,

$$\tau_s = -\frac{1}{v}\left(\frac{\partial v}{\partial p}\right)_s \tag{7.35}$$

The governing equations for inviscid, compressible flow are

Continuity:

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \, d\mathcal{V} + \oiint_S \rho \mathbf{V} \cdot \mathbf{dS} = 0 \tag{7.39}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{V} = 0 \tag{7.40}$$

Momentum:

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \mathbf{V} \, d\mathcal{V} + \oiint_S (\rho \mathbf{V} \cdot \mathbf{dS})\mathbf{V} = -\oiint_S p \, \mathbf{dS} + \iiint_{\mathcal{V}} \rho \mathbf{f} \, d\mathcal{V} \tag{7.41}$$

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \rho f_x \tag{7.42a}$$

$$\rho \frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \rho f_y \tag{7.42b}$$

$$\rho \frac{Dw}{Dt} = -\frac{\partial p}{\partial z} + \rho f_z \tag{7.42c}$$

Energy:

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \left(e + \frac{V^2}{2}\right) d\mathcal{V} + \oiint_S \rho \left(e + \frac{V^2}{2}\right) \mathbf{V} \cdot \mathbf{dS}$$

(*continued*)

$$= \iiint_{\mathcal{V}} \dot{q}\rho \, d\mathcal{V} - \oint_S p\mathbf{V} \cdot \mathbf{dS} + \iiint_{\mathcal{V}} \rho(\mathbf{f} \cdot \mathbf{V}) \, d\mathcal{V} \tag{7.43}$$

$$\rho \frac{D(e + V^2/2)}{Dt} = \rho\dot{q} - \nabla \cdot p\mathbf{V} + \rho(\mathbf{f} \cdot \mathbf{V}) \tag{7.44}$$

If the flow is steady and adiabatic, Equations (7.43) and (7.44) can be replaced by

$$h_0 = h + \frac{V^2}{2} = \text{const}$$

Equation of state (perfect gas):

$$p = \rho RT \tag{7.1}$$

Internal energy (calorically perfect gas):

$$e = c_v T \tag{7.6a}$$

Total temperature T_0 and total enthalpy h_0 are defined as the properties that would exist if (in our imagination) we slowed the fluid element at a point in the flow to zero velocity adiabatically. Similarly, total pressure p_0 and total density ρ_0 are defined as the properties that would exist if (in our imagination) we slowed the fluid element at a point in the flow to zero velocity isentropically. If a general flow field is adiabatic, h_0 is constant throughout the flow; in contrast, if the flow field is nonadiabatic, h_0 varies from one point to another. Similarly, if a general flow field is isentropic, p_0 and ρ_0 are constant throughout the flow; in contrast, if the flow field is nonisentropic, p_0 and ρ_0 vary from one point to another.

Shock waves are very thin regions in a supersonic flow across which the pressure, density, temperature, and entropy increase; the Mach number, flow velocity, and total pressure decrease; and the total enthalpy stays the same.

7.8 PROBLEMS

Note: In the following problems, you will deal with both the International System of Units (SI) (N, kg, m, s, K) and the English Engineering System (lb, slug, ft, s, °R). Which system to use will be self-evident in each problem. All problems deal with calorically perfect air as the gas, unless otherwise noted. Also, recall that 1 atm = 2116 lb/ft^2 = 1.01×10^5 N/m^2.

7.1 The temperature and pressure at the stagnation point of a high-speed missile are 934°R and 7.8 atm, respectively. Calculate the density at this point.

7.2 Calculate c_p, c_v, e, and h for

a. The stagnation point conditions given in Problem 7.1

b. Air at standard sea level conditions

(If you do not remember what standard sea level conditions are, find them in an appropriate reference, such as Reference 2.)

7.3 Just upstream of a shock wave, the air temperature and pressure are 288 K and 1 atm, respectively; just downstream of the wave, the air temperature and pressure are 690 K and 8.656 atm, respectively. Calculate the changes in enthalpy, internal energy, and entropy across the wave.

7.4 Consider the isentropic flow over an airfoil. The freestream conditions are $T_\infty = 245$ K and $p_\infty = 4.35 \times 10^4$ N/m^2. At a point on the airfoil, the pressure is 3.6×10^4 N/m^2. Calculate the density at this point.

7.5 Consider the isentropic flow through a supersonic wind-tunnel nozzle. The reservoir properties are $T_0 = 500$ K and $p_0 = 10$ atm. If $p = 1$ atm at the nozzle exit, calculate the exit temperature and density.

7.6 Consider air at a pressure of 0.2 atm. Calculate the values of τ_T and τ_s. Express your answer in SI units.

7.7 Consider a point in a flow where the velocity and temperature are 1300 ft/s and 480°R, respectively. Calculate the total enthalpy at this point.

7.8 In the reservoir of a supersonic wind tunnel, the velocity is negligible, and the temperature is 1000 K. The temperature at the nozzle exit is 600 K. Assuming adiabatic flow through the nozzle, calculate the velocity at the exit.

7.9 An airfoil is in a freestream where $p_\infty = 0.61$ atm, $\rho_\infty = 0.819$ kg/m^3, and $V_\infty = 300$ m/s. At a point on the airfoil surface, the pressure is 0.5 atm. Assuming isentropic flow, calculate the velocity at that point.

7.10 Calculate the percentage error obtained if Problem 7.9 is solved using (incorrectly) the incompressible Bernoulli equation.

7.11 Repeat Problem 7.9, considering a point on the airfoil surface where the pressure is 0.3 atm.

7.12 Repeat Problem 7.10, considering the flow of Problem 7.11.

7.13 Bernoulli's equation, Equation (3.13), (3.14), or (3.15), was derived in Chapter 3 from Newton's second law; it is fundamentally a statement that force = mass × acceleration. However, the terms in Bernoulli's equation have dimensions of energy per unit volume (check it out), which prompt some argument that Bernoulli's equation is an energy equation for incompressible flow. If this is so, then it should be derivable from the energy equation for compressible flow discussed in the present chapter. Starting with Equation (7.53) for inviscid, adiabatic compressible flow, make the appropriate assumptions for an incompressible flow and see what you need to do to obtain Bernoulli's equation.

CHAPTER 8

Normal Shock Waves and Related Topics

Shock wave: A large-amplitude compression wave, such as that produced by an explosion, caused by supersonic motion of a body in a medium.

From the American Heritage Dictionary of the English Language, 1969

PREVIEW BOX

When you have a supersonic flow, chances are that you also have shock waves. Shock waves are exceptionally thin regions—usually much thinner than the thickness of this page—across which the flow properties change drastically. Take pressure for example. The gas pressure in front of the shock wave may be 1 atm, and immediately behind it the pressure may be 20 atm. Imagine that you are a fluid element crossing the shock wave; at one moment you are at 1 atm pressure and a split second later you are at 20 atm. I do not know about you, but if it were me I would be shocked out of my mind. (Perhaps this is why we call them "shock" waves.)

This chapter is all about shock waves. Here we learn how to calculate the change in flow properties across a shock, and we examine the important physical aspects and consequences of shock waves. In this chapter, we focus on shock waves that are perpendicular to the flow—normal shock waves. What we learn here is directly transferable to the study of shock waves that make an oblique angle to the flow—oblique shock waves—discussed in the next chapter. A study of shock-wave phenomena is one of the most important aspects of learning about supersonic flows. So take this chapter very seriously. Besides, shock waves are rather exciting events in nature, so enjoy this excitement.

8.1 INTRODUCTION

The purpose of this chapter and Chapter 9 is to develop shock-wave theory, thus giving us the means to calculate the changes in the flow properties across a wave. These changes were discussed qualitatively in Section 7.6; make certain that you are familiar with these changes before continuing.

The focus of this chapter is on normal shock waves, as sketched in Figure 7.4*b*. At first thought, a shock wave that is normal to the upstream flow may seem to be a very special case—and therefore a case of little practical interest—but nothing could be further from the truth. Normal shocks occur frequently in nature. Two such examples are sketched in Figure 8.1; there are many more. The supersonic flow over a blunt body is shown at the left of Figure 8.1. Here, a strong bow shock wave exists in front of the body. (We study such bow shocks in Chapter 9.) Although this wave is curved, the region of the shock closest to the nose is essentially normal to the flow. Moreover, the streamline that passes through this normal portion of the bow shock later impinges on the nose of the body and controls the values of stagnation (total) pressure and temperature at the nose. Since the nose region of high-speed blunt bodies is of practical interest in the calculation of drag and aerodynamic heating, the properties of the flow behind the normal portion of the shock wave take on some importance. In another example, shown at the right of Figure 8.1, supersonic flow is established inside a nozzle (which can be a supersonic wind tunnel, a rocket engine, etc.) where the back pressure is high enough to cause a normal shock wave to stand inside the nozzle. (We discuss such "overexpanded" nozzle flows in Chapter 10.) The conditions under which this shock wave will occur and the determination of flow properties at the nozzle exit downstream of the normal shock are both important questions to be answered. In summary, for these and many other applications, the study of normal shock waves is important.

Finally, we will find that many of the normal shock relations derived in this chapter carry over directly to the analysis of oblique shock waves, as discussed

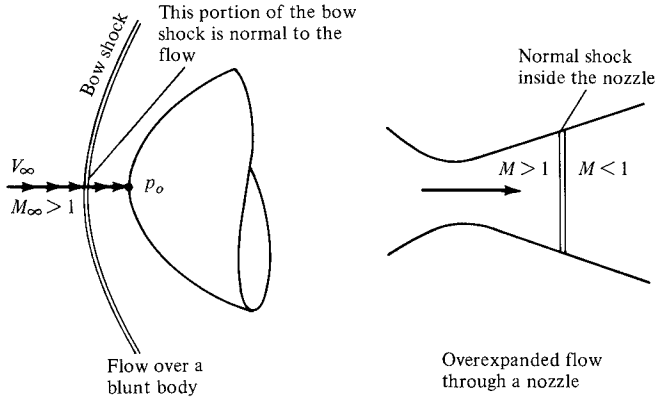


Figure 8.1 Two examples where normal shock waves are of interest.

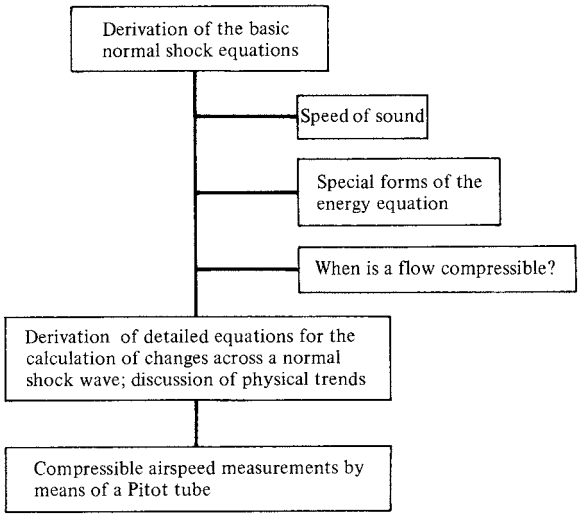


Figure 8.2 Road map for Chapter 8.

in Chapter 9. So once again, time spent on normal shock waves is time well spent.

The road map for this chapter is given in Figure 8.2. As you can see, our objectives are fairly short and straightforward. We start with a derivation of the basic continuity, momentum, and energy equations for normal shock waves, and then we employ these basic relations to obtain detailed equations for the calculation of flow properties across the shock wave. In addition, we emphasize the physical trends indicated by the equations. On the way toward this objective, we take three side streets having to do with (1) the speed of sound, (2) special forms of the energy equation, and (3) a further discussion of the criteria used to judge when a flow must be treated as compressible. Finally, we apply the results of this chapter to the measurement of airspeed in a compressible flow using a Pitot tube. Keep the road map in Figure 8.2 in mind as you progress through the chapter.

8.2 THE BASIC NORMAL SHOCK EQUATIONS

Consider the normal shock wave sketched in Figure 8.3. Region 1 is a uniform flow upstream of the shock, and region 2 is a different uniform flow downstream of the shock. The pressure, density, temperature, Mach number, velocity, total pressure, total enthalpy, total temperature, and entropy in region 1 are p_1, ρ_1, T_1, M_1, u_1, $p_{0,1}$, $h_{0,1}$, $T_{0,1}$, and s_1, respectively. The corresponding variables in region 2 are denoted by p_2, ρ_2, T_2, M_2, u_2, $p_{0,2}$, $h_{0,2}$, $T_{0,2}$, and s_2. (Note that we are denoting the magnitude of the flow velocity by u rather than V; reasons for this will become obvious as we progress.) The problem of the normal shock wave

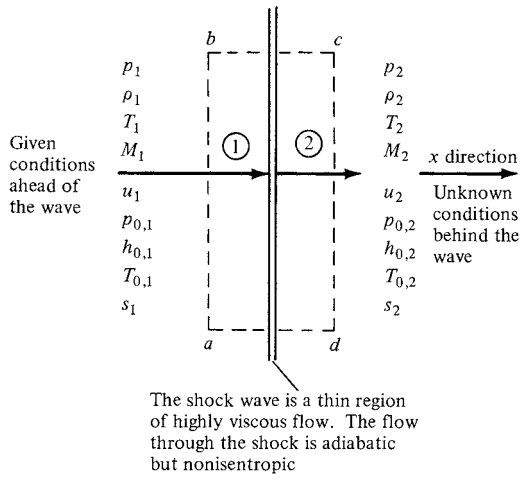


Figure 8.3 Sketch of a normal wave.

is simply stated as follows: given the flow properties upstream of the wave (p_1, T_1, M_1, etc.), calculate the flow properties (p_2, T_2, M_2, etc.) downstream of the wave. Let us proceed.

Consider the rectangular control volume *abcd* given by the dashed line in Figure 8.3. The shock wave is inside the control volume, as shown. Side *ab* is the edge view of the left face of the control volume; this left face is perpendicular to the flow, and its area is A. Side *cd* is the edge view of the right face of the control volume; this right face is also perpendicular to the flow, and its area is A. We apply the integral form of conservation equations to this control volume. In the process, we observe three important physical facts about the flow given in Figure 8.3:

1. The flow is steady, that is, $\partial/\partial t = 0$.
2. The flow is adiabatic, that is, $\dot{q} = 0$. We are not adding or taking away heat from the control volume (we are not heating the shock wave with a Bunsen burner, for example). The temperature increases across the shock wave, not because heat is being added, but rather, because kinetic energy is converted to internal energy across the shock wave.
3. There are no viscous effects on the sides of the control volume. The shock wave itself is a thin region of extremely high velocity and temperature gradients; hence, friction and thermal conduction play an important role on the flow structure inside the wave. However, the wave itself is buried inside the control volume, and with the integral form of the conservation equations, we are not concerned about the details of what goes on inside the control volume.
4. There are no body forces; $\mathbf{f} = 0$.

Consider the continuity equation in the form of Equation (7.39). For the conditions described above, Equation (7.39) becomes

$$\oiint_S \rho \mathbf{V} \cdot \mathbf{dS} = 0 \tag{8.1}$$

To evaluate Equation (8.1) over the face ab, note that $\mathbf{V}$ is pointing into the control volume whereas $\mathbf{dS}$ by definition is pointing out of the control volume, in the opposite direction of $\mathbf{V}$; hence, $\mathbf{V} \cdot \mathbf{dS}$ is negative. Moreover, ρ and $|\mathbf{V}|$ are uniform over the face ab and equal to ρ_1 and u_1, respectively. Hence, the contribution of face ab to the surface integral in Equation (8.1) is simply $-\rho_1 u_1 A$. Over the right face cd both $\mathbf{V}$ and $\mathbf{dS}$ are in the same direction, and hence $\mathbf{V} \cdot \mathbf{dS}$ is positive. Moreover, ρ and $|\mathbf{V}|$ are uniform over the face cd and equal to ρ_2 and u_2, respectively. Thus, the contribution of face cd to the surface integral is $\rho_2 u_2 A$. On sides bc and ad, $\mathbf{V}$ and $\mathbf{dS}$ are always perpendicular; hence, $\mathbf{V} \cdot \mathbf{dS} = 0$, and these sides make no contribution to the surface integral. Hence, for the control volume shown in Figure 8.3, Equation (8.1) becomes

$$-\rho_1 u_1 A + \rho_2 u_2 A = 0$$

or

$$\boxed{\rho_1 u_1 = \rho_2 u_2} \tag{8.2}$$

Equation (8.2) is the continuity equation for normal shock waves.

Consider the momentum equation in the form of Equation (7.41). For the flow we are treating here, Equation (7.41) becomes

$$\oiint_S (\rho \mathbf{V} \cdot \mathbf{dS})\mathbf{V} = -\oiint_S p \, \mathbf{dS} \tag{8.3}$$

Equation (8.3) is a vector equation. Note that in Figure 8.3, the flow is moving only in one direction (i.e., in the x direction). Hence, we need to consider only the scalar x component of Equation (8.3), which is

$$\oiint_S (\rho \mathbf{V} \cdot \mathbf{dS})u = -\oiint_S (p \, dS)_x \tag{8.4}$$

In Equation (8.4), $(p \, dS)_x$ is the x component of the vector $(p \, \mathbf{dS})$. Note that over the face ab, $\mathbf{dS}$ points to the left (i.e., in the negative x direction). Hence, $(p \, dS)_x$ is negative over face ab. By similar reasoning, $(p \, dS)_x$ is positive over the face cd. Again noting that all the flow variables are uniform over the faces ab and cd, the surface integrals in Equation (8.4) become

$$\rho_1(-u_1 A)u_1 + \rho_2(u_2 A)u_2 = -(-p_1 A + p_2 A) \tag{8.5}$$

or

$$\boxed{p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2} \tag{8.6}$$

Equation (8.6) is the momentum equation for normal shock waves.

Consider the energy equation in the form of Equation (7.43). For steady, adiabatic, inviscid flow with no body forces, this equation becomes

$$\oiint_S \rho \left(e + \frac{V^2}{2} \right) \mathbf{V} \cdot \mathbf{dS} = - \oiint_S p \mathbf{V} \cdot \mathbf{dS} \tag{8.7}$$

Evaluating Equation (8.7) for the control surface shown in Figure 8.3, we have

$$-\rho_1 \left(e_1 + \frac{u_1^2}{2} \right) u_1 A + \rho_2 \left(e_2 + \frac{u_2^2}{2} \right) u_2 A = -(-p_1 u_1 A + p_2 u_2 A)$$

Rearranging, we obtain

$$p_1 u_1 + \rho_1 \left(e_1 + \frac{u_1^2}{2} \right) u_1 = p_2 u_2 + \rho_2 \left(e_2 + \frac{u_2^2}{2} \right) u_2 \tag{8.8}$$

Dividing by Equation (8.2), that is, dividing the left-hand side of Equation (8.8) by $\rho_1 u_1$ and the right-hand side by $\rho_2 u_2$, we have

$$\frac{p_1}{\rho_1} + e_1 + \frac{u_1^2}{2} = \frac{p_2}{\rho_2} + e_2 + \frac{u_2^2}{2} \tag{8.9}$$

From the definition of enthalpy, $h \equiv e + pv = e + p/\rho$. Hence, Equation (8.9) becomes

$$\boxed{h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}} \tag{8.10}$$

Equation (8.10) is the energy equation for normal shock waves. Equation (8.10) should come as no surprise; the flow through a shock wave is adiabatic, and we derived in Section 7.5 the fact that for a steady, adiabatic flow, $h_0 = h + V^2/2 =$ const. Equation (8.10) simply states that h_0 (hence, for a calorically perfect gas T_0) is constant across the shock wave. Therefore, Equation (8.10) is consistent with the general results obtained in Section 7.5.

Repeating the above results for clarity, the basic normal shock equations are

Continuity: $$\rho_1 u_1 = \rho_2 u_2 \tag{8.2}$$

Momentum: $$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2 \tag{8.6}$$

Energy: $$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \tag{8.10}$$

Examine these equations closely. Recall from Figure 8.3 that all conditions upstream of the wave, ρ_1, u_1, p_1, etc., are known. Thus, the above equations are a system of three algebraic equations in four unknowns, ρ_2, u_2, p_2, and h_2. However, if we add the following thermodynamic relations

Enthalpy: $$h_2 = c_p T_2$$

Equation of state: $$p_2 = \rho_2 R T_2$$

we have five equations for five unknowns, namely, ρ_2, u_2, p_2, h_2, and T_2. In Section 8.6, we explicitly solve these equations for the unknown quantities behind

the shock. However, rather than going directly to that solution, we first take three side trips as shown in the road map in Figure 8.2. These side trips involve discussions of the speed of sound (Section 8.3), alternate forms of the energy equation (Section 8.4), and compressibility (Section 8.5)—all of which are necessary for a viable discussion of shock-wave properties in Section 8.6.

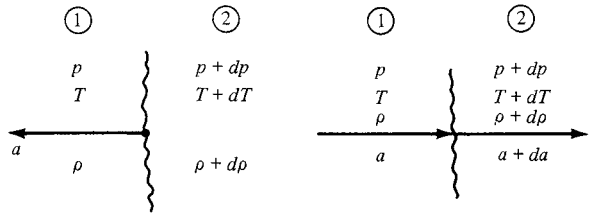
Finally, we note that Equations (8.2), (8.6), and (8.10) are not limited to normal shock waves; they describe the changes that take place in any steady, adiabatic, inviscid flow where only one direction is involved. That is, in Figure 8.3, the flow is in the x direction only. This type of flow, where the flow-field variables are functions of x only [$p = p(x)$, $u = u(x)$, etc.], is defined as *one-dimensional flow*. Thus, Equations (8.2), (8.6), and (8.10) are governing equations for one-dimensional, steady, adiabatic, inviscid flow.

8.3 SPEED OF SOUND

Common experience tells us that sound travels through air at some finite velocity. For example, you see a flash of lightning in the distance, but you hear the corresponding thunder at some later moment. What is the physical mechanism of the propagation of sound waves? How can we calculate the speed of sound? What properties of the gas does it depend on? The speed of sound is an extremely important quantity which dominates the physical properties of compressible flow, and hence the answers to the above questions are vital to our subsequent discussions. The purpose of this section is to address these questions.

The physical mechanism of sound propagation in a gas is based on molecular motion. For example, imagine that you are sitting in a room, and suppose that a firecracker goes off in one corner. When the firecracker detonates, chemical energy (basically a form of heat release) is transferred to the air molecules adjacent to the firecracker. These energized molecules are moving about in a random fashion. They eventually collide with some of their neighboring molecules and transfer their high energy to these neighbors. In turn, these neighboring molecules eventually collide with their neighbors and transfer energy in the process. By means of this "domino" effect, the energy released by the firecracker is propagated through the air by molecular collisions. Moreover, because T, p, and ρ for a gas are macroscopic averages of the detailed microscopic molecular motion, the regions of energized molecules are also regions of slight variations in the local temperature, pressure, and density. Hence, as this energy wave from the firecracker passes over our eardrums, we "hear" the slight pressure changes in the wave. This is *sound,* and the propagation of the energy wave is simply the propagation of a *sound wave* through the gas.

Because a sound wave is propagated by molecular collisions, and because the molecules of a gas are moving with an average velocity of $\sqrt{8RT/\pi}$ given by kinetic theory, then we would expect the velocity of propagation of a sound wave to be approximately the average molecular velocity. Indeed, the speed of sound is about three-quarters of the average molecular velocity. In turn, because the kinetic theory expression given above for the average molecular velocity depends only



(*a*) A sound wave propagating with velocity *a* into a stagnant gas.

(*b*) A stationary sound wave in a moving gas; the upstream velocity relative to the wave is *a*.

Figure 8.4 Moving and stationary sound waves; two analogous pictures, only the perspective is different.

on the *temperature* of the gas, we might expect the speed of sound to also depend on temperature only. Let us explore this matter further; indeed, let us now derive an equation for the speed of sound in a gas. Although the propagation of sound is due to molecular collisions, we do not use such a microscopic picture for our derivation. Rather, we take advantage of the fact that the macroscopic properties p, T, ρ, etc., change across the wave, and we use our macroscopic equations of continuity, momentum, and energy to analyze these changes.

Consider a sound wave propagating through a gas with velocity a, as sketched in Figure 8.4*a*. Here, the sound wave is moving from right to left into a stagnant gas (region 1), where the local pressure, temperature, and density are p, T, and ρ, respectively. Behind the sound wave (region 2), the gas properties are slightly different and are given by $p + dp$, $T + dT$, and $\rho + d\rho$, respectively. Now imagine that you hop on the wave and ride with it. When you look upstream, into region 1, you see the gas moving toward you with a relative velocity a, as sketched in Figure 8.4*b*. When you look downstream, into region 2, you see the gas receding away from you with a relative velocity $a + da$, as also shown in Figure 8.4. (We have enough fluid-dynamic intuition by now to realize that because the pressure changes across the wave by the amount dp, then the relative flow velocity must also change across the wave by some amount da. Hence, the relative flow velocity behind the wave is $a + da$.) Consequently, in Figure 8.4*b*, we have a picture of a stationary sound wave, with the flow ahead of it moving left to right with velocity a. The pictures in Figure 8.4*a* and *b* are analogous; only the perspective is different. For purposes of analysis, we use Figure 8.4*b*.

(*Note:* Figure 8.4*b* is similar to the picture of a normal shock wave shown in Figure 8.3. In Figure 8.3, the normal shock wave is stationary, and the upstream flow is moving left to right at a velocity u_1. If the upstream flow were to be suddenly shut off, then the normal shock wave in Figure 8.3 would suddenly propagate to the left with a wave velocity of u_1, similar to the moving sound wave shown in Figure 8.4*a*. The analysis of moving waves is slightly more subtle than the analysis of stationary waves; hence, it is simpler to begin a study of shock waves and sound waves with the pictures of stationary waves as shown

in Figures 8.3 and 8.4*b*. Also, please note that the sound wave in Figure 8.4*b* is nothing more than an infinitely weak normal shock wave.)

Examine closely the flow through the sound wave sketched in Figure 8.4*b*. The flow is one-dimensional. Moreover, it is adiabatic, because we have no source of heat transfer into or out of the wave (e.g., we are not "zapping" the wave with a laser beam or heating it with a torch). Finally, the gradients within the wave are very small—the changes dp, dT, $d\rho$, and da are infinitesimal. Therefore, the influence of dissipative phenomena (viscosity and thermal conduction) is negligible. As a result, the flow through the sound wave is both adiabatic and reversible—the flow is *isentropic*. Since we have now established that the flow is one-dimensional and isentropic, let us apply the appropriate governing equations to the picture shown in Figure 8.4*b*.

Applying the continuity equation, Equation (8.2), to Figure 8.4*b*, we have

$$\rho a = (\rho + d\rho)(a + da)$$

or

$$\rho a = \rho a + a\,d\rho + \rho\,da + d\rho\,da \tag{8.11}$$

The product of two differentials, $d\rho\,da$, can be neglected in comparison with the other terms in Equation (8.11). Hence, solving Equation (8.11) for a, we obtain

$$a = -\rho\frac{da}{d\rho} \tag{8.12}$$

Now consider the one-dimensional momentum equation, Equation (8.6), applied to Figure 8.4*b*:

$$p + \rho a^2 = (p + dp) + (\rho + d\rho)(a + da)^2 \tag{8.13}$$

Again ignoring products of differentials, Equation (8.13) becomes

$$dp = -2a\rho\,da - a^2\,d\rho \tag{8.14}$$

Solving Equation (8.14) for da, we have

$$da = \frac{dp + a^2\,d\rho}{-2a\rho} \tag{8.15}$$

Substituting Equation (8.15) into (8.12), we obtain

$$a = -\rho\frac{dp/d\rho + a^2}{-2a\rho} \tag{8.16}$$

Solving Equation (8.16) for a^2, we have

$$a^2 = \frac{dp}{d\rho} \tag{8.17}$$

As discussed above, the flow through a sound wave is isentropic; hence, in Equation (8.17), the rate of change of pressure with respect to density, $dp/d\rho$, is an

isentropic change. Hence, we can rewrite Equation (8.17) as

$$\boxed{a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s}} \tag{8.18}$$

Equation (8.18) is a fundamental expression for the speed of sound in a gas.

Assume that the gas is calorically perfect. For such a case, the isentropic relation given by Equation (7.32) holds, namely,

$$\frac{p_1}{p_2} = \left(\frac{\rho_1}{\rho_2}\right)^{\gamma} \tag{8.19}$$

From Equation (8.19), we have

$$\frac{p}{\rho^{\gamma}} = \text{const} = c$$

or

$$p = c\rho^{\gamma} \tag{8.20}$$

Differentiating Equation (8.20) with respect to ρ, we obtain

$$\left(\frac{\partial p}{\partial \rho}\right)_s = c\gamma\rho^{\gamma-1} \tag{8.21}$$

Substituting Equation (8.20) for the constant c in Equation (8.21), we have

$$\left(\frac{\partial p}{\partial \rho}\right)_s = \left(\frac{p}{\rho^{\gamma}}\right)\gamma\rho^{\gamma-1} = \frac{\gamma p}{\rho} \tag{8.22}$$

Substituting Equation (8.22) into (8.18), we obtain

$$\boxed{a = \sqrt{\frac{\gamma p}{\rho}}} \tag{8.23}$$

Equation (8.23) is an expression for the speed of sound in a calorically perfect gas. At first glance, Equation (8.23) seems to imply that the speed of sound would depend on both p and ρ. However, pressure and density are related through the perfect gas equation of state,

$$\frac{p}{\rho} = RT \tag{8.24}$$

Hence, substituting Equation (8.24) into (8.23), we have

$$\boxed{a = \sqrt{\gamma RT}} \tag{8.25}$$

which is our final expression for the speed of sound; it clearly states that the *speed of sound in a calorically perfect gas is a function of temperature only.* This is consistent with our earlier discussion of the speed of sound being a molecular phenomenon, and therefore it is related to the average molecular velocity $\sqrt{8RT/\pi}$.

The speed of sound at standard sea level is a useful value to remember; it is

$$a_s = 340.9 \text{ m/s} = 1117 \text{ ft/s}$$

Recall the definition of compressibility given in Section 7.3. In particular, from Equation (7.35) for the isentropic compressibility, repeated below,

$$\tau_s = -\frac{1}{v}\left(\frac{\partial v}{\partial p}\right)_s$$

and recalling that $v = 1/\rho$ (hence, $dv = -d\rho/\rho^2$), we have

$$\tau_s = -\rho\left[-\frac{1}{\rho^2}\left(\frac{\partial \rho}{\partial p}\right)_s\right] = \frac{1}{\rho(\partial p/\partial \rho)_s} \tag{8.26}$$

However, recall from Equation (8.18) that $(\partial p/\partial \rho)_s = a^2$. Hence, Equation (8.26) becomes

$$\tau_s = \frac{1}{\rho a^2}$$

or

$$a = \sqrt{\frac{1}{\rho \tau_s}} \tag{8.27}$$

Equation (8.27) relates the speed of sound to the compressibility of a gas. The lower the compressibility, the higher the speed of sound. Recall that for the limiting case of an incompressible fluid $\tau_s = 0$. Hence, Equation (8.27) states that the speed of sound in a theoretically incompressible fluid is infinite. In turn, for an incompressible flow with finite velocity V, the Mach number, $M = V/a$, is zero. Hence, the incompressible flows treated in Chapters 3 to 6 are theoretically zero-Mach-number flows.

Finally, in regard to additional physical meaning of the Mach number, consider a fluid element moving along a streamline. The kinetic and internal energies per unit mass are $V^2/2$ and e, respectively. Their ratio is [recalling Equations (7.6*a*), (7.10), and (8.25)]

$$\frac{V^2/2}{e} = \frac{V^2/2}{c_v T} = \frac{V^2/2}{RT/(\gamma - 1)} = \frac{(\gamma/2)V^2}{a^2/(\gamma - 1)} = \frac{\gamma(\gamma - 1)}{2}M^2$$

Hence, we see that the square of the Mach number is proportional to the ratio of kinetic energy to internal energy of a gas flow. In other words, the Mach number is a measure of the directed motion of the gas compared with the random thermal motion of the molecules.

EXAMPLE 8.1

Consider an airplane flying at a velocity of 250 m/s. Calculate its Mach number if it is flying at a standard altitude of (*a*) sea level, (*b*) 5 km, (*c*) 10 km.

■ Solution

(a) From Appendix D for the standard atmosphere, at sea level, $T_\infty = 288$ K.

$$a_\infty = \sqrt{\gamma RT} = \sqrt{(1.4)(287)(288)} = 340.2 \text{ m/s}$$

Hence,
$$M_\infty = \frac{V_\infty}{a_\infty} = \frac{250}{340.2} = \boxed{0.735}$$

(b) At 5 km, from Appendix D, $T_\infty = 255.7$.

$$a_\infty = \sqrt{(1.4)(287)(255.7)} = 320.5 \text{ m/s}$$
$$M_\infty = \frac{V_\infty}{a_\infty} = \frac{250}{320.2} = \boxed{0.78}$$

(c) At 10 km, from Appendix D, $T_\infty = 223.3$.

$$a_\infty = \sqrt{(1.4)(287)(223.3)} = 299.5 \text{ m/s}$$
$$M_\infty = \frac{V_\infty}{a_\infty} = \frac{250}{299.5} = \boxed{0.835}$$

Note: (1) The Mach number used here is the freestream Mach number. When we refer to the Mach number of an airplane or any other object in flight, it is the velocity of the object divided by the freestream speed of sound.

(2) The Mach number of the airplane in this example obviously depends on the altitude at which it is flying, because the speed of sound is different at different altitudes. In this example, an airplane velocity of 250 m/s corresponds to a Mach number of 0.735 at sea level, but a higher Mach number of 0.835 at an altitude of 10 km.

EXAMPLE 8.2

Consider the flow properties at the point in the flow described in Example 7.3, where the temperature is 320 K and the velocity is 1000 m/s. Calculate the Mach number at this point.

■ Solution

$$a = \sqrt{\gamma RT} = \sqrt{(1.4)(287)(320)} = 358.6 \text{ m/s}$$
$$M = \frac{V}{a} = \frac{1000}{358.6} = \boxed{2.79}$$

Note: This simple calculation is given here to demonstrate that Mach number is a local property of the flow; it varies from point-to-point throughout the flow field. This is in contrast to the freestream Mach number calculated in Example 8.1. A purpose of these two examples is to illustrate the two uses of Mach number.

EXAMPLE 8.3

Calculate the ratio of kinetic energy to internal energy at a point in an airflow where the Mach number is: (a) $M = 2$, and (b) $M = 20$.

■ Solution

$$(a)\ \frac{V^2/2}{e} = \frac{\gamma(\gamma-1)}{2}M^2 = \frac{(1.4)(0.4)}{2}(2)^2 = \boxed{1.12}$$

$$(b)\ \frac{V^2/2}{e} = \frac{\gamma(\gamma-1)}{2}M^2 = \frac{(1.4)(0.4)}{2}(20)^2 = \boxed{112}$$

Note: Examining these two results, we see that at Mach 2, the kinetic energy and internal energy are about the same, whereas at the large hypersonic Mach number of 20, the kinetic energy is more than a hundred times larger than the internal energy. This is one characteristic of hypersonic flows—high ratios of kinetic to internal energy.

EXAMPLE 8.4

Consider a point in a flow of air where the pressure and density are 0.7 atm and 0.0019 slug/ft^3, respectively. (a) Calculate the corresponding value of the isentropic compressibility. (b) From that value of the isentropic compressibility, calculate the speed of sound at the point in the flow.

■ Solution

(a) The isentropic compressibility, τ_s , is defined by Equation (7.35),

$$\tau_s = -\frac{1}{v}\left(\frac{\partial v}{\partial p}\right)_s \tag{7.35}$$

The relation between p and v for an isentropic process is given by Equation (7.32), which can be written in the form:

$$p = c\,\rho^\gamma = c\left(\frac{1}{v}\right)^\gamma \tag{E8.1}$$

where c is a constant. Solving Equation (E8.1) for v gives

$$v = c_1\,p^{-(1/\gamma)} \tag{E8.2}$$

where $c_1 = c^{(1/\gamma)}$, another constant value. Differentiating Equation (E8.2), we have

$$\left(\frac{\partial v}{\partial p}\right)_s = c_1\left(-\frac{1}{\gamma}\right)p^{-(1/\gamma)-1} \tag{E8.3}$$

From Equation (E8.2)

$$c_1 = vp^{1/\gamma}$$

Substituting this expression for c_1 into Equation (E8.3),

$$\left(\frac{\partial v}{\partial p}\right)_s = -\frac{1}{\gamma}\left(vp^{1/\gamma}\right)p^{-(1/\gamma)-1} = -\frac{v}{\gamma p}$$

Substituting this result into Equation (7.35), we have

$$\tau_s = -\frac{1}{v}\left(-\frac{v}{\gamma p}\right) = \frac{1}{\gamma p} \tag{E8.4}$$

Thus, for air at a pressure of 0.7 atm, Equation (E8.4) yields

$$\tau_s = \frac{1}{(1.4)(0.7)} = \boxed{1.02 \text{ atm}^{-1}}$$

(b) From Equation (8.27)

$$a = \sqrt{\frac{1}{\rho \tau_s}} \tag{8.27}$$

and using consistent units for τ_s gives

$$\tau_s = 1.02 \text{ atm}^{-1} \left(\frac{1\text{atm}}{2116 \text{ lb/ft}^2} \right)$$
$$= 4.82 \times 10^{-4} (\text{lb/ft}^2)^{-1}$$

we have

$$a = \sqrt{\frac{1}{\rho \tau_s}} = \sqrt{\frac{1}{0.0019(4.82 \times 10^{-4})}} = \boxed{1045 \text{ ft/s}}$$

Check:

$$T = \frac{p}{\rho R} = \frac{(0.7)(2116)}{(0.0019)(1716)} = 454.3 \ ^\circ\text{R}$$

From Equation (8.25)

$$a = \sqrt{\gamma RT} = \sqrt{(1.4)(1716)(454.3)} = 1045 \text{ ft/s}$$

The answer checks!

EXAMPLE 8.5

By the seventeenth century it was understood that sound propagates through air at some finite velocity. By the time Isaac Newton published his *Principia* in 1687, artillery tests had already shown that the sea-level speed of sound was approximately 1140 ft/s. Comparing that result with today's knowledge of the standard speed of sound at sea level, namely 1117 ft/s, shows that these early seventeenth century measurements were remarkably accurate. Armed with this experimental result for the speed of sound, Isaac Newton in his *Principia* made the first calculation of the speed of sound (See Reference 62). Here Newton correctly theorized that the speed of sound was related to the "elasticity" of the air, which is the reciprocal of the compressibility, τ. However, he incorrectly assumed that the changes of properties in a sound wave took place isothermally. Using Newton's assumption of isothermal changes through the sound wave, calculate the value obtained by Newton for sea-level speed of sound.

■ Solution

The isothermal compressibility is defined as

$$\tau_T = -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_T \tag{7.34}$$

From the equation of state, $v = RT/p$. Thus

$$\left(\frac{\partial v}{\partial p}\right)_T = -\frac{RT}{p^2}$$

Substituting this result into Equation (7.34), we have

$$\tau_T = -\frac{1}{v}\left(-\frac{RT}{p^2}\right) = \frac{RT}{(pv)p} = \frac{RT}{(RT)_p} = \frac{1}{p} \tag{E8.5}$$

Using this isothermal value for the compressibility in Equation (8.27) rather than the correct isentropic compressibility, we have (incorrectly)

$$a_T = \sqrt{\frac{1}{\rho\tau_T}} \qquad \text{(Newton's } \textit{incorrect} \text{ result)} \tag{E8.6}$$

At standard sea level, $p = 2116$ lb/ft^2 and $\rho = 0.002377$ slug/ft^3. Hence, the calculation of the speed of sound assuming isothermal conditions through the sound wave is, from Equations (E8.5) and (E8.6),

$$a_T = \sqrt{\frac{1}{\rho\tau_T}} = \sqrt{\frac{p}{\rho}} = \sqrt{\frac{2116}{0.002377}} = \boxed{943.5 \text{ ft/s}}$$

Isaac Newton reported in his *Principia* a value of 979 ft/s for the speed of sound, about 4 percent higher than the value calculated in this example. The difference is most likely due to the rather imprecise values for sea-level atmospheric properties known in Newton's time. *Note:* The isentropic compressibility is given by

$$\tau_s = \frac{1}{\gamma p} \tag{E8.4}$$

The isothermal compressibility is given by

$$\tau_T = \frac{1}{p} \tag{E8.5}$$

The two values differ by the factor γ. The speed of sound calculated from the isothermal compressibility will be smaller than that calculated from the isentropic compressibility by the factor $(\gamma)^{-1/2}$, or by 0.845. The isothermal calculation will yield a speed of sound about 15 percent lower than the correct value. It is interesting that Newton's calculated value of the speed of sound, as reported in his *Principia*, was about 15 percent lower than the measured value at that time of 1140 ft/s. Undaunted, Newton tried to explain away the differences as due to the presence of dust particles and water vapor in the atmosphere. Finally, a century later the French mathematician Laplace corrected Newton's error by correctly assuming that a sound wave was adiabatic, not isothermal. Therefore, by Napoleon's time in the 1820s, the process and relationship for the propagation of sound in a gas were fully understood.

8.3.1 Comments

In Examples 8.4 and 8.5 we dealt with the role of compressibility, τ, in the determination of the speed of sound. We found that both τ_T and τ_s are functions

of pressure. When τ_s is used in Equation (8.27) to obtain the speed of sound, density also appears in the equation, making it seem that $a = a(\rho, p)$. Indeed, for both Examples 8.4 and 8.5, we used values of both p and ρ to calculate the speed of sound. But keep in mind that p and ρ in the equation for speed of sound always appear in the form p/ρ [see Equation (8.23), for example], and from the perfect gas equation of state, $p/\rho = RT$. Hence, we emphasize again that the speed of sound in a perfect gas is a function of *temperature only*. If we have a gas, and we double the pressure of this gas while keeping the temperature constant, the speed of sound remains the same. If we halve the density, keeping the temperature constant, the speed of sound remains the same. Only in the case of an equilibrium chemically reacting gas and/or a gas where intermolecular forces are important (a gas that is *not* a perfect gas, as discussed in Section 7.2.1), does the speed of sound become a function of both temperature and pressure (see, for example, References 21 and 55).

EXAMPLE 8.6

(a) Consider a long tube with a length of 300 m. The tube is filled with air at a temperature of 320 K. A sound wave is generated at one end of the tube. How long will it take for the wave to reach the other end?

(b) If the tube is filled with helium at a temperature of 320 K, and a sound wave is generated at one end of the tube, how long will it take the sound wave to reach the other end? For a monatomic gas such as helium, $\gamma = 1.67$. Also, for helium R = 2078.5 J/(Kg · K).

■ Solution

(a) $$a = \sqrt{\gamma RT} = \sqrt{(1.4)(287)(320)} = 358.6 \text{ m/s}$$

Letting $l =$ length of the tube and $t =$ time for the sound wave to traverse the length l,

$$t = \frac{l}{a} = \frac{300}{358.6} = \boxed{0.837 \text{ s}}$$

(b) $$a = \sqrt{\gamma RT} = \sqrt{(1.67)(2078.5)(320)} = 1054 \text{ m/s}$$

$$t = \frac{l}{a} = \frac{300}{1054} = \boxed{0.285 \text{ s}}$$

Note: The speed of sound in helium is much faster than in air at the same temperature for two reasons: (1) γ is larger for helium and, more important, (2) helium has a molecular weight $M = 4$, which is much lighter than that for air with $M = 28$. Because $R = \Re/M$ where $\Re$ is the universal gas constant, the same value for all gases, then R for helium is much larger than for air.

8.4 SPECIAL FORMS OF THE ENERGY EQUATION

In this section, we elaborate upon the energy equation for adiabatic flow, as originally given by Equation (7.44). In Section 7.5, we obtained for a steady,

adiabatic, inviscid flow the result that

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \tag{8.28}$$

where V_1 and V_2 are velocities at any two points along a three-dimensional streamline. For the sake of consistency in our current discussion of one-dimensional flow, let us use u_1 and u_2 in Equation (8.28):

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \tag{8.29}$$

However, keep in mind that all the subsequent results in this section hold in general along a streamline and are by no means limited to just one-dimensional flows.

Specializing Equation (8.29) to a calorically perfect gas, where $h = c_p T$, we have

$$\boxed{c_p T_1 + \frac{u_1^2}{2} = c_p T_2 + \frac{u_2^2}{2}} \tag{8.30}$$

From Equation (7.9), Equation (8.30) becomes

$$\frac{\gamma R T_1}{\gamma - 1} + \frac{u_1^2}{2} = \frac{\gamma R T_2}{\gamma - 1} + \frac{u_2^2}{2} \tag{8.31}$$

Since $a = \sqrt{\gamma RT}$, Equation (8.31) can be written as

$$\boxed{\frac{a_1^2}{\gamma - 1} + \frac{u_1^2}{2} = \frac{a_2^2}{\gamma - 1} + \frac{u_2^2}{2}} \tag{8.32}$$

If we consider point 2 in Equation (8.32) to be a stagnation point, where the stagnation speed of sound is denoted by a_0, then, with $u_2 = 0$, Equation (8.32) yields (dropping the subscript 1)

$$\boxed{\frac{a^2}{\gamma - 1} + \frac{u^2}{2} = \frac{a_0^2}{\gamma - 1}} \tag{8.33}$$

In Equation (8.33), a and u are the speed of sound and flow velocity, respectively, at any given point in the flow, and a_0 is the stagnation (or total) speed of sound *associated* with that same point. Equivalently, if we have any two points along a streamline, Equation (8.33) states that

$$\frac{a_1^2}{\gamma - 1} + \frac{u_1^2}{2} = \frac{a_2^2}{\gamma - 1} + \frac{u_2^2}{2} = \frac{a_0^2}{\gamma - 1} = \text{const} \tag{8.34}$$

Recalling the definition of a^* given at the end of Section 7.5, let point 2 in Equation (8.32) represent sonic flow, where $u = a^*$. Then

$$\frac{a^2}{\gamma - 1} + \frac{u^2}{2} = \frac{a^{*2}}{\gamma - 1} + \frac{a^{*2}}{2}$$

or

$$\boxed{\frac{a^2}{\gamma - 1} + \frac{u^2}{2} = \frac{\gamma + 1}{2(\gamma - 1)} a^{*2}} \tag{8.35}$$

In Equation (8.35), a and u are the speed of sound and flow velocity, respectively, at any given point in the flow, and a^* is a characteristic value *associated* with that same point. Equivalently, if we have any two points along a streamline, Equation (8.35) states that

$$\frac{a_1^2}{\gamma - 1} + \frac{u_1^2}{2} = \frac{a_2^2}{\gamma - 1} + \frac{u_2^2}{2} = \frac{\gamma + 1}{2(\gamma - 1)} a^{*2} = \text{const} \tag{8.36}$$

Comparing the right-hand sides of Equations (8.34) and (8.36), the two properties a_0 and a^* associated with the flow are related by

$$\frac{\gamma + 1}{2(\gamma - 1)} a^{*2} = \frac{a_0^2}{\gamma - 1} = \text{const} \tag{8.37}$$

Clearly, these defined quantities, a_0 and a^*, are both constants along a given streamline in a steady, adiabatic, inviscid flow. If all the streamlines emanate from the same uniform freestream conditions, then a_0 and a^* are constants throughout the entire flow field.

Recall the definition of total temperature T_0, as discussed in Section 7.5. In Equation (8.30), let $u_2 = 0$; hence $T_2 = T_0$. Dropping the subscript 1, we have

$$\boxed{c_p T + \frac{u^2}{2} = c_p T_0} \tag{8.38}$$

Equation (8.38) provides a formula from which the defined total temperature T_0 can be calculated from the given actual conditions of T and u at any given point in a general flow field. Equivalently, if we have any two points along a streamline in a steady, adiabatic, inviscid flow, Equation (8.38) states that

$$c_p T_1 + \frac{u_1^2}{2} = c_p T_2 + \frac{u_2^2}{2} = c_p T_0 = \text{const} \tag{8.39}$$

If all the streamlines emanate from the same uniform freestream, then Equation (8.39) holds throughout the entire flow, not just along a streamline.

For a calorically perfect gas, the ratio of total temperature to static temperature T_0/T is a function of Mach number only, as follows. From Equations (8.38) and (7.9), we have

$$\frac{T_0}{T} = 1 + \frac{u^2}{2c_pT} = 1 + \frac{u^2}{2\gamma RT/(\gamma - 1)} = 1 + \frac{u^2}{2a^2/(\gamma - 1)}$$

$$= 1 + \frac{\gamma - 1}{2}\left(\frac{u}{a}\right)^2$$

Hence,

$$\boxed{\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2}M^2} \tag{8.40}$$

Equation (8.40) is very important; it states that only M (and, of course, the value of γ) dictates the ratio of total temperature to static temperature.

Recall the definition of total pressure p_0 and total density ρ_0, as discussed in Section 7.5. These definitions involve an *isentropic* compression of the flow to zero velocity. From Equation (7.32), we have

$$\frac{p_0}{p} = \left(\frac{\rho_0}{\rho}\right)^\gamma = \left(\frac{T_0}{T}\right)^{\gamma/(\gamma-1)} \tag{8.41}$$

Combining Equations (8.40) and (8.41), we obtain

$$\boxed{\frac{p_0}{p} = \left(1 + \frac{\gamma - 1}{2}M^2\right)^{\gamma/(\gamma-1)}} \tag{8.42}$$

$$\boxed{\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma - 1}{2}M^2\right)^{1/(\gamma-1)}} \tag{8.43}$$

Similar to the case of T_0/T, we see from Equations (8.42) and (8.43) that the total-to-static ratios p_0/p and ρ_0/ρ are determined by M and γ only. Hence, for a given gas (i.e., given γ), the ratios T_0/T, p_0/p, and ρ_0/ρ depend only on Mach number.

Equations (8.40), (8.42), and (8.43) are very important; they should be branded on your mind. They provide formulas from which the defined quantities T_0, p_0, and ρ_0 can be calculated from the actual conditions of M, T, p, and ρ at a given point in a general flow field (assuming a calorically perfect gas). They are so important that values of T_0/T, p_0/p, and ρ_0/ρ obtained from Equations (8.40), (8.42), and (8.43), respectively, are tabulated as functions of M in Appendix A for $\gamma = 1.4$ (which corresponds to air at standard conditions).

Consider a point in a general flow where the velocity is exactly sonic (i.e., where $M = 1$). Denote the static temperature, pressure, and density at this sonic

condition as T^*, p^*, and ρ^*, respectively. Inserting $M = 1$ into Equations (8.40), (8.42), and (8.43), we obtain

$$\boxed{\frac{T^*}{T_0} = \frac{2}{\gamma + 1}} \tag{8.44}$$

$$\boxed{\frac{p^*}{p_0} = \left(\frac{2}{\gamma + 1}\right)^{\gamma/(\gamma-1)}} \tag{8.45}$$

$$\boxed{\frac{\rho^*}{\rho_0} = \left(\frac{2}{\gamma + 1}\right)^{1/(\gamma-1)}} \tag{8.46}$$

For $\gamma = 1.4$, these ratios are

$$\frac{T^*}{T_0} = 0.833 \qquad \frac{p^*}{p_0} = 0.528 \qquad \frac{\rho^*}{\rho_0} = 0.634$$

which are useful numbers to keep in mind for subsequent discussions.

We have one final item of business in this section. In Chapter 1, we defined the Mach number as $M = V/a$ (or, following the one-dimensional notation in this chapter, $M = u/a$). In turn, this allowed us to define several regimes of flow, among them being

$M < 1$ (subsonic flow)
$M = 1$ (sonic flow)
$M > 1$ (supersonic flow)

In the definition of M, a is the local speed of sound, $a = \sqrt{\gamma RT}$. In the theory of supersonic flow, it is sometimes convenient to introduce a "characteristic" Mach number M^* defined as

$$M^* \equiv \frac{u}{a^*}$$

where a^* is the value of the speed of sound at sonic conditions, *not* the actual local value. This is the same a^* introduced at the end of Section 7.5 and used in Equation (8.35). The value of a^* is given by $a^* = \sqrt{\gamma RT^*}$. Let us now obtain a relation between the actual Mach number M and this defined characteristic Mach number M^*. Dividing Equation (8.35) by u^2, we have

$$\frac{(a/u)^2}{\gamma - 1} + \frac{1}{2} = \frac{\gamma + 1}{2(\gamma - 1)}\left(\frac{a^*}{u}\right)^2$$

$$\frac{(1/M)^2}{\gamma - 1} = \frac{\gamma + 1}{2(\gamma - 1)}\left(\frac{1}{M^*}\right)^2 - \frac{1}{2}$$

$$\boxed{M^2 = \frac{2}{(\gamma + 1)/M^{*2} - (\gamma - 1)}} \tag{8.47}$$

Equation (8.47) gives M as a function of M^*. Solving Equation (8.47) for M^{*2}, we have

$$\boxed{M^{*2} = \frac{(\gamma + 1)M^2}{2 + (\gamma - 1)M^2}} \tag{8.48}$$

which gives M^* as a function of M. As can be shown by inserting numbers into Equation (8.48) (try some yourself),

$$\begin{aligned}
&M^* = 1 && \text{if } M = 1 \\
&M^* < 1 && \text{if } M < 1 \\
&M^* > 1 && \text{if } M > 1 \\
&M^* \to \sqrt{\frac{\gamma + 1}{\gamma - 1}} && \text{if } M \to \infty
\end{aligned}$$

Therefore, M^* acts qualitatively in the same fashion as M except that M^* approaches a finite value when the actual Mach number approaches infinity.

In summary, a number of equations have been derived in this section, all of which stem in one fashion or another from the basic energy equation for steady, inviscid, adiabatic flow. Make certain that you understand these equations and become very familiar with them before progressing further. These equations are pivotal in the analysis of shock waves and in the study of compressible flow in general.

EXAMPLE 8.7

Repeat Example 7.3 using the equations from the present section.

■ Solution

In Example 8.2, the local Mach number was calculated to be $M = 2.79$. From Equation (8.40),

$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2}M^2 = 1 + \frac{0.4}{2}(2.79)^2 = 2.557$$

From Example 7.3, $T = 320$ K. Thus,

$$T_0 = 2.557T = (2.557)(320) = \boxed{818 \text{ K}}$$

From Equation (8.42),

$$\frac{p_0}{p} = \left[1 + \frac{\gamma - 1}{2}(M)^2\right]^{\frac{\gamma}{\gamma - 1}} = (2.557)^{\frac{1.4}{0.4}} = 26.7$$

From Example 7.3, $p = 1$ atm. Thus,

$$p_0 = 26.7p = 26.7(1) = \boxed{26.7 \text{ atm}}$$

These answers agree with the results obtained in Example 7.3. The technique used here for calculating T_0 and p_0 from the Mach number is, philosophically, more fundamental

than that used in Example 7.3. As we proceed with our discussions, you will find that Mach number is the major governing parameter for compressible flow.

Note: In this example, we used analytical equations to obtain the answers. For our subsequent examples we will use the tabulations in Appendix A, which are obtained from the analytical equations. These tabulations are a convenience that saves us from working through the equations each time.

EXAMPLE 8.8

Consider a point in an airflow where the local Mach number, static pressure, and static temperature are 3.5, 0.3 atm, and 180 K, respectively. Calculate the local values of p_0, T_0, T^*, a^*, and M^* at this point.

■ Solution

From Appendix A, for $M = 3.5$, $p_0/p = 76.27$ and $T_0/T = 3.45$. Hence,

$$p_0 = \left(\frac{p_0}{p}\right) p = 76.27(0.3 \text{ atm}) = \boxed{22.9 \text{ atm}}$$

$$T_0 = \frac{T_0}{T} T = 3.45(180) = \boxed{621 \text{ K}}$$

For $M = 1$, $T_0/T^* = 1.2$. Hence,

$$T^* = \frac{T_0}{1.2} = \frac{621}{1.2} = \boxed{517.5 \text{ K}}$$

$$a^* = \sqrt{\gamma R T^*} = \sqrt{1.4(287)(517.5)} = \boxed{456 \text{ m/s}}$$

$$a = \sqrt{\gamma R T} = \sqrt{1.4(287)(180)} = 268.9 \text{ m/s}$$

$$V = Ma = 3.5(268.9) = 941 \text{ m/s}$$

$$M^* = \frac{V}{a^*} = \frac{941}{456} = \boxed{2.06}$$

The above result for M^* can also be obtained directly from Equation (8.48):

$$M^{*2} = \frac{(\gamma + 1)M^2}{2 + (\gamma - 1)M^2} = \frac{2.4(3.5)^2}{2 + 0.4(3.5)^2} = 4.26$$

Hence, $M^* = \sqrt{4.26} = 2.06$, as obtained above.

EXAMPLE 8.9

In Example 3.1, we illustrated for an incompressible flow, the calculation of the velocity at a point on an airfoil when we were given the pressure at that point and the freestream velocity and pressure. (It would be useful to review Example 3.1 before going further.) The solution involved the use of Bernoulli's equation. Let us now examine the compressible flow analog of Example 3.1. Consider an airfoil in a freestream where $M_\infty = 0.6$ and $p_\infty = 1$ atm, as

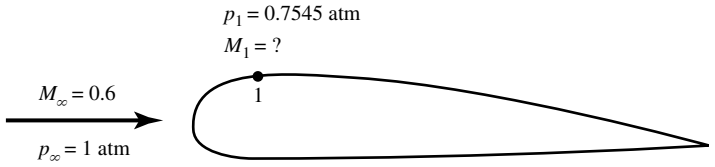


Figure 8.5 Figure for Example 8.2.

sketched in Figure 8.5. At point 1 on the airfoil the pressure is $p_1 = 0.7545$ atm. Calculate the local Mach number at point 1. Assume isentropic flow over the airfoil.

■ Solution

We cannot use Bernoulli's equation because the freestream Mach number is high enough that the flow should be treated as compressible. The free stream total pressure for $M_\infty = 0.6$ is, from Appendix A

$$p_{0,\infty} = \frac{p_{0,\infty}}{p_\infty} p_\infty = (1.276)(1) = 1.276 \text{ atm}$$

Recall that for an isentropic flow, the total pressure is constant throughout the flow. Hence,

$$p_{0,1} = p_{0,\infty} = 1.276 \text{ atm}$$

or

$$\frac{p_{0,1}}{p_1} = \frac{1.276}{0.7545} = 1.691$$

From Appendix A, for a ratio of total to static pressure equal to 1.69, we have

$$\boxed{M_1 = 0.9}$$

This is the local Mach number at point 1 on the airfoil in Figure 8.5.

EXAMPLE 8.10

Note that flow velocity did not enter the calculations in Example 8.9. For compressible flow, Mach number is a more fundamental variable than velocity; we will see this time-and-time again in the subsequent sections and chapters dealing with compressible flow. However, we can certainly calculate velocities for compressible flow problems, but in such cases we usually need to know something about the temperature level of the flow. For the conditions that prevail in Example 8.9, calculate the velocity at point 1 on the airfoil when the free stream temperature is 59°F.

■ Solution

We will need to deal with consistent units. Since 0°F is the same as 460°R,

$$T_\infty = 460 + 59 = 519°\text{R}$$

The flow is isentropic, hence, from Equation (7.32)

$$\frac{p_1}{p_\infty} = \left(\frac{T_1}{T_\infty}\right)^{\gamma/(\gamma-1)}$$

or $$T_1 = T_\infty \left(\frac{p_1}{p_\infty}\right)^{(\gamma-1)/\gamma} = 519 \left(\frac{0.7545}{1}\right)^{(1.4-1)/1.4} = 478.9^\circ\text{R}$$

From Equation (8.25), the speed of sound at point 1 is

$$a_1 = \sqrt{\gamma R T_1} = \sqrt{(1.4)(1716)(478.9)} = 1072.6 \text{ ft/s}$$

Hence, $$V_1 = M_1 a_1 = (0.9)(1072.6) = \boxed{965.4 \text{ ft/s}}$$

8.5 WHEN IS A FLOW COMPRESSIBLE?

As a corollary to Section 8.4, we are now in a position to examine the question, When does a flow have to be considered compressible, that is, when do we have to use analyses based on Chapters 7 to 14 rather than the incompressible techniques discussed in Chapters 3 to 6? There is no specific answer to this question; for subsonic flows, it is a matter of the degree of accuracy desired whether we treat ρ as a constant or as a variable, whereas for supersonic flow the qualitative aspects of the flow are so different that the density *must* be treated as variable. We have stated several times in the preceding chapters the rule of thumb that a flow can be reasonably assumed to be incompressible when $M < 0.3$, whereas it should be considered compressible when $M > 0.3$. There is nothing magic about the value 0.3, but it is a convenient dividing line. We are now in a position to add substance to this rule of thumb.

Consider a fluid element initially at rest, say, an element of the air around you. The density of this gas at rest is ρ_0. Let us now accelerate this fluid element isentropically to some velocity V and Mach number M, say, by expanding the air through a nozzle. As the velocity of the fluid element increases, the other flow properties will change according to the basic governing equations derived in Chapter 7 and in this chapter. In particular, the density ρ of the fluid element will change according to Equation (8.43):

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{1/(\gamma-1)} \tag{8.43}$$

For $\gamma = 1.4$, this variation is illustrated in Figure 8.6, where ρ/ρ_0 is plotted as a function of M from zero to sonic flow. Note that at low subsonic Mach numbers, the variation of ρ/ρ_0 is relatively flat. Indeed, for $M < 0.32$, the value of ρ deviates from ρ_0 by less than 5 percent, and for all practical purposes the flow can be treated as incompressible. However, for $M > 0.32$, the variation in ρ is larger than 5 percent, and its change becomes even more pronounced as M increases. As a result, many aerodynamicists have adopted the rule of thumb that the density variation should be accounted for at Mach numbers above 0.3; that is, the flow should be treated as compressible. Of course, keep in mind that all flows, even at the lowest Mach numbers, are, strictly speaking, compressible. Incompressible flow is really a myth. However, as shown in Figure 8.6, the

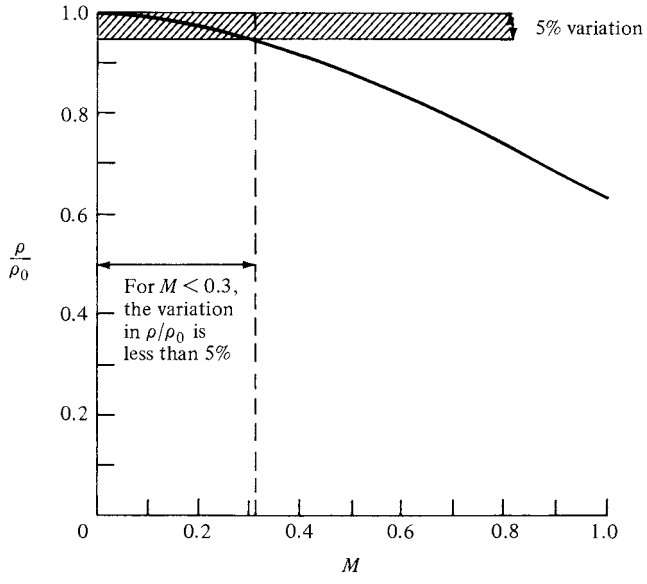


Figure 8.6 Isentropic variation of density with Mach number.

assumption of incompressible flow is very reasonable at low Mach numbers. For this reason, the analyses in Chapters 3 to 6 and the vast bulk of existing literature for incompressible flow are quite practical for many aerodynamic applications.

To obtain additional insight into the significance of Figure 8.6, let us ask how the ratio ρ/ρ_0 affects the change in pressure associated with a given change in velocity. The differential relation between pressure and velocity for a compressible flow is given by Euler's equation, Equation (3.12) repeated below:

$$dp = -\rho V\, dV \tag{3.12}$$

This can be written as

$$\frac{dp}{p} = -\frac{\rho}{p} V^2 \frac{dV}{V}$$

This equation gives the fractional change in pressure for a given fractional change in velocity for a compressible flow with local density ρ. If we now *assume* that the density is constant, say, equal to ρ_0 as denoted in Figure 8.6, then Equation (3.12) yields

$$\left(\frac{dp}{p}\right)_0 = -\frac{\rho_0}{p} V^2 \frac{dV}{V}$$

where the subscript zero implies the assumption of constant density. Dividing the last two equations, and assuming the same dV/V and p, we have

$$\frac{dp/p}{(dp/p)_0} = \frac{\rho}{\rho_0}$$

Hence, the degree by which ρ/ρ_0 deviates from unity as shown in Figure 8.6 is related to the same degree by which the fractional pressure change for a given dV/V is predicted. For example, if $\rho/\rho_0 = 0.95$, which occurs at about $M = 0.3$ in Figure 8.5, then the fractional change in pressure for a compressible flow with local density ρ as compared to that for an incompressible flow with density ρ_0 will be about 5 percent different. Keep in mind that the above comparison is for the local fractional change in pressure, the actual integrated pressure change is less sensitive. For example, consider the flow of air through a nozzle starting in the reservoir at nearly zero velocity and standard sea level values of $p_0 = 2116$ lb/ft^2 and $T_0 = 510^\circ$R, and expanding to a velocity of 350 ft/s at the nozzle exit. The pressure at the nozzle exit will be calculated assuming first incompressible flow and then compressible flow.

Incompressible flow: From Bernoulli's equation,

$$p = p_0 - \tfrac{1}{2}\rho V^2 = 2116 - \tfrac{1}{2}(0.002377)(350)^2 = \boxed{1970 \text{ lb/ft}^2}$$

Compressible flow: From the energy equation, Equation (8.30), with $c_p = 6006$[(ft) (lb)/slug$^\circ$R] for air,

$$T = T_0 - \frac{V^2}{2c_p} = 519 - \frac{(350)^2}{2(6006)} = 508.8^\circ\text{R}$$

From Equation (7.32),

$$\frac{p}{p_0} = \left(\frac{T}{T_0}\right)^{\gamma/(\gamma-1)} = \left(\frac{508.8}{519}\right)^{3.5} = 0.9329$$

$$p = 0.9329 p_0 = 0.9329(2116) = \boxed{1974 \text{ lb/ft}^2}$$

Note that the two results are almost the same, with the compressible value of pressure only 0.2 percent higher than the incompressible value. Clearly, the assumption of incompressible flow (hence, the use of Bernoulli's equation) is certainly justified in this case. Also, note that the Mach number at the exit is 0.317 (work this out for yourself). Hence, we have shown that for a flow wherein the Mach number ranges from zero to about 0.3, Bernoulli's equation yields a reasonably accurate value for the pressure—another justification for the statement that flows wherein $M < 0.3$ are essentially incompressible flows. On the other hand, if this flow were to continue to expand to a velocity of 900 ft/s, a repeat of the above calculation yields the following results for the static pressure at the end of the expansion:

Incompressible (Bernoulli's equation): $p = 1153$ lb/ft^2

Compressible: $p = 1300$ lb/ft^2

Here, the difference between the two sets of results is considerable—a 13 percent difference. In this case, the Mach number at the end of the expansion is 0.86. Clearly, for such values of Mach number, the flow must be treated as compressible.

In summary, although it may be somewhat conservative, this author suggests on the strength of all the above information, including Figure 8.6, that flows wherein the local Mach number exceeds 0.3 should be treated as compressible. Moreover, when $M < 0.3$, the assumption of incompressible flow is quite justified.

8.6 CALCULATION OF NORMAL SHOCK-WAVE PROPERTIES

Consider again the road map given in Figure 8.2. We have finished our three side trips (Sections 8.3 to 8.5) and are now ready to get back on the main road toward the calculation of changes of flow properties across a normal shock wave. Return again to Section 8.2, and recall the basic normal shock equations given by Equations (8.2), (8.6), and (8.10):

Continuity:
$$\rho_1 u_1 = \rho_2 u_2 \tag{8.2}$$

Momentum:
$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2 \tag{8.6}$$

Energy:
$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \tag{8.10}$$

In addition, for a calorically perfect gas, we have

$$h_2 = c_p T_2 \tag{8.49}$$

$$p_2 = \rho_2 R T_2 \tag{8.50}$$

Return again to Figure 8.3, and recall the basic normal shock-wave problem: given the conditions in region 1 ahead of the shock, calculate the conditions in region 2 behind the shock. Examining the five equations given above, we see that they involve five unknowns, namely, ρ_2, u_2, p_2, h_2, and T_2. Hence, Equations (8.2), (8.6), (8.10), (8.49), and (8.50) are sufficient for determining the properties behind a normal shock wave in a calorically perfect gas. Let us proceed.

First, dividing Equation (8.6) by (8.2), we obtain

$$\frac{p_1}{\rho_1 u_1} + u_1 = \frac{p_2}{\rho_2 u_2} + u_2$$

$$\frac{p_1}{\rho_1 u_1} - \frac{p_2}{\rho_2 u_2} = u_2 - u_1 \tag{8.51}$$

Recalling from Equation (8.23) that $a = \sqrt{\gamma p/\rho}$, Equation (8.51) becomes

$$\frac{a_1^2}{\gamma u_1} - \frac{a_2^2}{\gamma u_2} = u_2 - u_1 \tag{8.52}$$

Equation (8.52) is a combination of the continuity and momentum equations. The energy equation, Equation (8.10), can be used in one of its alternate forms, namely, Equation (8.35), rearranged below, and applied first in region 1 and then